Mathematical Physics 1 & 2: physics part

Dejan Gajic

Contents

1 Introduction **3**
1.1 Some useful literature . . . 3

2 Classical mechanics **5**
2.1 Newton's laws of motion . . . 5
2.2 Principles of Relativity . . . 12
2.3 Conservative forces . . . 21
2.4 Calculus of variations . . . 24
2.5 Lagrangian mechanics . . . 27
2.6 Hamiltonian mechanics . . . 29
2.7 Symplectic manifolds . . . 35
2.8 Lagrangian mechanics on manifolds . . . 43

3 Lagrangian field theory **48**
3.1 Preliminaries: metric, connections and volume forms . . . 48
3.2 Lagrangians and field manifolds . . . 57
3.3 Noether's theorem . . . 61
3.4 The Einstein–Hilbert action . . . 67

4 Gauge theories **74**
4.1 Gauge transformations in electromagnetism . . . 74
4.2 Gauge transformations for Lie groups . . . 75
4.3 Yang–Mills theory . . . 79

5 Quantum mechanics **82**
5.1 The kinematic postulates of quantum mechanics . . . 82
5.2 Position and momentum operators . . . 89
5.3 The dynamic postulates of quantum mechanics . . . 91
5.4 The uncertainty principle . . . 96
5.5 The self-adjointness of Schrödinger operators . . . 98
5.6 The free Schrödinger equation . . . 102
5.7 Multiple particles and mixed states . . . 105

A Deriving Poincaré transformations **113**

B Geometry **116**
B.1 Geodesics and normal coordinates . . . 116
B.2 Curvature . . . 120

1 Introduction

These lecture notes cover the "physics" part of the Mathematical Physics 1 & 2 courses at Leipzig University. We will touch on the following topics in this course: *classical mechanics* (Newtonian, Lagrangian and Hamiltonian), *Lagrangian field theory*, *Lorentzian geometry and general relativity* and *quantum mechanics*.

Since these topics all concern objects and concepts that are entirely well-defined mathematically, we can cover them with a similar style and precision as topics in mathematics. In contrast with typical physics texts on these topics, we will therefore put an emphasis on *mathematical rigour*. Roughly, this means that, we will always try to keep track of the *domains* and *codomains* are of all the maps that we consider, and we will make any underlying additional assumptions explicit. The tools and mathematical concepts that are necessary for this kind of approach will be covered in the "mathematics" part of this course.

Throughout the notes, there will be exercises, which are intended to help you interact with the text in an active manner. They are optional and are supplemented by problem sheets.

If you spot any typos or other errors, please let me know!

1.1 Some useful literature

Below you can find a list of literature that complements these lecture notes.

Classical mechanics

- Arnol'd, Vladimir Igorevich. *Mathematical methods of classical mechanics.* Vol. 60. Springer Science & Business Media, 1978.
- Thirring, Walter. *Classical mathematical physics: dynamical systems and field theories.* Springer Science & Business Media, 1997.

Lagrangian field theory

- Thirring, Walter. *Classical mathematical physics: dynamical systems and field theories.* Springer Science & Business Media, 1997.
- Frankel, Theodore. *The geometry of physics: an introduction.* Cambridge University Press, 1997.

Gauge theories

- Frankel, Theodore. *The geometry of physics: an introduction.* Cambridge University Press, 1997.

Lorentzian geometry and general relativity

- Wald, Robert M. General relativity. University of Chicago press, 1984.
- Hawking, Stephen W., and Ellis, George F.R. *The large scale structure of space-time.* Cambridge University Press, 1973.
- Reall, Harvey. *Mathematical Tripos Part III: General relativity.* https://www.damtp.cam.ac.uk/user/hsr1000/part3_gr_lectures.pdf, 2022
- Lee, John M. *Introduction to Riemannian manifolds.* Springer, 1997.

Quantum mechanics

- Hall, Brian C. *Quantum theory for mathematicians.* Springer Publication, 2013.
- Reed, Michael, and Simon, Barry. *Methods of Modern Mathematical Physics, Volume IV: Analysis of Operators.* Academic Press, New York, 1978

2 Classical mechanics

2.1 Newton's laws of motion

We first define the main mathematical objects of interest in classical mechanics.

Definition 2.1. *We will refer to the set* $\mathbb{R}^{3+1} := \mathbb{R}^4$ *as* spacetime. *Equipping* $\mathbb{R}^{3+1}$ *with Cartesian coordinates labelled* (t, x^1, x^2, x^3)*, we can decompose* $\mathbb{R}^{3+1}$*:*

$$\mathbb{R}^{3+1} = \mathbb{R}_t \times \mathbb{R}^3_{\mathbf{x}},$$

with $t \in \mathbb{R}$ *interpreted as a* time coordinate *and* $\mathbf{x} := (x^1, x^2, x^3) \in \mathbb{R}^3$ *interpreted as* spatial coordinates*. At the same time, we will use the notation* t *and* x^i *to denote the* maps $t, x^i : \mathbb{R}^{3+1} \to \mathbb{R}$ *satisfying:* $t(s^0, s^1, s^2, s^3) = s^0$ *and* $x^i(s^0, s^1, s^2, s^3) = s^i$ *for all* $(s^0, s^1, s^2, s^3) \in \mathbb{R}^{3+1}$*.*

We can describe (subsets of) $\mathbb{R}^{3+1}$ *using different coordinates. Let* $U, V \subseteq \mathbb{R}^{3+1}$ *be open, then a map*

$$\Phi := (t', \mathbf{x}') := (t', (x')^1, (x')^2, (x')^3) : U \to V.$$

is a coordinate chart *on* U *if it is a smooth diffeomorphism. We will also denote* $\underline{\Phi} = ((x')^1, (x')^2, (x')^3)$*.*

The above all generalizes when $\mathbb{R}^3$ *is replaced by* $\mathbb{R}^n$ *for general* $n \in \mathbb{N}_1$*.*

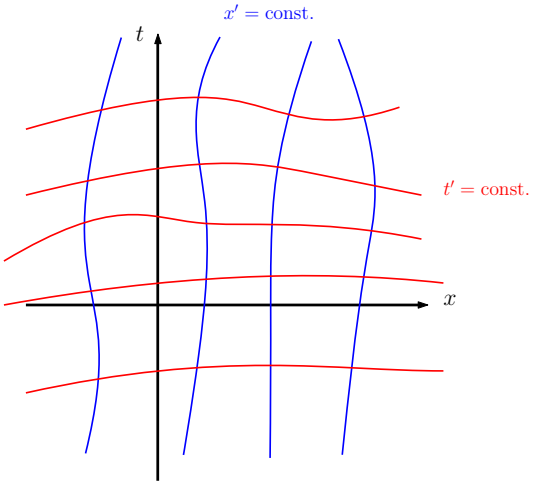


Figure 1: Illustration of the lines of constant t' and constant x' in $\mathbb{R}^{1+1}$.

The notation $\mathbb{R}^{3+1}$, rather than $\mathbb{R}^4$, might seem a bit redundant. We can use it when we want to emphasize not only that there is one time direction and there are three spatial directions in our $\mathbb{R}^4$, but also that we have not committed to describing the set $\mathbb{R}^4$ using the standard Cartesian coordinates (t, x^1, x^2, x^3) (corresponding to the coordinate chart $\Phi = \mathrm{id}$, the identity map), but we allow for the consideration of a non-trivial choice of coordinates (t', y^1, y^2, y^3). It may for example be convenient to describe points in (a subset of) $\mathbb{R}^{3+1}$ using spherical coordinates (t, r, θ, φ). We might also want to consider a different choice of global, Cartesian coordinates, corresponding to a rotation of points in $\mathbb{R}^3$: $\Phi = \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix}$, with $R \in SO(3)$.

Much of this course (and the study of equations of physics in general) revolves around understanding how the description of physical systems changes under a change of coordinates! Later, we will work with mathematical objects called *manifolds* $\mathcal{M}$ in order to describe points in spacetime when we want to be agnostic about our choice of coordinates, or in cases where we are not even *able* to describe all points using a single choice of coordinates.

Definition 2.2. *The* motion of a system of $N \in \mathbb{N}$ point particles *is the map:*

$$\gamma : \mathbb{R}_t \supseteq I \to \mathbb{R}^{3N}_{\mathbf{x}},$$
$$\gamma(t) = \begin{pmatrix} \mathbf{x}_1(t) \\ \mathbf{x}_2(t) \\ \vdots \\ \mathbf{x}_N(t)) \end{pmatrix},$$
$$\mathbf{x}_i(t) \in \mathbb{R}^3 \quad \forall 1 \leq i \leq N.$$

We will consider also the corresponding spacetime motion $\gamma = (\mathrm{id}, \underline{\gamma}) : I \to I \times \mathbb{R}^{3N}$, *with* $\mathrm{id} : I \to I$ *denoting the identity function, so that* $(\mathrm{id}, \underline{\gamma})(t) = (t, \underline{\gamma}(t))$.

A point particle is used to model a body of very small size compared to the distance scale under consideration and the distance between different bodies, whose only information we keep track of is the position of its centre of mass in space as a function of time.

When there is no ambiguity regarding which time coordinate to consider, we will often make use of the following standard notation:

$$\dot{(\cdot)} = \frac{d(\cdot)}{dt},$$
$$\ddot{(\cdot)} = \frac{d^2(\cdot)}{dt^2}.$$

Given a vector $(t, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n$, we will also denote $x^0 = t$ and $x^i = \mathbf{x}^i$ for $1 \leq i \leq n$ and $x = (t, \mathbf{x})$. We will also denote with $e_\mu \in \mathbb{R}^{n+1}$ the vector with all zeroes apart from one 1 in the μ-th slot. The set $\{e_\mu\}_{0 \leq \mu \leq n}$ forms a basis of $\mathbb{R}^{n+1}$.

Consider a coordinate chart $\Phi = (t', \underline{\Phi}) : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ and the spacetime motion of a particle $\gamma : I \to \mathbb{R}^{n+1}$. Let $\tau : I \to \mathbb{R}$ be defined as $\tau(t) = t'(\gamma(t))$. If $\frac{d\tau}{dt}(t) \neq 0$ for all $t \in I$, then the inverse $\tau^{-1} : \tau(\mathbb{R}) \to I$ is well-defined and we can treat $\gamma \circ \tau^{-1}$ as a *reparametrization* of γ. The path

$$t' \mapsto (\Phi \circ \gamma \circ \tau^{-1})(t') = \Big(t', \underline{\Phi}\big(\gamma(\tau^{-1}(t'))\big)\Big)$$

then describes the spacetime motion of the particle with respect to the coordinate chart Φ.

The postulates of Newton's theory of mechanics can be structured as a set of laws governing the motion of systems of N point particles interacting with each other and possibly also with an external environment, known as *Newton's laws of motion* (Newton 1687).

Newton's First Law There exists a coordinate chart $\Phi = (t', \underline{\Phi})$ on $\mathbb{R}^{3+1}$ with respect to which the spacetime motion $\gamma : \mathbb{R} \supseteq I \to \mathbb{R}^{3+1}$ of any particle interacting with nothing (a free particle) satisfies:

$$\frac{d^2}{dt'^2}(\Phi \circ \gamma \circ \tau^{-1})(t') = 0. \tag{2.1}$$

In practice, it would be impossible to arrange a body to experience no interaction with anything else. For example, there will always be a non-trivial gravitational attraction between the many different bodies in the universe.

What Newton's First Law is really saying is that there exists a *preferred* choice of coordinate chart $\Phi : \mathbb{R}^{3+1} \to \mathbb{R}^4$ that has the property that if we were to remove everything apart from a

single particle, that particle should move in straight line with constant speed when measured with respect to the time coordinate of Φ. For convenience, we may assume that the identity map $\Phi = \mathrm{id}$ is such a preferred coordinate chart.

In practice and in the context of classical mechanics, we typically declare coordinates with respect to which either a point on the surface of the earth, the centre of mass of the earth, the sun or the stars are stationary (depending on the scale of the system under consideration) as our preferred coordinate chart.

Coordinate charts compatible with Newton's First Law (i.e. satisfying (2.1)) are certainly not *unique*.[1] In the following lemma, we show that the *only* coordinate charts compatible with Newton's first law are the affine maps (this means: $\Phi - \Phi(0)$ is a linear map) for which moreover t' only depends on t and not on $\mathbf{x}$.

Proposition 2.1. *Let $\Phi = (t', \underline{\Phi}) : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a coordinate chart. Consider all paths $\gamma : \mathbb{R} \to \mathbb{R}^{n+1}$ that satisfy $\frac{d^2\gamma}{dt^2}(t) = 0$ and assume that $\tau(t) = t'(\gamma(t))$ satisfies $\frac{d\tau}{dt}(t) \neq 0$ for all $t \in \mathbb{R}$.*[2]

If also $\frac{d^2(\Phi\circ\gamma\circ\tau^{-1})}{dt'^2}(t') = 0$, then:

(i) *Φ is affine,*

(ii) *$t' : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ is independent of $\mathbf{x} \in \mathbb{R}^n$.*

Proof. Consider the path $\gamma(t) = (t^0 + v^0 t, \mathbf{x}_0 + t\mathbf{v})$ with $v^0 \neq 0$ and $\mathbf{x}_0, \mathbf{v} \in \mathbb{R}^n$. Denote $x = (t^0, \mathbf{x})$ and $v = (v^0, \mathbf{v})$, so that $\gamma(t) = x + tv$.

"(ii)": Note that by the chain rule for differentiation:

$$a(t) := \frac{d\tau}{dt}(t) = \sum_{\mu=0}^{n} (\partial_\mu t')(\gamma(t)) v^\mu.$$

By assumption, a must a non-zero function. for all v. Let $\mathbf{v} = 0$. Then we must have that $(\partial_t t')(\gamma(t)) \neq 0$ for all $t \in \mathbb{R}$.

Now suppose that $(\partial_i t')(\gamma(t)) \neq 0$ for some $i \in \{1, \dots, n\}$ and $t \in \mathbb{R}$. Then we can choose $\mathbf{v}$ so that $a(t) = 0$, which contradicts our assumption on a. Since the above is true for all x and v, this means that t' can only depend on t, so we can write $a(t) = \frac{dt'}{dt}$.

"(i)": We can express:

$$\Phi(\gamma(t)) = (t'(\gamma(t)), \underline{\Phi}(v^0 t, \mathbf{x}_0 + t\mathbf{v})),$$

so by the chain rule, we obtain:

$$\frac{d(\Phi \circ \gamma)}{dt}(t) = \Big(a(t), \sum_\mu v^\mu \partial_\mu \underline{\Phi}(\gamma(t))\Big).$$

Then, considering the parameter $\tau : \mathbb{R} \to \mathbb{R}$ and its inverse $\tau^{-1} : \mathbb{R} \to \mathbb{R}$, we can apply the chain rule again and use that $\frac{d\tau^{-1}}{dt'} = \frac{1}{a(\tau^{-1}(t'))}$ to obtain:

$$\frac{d(\Phi \circ \gamma \circ \tau^{-1})}{dt'}(t') = \Big(1, \frac{1}{a(\tau^{-1}(t'))} \sum_\mu v^\mu \partial_\mu \underline{\Phi}(\gamma(\tau^{-1}(t')))\Big).$$

[1] Exercise: Let $\Phi = (t', \underline{\Phi})$ be an affine map such that $t'(t) = at + b$ for some $a, b \in \mathbb{R}$ with $a \neq 0$. Show that Φ is compatible with Newton's first law. Show that we cannot drop the assumption $t'(t) = at + b$. Hint: Consider $\gamma(t) = (t, \mathbf{x}_0 + t\mathbf{v}) \in \mathbb{R}^{n+1}$ and suppose that $t'(t, \mathbf{x}) = t + \langle \mathbf{w}, \mathbf{x} \rangle$ for some $\mathbf{w} \in \mathbb{R}^3 \setminus \{0\}$. What happens to $\frac{d\tau}{dt}(t)$ as we vary $\mathbf{v}$?

[2] Suppose that $\gamma(t) = (t^0 + v^0 t, \mathbf{x}_0 + t\mathbf{v})$. By performing an affine transformation $t \mapsto t^0 + v^0 t =: \tilde{t}$ and a rescaling of $\mathbf{v}$ and shifting of $\mathbf{x}_0$, this path can be put in the form $\tilde{\gamma}(\tilde{t}) = (\tilde{t}, \tilde{\mathbf{x}}_0 + \tilde{t}\tilde{\mathbf{v}})$, which can be interpreted as the spacetime motion of a particle.

After yet another application of the chain rule:

$$\frac{d^2(\Phi\circ\gamma\circ\tau^{-1})}{dt'^2}(t') = \frac{1}{a^2(\tau^{-1}(t'))}\Big(0, \sum_{\mu,\nu} v^\mu v^\nu \partial_\nu\partial_\mu \underline{\Phi}(\gamma(\tau^{-1}(t'))) - \frac{\dot{a}}{a}(\tau^{-1}(t')\sum_\mu v^\mu \partial_\mu \underline{\Phi}(\gamma(\tau^{-1}(t'))\Big).$$

Now we demand that the left-hand side vanishes for all choices of v and x with $v^0 \neq 0$. This implies that for all x, v with $v^0 \neq 0$ at $t = 0$:

$$\sum_{\mu,\nu} v^\mu v^\nu \left[\partial_\nu\partial_\mu \underline{\Phi}\right](x) - \sum_\mu v^\mu \left[\frac{\dot{a}}{a}(0)\partial_\mu \underline{\Phi}\right](x) = 0$$

Note that the above expression is a second-order polynomial in $v^0, v^1, \ldots, v^n$. This means that the terms inside the square brackets must each vanish independently for all μ, ν. Indeed, just take two derivatives in the components v^μ to conclude that everything inside the first square bracket must vanish, which then implies that also the terms inside the second square bracket must vanish. Therefore, $(D^2\underline{\Phi})(x) = 0$, so $\underline{\Phi}$ must be affine.

Suppose that $\dot{a}(0) \neq 0$. Then $\partial_\mu \underline{\Phi}(x) = 0$ for all $0 \leq \mu \leq 3$, but then $(D\Phi)(x)$ cannot be invertible at x, which is a contradiction with Φ being a diffeomorphism. We must therefore have that $\dot{a}(0) = 0$. In fact, since the terms inside the square brackets must vanish also at $\gamma(t)$ for all t, we conclude by the above that $\dot{a}(t) = 0$ for all t and hence Φ must be affine. □

By the above proposition, the restriction to coordinate charts compatible with Newton's First Law implies (in particular) the restriction to time functions $t' = a + bt$. In other words, up to a shift (for example, a change of time zone) and a rescaling by a constant (change of units, for example hours instead of seconds), all observers associated to coordinate charts which are compatible with Newton's First Law agree on the time! We did not *assume* this, but it is a consequence of demanding the First Law to hold.

We consider the following subset of affine coordinate charts with $t' = at + b$:

Definition 2.3. Galilean coordinate charts, *or* Galilean (coordinate) transformations *are affine maps* $g = (t', \underline{\Phi}) : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ *that preserve time intervals in* $\mathbb{R}_t$ *and norms in* $\mathbb{R}^n_{\mathbf{x}}$*, i.e. for all* $(t_1, \mathbf{x}_1), (t_2, \mathbf{x}_2) \in \mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$*:*

$$|t'(t_1, \mathbf{x}_1) - t'(t_2, \mathbf{x}_2)| = |t_1 - t_2|,$$
$$|\underline{\Phi}(t_1, \mathbf{x}_1) - \underline{\Phi}(t_1, \mathbf{x}_2)| = |\mathbf{x}_1 - \mathbf{x}_2|.$$

We denote with Gal(n) *the set of all Galilean transformations on* $\mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$.

We now introduce the concept of a reference frame in $\mathbb{R}^{n+1}$.

Definition 2.4.

(i) *Let* $\gamma : \mathbb{R}_t \to \mathbb{R}^{n+1}$*. Let* $e'_\mu : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ *be smooth, such that* $\{e_\mu(x)\}$ *form a linearly independent set at each* $x \in \mathbb{R}^{n+1}$*, with* $e'_0 \circ \gamma = \dot{\gamma}(t)$*. Then* $\{e'_\mu \circ \gamma\}$ *is called a* reference frame along γ.

(i) *Let* $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ *be a diffeomorphism and consider* $D\Phi^{-1} : \mathbb{R}^{n+1} \to \mathcal{L}(\mathbb{R}^{n+1}, \mathbb{R}^{n+1})$*. Define* $X_\mu : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$*, with* $X_\mu(x) := (D\Phi^{-1})(\Phi(x))(e_\mu)$*. Consider* $(\gamma \circ \tau^{-1})(t') := \Phi^{-1}(t', 0)$*, assuming that* $\dot{\tau}(t) = \frac{d}{dt}(t' \circ \gamma) \neq 0$*. Let* e'_0 *be a smooth extension of* $\dot{\gamma} : \gamma(\mathbb{R}) \to \mathbb{R}^{n+1}$ *and let* $e'_i = X_i$.[3] *Then* $\{e'_\mu \circ \gamma\}$ *is a reference frame along* γ*, which we will call the* reference frame associated to Φ.[4]

(i) *The reference frames associated to Galilean transformations are called* Galilean inertial reference frames.

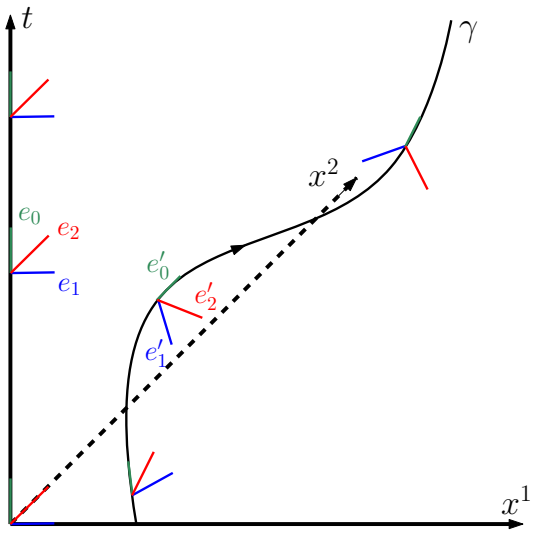


Figure 2: An illustration of a reference frame along a curve $\gamma : \mathbb{R} \to \mathbb{R}^{n+1}$ in the case $n = 2$.

In the case $n = 3$, we can think of γ in the definition above as representing the spacetime path of an *observer* who assigns coordinates to spacetime using the coordinate chart Φ, with respect to which they are standing still at the origin $0 \in \mathbb{R}^3$. We can think of e'_i, $i \in \{1, 2, 3\}$ as measuring rods attached to the observer, which are allowed to change in time. The value of t' can be thought of as the time indicated on a clock carried by the observer, which is allowed to be very unreliable and run faster or slower, depending on the position of the observer.

In the special case of Galilean coordinate charts, $\frac{d^2\gamma}{dt^2}(t) = 0$, so the observer moves at constant velocity with respect to the standard coordinates on $\mathbb{R}^{n+1}$ and, as we will see, $\{e'_i\}$ form an orthonormal frame with respect to the standard, Euclidean metric on $\mathbb{R}^n_{\mathbf{x}}$, which is moreover non-rotating.

Note that a rescaling of the form $(t, x^1, \dots, x^n) \mapsto (\lambda_0 t, \lambda_1 x^1, \dots, \lambda_n x^n)$ with $\lambda_\mu \in \mathbb{R}$ simply rescales each e'_μ by a constant and can be interpreted as a change of units along the measuring rods/clock of the observer.

Later (Proposition 2.5), we will see that Galilean inertial reference frames encompass, up to a constant rescaling (or change of units), *all* reference frames of coordinate charts compatible with Newton's First Law.

Definition 2.5. *Given* masses $m_i \in \mathbb{R}_+$ *and suitably regular functions* $\mathbf{F}_i : \mathbb{R}_t \times \mathbb{R}^{3N}_{\mathbf{x}} \times \mathbb{R}^{3N}_{\mathbf{x}} \to \mathbb{R}^3_{\mathbf{x}}$, *with* $i = 1, \dots, N$, *the* Newton equations *comprise the following system of ODE (ordinary differential equations):*

$$m_i \ddot{\mathbf{x}}_i(t) = \mathbf{F}_i(t, \gamma(t), \dot{\gamma}(t)) \quad 1 \leq i \leq N. \tag{2.2}$$

We interpret $\mathbf{x}_i(t)$ as the position vector of the i-th particle at time t and the function $\mathbf{F}_i$ as a *force* acting on the i-th particle, which it is is fixed by the choice of physical model. We can now state Newton's Second Law ("force equals mass times acceleration"):

[3]Note that $e'_0(t) = \frac{d\gamma}{dt}(t) = \frac{d\tau}{dt}(t)(X_0 \circ \gamma)(t)$.

[4]Exercise: Determine the reference frame associated to the spherical coordinate chart $\Phi = (t, r, \theta, \varphi)$.

Newton's Second Law The acceleration of the i-th particle in a system of N particles with masses m_i experiencing a force $\mathbf{F}_i$ is given by (2.2),
with respect to a choice of coordinate chart compatible with Newton's First Law.

Fixing the physical model under consideration amounts to fixing the functions $\mathbf{F}_i$. In general, the functions $\mathbf{F}_i$ will depend on the choice of preferred coordinate chart in Newton's First Law. For certain forces, (2.2) will remain valid with the same map $\mathbf{F}_i$ when we consider *any* Galilean coordinate chart. We say that such forces satisfy the "Principle of Galilean Relativity". In such physical models, there is no *preferred* Galilean coordinate chart. Clearly, the choice $\mathbf{F}_i = 0$ for all i is an example of such a physical model (describing a collection of free particles), by the fact that all Galilean coordinate charts are compatible with Newton's First Law. We will discuss this more in the next section.

If $\mathbf{F}_i$ does not satisfy the Principle of Galilean Relativity, or if we consider coordinate charts that are not Galilean (for example, associated to accelerated or rotating observers), we have to modify the $\mathbf{F}_i$ appearing on the right-hand side of (2.2).

Newton's *Principle of Determinacy* is the statement that the motion of a system of N particles is uniquely determined by the specification of their initial positions and velocities. Continuous dependence on initial data roughly says that small changes in initial positions and velocity should only lead to small changes in position and velocity at finite, later times. From the perspective of the ODE (2.2), this becomes a theorem:

Theorem 2.2 (Well-posedness of the Newton equations)**.**

(i) *Let $I \subseteq \mathbb{R}$ be an open interval, with $0 \in I$. Let $U \subseteq \mathbb{R}^{6N}$ be open. Assume that $\mathbf{F}_i : I \times U \to \mathbb{R}^3$ are continuous in the first variable for all $1 \leq i \leq N$ and that there exists a constant $C > 0$, such that for all $t \in I$, $x, y \in \mathbb{R}^{6N}$ and $1 \leq i \leq N$:*[5]

$$||\mathbf{F}_i(t, x) - \mathbf{F}_i(t, y)|| \leq C||x - y||. \tag{2.3}$$

Given initial data $(\gamma_0, \gamma_1) \in U$, *there exists a $T = T(\gamma_0, \gamma_1) > 0$ and a unique C^1 solution $\gamma : I \supset (-T, T) \to \mathbb{R}^{3N}$ to (2.2) satisfying*

$$(\gamma, \dot{\gamma})(0) = (\gamma_0, \gamma_1).$$

(ii) *If we can take $I = \mathbb{R}$ and $U = \mathbb{R}^{6N}$ in the assumptions in* (i),[6] *then we can take $T = \infty$ and γ is well-defined on $\mathbb{R}$, i.e. the solution is global.*

(iii) *For all $\epsilon > 0$, there exists a $\delta > 0$, such that if two data sets (γ_0, γ_1) and $(\tilde{\gamma}_0, \tilde{\gamma}_1)$ satisfy*

$$|(\gamma_0 - \tilde{\gamma}_0, \gamma_1 - \tilde{\gamma}_1)| \leq \delta,$$

then for all $t \in (-\tilde{T}, \tilde{T})$ with $\tilde{T} < T$:

$$|(\gamma - \tilde{\gamma}, \dot{\gamma} - \dot{\tilde{\gamma}})(t)| \leq \epsilon,$$

Proof. Existence and uniqueness follows directly from the Picard–Lindelöf Theorem for local existence and uniqueness of ODE, after rewriting the system as a first order system with $\mathbf{y}_i := \dot{\mathbf{x}}_i$, so $(\dot{\mathbf{x}}_i, \dot{\mathbf{y}}_i) = (\mathbf{x}_i, \mathbf{F}_i(\mathbf{x}_i, \mathbf{y}_i))$. Continuous dependence of data in (iii) follows from a Grönwall inequality. □

[5] When there exist I and U such that (2.3) holds, we say that $\mathbf{F}_i$ is *locally Lipschitz continuous in the last $6N$ variables*, with a Lipschitz constant that is independent of t.

[6] This means that F_i is globally Lipschitz continuous (with a Lipschitz constant that is independent of t).

Observe that for general $\mathbf{F}_i$ satisfying the assumptions of Theorem 2.2(i), *global* existence and uniqueness are not guaranteed, nor is it possible to write down an expression for γ in closed form. We will, however, encounter important examples for $\mathbf{F}_i$ where we *are* able to make sense of solutions globally (and sometimes even in closed form).

Example 2.1 (Failure of global existence)**.** *Consider a 1-particle system on $\mathbb{R}$ with mass $m = 1$ and $F : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ with $F(x, v) = 2xv$ and $x(0) = x_0$, $\dot{x}(0) = \dot{x}_0$. Then (2.2) gives:*

$$\ddot{x} = 2x\dot{x},$$

which is equivalent to

$$\dot{x} = A + x^2,$$

with $A = \dot{x}_0 - x_0^2$. Consider initial data such that $A = 0$.

When $\dot{x}_0 \neq 0$, this is equivalent to the integral equation

$$\int_{x_0}^{x(t)} y^{-2} dy = \int_0^t dt.$$

which has solutions $x(t) = \frac{1}{x_0^{-1} - t}$ that are only well-defined for $t < x_0^{-1}$ or $t > x_0^{-1}$. Assuming additionally that the initial data satisfy $x_0 > 0$ therefore implies that the solutions will only exist up to $t = x_0^{-1}$.

EXERCISE: *What can you say when the initial data satisfy $\dot{x}_0 = 0$?*

If there do not exist I and U such that (2.3) holds, then we can even have failure of local (in time) uniqueness of solutions.[7]

Example 2.2 (Failure of local uniqueness)**.** *Consider a 1-particle system on $\mathbb{R}$ with mass $m = 1$ and $F : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ with $F(x, v) = v^{\frac{1}{3}}$ and $x(0) = x_0 \in \mathbb{R}$, $\dot{x}(0) = 0$. Then (2.2) gives:*

$$\ddot{x} = \dot{x}^{\frac{1}{3}}.$$

The following functions are solutions all to (2.2) arising from the initial data $(x(0), \dot{x}(0)) = (x_0, 0)$: $x(t) = x_0$, $x(t) = x_0 \pm \frac{2}{5} \left(\frac{2}{3}\right)^{\frac{3}{2}} t^{\frac{5}{2}}$.

Newton's Third Law may be thought of as the requirement of additional restrictions on $\mathbf{F}_i$, for example. We will not impose the Third Law by default, but only when specifically indicated.

Newton's Third Law The forces $\mathbf{F}_i$ in (2.2) have the following structure: for all $1 \leq i, j \leq N$, with $i \neq j$, there exists functions $f_{ij} : \mathbb{R} \times \mathbb{R}^{12} \to \mathbb{R}$ satisfying $f_{ji}(t, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}) = f_{ij}(t, \mathbf{b}, \mathbf{a}, \mathbf{d}, \mathbf{c})$ for any $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathbb{R}^3$ and $\mathbf{F}_{\text{ext},i} : \mathbb{R} \times \mathbb{R}^3 \to \mathbb{R}^3$, such that $\mathbf{F}_i$ can be split up as follows:

$$\mathbf{F}_i(t, \mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N) = \sum_{1 \leq j \leq N, j \neq i} f_{ij}(t, \mathbf{x}_i, \mathbf{x}_j, \mathbf{v}_i, \mathbf{v}_j)(\mathbf{x}_j - \mathbf{x}_i) + \mathbf{F}_{\text{ext},i}(t, \mathbf{x}_i, \mathbf{v}_i).$$

The parts $\mathbf{F}_{ij}(t, \mathbf{x}_i, \mathbf{x}_j, \mathbf{v}_i, \mathbf{v}_j) := f_{ij}(t, \mathbf{x}_i, \mathbf{x}_j, \mathbf{v}_i, \mathbf{v}_j)(\mathbf{x}_j - \mathbf{x}_i)$ are called "forces of interaction" and $\mathbf{F}_{\text{ext},i}$ is called an "external force". Note that $\mathbf{F}_{ji}(t, \mathbf{x}_j, \mathbf{x}_i, \mathbf{v}_j, \mathbf{v}_i) = -\mathbf{F}_{ij}(t, \mathbf{x}_i, \mathbf{x}_j, \mathbf{v}_i, \mathbf{v}_j)$, so two particles carry out forces on each other of the same size, but with opposite direction.

[7] One can show that local existence of solutions holds with the weaker assumption of local continuity, like in the example $F(x, v) = v^{\frac{1}{3}}$.

2.2 Principles of Relativity

We return to the Galilean coordinates or Galilean transformations from the previous section. Recall first the definition of a group:

Definition 2.6. *A* group *is a set G equipped with a binary operation $\odot : G \times G \to G$ that satisfies the following properties:*

1. *(Associativity.) For all $a, b, c \in G$*
$$(a \odot b) \odot c = a \odot (b \odot c).$$
2. *(Existence of an identity element.) There exists an $e \in G$ such that for all $a \in G$:*
$$e \odot a = a \odot e = a.$$
3. *(Existence of an inverse.) For all $a \in G$, there exists an element $b \in G$ such that*
$$a \odot b = b \odot a = e.$$
EXERCISE: *Show that the identity and inverse element of a group are unique. We denote $a^{-1} = b$.*

A subgroup *of G is a subset $H \subseteq G$, such that $\odot(H \times H) = H$ and for all $a \in H$, $a^{-1} \in H$ (H is closed under group operations and taking the inverse).*

A group isomorphism is a bijection between two groups $(G, \odot)$ and $(G', \otimes)$, $f : G \to G'$, such that for all $g, h \in G$: $f(g \odot h) = f(g) \otimes f(h)$.

*A (*left*)* group action *on a set X is a map $f : G \times X \to X$ satisfying: for all $g, h \in G$ and $x \in X$*

1. *$f(g \odot h, x) = f(g, f(h, x))$,*
2. *$f(e, x) = x$.*

We will make use of the shorthand notation $g \cdot x = f(g, x)$.

Example 2.3. *One can easily verify that $(\mathbb{R}^n, +)$ and $(GL(\mathbb{R}^n), \cdot)$ (invertible linear maps from $\mathbb{R}^n$ to $\mathbb{R}^n$, with function composition or matrix multiplication as the binary operation) are groups. Consider*

$$O(n) = \{A \in GL(\mathbb{R}^n) \mid AA^T = \mathbf{1}_n\},$$
$$SO(n) = \{A \in O(n) \mid \det A = 1\}.$$

Then $O(n)$ is a subgroup of $GL(\mathbb{R}^n)$ (the orthogonal group) and $SO(n)$ is a subgroup of $O(n)$ (the special orthogonal group). We obtain the corresponding group actions on $\mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$:

1. *(Spacetime translations) Consider the group $(\mathbb{R}^{n+1}, +)$. Translations in $\mathbb{R}^{n+1}$ correspond to the group action:*
$$b \cdot x = x + b.$$
The dimension of this group (as a vector subspace of $\mathbb{R}^{n+1^2}$) is $n + 1$.
2. *(Galilean boosts) Consider the group $(\mathbb{R}^n, +)$. Boosts on $\mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$ correspond to the group action:*
$$\mathbf{v} \cdot (t, \mathbf{x}) = (t, \mathbf{x} + t\mathbf{v}) = \begin{pmatrix} 1 & 0 \\ \mathbf{v} & \mathbf{1}_n \end{pmatrix} \begin{pmatrix} t \\ \mathbf{x} \end{pmatrix}.$$
The dimension of this group (as a vector subspace of $\mathbb{R}^{(n+1)^2}$) is n.

3. *(Rotations) Consider the group $(SO(n), \cdot)$, with $O(n) = \{A \in GL(\mathbb{R}^n) \,|\, AA^T = \mathbf{1}\}$ and $SO(n) = \{A \in O(n) \,|\, \det A = 1\}$. Motivated by the case $n = 3$, we refer to the group elements as "rotations". Rotations act on $\mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$ via the group action:*

$$R \cdot (t, \mathbf{x}) = (t, R\mathbf{x}).$$

4. *(Reflections) Consider the group $(\mathbb{Z}_2 = \{-1, 1\}, \cdot)$. Reflections through the x^i-plane on $\mathbb{R}_t \times \mathbb{R}^n_{\mathbf{x}}$ correspond to the group action:*

$$g \cdot (t, \mathbf{x}) = (t, x^1, \ldots, x^{i-1}, gx^i, x^i, \ldots, x^n).$$

Reflections in time (time reversion) correspond to the group action

$$g \cdot (t, \mathbf{x}) = (gt, \mathbf{x}).$$

EXERCISE*: Show that the dimension of $O(n)$, as a submanifold of $\mathbb{R}^{n^2}$ and $SO(n)$ is $\frac{n(n-1)}{2}$ (hint: what is the dimension of the corresponding tangent space?).*
EXERCISE*: Show that the any element of $O(n)$ is the composition of a rotation (an element of $SO(n)$) with a reflection through an x^i-plane, which corresponds to the diagonal matrix with a coefficient -1 in the ith row and all other entries equal to 1.*
EXERCISE*: Verify that the group actions corresponding to translations, boosts, rotations and reflections are Galilean transformations.*

It turns out that translations, boosts, rotations and reflections in fact encompass *all* Galilean transformations.

Proposition 2.3. *The set of Galilean transformations* Gal(n) *can be characterized as follows:*

$$\mathrm{Gal(n)} = \left\{ g : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1} \,|\, g = T_b \circ \begin{pmatrix} \epsilon & 0 \\ \mathbf{v} & A \end{pmatrix},\ b \in \mathbb{R}^{n+1}, \mathbf{v} \in \mathbb{R}^n, A \in O(n), \epsilon \in \{1, -1\} \right\},$$

with $T_b(x) = x + b$. Furthermore, Gal(n) *is a group with respect to composition of maps.*

Proof. We will show that

$$\mathrm{Gal(n)} \subseteq \left\{ g : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1} \,|\, g = T_b \circ \begin{pmatrix} \epsilon & 0 \\ \mathbf{v} & A \end{pmatrix},\ b \in \mathbb{R}^{n+1}, \mathbf{v} \in \mathbb{R}^n, A \in O(n), \epsilon \in \{1, -1\} \right\}.$$

The inclusion $\supseteq$ follows from the example above. We can write

$$g(t, \mathbf{x}) = \begin{pmatrix} g^0 \\ \mathbf{g} \end{pmatrix}(t, \mathbf{x}) = \begin{pmatrix} g^0 \\ \mathbf{g} \end{pmatrix}(0, 0) + \begin{pmatrix} f^0(t, \mathbf{x}) \\ \mathbf{f}(t, \mathbf{x}) \end{pmatrix}$$

with $f^\mu(0, 0) = 0$. Hence, we can set $b = g(0, 0)$. By the assumption that g is affine, $\mathbf{f}$ must be linear in $(t, \mathbf{x})$, we can further write:

$$\mathbf{f}(t, \mathbf{x}) = A(\mathbf{x}) + t\mathbf{v},$$

for some constant $\mathbf{v} \in \mathbb{R}^n$ and a linear map $A : \mathbb{R}^n \to \mathbb{R}^n$.

We now appeal to the preservation of time intervals to conclude that:

$$|f^0(t, \mathbf{x})| = |g^0(t, \mathbf{x}) - g^0(0, 0)| = |t - 0| = |t|,$$

so $f^0(t, \mathbf{x}) = \pm t$. By continuity of f^0 (which follows from linearity), we must have that either $f^0(t, \mathbf{x}) = +t$ for all $(t, \mathbf{x})$ or $f^0(t, \mathbf{x}) = -t$ for all $(t, \mathbf{x})$.

To conclude that $A \in O(n)$, it remains to show that $A^T A = \mathbf{1}_n$ (note that this also implies that $A \in GL(\mathbb{R}^n)$). We first show that $\langle A\mathbf{x}, A\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$. Indeed, by preservation of norm and linearity of A we have that

$$\langle A\mathbf{x}, A\mathbf{y}\rangle = \frac{1}{4}(|A\mathbf{x} + A\mathbf{y}|^2 - |A\mathbf{x} - A\mathbf{y}|^2) = \frac{1}{4}(|\mathbf{x} + \mathbf{y}|^2 - |\mathbf{x} - \mathbf{y}|^2) = \langle \mathbf{x}, \mathbf{y}\rangle.$$

This follows by considering basis vectors $\mathbf{e}_i$ in $\mathbb{R}^n$ and using the preservation of the Euclidean inner product to conclude that for all $i, j \in \{1, \ldots, n\}$:

$$(A^T A)_{ij} = \langle \mathbf{e}_i, A^T A(\mathbf{e}_j)\rangle = \langle A(\mathbf{e}_i), A\mathbf{e}_j\rangle = \langle \mathbf{e}_i, \mathbf{e}_j\rangle = \delta_{ij},$$

with δ_{ij} the Kronecker delta.

Let us now check the group property of Gal(n). First, we need to show that the composition of two Galilean transformations is a Galilean transformation.[8] Let $b_1, b_2 \in \mathbb{R}^4$, $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$, $A, B \in O(n)$ and $\epsilon_1, \epsilon_2 \in \{-1, 1\}$. Then for any $(t, \mathbf{x}) \in \mathbb{R}^4$:

$$\begin{aligned}\left(T_{b_2} \circ \begin{pmatrix} \epsilon_2 & 0 \\ \mathbf{v}_2 & B \end{pmatrix}\right) \circ \left(T_{b_1} \circ \begin{pmatrix} \epsilon_1 & 0 \\ \mathbf{v}_1 & A \end{pmatrix}\right)(t, \mathbf{x}) &= \left(T_{b_2} \circ \begin{pmatrix} \epsilon_2 & 0 \\ \mathbf{v}_2 & B \end{pmatrix}\right)\left(\begin{pmatrix} \epsilon_1 & 0 \\ \mathbf{v}_1 & A \end{pmatrix}\begin{pmatrix} t \\ \mathbf{x} \end{pmatrix} + b_1\right) \\ &= \begin{pmatrix} \epsilon_2\epsilon_1 & 0 \\ \epsilon_1\mathbf{v}_2 + B\mathbf{v}_1 & BA \end{pmatrix}\begin{pmatrix} t \\ \mathbf{x} \end{pmatrix} + \begin{pmatrix} \epsilon_2 & 0 \\ \mathbf{v}_2 & B \end{pmatrix} b_1 + b_2,\end{aligned}$$

Since $O(n)$ is a group, we have that $BA \in O(n)$, so the very right-hand side above is a Galilean transformation. Associativity follows directly from associativity of map composition. The identity element is:

$$T_0 \circ \begin{pmatrix} 1 & 0 \\ 0 & \mathbf{1} \end{pmatrix}$$

Finally, by the above expression for the composition of two Galilean transformations, we can construct the inverse of $T_{b_1} \circ \begin{pmatrix} \epsilon_1 & 0 \\ \mathbf{v}_1 & A \end{pmatrix}$ as $T_{b_2} \circ \begin{pmatrix} \epsilon_2 & 0 \\ \mathbf{v}_2 & B \end{pmatrix}$, with $\epsilon_2 = \epsilon_1$, $B = A^{-1}$, $\mathbf{v}_2 = -\epsilon_1 A^{-1}\mathbf{v}_1$ and $b_2 = -\begin{pmatrix} \epsilon_2 & 0 \\ \mathbf{v}_2 & B \end{pmatrix} b_1$. □

We state without proof the folllowing proposition, which we need to interpret the Galilean group as a *Lie group*.

Proposition 2.4. *There exists a group isomorphism between* Gal(n) *and the group*

$$\left\{ \begin{pmatrix} \epsilon & 0 & b^0 \\ \mathbf{v} & A & \mathbf{b} \\ 0 & 0 & 1 \end{pmatrix} \in GL(\mathbb{R}^{n+2}),\ b \in \mathbb{R}^{n+1}, \mathbf{v} \in \mathbb{R}^n, A \in O(n), \epsilon \in \{-1, 1\} \right\},$$

so Gal(n) *is a matrix Lie group (a closed subset of* $GL(\mathbb{R}^{n+2})$*).*

The dimension of Gal(n) *is* $\frac{1}{2}n(n-1) + (n+1) + n$.

The proposition below proves that, up to a choice of space and time units, Galilean coordinate charts are precisely the coordinate charts compatible with Newton's First Law.

[8] This step can alternatively also be done directly from the definition of Gal(n).

Proposition 2.5.

(i) *Any affine coordinate transformation* $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ *satisfying* $\Phi^0(t, \mathbf{x}) = at+b$ *for some* $a, b \in \mathbb{R}$ *with* $a \neq 0$ *can be decomposed as* $\Phi = \Phi(0)+g\circ D\circ \begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$, *with* g *a Galilean transformation,* D *a diagonal* $(n+1)^2$*-matrix with entries* $\lambda_\mu \in (0, \infty)$, *where* $0 \leq \mu \leq n$ *and* $B \in O(n)$.

(ii) *Let* $\{e_{\mu'}\}$ *be the reference frame associated to the above coordinate chart* $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$. *Then there exist* $\lambda_\mu \in (0, \infty)$, *so that* $\{E_\mu\}$, *with* $E_\mu = \lambda_\mu e'_\mu$ *form a Galilean inertial frame.*

Proof. "(i)": We first rescale $t \mapsto \frac{1}{a}t$, which determines the 00th entry of D. Then we decompose $\Phi = (t', \underline{\Phi})$ and write $\underline{\Phi}(t, \mathbf{x}) = A\mathbf{x}+t\mathbf{v}$, with A an invertible n^2-matrix. By the polar decomposition theorem from linear algebra, we can write any invertible n^2 matrix A as $A = RP$, with $R \in O(n)$ an orthogonal matrix and P a symmetric, nonnegative-definite matrix.[9] By the spectral theorem in linear algebra, we can write $P = B^T DB$, with D a diagonal matrix with positive entries (no zeroes, by invertibility of Φ) and $B \in O(n)$ another orthogonal matrix. Therefore, $A = (RB^T)DB$.

"(ii)": Note that $(D\Phi)(x) = g \circ D \circ \begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$. Note that the reference frame associated to $D \circ \begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$ is $\lambda_\mu f_\mu$, with $\{f_\mu\}$ a Galilean inertial frame, since $\begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$ is a Galilean coordinate transformation. Then, g is also a Galilean coordinate transformation, so it simply rotates (and reflects): $E_\mu = \lambda_\mu A f_\mu$ for some $A \in O(n)$. □

We will now investigate how the Newton equations change under Galilean transformations.

Definition 2.7. *We say the a force* $\mathbf{F} : \mathbb{R} \times \mathbb{R}^{3N} \times \mathbb{R}^{3N} \to \mathbb{R}^3$ *satisfies the* principle of Galilean Relativity *if for any Galilean coordinate transformation* $\Phi = T_b \circ \begin{pmatrix} \epsilon & 0 \\ \mathbf{v} & A \end{pmatrix} = \mathbb{R}^{3+1} \to \mathbb{R}^{3+1}$ *with* $\epsilon \in \{-1, 1\}$, $b \in \mathbb{R}^{3+1}$, $\mathbf{v} \in \mathbb{R}^3$, $A \in O(3)$ *and any* $(t, \mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N)$*:*

$$\mathbf{F}(\epsilon t + b^0, A\mathbf{x}_1 + \mathbf{b} + t\mathbf{v}, \ldots, A\mathbf{x}_N + \mathbf{b} + t\mathbf{v}, A\mathbf{v}_1 + \mathbf{v}, \ldots, A\mathbf{v}_1 + \mathbf{v}) = A\mathbf{F}(t, \mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N).$$

We say that $\mathbf{F}$ *is invariant under* Gal(3). *We can similarly consider forces that are invariant under a subgroup of* Gal(3) *(e.g. translations, rotations).*

In the language of *pushforwards* and *pullbacks*, which we will encounter later, $F := (0, \mathbf{F}) : \mathbb{R}^{3+1} \times \mathbb{R}^{3+1} \to \mathbb{R}^{3+1}$ satisfies the Principle of Galilean Relativity if: $\Phi_*(F(x, v)) = F(\Phi(x), \Phi_*(v))$. Exercise: Show that the forces appearing on the right-hand side of (2.2) satisfy the Principle of Galilean Relativity if and only if (2.2) remains valid for t replaced with $\epsilon t + b^0$ and $\mathbf{x}_i$ replaced with $\mathbf{y}_i = A\mathbf{x}_i + \mathbf{b} + t\mathbf{v}$ for all $A \in O(n)$, $b \in \mathbb{R}^{n+1}$ and $\mathbf{v} \in \mathbb{R}^n$. In other words, (2.2) holds with the same functions $\mathbf{F}_i$ even when expressed with respect to a different inertial coordinate chart.

The principle of Galilean Relativity can be interpreted as saying that physical models remain the same when considered with respect to different Galilean inertial coordinate charts. We can interpret this as follows: two observers will agree on the motion described by particles determined by forces satisfying the principle of Galilean Relativity, with a prescribed initial position and velocity, if:

- the observers have a constant relative velocity,
- the observers are rotated with respect to each other,

[9] Consider $A^T A$. This is a symmetric (and hence diagonalizable) matrix, with positive eigenvalues, so $P := \sqrt{A^T A}$ is well-defined and also symmetric and non-negative-definite. Then $R^T R = (P^{-1})^T A^T A P^{-1} = P^{-1} P^2 P^{-1} = \mathbf{1}$, so R is indeed orthogonal.

- one observer looks at the particles through a mirror ("through the looking glass"),
- one observer moves backwards in time compared to the other.

Observers will, for example, not agree if they experience a non-trivial relative acceleration or if they rotate at a non-zero relative angular velocity.

The principle of Galilean Relativity is relevant when considering closed systems of N particles that only interact with each other, rather than subsystems interacting also with their environment. For example, a particle experiencing friction when moving through the air does not describe a closed system and the corresponding force does not satisfy the principle of Galilean Relativity, so it matters which inertial coordinate chart we choose to describe it.

Imposing the Principle of Galilean Relativity introduces restrictions of what the forces $\mathbf{F}_i$ are allowed to look like.

Proposition 2.6. *Let $\mathbf{F} : \mathbb{R}_t \times \mathbb{R}^{3N} \times \mathbb{R}^{3N} \to \mathbb{R}^3$ be a force that is invariant under time translations. Then $\mathbf{F}(t, \cdot) = \mathbf{F}(0, \cdot)$ for all $t \in \mathbb{R}$.*

Proof. By the invariance property of $\mathbf{F}$: for all $t, s \in \mathbb{R}$, $\mathbf{x}_i, \mathbf{v}_i \in \mathbb{R}^3$, $i \in \{1, \dots, N\}$:

$$\mathbf{F}(t + s, \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N) = \mathbf{F}(t, \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N).$$

We can take $t = 0$ to conclude that $\mathbf{F}(s, \cdot) = \mathbf{F}(0, \cdot)$ for all $s \in \mathbb{R}$. □

We already saw that in the case of a single particle, $\mathbf{F} \equiv 0$ satisfies the Principle of Galilean Relativity by Newton's First Law. Now we will prove the converse, which is: in the case of a single particle, the *only* force $\mathbf{F}$ that satisfies the Principle of Galilean Relativity is $\mathbf{F} = 0$.

Proposition 2.7. *If $\mathbf{F} : \mathbb{R} \times \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ satisfies the Principle of Galilean Relativity, then $\mathbf{F} \equiv 0$.*

Proof. Under spacetime translations and with boosts, $(t', \mathbf{y}) = (t + s, \mathbf{x} + \mathbf{b} + t\mathbf{w})$, we have that:

$$\mathbf{F}(t + s, \mathbf{x} + \mathbf{b} + t\mathbf{w}, \mathbf{v} + \mathbf{w}) = \mathbf{F}(t, \mathbf{x}, \mathbf{v}).$$

We set $t = 0$, $\mathbf{x} = 0$ and $\mathbf{v} = 0$, to conclude that $\mathbf{F}(s, \mathbf{b}, \mathbf{w})) = \mathbf{F}(0, 0, 0) =: \mathbf{F}_0$ for all $s \in \mathbb{R}$, $\mathbf{b} \in \mathbb{R}^3$ and $\mathbf{w} \in \mathbb{R}$.

Finally, consider an arbitrary rotation $R \in SO(3)$. Then we also have:

$$\mathbf{F}_0 = R\mathbf{F}_0.$$

Since the only vector that is invariant under all rotations in the zero vector, we conclude that $\mathbf{F}_0 = 0$. □

When considering a system consisting of multiple particles, we derive the following constraints on the forces:

Proposition 2.8. *If $\mathbf{F} : \mathbb{R} \times \mathbb{R}^{3N} \times \mathbb{R}^{3N} \to \mathbb{R}^3$ satisfies the Principle of Galilean Relativity, then there exist maps $\mathbf{f}_i : \mathbb{R}^{3(\frac{1}{2}N(N-1))} \times \mathbb{R}^{3(\frac{1}{2}N(N-1))} \to \mathbb{R}^3$ such that*

$$\mathbf{F}_i(t, \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N) = \mathbf{f}_i(\mathbf{X}, \mathbf{V}),$$

with $\mathbf{X} = \{\mathbf{X}_{ij}\}_{1 \le i,j \le N}$ and $\mathbf{V} = \{\mathbf{V}_{ij}\}_{1 \le i,j \le N}$, with: $\mathbf{X}_{ij} = \mathbf{x}_i - \mathbf{x}_j$ and $\mathbf{V}_{ij} = \mathbf{v}_i - \mathbf{v}_j$, respectively.

For any $A \in O(3)$

$$A\mathbf{f}_i(\mathbf{X}, \mathbf{V}) = \mathbf{f}_i(A \cdot \mathbf{X}, A \cdot \mathbf{V}),$$

where $(A \cdot \mathbf{X})_{ij} := A(\mathbf{x}_i - \mathbf{x}_j)$ and $A \cdot \mathbf{V} := A(\mathbf{v}_i - \mathbf{v}_j)$.

Proof. By Proposition 2.6, $\mathbf{F}_i$ must be time-independent. By invariance under translations in $\mathbb{R}^3$, $\mathbf{F}_i$ must depend only on relative spatial coordinates $\mathbf{x}_j - \mathbf{x}_k$. By invariance under boosts, $\mathbf{F}_i$ must also only depend on relative velocities $\mathbf{v}_j - \mathbf{v}_k$. Finally, we appeal to invariance under $O(3)$ to conclude that:

$$\mathbf{f}_i(A\cdot\mathbf{X}, A\cdot\dot{\mathbf{X}}) = \mathbf{F}(t, A\mathbf{x}_1, \ldots, A\mathbf{x}_N, A\mathbf{v}_1, \ldots, A\mathbf{v}_N) = A\mathbf{F}(t, \mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N) = A\mathbf{f}_i(\mathbf{X}, \dot{\mathbf{X}}).$$

□

Remark 2.1. *Examples of physical theories satisfying the principle of Galilean Relativity in the sense that solutions to the corresponding equations arising from the same initial data agree when expressed with respect to any different choice of Galilean coordinates: Newtonian gravity, (non-relativistic) Euler's equations of fluid mechanics. Counter-example: The Maxwell equations. The Navier–Stokes equations of fluid mechanics are not invariant under time reversal, but they are invariant under all other Galilean transformations.*

Definition 2.8. *We define the Lorentzian inner product of $x, y \in \mathbb{R}^{n+1}$ as follows:*

$$m(x, y) = -x^0 y^0 + \sum_{i=1}^{n} x^i y^i.$$

Note that m is a symmetric and non-degenerate bilinear form, just like a standard inner product, but $m(x, x)$ is not positive definite, so $\sqrt{m(x, x)}$ does not define a norm.

Definition 2.9. Poincaré coordinate charts*, or* Poincaré (coordinate) transformations *are affine maps $g : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$, with $n \geq 1$, that preserve Lorentzian length, i.e.*

$$m(g(x) - g(y), g(x) - g(y)) = m(x - y, x - y) \quad \forall x, y \in \mathbb{R}^{n+1}.$$

We denote the set of of Poincaré transformations in $\mathbb{R}^{n+1}$ by $P(n; 1)$ and their corresponding reference frames as Poincaré inertial frames.

We now give a little bit of motivation for the definition of m, by imposing the condition that our coordinate charts should have the property that the speed of light c stays constant. Indeed, let $(t_1, \mathbf{x}_1) \in \mathbb{R}^{n+1}$ denote the spacetime event of emission of a "light signal" (for the purposes of this argument, this is a particle travelling at a constant velocity with speed c) and $(t_2, \mathbf{x}_2) \in \mathbb{R}^{n+1}$ the spacetime event of absorption of the light signal. Then the spatial distance between these two spacetime events is $\sqrt{\sum_{i=1}(x_1^i - x_2^i)^2}$, but it should also be $c(t_2 - t_1)$ (speed × time). Hence:

$$-c^2(t_2 - t_1)^2 + \sum_{i=1}(x_1^i - x_2^i)^2 = 0.$$

If we describe the same event with respect to different spacetime coordinates $(t', \mathbf{y})$ for which the speed of the light signal stays c, we should similarly have that:

$$-c^2(t_2 - t_1)^2 + \sum_{i=1}(x_1^i - x_2^i)^2 = 0 = -c^2(t_2' - t_1')^2 + \sum_{i=1}(y_1^i - y_2^i)^2.$$

Without loss of generality, we can set $c = 1$ (i.e. we consider physical units for which the speed of light gives the value 1, e.g lightseconds instead of meters). Hence, the Lorentzian length is preserved for spacetime intervals $x_1 - x_2$, which satisfy $m(x_1 - x_2, x_1 - x_2) = 0$. One can further motivate the affine nature of the maps g and argue that the Lorentzian length should also be preserved even if $m(x_1 - x_2, x_1 - x_2) \neq 0$, under the additional assumptions of homogeneity and isotropy of spacetime, but we will leave the full details of the argument to Appendix A.

Example 2.4. *The following group actions on $\mathbb{R}^{n+1}$ are examples of Poincaré transformations:*

1. *(Translations) The translation group $(\mathbb{R}^{n+1}, +)$ with group action*

$$b \cdot x = x + b.$$

2. *(Rotations and reflections in space) The group $(O(n), \cdot)$ with group action*

$$A \cdot x = (x^0, A\mathbf{x}).$$

Let $\mathbf{n}$ be a unit norm vector, $v \in [0, 1)$ and $\gamma_L = \frac{1}{\sqrt{1-v^2}}$. Then following map is also a Poincaré transformation:

$$x \mapsto (\gamma_L(x^0 - v\mathbf{n} \cdot \mathbf{x}), \mathbf{x}_\perp + \gamma_L(\mathbf{x}_{||} - vx^0\mathbf{n})),$$

where $\mathbf{x}_{||} = \langle \mathbf{x}, \mathbf{n} \rangle \mathbf{n}$ and $\mathbf{x}_\perp = \mathbf{x} - \mathbf{x}_{||}$. This map is called a Lorentz boost *in the $\mathbf{n}$–direction.*
EXERCISE: *Show that boosts do not correspond to group actions, by considering a composition of two boosts.*

We say a physical theory satisfies the *Principle of Special Relativity* if it is "invariant under Poincaré transformations". That is to say, observers related by Poincaré transformations agree about the solutions to the corresponding physical equations, given the same initial data. We will later explore what this means precisely in the context of field theories.

EXERCISE: Show that the force term on the right-hand side of Newton's equations of gravity can change when we perform a coordinate change corresponding to a Poincaré transformation.

We will later see that Maxwell's equations of electromagnetism *are* invariant under Poincaré transformations.

Proposition 2.9. *We can write:*

$$P(n; 1) = \left\{ g : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1} \,|\, g = T_b \circ \Lambda,\ b \in \mathbb{R}^{n+1}, \Lambda \in O(n; 1) \right\},$$

where

$$O(n; 1) := \{\Lambda \in GL(\mathbb{R}^{n+1}) \,|\, m(\Lambda x, \Lambda y) = m(x, y)\}$$

is called the set of Lorentz transformations.

Furthermore, for any $\Lambda \in O(n; 1)$, there exist $A \in O(n)$, $R \in SO(n)$, $\psi \in \mathbb{R}$ and $\epsilon \in \{-1, 1\}$, such that

$$\Lambda = \begin{pmatrix} \epsilon & 0 \\ 0 & A \end{pmatrix} L_\psi \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix},$$

with

$$L_\psi = \begin{pmatrix} \cosh\psi & 0 & \dots & 0 & \sinh\psi \\ 0 & & & & 0 \\ \vdots & & \mathbf{1}_{n-1} & & \vdots \\ 0 & & & & 0 \\ \sinh\psi & 0 & \dots & 0 & \cosh\psi \end{pmatrix}.$$

The set $P(n; 1)$ forms a group under function composition.

Proof. Problem Sheet 3. □

Remark 2.2. *We can express*

$$\cosh\psi = \frac{1}{\sqrt{1-v^2}},$$

$$\sinh\psi = \frac{v}{\sqrt{1-v^2}},$$

with $v^2 < 1$ to put L_ψ in the more familiar form of a Lorentz boost in the direction of the nth coordinate axis.

Proposition 2.10. *There exists a group isomorphism between $P(n;1)$ and*

$$\left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \in GL(\mathbb{R}^{n+2}),\ b \in \mathbb{R}^{n+1}, A \in O(n;1) \right\},$$

and $P(n;1)$ in a matrix Lie group.

Proof. EXERCISE. □

Remark 2.3. *Consider the subgroup $O(n;1)$ of Lorentz transformations.*

- *By restricting to $\Lambda \in O(n;1)$ with $\det\Lambda = 1$, we obtain the subgroup $SO(n;1)$ called the* proper Lorentz group.
- *By restricting to $\Lambda \in O(n;1)$ with $\Lambda_{00} > 0$, we obtain the* orthochonous Lorentz group $O(n;1)^+$.
- *By restricting to $\Lambda \in SO(n;1)$ with $\Lambda_{00} > 0$, we obtain the* proper, orthochronous Lorentz group $SO(n;1)^+$.

When viewed as a topological space, $O(n;1)$ has four connected components and the component $SO(n;1)^+$ is one of these, namely the one that contains the identity element.

We saw that, up to a constant shift and constant rescaling, there is an absolute time parameter t that we can consider along spacetime paths $\gamma : I \to \mathbb{R}^{n+1}$ when restricting to Galilean coordinate charts (or more generally, coordinate charts compatible with Newton's First Law). We can use this time parameter to define the acceleration of γ in an unambiguous manner (up to a choice of time units). For forces satisfying the Principle of Galilean Relativity, there is therefore no preferred Galilean coordinate chart when studying the Newton equations (2.2).

Now replace Galilean coordinate charts with Poincaré coordinate charts. Then it can be shown that all Poincaré are compatible with Newton's First Law, if we restrict to motions of particles with constant velocities satisfying $|\mathbf{v}|^2 < 1$ (which means: free particles travelling at speeds less than the speed of light $c = 1$) (see Problem Sheet 3).

In order to formulate a version of Newton's Second Law, we need to consider an analogue of the Newton equations (2.2) that is invariant when we consider a Poincaré coordinate transformation. The first problem we face is that there is no longer an absolute time parameter t in $\mathbb{R}^{n+1}$ (up to a constant shift and constant rescaling) that we can consider along spacetime paths.

Indeed, let $\gamma = (\mathrm{id}, \underline{\gamma}) : I \to \mathbb{R}^{n+1}$ be the spacetime motion of a particle and let $\Phi = L_\psi$ be a Lorentz boost. Then $\tau(t) = t'(\gamma(t)) = \cosh\psi t + \sinh\psi\gamma^n(t)$. Since $\gamma^n(t)$ can be an arbitrary function, the function $\tau : I \to \mathbb{R}$ can be arbitrarily complicated and $\frac{d^2}{dt'^2}\gamma(\tau^{-1}(t'))$ can look very different from $\frac{d^2}{dt^2}\gamma(t)$.

We would like to consider time parameters τ along a spacetime path that, up to a shift and rescaling, all Poincaré coordinate charts agree on, so that the acceleration of a path can be defined unambigously.

A time parameter which has this property is the *proper time*.

Definition 2.10. *The proper time τ along a spacetime path $\gamma : [a,b] \to \mathbb{R}^{n+1}$ satisfying $m(\dot\gamma,\dot\gamma)(t) < 0$ for all $t \in [a,b]$ is defined as:*

$$\tau[\gamma;a](t) := \int_a^t \sqrt{-m(\dot\gamma(s),\dot\gamma(s))}\, ds.$$

One can show that the proper time is the only parameter, up the constant shifts and rescaling, that is invariant under Poincaré coordinate transformations and such that $\frac{d\tau}{dt}[\gamma; a](t)$ depends only on $\gamma(t)$, $\frac{d\gamma}{dt}(t)$ (locality). Proper time may be thought of as an analogue of *arc length* in the setting of paths in $\mathbb{R}^n$ equipped with the Euclidean inner product.

Since we now have a natural time parameter, we can define a natural velocity and momentum associated to the particle:

Definition 2.11. *The* four-velocity $U : I \to \mathbb{R}^{3+1}$ *associated to a spacetime path* $\gamma : I \to \mathbb{R}^{3+1}$ *is defined as:*

$$U(\tau) := \frac{d\gamma}{d\tau}(\tau).$$

If $\gamma : I \to \mathbb{R}^{3+1}$ *is the spacetime motion of a particle with mass* $\mathfrak{m}$*, then the* four-momentum $P : I \to \mathbb{R}^{3+1}$ *of the particle is* $P = \mathfrak{m}U$*.*

Let $\gamma(t) = (t, \mathbf{x}(t))$. Then $\frac{d\tau}{dt} = \sqrt{-m(\dot\gamma(t), \dot\gamma(t)} = \sqrt{1 - |\dot{\mathbf{x}}|^2(t)}$, so

$$P(\tau(t)) = \frac{\mathfrak{m}}{\sqrt{1 - |\dot{\mathbf{x}}|^2(t)}} \left(1, \frac{d\mathbf{x}}{dt}(t)\right).$$

Given $F : \mathbb{R} \times \mathbb{R}^{3+1} \to \mathbb{R}^{3+1} \to \mathbb{R}^{3+1}$, we can state the following analogue of the Newton equations (2.2):

$$\mathfrak{m}\frac{d^2\gamma}{d\tau^2}(\tau) = F\left(\tau, \gamma(\tau), \frac{d\gamma}{d\tau}(\tau)\right), \tag{2.4}$$

or equivalently, $\frac{dP}{d\tau}(\tau) = F(\gamma(\tau), U(\tau))$. We will refer to F as a "force", in analogy with the Galilean setting. Note that $m(U, U) = -1$, by construction so:

$$0 = \frac{d}{d\tau}\left(m(U, U)\right) = 2m\left(\frac{dU}{d\tau}, U\right) = \frac{2}{\mathfrak{m}} m(F, U).$$

Hence, F is orthogonal to U with respect to the Lorentzian inner product m. In other words, the force F has only spatial components when measured with respect to an observer moving along γ.

Definition 2.12. *The force* F *satisfies the Principle of Special Relativity if for any* $\Phi \in P(3; 1)$*:*

$$\Phi_*(F(\tau, x, v)) = F(\tau, \Phi(x), \Phi_*(v)).$$

If F satisfies the Principle of Special Relativity, then:

$$\mathfrak{m}\frac{d^2(\Phi \circ \gamma)}{d\tau^2}(\tau) = F\left(\tau, (\Phi \circ \gamma)(\tau), \frac{d(\Phi \circ \gamma)}{d\tau}(\tau)\right).$$

In other words, two observers related via a Poincaré coordinate transformation will agree on the motion of a particle, provided they agree on its initial position and velocity.

An example of a force satisfying the Principle of Special Relativity is the Lorentz force in special relativity: $F^\mu = \mathfrak{q} F^{\mu\alpha} m_{\alpha\beta} U^\beta$, with $m_{00} = -1$, $m_{0i} = 0$, $m_{ij} = \delta_{ij}$, $\mathfrak{q}$ the electric charge of the particle, and with $F^{\alpha\beta}$ the components of a tensor field that is related to the *Faraday tensor*. They are equal to appropriate components of electric and magnetic fields satisfying the Maxwell equations. We will introduce the Faraday tensor later in the course.

2.3 Conservative forces

We will now consider (2.2) with some additional restrictions.

Definition 2.13. *The forces* $\mathbf{F}_i : \mathbb{R}_t \times U \times \mathbb{R}^{3N}_{\mathbf{x}} \to \mathbb{R}^3$*, with* $i \in \{1, \dots, N\}$ *and* $U \subseteq \mathbb{R}^{3N}_{\mathbf{x}}$ *open, are* conservative *if they satisfy the following assumption: for all* $1 \leq i \leq N$

$$\mathbf{F}_i(t, \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N) = -(\nabla_{\mathbf{x}_i} U)(\mathbf{x}_1, \dots, \mathbf{x}_N), \tag{2.5}$$

for some C^2 *function* $U : \mathbb{R}^{3N} \to \mathbb{R}$*, which we call the* potential*. Here* $\nabla_{\mathbf{x}_i}$ *denotes the gradient with respect to* $\mathbf{x}_i$*. Note that, in particular, we are assuming that the functions* $\mathbf{F}_i$ *are independent of* t *and the velocity variables* $\mathbf{v}_j$*.*

Remark 2.4. *An equivalent way of characterizing conservative forces in the* $N = 1$ *case is as* C^1 *functions of the form* $\mathbf{F}(\mathbf{x})$ *that are additionally path independent, i.e. for any piecewise* C^1 *closed loop* $\underline{\gamma} : \mathbb{S}^1 \to \mathbb{R}^3$

$$\int_{\mathbb{S}^1} \langle \mathbf{F}(\underline{\gamma}(t)), \dot{\underline{\gamma}}(t) \rangle \, dt = 0.$$

Yet another equivalent way of characterizing the conservative property of $\mathbf{F}$ *in the* $N = 1$ *case is:* $\mathbf{F}$ *is a* C^1 *function that depends only on the position variable and*

$$\nabla \times \mathbf{F} \equiv 0.$$

Example 2.5 (Particle in one dimension)**.** *Consider Newton's equation on* $I \subset \mathbb{R}$ *with* $F(t, x, v) = f(x)$ *for some* C^1 *function* $f : \mathbb{R} \to \mathbb{R}$*. Then we can write*

$$m\ddot{x}(t) = f(x(t)) = -U'(x(t)),$$

with $U(x) = -\int_{x_0}^{x} f(x') \, dx'$*, with* $x_0 \in I$ *arbitrary.*

Example 2.6 (Newton's theory of gravity)**.**

$$m_i \ddot{\mathbf{x}}_i = -(\nabla_{\mathbf{x}_i} U)(\mathbf{x}_1, \dots, \mathbf{x}_N),$$
$$U(\mathbf{x}_1, \dots, \mathbf{x}_N) = -\sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{G m_i m_j}{|\mathbf{x}_i - \mathbf{x}_j|},$$

with $G \in (0, \infty)$ *the gravitational constant.*

2.3.1 Energy

An important and very useful property of solutions to Newton's equations with conservative forces is that they admit a conserved energy.

Definition 2.14. *The* energy associated to $\underline{\gamma} : \mathbb{R} \supseteq I \to \mathbb{R}^{3N}$ *is defined as follows:*

$$E_{\underline{\gamma}}(t) = \sum_{i=1}^{N} \frac{1}{2} m_i |\dot{\mathbf{x}}_i|^2(t) + U(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)).$$

We refer to $T(\mathbf{x}_1(t), \dots, \mathbf{x}_2(t)) := \sum_{i=1}^{N} \frac{1}{2} m_i |\dot{\mathbf{x}}_i|^2(t)$ *as the total* kinetic energy *and* $U(\mathbf{x}_1(t), \dots, \mathbf{x}_2(t))$ *as the total* potential energy *of the system of* N *particles.*

The energy has the property that it is conserved in time.

Proposition 2.11. *Let $\underline{\gamma} : \mathbb{R} \supseteq I \to \mathbb{R}^{3N}$ be a solution to (2.2) with conservative forces $\mathbf{F}_i$. Then $\frac{d}{dt} E_{\underline{\gamma}}(t) = 0$.*

Proof. By the chain rule, we have that

$$\begin{aligned}\frac{d}{dt} E_{\underline{\gamma}}(t) &= \sum_{i=1}^{N} \frac{1}{2} m_i \frac{d}{dt} |\dot{\mathbf{x}}_i|^2(t) + \frac{d}{dt} U(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)) \\ &= \sum_{i=1}^{N} m_i \langle \ddot{\mathbf{x}}_i, \dot{\mathbf{x}}_i \rangle(t) + \sum_{i=1}^{N} \langle \nabla_{\mathbf{x}_i} U(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)), \dot{\mathbf{x}}_i(t) \rangle \\ &= \sum_{i=1}^{N} \langle m_i \ddot{\mathbf{x}}_i(t) + \nabla_{\mathbf{x}_i} U(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)), \dot{\mathbf{x}}_i(t) \rangle = 0. \quad \square\end{aligned}$$

We now briefly return to the analogue of the Newton equation of a single particle in special relativity (2.4), which features the four-momentum P.

If we remove the restriction to units with respect to which the speed of light satisfies $c = 1$ and we consider $\gamma(t) = (ct, \mathbf{x}(t))$, we obtain $P(\tau(t)) = \frac{\mathfrak{m}}{\sqrt{1 - c^{-2}|\dot{\mathbf{x}}|^2(t)}} \left(c, \frac{d\mathbf{x}}{dt}(t)\right)$. Let $v = \sup_{t \in I} |\dot{\mathbf{x}}(t)|$. Then, by Taylor's theorem:

$$\begin{aligned}cP^0(\tau(t)) &= \mathfrak{m}c^2 + \frac{1}{2}\mathfrak{m}|\dot{\mathbf{x}}(t)|^2 + O(c^{-3}v^3), \\ P^i(\tau(t)) &= \mathfrak{m}\frac{dx^i}{dt}(t) + O(c^{-2}v^2)\end{aligned}$$

where $O(c^{-k}v^k)$ denotes terms that can be uniformly bounded by $c^{-k}v^k$. We recognize $\frac{1}{2}m|\dot{\mathbf{x}}(t)|^2$ as the kinetic energy of a particle with mass $\mathfrak{m}$ and $\mathfrak{m}\frac{d\mathbf{x}}{dt}$ as its linear momentum.

For a free particle, corresponding to $F = 0$, $P(\tau)$ is conserved in τ. We therefore interpret (P^1, P^2, P^3) as a generalization of the linear momentum and cP^0 as the generalization of the energy of the free particle. We define $E_0 := \mathfrak{m}c^2$ to be the *rest energy* of the particle and $cP^0 - E_0$ as the (relativistic) kinetic energy of the particle. Note that if $v \ll c$, the rest energy is much bigger than the kinetic energy.

2.3.2 Central forces

Definition 2.15. *A $SO(3)$-invariant (rotation-invariant) force $\mathbf{F} : \mathbb{R}^3_{\mathbf{x}} \setminus \{0\} \to \mathbb{R}^3$ is also called a* central force.

Proposition 2.12. *Let $\mathbf{F} : \mathbb{R}^3_{\mathbf{x}} \setminus \{0\} \to \mathbb{R}^3$ be a continuously differentiable central force. Then $\mathbf{F}$ is conservative.*

Proof. We can decompose $\mathbb{R}^3 \ni \mathbf{F}(\mathbf{x}) = F^r(\mathbf{x})\mathbf{e}_r(\mathbf{x}) + F^\theta(\mathbf{x})\mathbf{e}_\theta(\mathbf{x}) + F^\varphi(\mathbf{x})\mathbf{e}_\varphi(\mathbf{x})$, with $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$ a right-handed orthonormal basis of vector fields on $\mathbb{R}^3 \setminus \{0\}$ such that the vector field $\mathbf{e}_r$ is defined as follows $\mathbf{e}_r(\mathbf{x}) = \frac{\mathbf{x}}{|\mathbf{x}|}$. By assumption, $\mathbf{F}$ is invariant under rotations $R \in SO(3)$, so $R\mathbf{F}(\mathbf{x}) = \mathbf{F}(R\mathbf{x})$. By taking R to be a 180° rotation around the axis through $\mathbf{e}_r(\mathbf{x})$ (so, in particular $R\mathbf{e}_r(\mathbf{x}) = \mathbf{e}_r(\mathbf{x})$ and $R\mathbf{x} = \mathbf{x}$), we obtain

$$F^r(\mathbf{x})\mathbf{e}_r(\mathbf{x}) - F^\theta(\mathbf{x})\mathbf{e}_\theta(\mathbf{x}) - F^\varphi(\mathbf{x})\mathbf{e}_\varphi(\mathbf{x}) = R\mathbf{F}(\mathbf{x}) = \mathbf{F}(R\mathbf{x}) = \mathbf{F}(\mathbf{x}) = F^r(\mathbf{x})\mathbf{e}_r(\mathbf{x}) + F^\theta(\mathbf{x})\mathbf{e}_\theta(\mathbf{x}) + F^\varphi(\mathbf{x})\mathbf{e}_\varphi(\mathbf{x}).$$

Hence $F^\theta = F^\varphi \equiv 0$. Now let R be a general rotation, then $F^r(\mathbf{x})R\mathbf{e}_r(\mathbf{x}) = F^r(R\mathbf{x})\mathbf{e}_r(R\mathbf{x}) = F^r(R\mathbf{x})R\mathbf{e}_r(\mathbf{x})$, so $F^r(R\mathbf{x}) = F^r(\mathbf{x})$. This means that we can write:

$$\mathbf{F}(\mathbf{x}) = F^r(|\mathbf{x}|)\mathbf{e}_r(\mathbf{x}).$$

Now, for arbitrary $r_0 > 0$, define $U : \mathbb{R}^3_{\mathbf{x}} \setminus \{0\} \to \mathbb{R}$ as follows:

$$U(\mathbf{x}) = -\int_{r_0}^{|\mathbf{x}|} F^r(r)\, dr. \qquad \square$$

Let $t \mapsto \mathbf{x}(t)$ denote a C^2 path in $\{z = 0\} \subset \mathbb{R}^3$. Let $\mathbf{e}_r(\mathbf{x}) = \frac{\mathbf{x}}{r}$. In polar coordinates, we can write $\mathbf{e}_r(\mathbf{x}) = (\cos\varphi, \sin\varphi, 0)^T$. We denote with $\mathbf{e}_\varphi(\mathbf{x})$ the corresponding orthogonal, unit norm vector field that points in the direction of increasing φ. Then $\mathbf{e}_\varphi(\mathbf{x}) = (-\sin\varphi, \cos\varphi, 0)^T$. Let $\mathbf{e}_z$ denote the constant unit norm vector field pointing in the x^3-direction in $\mathbb{R}^3$. Then $\{\mathbf{e}_r(\mathbf{x}), \mathbf{e}_\varphi(\mathbf{x}), \mathbf{e}_z(\mathbf{x})\}$ form an orthonormal basis of vector fields on $\{x^3 = 0\} \setminus \{0\}$, in particular along the path $t \mapsto \mathbf{x}(t)$.

Note that

$$\partial_r \mathbf{e}_r(\mathbf{x}) = \partial_r \mathbf{e}_\varphi(\mathbf{x}) = 0,\ \partial_\varphi \mathbf{e}_r(\mathbf{x}) = \mathbf{e}_\varphi(\mathbf{x}) \text{ and } \partial_\varphi \mathbf{e}_\varphi(\mathbf{x}) = -\mathbf{e}_r(\mathbf{x}).$$

By construction, we have that $\mathbf{x}(t) = r(t)\mathbf{e}_r(\mathbf{x}(t))$, with $r(t) := |\mathbf{x}(t)|$. We also denote with $\varphi(t) \in [0, 2\pi)$ the unique angle satisfying $r(t)(\cos\varphi(t), \sin\varphi(t), 0)^T = \mathbf{x}(t)$. By the chain rule:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \dot{r}(t)\mathbf{e}_r(\mathbf{x}(t)) + r(t)\dot{\mathbf{e}}_r(\mathbf{x}(t)) \\ &= \dot{r}(t)\mathbf{e}_r(\mathbf{x}(t)) + r(t)\dot{\varphi}\partial_\varphi \mathbf{e}_r(\mathbf{x}(t)) \\ &= \dot{r}(t)\mathbf{e}_r(\mathbf{x}(t)) + r(t)\dot{\varphi}\mathbf{e}_\varphi(\mathbf{x}(t)). \end{aligned} \tag{2.6}$$

We can express the angular momentum $\mathbf{L} = \mathbf{x} \times m\dot{\mathbf{x}}$ as follows using (2.6):

$$\mathbf{L} = \mathbf{x} \times m\dot{\mathbf{x}} = mr^2\dot{\varphi}\mathbf{e}_r \times \mathbf{e}_\varphi = mr^2\dot{\varphi}\mathbf{e}_z. \tag{2.7}$$

Proposition 2.13 (Central force in 2 dimensions)**.** *Consider Newton's equation in $\mathbb{R}^3$ with a central force that is restricted to the plane $\{x^3 = 0\}$:*

$$m\ddot{\mathbf{x}} = \mathbf{F} = -\nabla U,$$

with $U : \mathbb{R}^3 \setminus \{0\} \to \mathbb{R}$ satisfying $U(x^1, x^2, x^3) = u(\sqrt{(x^1)^2 + (x^2)^2})$, where $u : (0, \infty) \to \infty$ is a C^2 function.

Then $r(t)$ satisfies the Newton equations (2.2) on $(0, \infty)$ with respect to a force corresponding to the potential

$$V(r) = u(r) + \frac{L^2}{2mr^2},$$

with $L = |\mathbf{L}|$ and $\mathbf{L}$ conserved in time. The potential V is called the effective potential.

Proof. Note that

$$\ddot{\mathbf{x}} = \frac{d}{dt}(\dot{r}\mathbf{e}_r + r\dot{\varphi}\mathbf{e}_\varphi) = (\ddot{r} - r\dot{\varphi}^2)\mathbf{e}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi})\mathbf{e}_\varphi.$$

Using that $\mathbf{F}$ is central, we have that with respect to polar coordinates (r, φ), U depends only on r and $\mathbf{F} = -u'(r)\mathbf{e}_r$, so

$$\begin{aligned} m\ddot{r}(t) - mr\dot{\varphi}^2(t) &= -u'(r(t)) \\ 2m\dot{r}\dot{\varphi} + mr\ddot{\varphi} &= 0. \end{aligned}$$

Note that the second equation is equivalent to $\frac{d}{dt}(mr^2\dot{\varphi}) = 0$. Since $\mathbf{L} = mr^2\dot{\varphi}\mathbf{e}_z$, this implies conservation of angular momentum $\mathbf{L}(\mathbf{x}(t))$ in time. We therefore have that $\dot{\varphi} = \frac{\pm L}{mr^2}$. Filling this in, we obtain:

$$m\ddot{r}(t) = -u'(r(t)) + \frac{L^2}{mr^3(t)} = -V'(r(t)). \qquad \square$$

2.4 Calculus of variations

We consider maps of the form:

$$\mathcal{S} : C^1([a,b];\mathbb{R}^n) \to \mathbb{R},$$

where $C^1([a,b];\mathbb{R}^n)$ is the space of continuous functions from the interval $[a,b]$ to $\mathbb{R}^n$. Maps of the form $X \to \mathbb{R}$ or $X \to \mathbb{C}$, with X an appropriate function space, are called *functionals*.

To do calculus of variations, we need to study how $\mathcal{S}$ changes when vary γ, while keeping its endpoints fixed, i.e. we consider

$$\mathcal{S}(\gamma + sh),$$

with $s \in \mathbb{R}$ and $h \in C_0^1((a,b);\mathbb{R}^n)$ (h is C^1 and compactly supported on (a,b)).

Definition 2.16. *An* action *on $[a,b]$ is a functional $\mathcal{S} : C^1([a,b];U_x) \to \mathbb{R}$, with $U_x \subset \mathbb{R}^n$ open, of the following form:*

$$\mathcal{S}(\gamma) = \int_a^b \mathcal{L}(t,\gamma(t),\dot\gamma(t))\,dt, \tag{2.8}$$

with $\mathcal{L} \in C^0([a,b] \times U_x \times V_v)$, where $V_v \subset \mathbb{R}^n$ is open. We will refer to $\mathcal{L}$ as a Lagrangian.

We denote the partial derivative of $\mathcal{L}$ with respect to the 2nd until $n+1^{th}$ variables with $\partial_{x^i}\mathcal{L}$, with respect to the last n variables with $\partial_{v^i}\mathcal{L}$ and with respect to the first variable with $\partial_t\mathcal{L}$. When considering a path $\gamma : [a,b] \to \mathbb{R}^n$ with derivative $\dot\gamma$, we can apply the chain rule to determine derivatives of the composition $\widetilde{\mathcal{L}} = \mathcal{L} \circ (\mathrm{id}, \gamma, \dot\gamma)$. For example,

$$\frac{d}{dt}(\widetilde{\mathcal{L}} \circ (\mathrm{id},\gamma,\dot\gamma))(t) = \sum_{i=1}^n \left[\partial_t\mathcal{L} + \partial_{x^i}\mathcal{L}\dot\gamma^i(t) + \partial_{v^i}\mathcal{L}\ddot\gamma^i(t)\right](t,\gamma(t),\dot\gamma(t)).$$

Theorem 2.14. *Let $\mathcal{S}$ satisfy (2.8) with $\mathcal{L} \in C^2([a,b]\times U_x \times V_v)$ and let $h \in C_0^1((a,b);\mathbb{R}^n)$. Then*

$$\frac{d}{ds}\Big|_{s=0}\mathcal{S}(\gamma+sh) = \int_a^b \left[\frac{\partial\mathcal{L}}{\partial x^i} \circ (\mathrm{id},\gamma,\dot\gamma) - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial v^i} \circ (\mathrm{id},\gamma,\dot\gamma)\right)\right](t) h^i(t)\,dt.$$

Here we use the Einstein summation convention*: we omit the sum symbol "$\Sigma_{i=1}^n$" when there are repeated indices, i.e. $v^i w_i = \sum_{i=1}^n v^i w_i$.*

Proof. First of all, since $U_x \subset \mathbb{R}^n$ is open $\gamma(t) + sh(t) \in U_x$ and $\dot\gamma(t) + s\dot h(t) \in V_v$ for all $t \in [a,b]$ if we take $|s|$ sufficiently small. We have that:

$$\begin{aligned}
\frac{d}{ds}\Big|_{s=0}\mathcal{S}(\gamma+sh) &= \lim_{s\to 0}\frac{1}{s}\int_a^b \mathcal{L}(t,\gamma(t)+sh(t),\dot\gamma(t)+s\dot h(t)) - \mathcal{L}(t,\gamma(t),\dot\gamma(t))\,dt \\
&= \int_a^b \frac{\partial\mathcal{L}}{\partial x^i}(t,\gamma(t),\dot\gamma(t))h^i(t) + \frac{\partial\mathcal{L}}{\partial v^i}(t,\gamma(t),\dot\gamma(t))\dot h^i(t)\,dt \\
&= \int_a^b \frac{\partial\mathcal{L}}{\partial x^i}(t,\gamma(t),\dot\gamma(t))h^i(t) - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial v^i} \circ (\mathrm{id},\gamma,\dot\gamma)\right)(t)h^i(t)\,dt + \cancel{\frac{\partial\mathcal{L}}{\partial v^i}(t,\gamma(t),\dot\gamma(t))h^i(t)\Big|_{t=a}^{t=b}}
\end{aligned}$$

□

In the literature, you will sometimes encounter the notation "$\delta\mathcal{S}$" or "$\frac{\delta\mathcal{S}}{\delta h}$" for $\frac{d}{ds}|_{s=0}\mathcal{S}(\gamma+sh)$.

Definition 2.17. *An* extremal of an action $\mathcal{S}$ *is a curve $\gamma \in C^1([a,b];U_x)$ such that*

$$\frac{d}{ds}\Big|_{s=0}\mathcal{S}(\gamma+sh) = 0$$

for all $h \in C_0^1((a,b);\mathbb{R}^n)$.

An extremal curve may also minimize the functional, i.e. $\mathcal{S}(\delta) \geq \mathcal{S}(\gamma)$ with $\delta(a) = \gamma(a)$ and $\delta(b) = \gamma(b)$, for all $\delta \in C^1([a,b]; U_x)$, but this need not always be the case. A curve γ is a *local minimum* if

$$\frac{d^2}{ds^2}\Big|_{s=0} \mathcal{S}(\gamma + sh) > 0$$

for all $h \in C_0^2((a,b); \mathbb{R}^n)$.

Lemma 2.15. *Let $f \in C^0([a,b]; \mathbb{R})$, such that for every $\phi \in C_0^\infty((a,b); \mathbb{R})$:*

$$\int_a^b f(t)\phi(t)\, dt = 0.$$

Then $f \equiv 0$.

Proof. We will prove the lemma by contradiction. Suppose there exists a $t_0 \in [a,b]$ such that $c := f(t_0) \neq 0$. Without loss of generality, assume that $c > 0$ (otherwise, just consider $-f$ instead of f). Then, by continuity of f, there exists a $\delta > 0$, such that

$$f(t) > \frac{c}{2}$$

in $(t_0 - \delta, t_0 + \delta)$.

To reach a contradiction, will will construct a ϕ that vanishes in $[a,b] \setminus (t_0 - \delta, t_0 + \delta)$ and is positive on $(t_0 - \delta, t_0 + \delta)$.

We consider first the bump function or mollifier $\eta \in C^\infty(\mathbb{R})$:

$$\eta(x) = \begin{cases} e^{-\frac{1}{1-|x|^2}} & \text{if } |x| < 1, \\ 0 & \text{if } |x| \geq 1, \end{cases}$$

Let $\phi(t) = \eta(\frac{2}{\delta}(t - t_0))$. Then, by continuity, there exists an $0 < \epsilon < \frac{\delta}{2}$, such that $\phi(t) > \frac{1}{2}\phi(t_0) = \frac{1}{2e}$ for all $t \in (t_0 - \epsilon, t_0 + \epsilon)$, so

$$\int_a^b f(t)\phi(t)\, dt = \int_{t_0 - \frac{\delta}{2}}^{t_0 + \frac{\delta}{2}} f(t)\phi(t)\, dt \geq \int_{t_0 - \epsilon}^{t_0 + \epsilon} f(t)\phi(t)\, dt \geq \frac{c\epsilon}{2e} > 0.$$

But this is in contradiction with $\int_a^b f(t)\phi(t)\, dt = 0$, so it must follow that $f \equiv 0$. □

Corollary 2.16. *A curve $\gamma \in C^1([a,b]; U_x)$ is an extremal of an action $\mathcal{S}$ if and only if*

$$\frac{\partial \mathcal{L}}{\partial x^i}(t, \gamma(t), \dot{\gamma}(t)) - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v^i}(t, \gamma(t), \dot{\gamma}(t)) = 0 \quad \text{for all } 1 \leq i \leq n \text{ and } t \in [a,b]. \tag{2.9}$$

Proof. Follows by combining Theorem 2.14 with Lemma 2.15, where we consider paths h with $h^i(t) = \phi(t)$, $\phi \in C_0^\infty([a,b]; \mathbb{R})$, and $h^j(t) = 0$ for $j \neq i$. □

The equations (2.9) are called the *Euler–Lagrange equations* corresponding to $\mathcal{L}$.

Example 2.7. *The arc length of paths $\underline{\gamma} \in C^1([a,b]; \mathbb{R}^n)$ is given by the action:*

$$\mathcal{S}(\underline{\gamma}) = \int_a^b \sqrt{|\dot{\underline{\gamma}}|^2(t)}\, dt.$$

Hence, if we assume that $\dot{\gamma}(t) \neq 0$ for all $t \in [a,b]$, then $\mathcal{S}$ is an action with Lagrangian $\mathcal{L} : \mathbb{R}_t \times \mathbb{R}^n_x \times (\mathbb{R}^n \setminus \{0\})_v \to \mathbb{R}$, with $\mathcal{L}(t, \mathbf{x}, \mathbf{v}) = \sqrt{|\mathbf{v}|^2}$, which is C^2. The solution to the Euler–Lagrange equations must correspond to extrema of the length. In Problem Sheet 4, you will show

that solutions to the Euler–Lagrange equation are straight lines or constant paths and they are in fact minima of the action.

The proper time of $\gamma \in C^1([a,b];\mathbb{R}^{n+1})$ *with* $m(\dot\gamma(t),\dot\gamma(t)) < 0$ *for all* $t \in [a,b]$ *is given by the action:*

$$\mathcal{S}(\gamma) = \int_a^b \sqrt{-m(\dot\gamma(t),\dot\gamma(t))}\, dt.$$

EXERCISE*: Show that extremals of proper time are maxima.*

When we considered Newton's equation, we saw that the equation does not in general remain valid with the same force functions $\mathbf{F}_i$ when performing a coordinate transformation. The change in the force function can be very complicated. Indeed, if this invariance property is satisfied with respect to Galilean coordinate transformations, we said that the force satisfies the Principle of Galilean Relativity.

The proposition below illustrates that in the case of the Euler–Lagrange equations, the change in the Lagrangian function is not so complicated. First, we will redefine slightly the domain of our Lagrangian, to allow for time-dependent coordinate transformations: $\mathcal{L} : U \times V \to \mathbb{R}$, where now $U, V \subseteq \mathbb{R}^{n+1}$ are open. Then the action takes the form: for $\gamma \in C^1([a,b];U)$:

$$\mathcal{S}(\gamma) = \int_a^b \mathcal{L}(\gamma(\tau),\dot\gamma(\tau))\, d\tau.$$

In this setting, the Euler–Lagrange equations take the following form: for all $0 \le \mu \le n$

$$\frac{\partial \mathcal{L}}{\partial x^\mu}(\gamma(\tau),\dot\gamma(\tau)) - \frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial v^i}(\gamma(\tau),\dot\gamma(\tau)) = 0. \tag{2.10}$$

EXERCISE: Prove that (2.10) are equivalent to $\frac{d}{ds}|_{s=0}\mathcal{S}(\gamma + sh) = 0$ for all $h \in C_0^\infty(\mathbb{R}^{n+1})$. If $\tau = t$, $V = \mathbb{R} \times W$, with $W \subseteq \mathbb{R}^n$ and if we restrict to spacetime paths satisfying $\gamma^0(\tau) = \tau$ (and $h^0(\tau) = \tau$)), then (2.10) are equivalent to (2.9) with respect to the Lagrangian $\mathcal{L}(\cdot, 1, \cdot) : U \times W \to \mathbb{R}$, i.e. we restrict to $v^0 = 1$ in the Lagrangian.

Proposition 2.17. *Let* $U, \widetilde{U} \subseteq \mathbb{R}^{n+1}$ *be open and let* $\Phi : U \to \widetilde{U}$ *be a coordinate chart, where we denote* $\Phi^\mu = \tilde{x}^\mu$*. Let* $V \subset \mathbb{R}^{n+1}$ *be open and denote*

$$\widetilde{V} := \left\{ \tilde{v} \in \mathbb{R}^{n+1} \,|\, \tilde{v} = D\Phi(x)v,\ (x,v) \in U \times V \right\}.$$

Consider the Lagrangians $\mathcal{L} : U \times V \to \mathbb{R}$ *and* $\widetilde{\mathcal{L}} : U \times \widetilde{V} \to \mathbb{R}$*, which is defined as follows:*

$$\widetilde{\mathcal{L}}(\tilde{x},\tilde{v}) = \mathcal{L}\left(\Phi^{-1}(\tilde{x}), D\Phi^{-1}(\tilde{x})\tilde{v}\right)$$

In particular, for any $\gamma \in C^1([a,b];U)$*:*

$$\widetilde{\mathcal{L}} \circ \left((\Phi\circ\gamma), \frac{d}{d\tau}(\Phi\circ\gamma)\right) = \mathcal{L} \circ \left(\gamma, \frac{d\gamma}{d\tau}\right).$$

Then (2.10) *hold with respect to* $\mathcal{L} : U \times V \to \mathbb{R}$ *along* $\gamma \in C^1([a,b];U)$ *if and only if for all* $1 \le i \le n$*:*

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial \tilde{x}^\mu}\left((\Phi\circ\gamma)(\tau), \frac{d}{dt}(\Phi\circ\gamma)(\tau)\right) - \frac{d}{d\tau}\frac{\partial \widetilde{\mathcal{L}}}{\partial \tilde{v}^\mu}\left((\Phi\circ\gamma)(\tau), \frac{d}{d\tau}(\Phi\circ\gamma)(\tau)\right) = 0.$$

Proof. This will follows from a more general discussion of Lagrangians on manifolds later in the course. □

For the sake of notational convenience, one sometimes drop the tilde in $\widetilde{\mathcal{L}}$ when changing coordinates. We will later also see that, in the context of spacetime manifolds $\mathcal{M}$, it is more natural to define Lagrangians as maps $\mathcal{L} : T\mathcal{M} \supseteq \mathcal{U} \to \mathbb{R}$, with $T\mathcal{M}$ the tangent bundle of the manifold $\mathcal{M}$ and $\mathcal{U}$ an open subset. Different *coordinate representations* of $\mathcal{L}$ are then related by the expression for $\widetilde{\mathcal{L}}$ in Proposition 2.17. In particular, we will consider manifolds of the form: $\mathcal{M} = \mathbb{R} \times \Sigma$, with Σ an n-dimensional manifold.

Example 2.8. *Let $\Psi : \mathbb{R}_t \times (0,\infty)_r \times (0,\pi)_\theta \times (0,2\pi)_\varphi \to \mathbb{R}_t \times \mathbb{R}^3_{\mathbf{x}}$, be defined as follows:*

$$\begin{aligned}
\Psi^0(t,r,\theta,\varphi) &= t,\\
\Psi^1(t,r,\theta,\varphi) &= r\cos\varphi\sin\theta\\
\Psi^2(t,r,\theta,\varphi) &= r\sin\varphi\sin\theta,\\
\Psi^3(t,r,\theta,\varphi) &= r\cos\theta.
\end{aligned}$$

Then $\Psi : \widetilde{U} := \mathbb{R}_t \times (0,\infty)_r \times (0,\pi)_\theta \times (0,2\pi)_\varphi \to \Psi(\mathbb{R}_t \times (0,\infty)_r \times (0,\pi)_\theta \times (0,2\pi)_\varphi) =: U$ is a diffeomorphism and $\Phi = \Psi^{-1} : U \to V$ corresponds to the spherical coordinate chart.

Consider the Lagrangian $\mathcal{L}(x,v) = \frac{1}{2}m|\mathbf{v}|^2 - u(|\mathbf{x}|)$ with $u \in C^2((0,\infty))$. Then

$$\widetilde{\mathcal{L}}((t,r,\theta,\varphi),w) = \mathcal{L}(\Psi(t,r,\theta,\varphi),(D\Psi)(t,r,\theta,\varphi)w) = \frac{1}{2}m(D\Psi(t,r,\theta,\varphi)w)^i(D\Psi(t,r,\theta,\varphi)w)_i - U(r)$$

where

$$(D\Psi)(t,r,\theta,\varphi)w = \begin{pmatrix} \partial_\mu\Psi^0(t,r,\theta,\varphi)w^\mu \\ \partial_\mu\Psi^1(t,r,\theta,\varphi)w^\mu \\ \partial_\mu\Psi^2(t,r,\theta,\varphi)w^\mu \\ \partial_\mu\Psi^3(t,r,\theta,\varphi)w^\mu \end{pmatrix} = \begin{pmatrix} w^0 \\ \cos\varphi\sin\theta w^r - r\sin\varphi\sin\theta w^\varphi + r\cos\varphi\cos\theta w^\theta \\ \sin\varphi\sin\theta w^r + r\cos\varphi\sin\theta w^\varphi + r\sin\varphi\cos\theta w^\theta \\ \cos\theta w^r - r\sin\theta w^\theta \end{pmatrix}$$

so, one can easily verify that:

$$(D\Psi(t,r,\theta,\varphi)w)^i(D\Psi(t,r,\theta,\varphi)w)_i = (w^r)^2 + r^2(w^\theta)^2 + r^2\sin^2\theta(w^\varphi)^2,$$

so

$$\widetilde{\mathcal{L}}((t,r,\theta,\varphi),w) = \frac{1}{2}m\left[(w^r)^2 + r^2(w^\theta)^2 + r^2\sin^2\theta(w^\varphi)^2\right] - u(r).$$

Let $\gamma \in C^1([a,b],U)$ with $\gamma^0(t) = t$. Writing $r(t) = (\Phi\circ\gamma)^1(t)$, $\theta(t) = (\Phi\circ\gamma)^2(t)$ and $\varphi(t) = (\Phi\circ\gamma)^3(t)$, we therefore obtain:

$$\widetilde{\mathcal{L}}(\gamma(t),\dot\gamma(t)) = \frac{m}{2}\left[\dot r^2(t) + r^2\dot\theta^2(t) + r^2(t)\sin^2\theta(t)\dot\varphi^2(t)\right] - u(r(t)).$$

2.5 Lagrangian mechanics

In the theorem below we show that we can arrive at Newton's equations by considering the Euler–Lagrange equations with respect to an appropriate Lagrangian.

Theorem 2.18 (Hamilton's principle of least action[10])**.** *Consider an N-particle system with conservative forces:*

$$m_A\ddot{\mathbf{x}}_A(t) = -(\nabla_{\mathbf{x}_A}V)(\mathbf{x}_1(t),\dots,\mathbf{x}_N(t)) \quad 1 \le A \le N, \tag{2.11}$$

[10]Since the extremal of the action need not be a minimum, the principle of least action is somewhat of a misnomer. A more fitting name would have been the principle of stationary action. The first formulation of this principle goes back to Maupertuis.

for some $V \in C^2(U)$, with $U \subset \mathbb{R}^{3N}$ open. Then a path $\underline{\gamma} : \mathbb{R} \supseteq I \ni t \mapsto (\mathbf{x}_1, \dots, \mathbf{x}_N)^T(t)$, with $I \subseteq \mathbb{R}$ an open interval, is a solution to (2.11) if and only it is an extremal of an action $\mathcal{S}$ on I with the Lagrangian $\mathcal{L} : \mathbb{R}^{3N} \times \mathbb{R}^{3N} \to \mathbb{R}$ defined as follows:

$$\mathcal{L}(t, \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N) = \frac{1}{2} \sum_{A=1}^{n} m_A |\mathbf{v}_A|^2 - V(\mathbf{x}_1, \dots, \mathbf{x}_N).$$

The terms $T(\dot{\mathbf{x}}_1(t), \dots, \dot{\mathbf{x}}_N(t)) := \frac{1}{2} \sum_{A=1}^{N} m_A |\dot{\mathbf{x}}_A|^2(t)$ are the kinetic energy of the system and $V(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t))$ isthe potential energy.

Proof. Consider a path $\underline{\gamma} : I \to U$. Then $\underline{\gamma}$ is an extremal between $\underline{\gamma}(t_1)$ and $\underline{\gamma}(t_2)$, with $t_1, t_2 \in I$ arbitrary, if and only if the E–L equations

$$\left[\frac{\partial \mathcal{L}}{\partial x_A^j}(t, \underline{\gamma}(t), \dot{\underline{\gamma}}(t)) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial v_A^j}(t, \underline{\gamma}(t), \dot{\underline{\gamma}}(t) \right) \right] = 0$$

hold on I for all $1 \leq j \leq 3$ and $A = 1, \dots, N$. Let

$$\mathcal{L}(\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{v}_1, \dots, \mathbf{v}_N) = \frac{1}{2} \sum_{A=1}^{N} m_A |\mathbf{v}_A|^2 - U(\mathbf{x}_1, \dots, \mathbf{x}_N).$$

Then the E–L equations are equivalent to:

$$-\partial_{x_A^j} U(\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)) - \frac{d}{dt}(m_A \dot{x}_A^j(t)) = 0,$$

which is equivalent to (2.11). □

The Lagrangian approach to mechanics has several key advantages over the Newtonian approach:

1. From Proposition 2.17, it follows that the Euler-Lagrange equations are "covariant" (maintain the same form) under very general coordinate changes. This is in contrast with Newton's equations (2.11), which are only covariant under Galilean coordinate transformations if the principle of Galilean Relativity holds. This means it is easier to work in coordinates that are naturally adapted to the physical problem under consideration, which need not be Galilean, without having to change the form of the equations.

2. Due to its coordinate independence, the Langrangian approach can be considered on manifolds (which cannot be equipped with a global coordinate chart). This makes it more natural to consider the motion of particles constrained to a submanifold.

3. The Lagrangian approach generalizes to the setting of equations of motions of *fields*, e.g. electromagnetic fields, and it provides moreover a natural choice for the definition of energy and momentum of fields.

Consider a coordinate chart $\Phi : \mathbb{R}^{3+1} \supseteq U \to \widetilde{U} \supseteq \mathbb{R}^{3+1}$. In the context of the motion of an N-particle system: $\underline{\gamma} : I \to U^{3N}$, we will refer to $q_A^i(t) := \Phi^i(t, \underline{\gamma}(t))$ as the *generalized coordinates* associated to the A-th particle at time t. Similarly, we will refer to $(p_A)_i(t) := \frac{\partial \widetilde{\mathcal{L}}}{\partial \dot{\widetilde{v}}_A^i}(t, \underline{\gamma}(t), \dot{\underline{\gamma}}(t))$, with $\widetilde{\mathcal{L}}$ the Lagrangian from Proposition 2.17, as *generalized momentum coordinates* of the Ath particle at time t. Finally, $(f_A)_i(t) = \frac{\partial \mathcal{L}}{\partial q_A^i}(t, \underline{\gamma}(t), \dot{\underline{\gamma}}(t))$ are the *generalized force coordinates* of the Ath particle at time t. The Euler–Lagrange equations then take the form:

$$\dot{(p_A)}_i(t) = (f_A)_i(t).$$

Example 2.9 (Free particle). *For a free particle, $V = 0$ and $T(\mathbf{v}) = \frac{1}{2}m|\mathbf{v}|^2$, the generalized coordinates are $q^i(t) = x^i(t)$, the generalized momenta are $p_i(t) = m\dot{x}^i(t)$ and the generalized forces are $f_i(t) = 0$.*[11] *Note that $\dot{p}_i = 0$, so the generalized momenta are therefore conserved in time and correspond to the components of the* linear momentum.

In the problem sheets, you will see that the solutions to the E–L equations corresponding to $\mathcal{L}(t, \mathbf{x}, \mathbf{v}) = \frac{1}{2}m|\mathbf{v}|^2$ must correspond to minimizers of the length between any two fixed points (and are therefore straight lines).

Example 2.10 (Particle with a central force). *Consider the inverse map of the coordinate chart on $\mathbb{R}^2$: $\Psi : (0,\infty)_r \times (0, 2\pi)_\varphi \to \mathbb{R}^2$, with $\Psi^1(r, \varphi) = r\cos\varphi$ and $\Psi^2(r, \varphi) = r\sin\varphi$. Denote $\Phi = \Psi^{-1}$. Consider the Lagrangian $\mathcal{L} : \mathbb{R} \times \mathbb{R}^2 \setminus \{0\} \times \mathbb{R}^2$, $\mathcal{L}(t, \mathbf{x}, \mathbf{v}) = \frac{1}{2}m|\mathbf{v}|^2 - u(|\mathbf{x}|)$. Then we can express:*

$$\widetilde{\mathcal{L}}(t, (r, \varphi), 1, w) = \mathcal{L}(\Psi(r, \varphi), D\Psi(r, \varphi)w) = \frac{1}{2}m((w^r)^2 + r^2(w^\varphi)^2) - u(r).$$

The generalized coordinates along a path $t \mapsto \underline{\gamma}(t)$ are $q^1(t) = \Phi^1(\underline{\gamma}(t)) =: r(t)$, $q^2(t) = \Phi^2(\underline{\gamma}(t)) =: \varphi(t)$. The generalized momenta are $p_r(t) = m\dot{r}(t)$ and $p_\varphi(t) = mr^2\dot{\varphi}(t)$ and they satisfy the equations:

$$\begin{aligned}\dot{p}_r(t) &= mr(t)\dot{\varphi}^2(t) - u'(r(t)),\\ \dot{p}_\varphi(t) &= 0.\end{aligned}$$

By the second equation, we immediately obtain conservation of p_φ in time, so we can write $mr^2\dot{\varphi} = L$, for some $L \in \mathbb{R}$. This corresponds precisely to the conservation of angular momentum in time; see (2.7)). Plugging this into the first equation gives:

$$m\ddot{r} = \frac{L^2}{mr^3} - u'(r).$$

Motivated by (2.10), we say introduce the following definition:

Definition 2.18. *Let $\Phi : \mathbb{R}^{n+1} \supseteq U \to V \subseteq \mathbb{R}^{n+1}$ be a coordinate chart. A coordinate function $\tilde{x}^i = \Phi^i$ is called* cyclic *if $\widetilde{\mathcal{L}}$, as defined in Proposition 2.17, is independent of $\tilde{x}^i$.*

It follows immediately by the Euler–Lagrange equations that the generalized momentum p_i corresponding to a cyclic generalized coordinate is conserved in time. This may be thought of as a generalization of the conservation of linear and angular momentum in Example 2.9 and 2.10, respectively.

2.6 Hamiltonian mechanics

We will now consider yet another reformulation of Newtonian mechanics called *Hamiltonian mechanics*. This approach can be motivated by its connection with quantum mechanics that we will see later in the course. It is also more convenient for making statements about complicated mechanical systems, like those arising in celestial mechanics, and it connects to the more general theory of dynamical systems. Hamiltonian mechanics moreover provides a natural path to statistical descriptions of motions of N particles.

Our first aim is to describe a solution to the Euler–Lagrange equations (2.9) $t \mapsto \underline{\gamma}(t) \in \mathbb{R}^n$ with a focus on the generalized momentum coordinates $p_i(t)$. The main object will be the Hamiltonian,

$$H(t, q^1, \dots, q^n, p_1, \dots, p_n) = \sup_{v \in \mathbb{R}^n} \langle v, p\rangle - \mathcal{L}(t, q^1, \dots, q^n, v^1, \dots, v^n),$$

which we will define precisely soon. In order to obtain an H that is well-defined, we will need to introduce the concept of *convexity* of subsets of $\mathbb{R}^n$ and of functions defined on convex subspaces.

[11] You may ignore the placement of indices (superscript/subscript) for now.

Definition 2.19. *A function $f : \mathbb{R} \supseteq I \to \mathbb{R}$, with I an interval, is* convex *if for all $x, y \in I$ and $t \in [0, 1]$*

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

The function is strictly convex *if the inequality is strict for $t \in (0, 1)$. Similarly, we say a function is (strictly) concave if the inequality is reversed.*

We can extend the notions of convexity/concavity to $f : \mathbb{R}^n \supseteq X \to \mathbb{R}$, with X a convex subset, i.e. if $x, y \in X$ then the line segment connecting x, y lies in X.

One can show that all convex functions on open convex subsets of $\mathbb{R}^n$ are locally Lipschitz continuous.

EXERCISE: Convince yourself of this when $n = 1$ by considering $x < w < z < y$ and comparing $\frac{f(z)-f(w)}{z-w}$ with $\frac{f(y)-f(x)}{y-x}$.

Definition 2.20. *Let $X \subseteq \mathbb{R}^n$ and $f : X \to \mathbb{R}$ be convex. Then the* Legendre transform *of f is defined as the function $f^* : \mathbb{R}^n \supseteq P \to \mathbb{R}$, with*

$$f^*(p) = \sup_{x \in X}(\langle x, p\rangle - f(x)),$$

with $P = \{y \in \mathbb{R}^n \mid \sup_{x \in X}(\langle x, y\rangle - f(x)) < \infty\}$.

In the proposition below we state some useful properties of the Legendre transform.

Proposition 2.19. *Let $X \subseteq \mathbb{R}^n$ be open and convex. Let $f \in C^2(X; \mathbb{R})$.*

(i) *f is convex if and only if $(D^2 f)(x)$ is non-negative definite at every $x \in X$, i.e. $(D^2 f)(x)(v, v) \geq 0$ for all $v \in \mathbb{R}^n$. It is strictly convex if and only if $(D^2 f)(x)$ is positive-definite at every $x \in X$, i.e. $(D^2 f)(x)(v, v) > 0$ for all $v \in \mathbb{R}^n$.*

(ii) *If f is strictly convex, then $(Df)^{-1} : \mathbb{R}^n \supseteq Df(X) \to X$ is well-defined and the Legendre transform $f^* : P \to \mathbb{R}$ with $P = Df(X)$ satisfies:*

$$f^*(p) = \langle (Df)^{-1}(p), p\rangle - f((Df)^{-1}(p)).$$

Furthermore, $(Df)^{-1}(p) = (Df^*)(p)$, or equivalently $(Df^*)^{-1}(x) = Df(x)$.

(iii) *If f is strictly convex, then $Df(X)$ is open and convex and f^* is convex. Furthermore, Legendre transform is an involution, i.e. $(f^*)^* = f$.*

Proof. (non-examinable)"(i)": We will first prove the statement for $X = [0, 1]$ and $n = 1$. Consider $\phi : [0, 1] \to \mathbb{R}$. Suppose that $\phi'' > 0$. Let $x, y \in [0, 1]$ with $x < y$ and let $z = (1-t)x + ty$. By the fundamental theorem of calculus:

$$\phi(z) - \phi(x) = \int_x^z \phi'(s)\, ds \leq \int_x^z \phi'(y)\, ds = \phi'(y)(x - z),$$
$$\phi(y) - \phi(z) = \int_z^y \phi'(s)\, ds \geq \int_y^z \phi'(y)\, ds = \phi'(y)(y - z).$$

Hence,

$$(1-t)(\phi(z) - \phi(x)) - t(\phi(y) - \phi(z)) \leq 0.$$

Using that $z = (1-t)x + ty$ and rearranging terms, we obtain:

$$\phi((1-t)x + ty) \leq (1-t)\phi(x) + t\phi(y).$$

If $\phi'' > 0$, then the above inequality becomes strict.

Now suppose that ϕ is C^2 and convex. By the inequalities above following from the fundamental theorem of calculus: for all $x < z < y$

$$\frac{\phi(x) - \phi(z)}{x - z} \leq \frac{\phi(y) - \phi(z)}{y - z},$$

which implies that $\phi'(x) \leq \phi'(y)$ for all $x < y$. This in turn implies that $\phi'' \geq 0$. The strictly convex case implies all inequalities are strict, so $\phi'' > 0$.

Now consider a general $f : X \to \mathbb{R}^n$ and define $\phi : [0,1] \to \mathbb{R}$ by $\phi(t) = f(x + t(y - x))$. Note that if ϕ is convex, then:

$$f((1-t)x + t(x-y)) = \phi(t) = \phi((1-t)\cdot 0 + t \cdot 1) \leq (1-t)\phi(0) + t\phi(1) = (1-t)f(x) + tf(y).$$

Similarly, if f is convex then for all $0 \leq t_1 \leq t_2 \leq 1$:

$$\begin{aligned}\phi((1-s)\cdot t_1 + st_2) = f(x + [(1-s)\cdot t_1 + st_2](y-x)) &= f((1-s)[x + t_1(y-x)] + s[x + t_2(y-x)]) \\ &\leq (1-s)f(x + t_1(y-x)) + sf(x + t_2(y-x)) = (1-s)\phi(t_1) + s\phi(t_2),\end{aligned}$$

so ϕ must be convex. Hence, f is (strictly) convex iff for all $x, y \in X$ ϕ is (strictly) convex.

By the chain rule, we have for all $t \in (0,1)$:

$$\begin{aligned}\phi'(t) &= (Df)(x + t(y-x))(y-x), \\ \phi''(t) &= (D^2 f)(x + t(y-x))(y-x, y-x).\end{aligned}$$

Hence, ϕ is C^2 in $(0,1)$ and $\phi'' \geq 0$ for all $x, y \in X$ iff f is non-negative definite. Since ϕ is convex iff $\phi'' \geq 0$ and ϕ is strictly convex iff $\phi'' > 0$, the desired properties of f follow.

"(ii)": We need to show that $Df : X \to \mathbb{R}^n$ is injective. Note that by the above, for all $x \neq y$:

$$(Df(y) - Df(x))(x - y) = \phi'(1) - \phi'(0) > 0,$$

where the final inequality follows from strict convexity of ϕ. If $Df(x) = Df(y)$, then the left-hand side must be zero, which is a contradiction, so Df must be injective.

Now let $g_p(x) = \langle p, x\rangle - f(x)$. Then $Dg_p(x) = p - Df(x)$. So $Dg_p(x) = 0$ iff $x = (Df)^{-1}(p) =: x_p$. Note moreover that $(D^2 g_p)(x) = -(D^2 f)(x)$. Since f is strictly convex, $(D^2 f)(x)$ is positive-definite, so $(D^2 g_p)(x)$ must be negative definite and g_p must be strictly concave. It therefore follows as above that $Dg_p : X \to \mathbb{R}^n$ must be injective, so x_p is the unique point for which $Dg_p(x) = 0$. Furthermore, for $\phi(t) = g_p(x_p + t(y - x_p))$, $\phi'(t) < 0$ implies that we cannot have $g_p(y) \geq g_p(x_p)$, so x_p is a global maximum for g_p, which concludes the desired formula for $f^*(p)$.

Finally, write $x_*(p) = (Df)^{-1}(p)$. Then:

$$\partial_{p_i} f^*(p) = \partial_{p_i}(p_j x_*^j(p) - f(x_*(p))) = x_*^i(p) + p_j \partial_{p_i}(x_*)^j(p) - \overbrace{\partial_{x^j} f(x_*(p))}^{=p_j} \partial_{p_i}(x_*)^j(p) = x_*^i(p),$$

so $(Df^*)(p) = x_*(p) = (Df)^{-1}(p)$.

"(iii)": If f is strictly convex, then $(D^2 f)(x)$ is positive-definite, which means that $(Df)^{-1}$ is C^1 by the inverse function theorem. Therefore openness of $Df(X)$ follows from continuity of $(Df)^{-1}$. Let $p_1, p_2 \in P = Df(X)$. Then

$$\begin{aligned}f^*(tp_1 + (1-t)p_2) &= \sup_{x \in X} \langle (1-t)p_1 + tp_2, x\rangle - f(x) \\ &= \sup_{x \in X} (1-t)(\langle p_1, x\rangle - f(x)) + t(\langle p_2, x\rangle - f(x)) \\ &\leq (1-t)\sup_{x \in X}(\langle p_1, x\rangle - f(x)) + t\sup_{x \in X}(\langle p_2, x\rangle - f(x)) \\ &= (1-t)f^*(p) + tf^*(p_2) < \infty.\end{aligned}$$

This implies that $(1-t)p_1+tp_2 \in P$, which means P is convex, and also shows that $f^*((1-t)p_1 + tp_2) \leq (1-t)f^*(p) + tf^*(p_2)$, which means that f^* is also convex. We are therefore allowed to take the Legendre transform of f^*:

$$(f^*)^*(q) = \langle q, (Df^*)^{-1}(q)\rangle - f^*((Df^*)^{-1}(q)).$$

Note moreover that $(Df^*)^{-1}(q) = Df(q)$, so

$$(f^*)^*(q) = \langle q, Df(q)\rangle - f^*(Df(q)) = \langle q, Df(q)\rangle - (\langle q, Df(q)\rangle - f(q)) = f(q).$$

□

Definition 2.21. *Consider a Lagrangian $\mathcal{L} : \mathbb{R}_t \times U \times \mathbb{R}^n_{\mathbf{v}} \to \mathbb{R}$ that is convex with respect to the velocity variable, with $U \subseteq \mathbb{R}^n$ open. Then the* Hamiltonian *$H : \mathbb{R} \times U \times P \to \mathbb{R}$ is defined as the Legendre transform $\mathcal{L}^*$ with respect to the velocity variable in $\mathbb{R}^n$:*

$$H(t,q,p) = \sup_{v\in\mathbb{R}^n} \langle v, p\rangle - \mathcal{L}(t,q,v),$$

with $P \subseteq \mathbb{R}^n$ a convex domain on which the Legendre transform is well-defined.

Suppose that $\mathcal{L}$ is C^2 and strictly convex in the velocity variable. Then by Proposition 2.19(ii):

$$H(t,q,p) = \langle v_*(t,q,p), p\rangle - \mathcal{L}(t,q,v_*(t,q,p)),$$

with $v_*(t,q,p) := (D\mathcal{L}(t,q,\cdot))^{-1}(p)$. Furthermore, we can take:

$$P = \bigcap_{(t,q)\in\mathbb{R}\times U} (D\mathcal{L}(t,q,\cdot))(\mathbb{R}^n).$$

Since each $D\mathcal{L}(t,q,\cdot))(\mathbb{R}^n)$ is convex, the above intersection must also be convex (Exercise: Check this.).

Theorem 2.20. *Let $\mathcal{L} : \mathbb{R}\times U\times\mathbb{R}^n \to \mathbb{R}$ be a C^2 Lagrangian that is strictly convex in the velocity variable, with $U \subseteq \mathbb{R}^n$ open. Then the n-dimensional system of Euler–Lagrange equations (2.9) along a path $t \mapsto q(t) \in U$,*

$$\frac{\partial\mathcal{L}}{\partial q^i}(t,q(t),\dot q(t)) - \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial v^i}(t,q(t),\dot q(t)) = 0,$$

are equivalent to the following $2n$-dimensional system of equations involving the associated Hamiltonian $H : \mathbb{R}\times U\times P\to\mathbb{R}$, with $P = \bigcap_{(t,q)\in\mathbb{R}\times U}(D\mathcal{L}(t,q,\cdot))(\mathbb{R}^n)$:

$$\dot p_i(t) = -\frac{\partial H}{\partial q^i}(t,q(t),p(t)), \tag{2.12}$$

$$\dot q^i(t) = \frac{\partial H}{\partial p_i}(t,q(t),p(t)), \tag{2.13}$$

provided $p_i(t) = \frac{\partial\mathcal{L}}{\partial v^i}(t,q(t),\dot q(t))$, i.e. $t\mapsto p_i(t)$ correspond to generalized momentum coordinates along $t\mapsto q(t)$ associated to $\mathcal{L}$.

Equations (2.12) and (2.13) are called the Hamilton equations.

Proof. Let $t\mapsto q(t)$ be a path. Define $p_i(t) = \frac{\partial\mathcal{L}}{\partial v^i}(t,q(t),\dot q(t))$. Define $v_*(t,q,p) := (\nabla_p H)(t,w,p)$. By Proposition 2.19(ii), we can express: $p = (\nabla_v\mathcal{L})(t,q,v_*(t,q,p))$. Applying this identity to $(q(t),p(t))$, we obtain $p_i(t) = \frac{\partial\mathcal{L}}{\partial v^i}(t,q(t),v_*(t,q(t),p(t)))$.

Combining the above, we obtain:

$$p_i(t) = \frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), v_*(t, q(t), p(t))) = \frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), \nabla_p H(t, q(t), p(t)))$$

Then we must have that

$$\frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), \dot{q}(t)) = \frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), \nabla_p H(t, q(t), p(t))). \tag{2.14}$$

Since the derivative of the map $v \mapsto \mathcal{L}(t, q, v)$ is invertible for all t and q, by Proposition 2.19, it is injective and the above identity implies that

$$\dot{q}^i(t) = v_*^i(t, q(t), p(t)) = \frac{\partial H}{\partial p_i}(t, q(t), p(t)).$$

Hence, the Hamilton equations for q^i must hold.

Since $H(t, q, p) = \langle v_*(t, q, p), p\rangle - \mathcal{L}(t, q, v_*(t, q, p))$, we also have by the chain rule that

$$\begin{aligned}\frac{\partial H}{\partial q^i}(t, q, p) &= \frac{\partial v_*^j}{\partial q^i}(t, q, p)p_j - \frac{\partial \mathcal{L}}{\partial q^i}(t, q, v_*(t, q, p)) - \frac{\partial \mathcal{L}}{\partial v^j}(t, q, v_*(t, q, p))\frac{\partial v_*^j}{\partial q^i}(t, q, p)\\ &= -\frac{\partial \mathcal{L}}{\partial q^i}(t, q, \nabla_p H(t, q, p)) + \frac{\partial^2 H}{\partial q^i \partial p^j}(t, q, p)\left[p_j - \frac{\partial \mathcal{L}}{\partial v^j}(t, q, \nabla_p H(t, q, p))\right]\\ &= -\frac{\partial \mathcal{L}}{\partial q^i}(t, q, \nabla_p H(t, q, p)),\end{aligned}$$

where we used that the term in square brackets vanishes as a result of the relation between p and $\nabla_v \mathcal{L}(t, q, v)$.

Now we evaluate the very left-hand side and very right-hand side of the above equation along the path $t \mapsto (q(t), p(t))$ and apply (2.14) to obtain:

$$\frac{\partial H}{\partial q^i}(t, q(t), p(t)) = -\frac{\partial \mathcal{L}}{\partial q^i}(t, q(t), \nabla_p H(t, q(t), p(t))) = -\frac{\partial \mathcal{L}}{\partial q^i}(t, q(t), \dot{q}(t)).$$

If we now suppose that $t \mapsto q(t)$ satisfies the Euler–Lagrange equations, then

$$\frac{\partial H}{\partial q^i}(t, q(t), p(t)) = -\frac{\partial \mathcal{L}}{\partial q^i}(t, q, \dot{q}(t)) = -\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), \dot{q}(t)) = -\dot{p}_i(t),$$

and we conclude that the Hamilton equations hold.

Conversely, suppose that the Hamilton equations hold, then we obtain

$$\dot{p}_i(t) = -\frac{\partial H}{\partial q^i}(t, q(t), p(t)) = \frac{\partial \mathcal{L}}{\partial q^i}(t, q, \dot{q}(t)),$$

which is equivalent to the Euler–Lagrange equations, using that $p_i(t) = \frac{\partial \mathcal{L}}{\partial v^i}(t, q(t), \dot{q}(t))$. □

Along a solution $t \to q(t)$ to the Euler–Lagrange equations corresponding to a system of N particles satisfying the Newton equations with a conservative force, the function $t \mapsto H(t, q(t), p(t))$ with $p_i(t) = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}(t, q(t), \dot{q}(t))$ gives the value of the total energy of the system at time t:

Proposition 2.21. *Let $\mathcal{L} : \mathbb{R} \times U \times \mathbb{R}^n \to \mathbb{R}$, with U open and $\mathcal{L}(t, q, v) = T(v) - V(q) = \frac{1}{2}|v|^2 - V(q)$ and $V \in C^2(U)$. Then:*

$$H(t, q, p) = \frac{|p|^2}{2m} + V(q) = T(v_*(t, q, p)) + V(q).$$

In particular, if $t \mapsto (q(t), p(t))$ is a solution to (2.12)–(2.13), then $H(q(t), p(t)) = E_q(t)$, i.e. the Hamiltonian evaluated along the motion agrees with its energy.

Proof. One can easily see that $\mathcal{L}$ is C^2 and $D^2\mathcal{L}(t,q,\cdot)$ is positive-definite, so $\mathcal{L}$ is strictly convex in v and $v_*(t,q,p)$ is well-defined. Furthermore, $D\mathcal{L}(t,q,\cdot)(\mathbb{R}^n) = \mathbb{R}^n$ for all $(t,q) \in \mathbb{R} \times U$. Hence, the Hamiltonian $H : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is well-defined and satisfies:

$$\begin{aligned} H(t,q,p) = p_i(v_*(t,q,p))^i - (T(v_*(q,p)) - U(q)) &= \frac{\partial \mathcal{L}}{\partial v^i}(t,q,v_*(t,q,p))(v_*(t,q,p))^i - \frac{1}{2}|v_*(t,q,p)|^2 + V(q) \\ &= T(v_*(t,q,p)) + U(q) = \frac{1}{2}|p|^2 + V(q). \end{aligned}$$

If $t \mapsto (q(t), p(t))$ satisfy (2.12)–(2.13), then $v_*(t, q(t), p(t)) = \dot{q}(t)$ from which $H(t, q(t), p(t)) = T(\dot{q}(t)) + V(q(t)) = E_q(t)$ follows. □

Conservation of energy (Proposition 2.11) takes the following form in the Hamiltonian setting:

Proposition 2.22. *Let $H : \mathbb{R} \times U \times P \to \mathbb{R}$, with $U, P \subseteq \mathbb{R}^n$ open, be C^1 with $\partial_t H(t,q,p) = 0$. If $t \mapsto (q(t), p(t))$ is a solution to the Hamilton equations (2.12)–(2.13) with respect to H, then the map $t \mapsto H(t, q(t), p(t))$ is constant.*

Proof.

$$\begin{aligned} \frac{d(H \circ (\mathrm{id}, q, p))}{dt}(t) = \frac{\partial H}{\partial q^i}(t, q(t), p(t))\dot{q}^i(t) + \frac{\partial H}{\partial p_i}(t, q(t), p(t))\dot{p}_i(t) + \partial_t H(t, q(t), p(t)) \\ = -\dot{p}_i(t)\dot{q}^i(t) + \dot{p}_i(t)\dot{q}^i(t) = 0. \end{aligned}$$

□

Note that the above proposition is *more general* than Proposition 2.11, since it also applies to time-independent Hamiltonians that are <u>not</u> of the form $\frac{|p|^2}{2} + V(q)$!

In the proposition below, we show that we can reduce the dimension of the system of Hamilton equations (the degrees of freedom) from $2n$ to $2(n-1)$, if we can identify a cyclic coordinate. This is a characteristic feature of the Hamiltonian formalism.

Proposition 2.23. *Let $H : \mathbb{R} \times U \times P \to \mathbb{R}$ be C^1, with $U, P \subseteq \mathbb{R}^n$ open, such that H is independent of the coordinate q^1. Then a solution to the corresponding Hamilton equations $t \mapsto (q(t), p(t))$ satisfies $p_1(t) = p_1(0)$ for all t and $t \mapsto (q^2, \dots, q^n, p_2, \dots p_n)(t)$ satisfies the Hamilton equations with respect to the reduced Hamiltonian $H_c : \mathbb{R} \times U_c \times P_c \to \mathbb{R}$ with $c = p_1(0)$ and $U_c = \{(q^2, \dots, q^n) \,|\, q \in U\}$, $P_c = \{(p_2, \dots, p_n) \,|\, p \in P\}$, which is defined as follows:*

$$H_c(t, q^2, \dots, q^n, p^2, \dots p^n) := H(t, q_1(0), q^2, \dots, q^n, c, p_2, \dots p_n).$$

Proof. It follows immediately from (2.12)–(2.13) that:

$$\begin{aligned} \dot{p_1}(t) &= -\frac{\partial H}{\partial q^1}(q(t), p(t)) = 0, \\ \dot{q_1}(t) &= \frac{\partial H}{\partial p_1}(q(t), p(t)), \end{aligned}$$

so $t \mapsto p_1(t)$ is constant and $p_1(t) = p_1(0)$ can be treated as a parameter. The Hamilton equations for H_c

$$\begin{aligned} \frac{\partial H_c}{\partial q^i} &= \frac{\partial H}{\partial q^i} = -\dot{p}_i \quad 2 \le i \le n, \\ \frac{\partial H_c}{\partial p_i} &= \frac{\partial H}{\partial p_i} = \dot{q}_i \quad 2 \le i \le n. \end{aligned}$$

After solving the above equations, we simply integrate the equation for $\dot{q}_1$ to obtain:

$$q_1(t) = q_1(0) + \int_0^t \frac{\partial H}{\partial p_1}(t, q_1(0), q_2(t), \dots, q_n(t), p_1(0), p_2(t), \dots, p_n(t))\, dt. \qquad \square$$

We refer to the $2n$-dimensional space of coordinates and momenta $(q^1, \dots, q^n, p_1, \dots p_n)$ as *phase space*. In the context of Hamiltonian mechanics on manifolds arising from Lagrangian mechanics, it is given by the cotangent bundle:

$$T^*\mathcal{M}.$$

The reason why the domain of H is taken to be $T^*\mathcal{M}$ and not for example $T\mathcal{M}$ is the following. We can generalize a (time independent) Lagrangian to be a map

$$\mathcal{L} : T\mathcal{M} \to \mathbb{R}.$$

The momentum map at $q \in \mathcal{M}$ can then be defined as follows: consider the following restriction of $\mathcal{L}$: $\mathcal{L}|_{T_q\mathcal{M}} : T_q\mathcal{M} \to \mathbb{R}$. Then we can define:

$$p(q)(\cdot) = d(\mathcal{L}|_{T_q\mathcal{M}})_v : T_q\mathcal{M} \cong T_v(T_q\mathcal{M}) \to \mathbb{R},$$

which is an element of $T_q^*\mathcal{M}$.

In coordinates $(q^1, \dots, q^n, v^1, \dots, v^1)$ in a neighbourhood of a point $(q, v) \in T\mathcal{M}$, we therefore obtain the relation:

$$p_i(q)(v) = \frac{\partial \mathcal{L}}{\partial v^i}(q, v).$$

The above definition of the momentum makes sense of the following phrase that is commonly encountered in the physics literature: "velocity is a vector because it transforms like a vector, and momentum is a covector because it transforms like a covector".

Note that we can therefore express the Legendre transform as follows:

$$H(q, p) = p(q)(v_*(q, p)) - \mathcal{L}(q, p),$$

using that $p(q) \in T_q\mathbb{R}^n$.

It is possible to study the Hamilton equations on more general even-dimensional manifolds. We will explore this in the next section.

2.7 Symplectic manifolds

A more geometric approach to studying Hamiltonian systems can be obtained by considering *symplectic manifolds*, which are even-dimensional manifolds $\mathcal{M}$ equipped with a symplectic form, which is a closed, non-degenerate 2-form $\omega \in \Omega^2(\mathcal{M})$.

This means: 1) at each $x \in \mathcal{M}$, ω_x is an anti-symmetric bilinear form, 2) let $v, w \in T_x\mathcal{M}$, if $\omega_x(v, w) = 0$ for all $w \in T_x\mathcal{M}$ and $x \in \mathcal{M}$, then $v = 0$ (non-degeneracy, and 3) $d\omega = 0$ (closedness).

The non-degeneracy and bilinearity of ω implies that for all vector fields $X \in \mathcal{T}(\mathcal{M})$, there exists a unique one form, denotes $X^\flat \in \Omega^1(\mathcal{M})$, such that $X^\flat(Y) = \omega(X, Y)$. The map $\flat : \mathcal{T}(\mathcal{M}) \to \Omega(\mathcal{M})$ is actually bijective, with inverse $\sharp : \Omega^1(\mathcal{M}) \to \mathcal{T}(\mathcal{M})$. In the setting of Riemannian and Lorentzian metrics later in the course, we will refer to analogous maps $\flat$ and $\sharp$ as *musical isomorphisms*.

Example 2.11. *The space $(\mathbb{R}^{2n}, \omega)$ with coordinates q^i and p_i, $i = 1, \dots, n$, and $\omega = \sum_{i=1}^n dq^i \wedge dp_i$ is a symplectic manifold. Given a vector field $X = \sum_{i=1}^n X^i \frac{\partial}{\partial q^i} + \tilde{X}^i \frac{\partial}{\partial p_i}$, we have that*

$$\omega(X, Y) = \sum_{i=1}^n dq^i(X)dp_i(Y) - dq^i(Y)dp_i(X) = \sum_{i=1}^n X^i dp_i(Y) - \tilde{X}^i dq^i(Y).$$

Since, $X^\flat(Y) = \omega(X, Y)$, we must have: $X^\flat = \sum_{i=1}^n -\tilde{X}^i dq^i + X^i dp_i$.

On any even-dimensional manifold, there exists coordinates so that locally ω takes the same form as in the example above. We will state the relevant theorem without proof.

Theorem 2.24. *(Darboux's theorem) Let $(\mathcal{M}, \omega)$ be a $2n$-dimensional symplectic manifold. At each point $x \in \mathcal{M}$, there exists a neighbourhood U_x which can be covered by a coordinate chart $(q^1, \dots, q^n, p_1, \dots, p_n)$ with respect to which ω takes the following form in U_x:*

$$\omega = \sum_{i=1}^{n} dq^i \wedge dp_i.$$

These coordinates are called: Darboux coordinates, symplectic coordinates *or* canonical coordinates.

Consider a function $H \in C^\infty(\mathcal{M})$, which we will call the Hamiltonian. A particularly important vector field on a symplectic manifold is the *Hamiltonian vector field* $X_H := (dH)^\sharp$. We can locally express with respect to Darboux coordinates (q^i, p_i): $dH = \sum_{i=1}^n \partial_{q^i} H dq^i + \partial_{p_i} H dp_i$. So by Darboux's Theorem combined with Example 2.11, we can express in these coordinates:

$$X_H = \sum_{i=1}^{n} (\partial_{p_i} H)\partial_{q^i} - (\partial_{q^i} H)\partial_{p_i}.$$

We refer to the triple $(\mathcal{M}, \omega, H)$ as a *Hamiltonian system.*

Recall that orbits of a vector field $X \in \mathcal{T}(\mathcal{M})$ are curves $\gamma : I \to \mathcal{M}$ such that $T_{\gamma(t)}\mathcal{M} \ni \dot\gamma(t) = X_{\gamma(t)}$. Now, we can give a completely geometric characterization of solutions to Hamilton's equations as orbits $t \mapsto \gamma(t)$ of the Hamiltonian vector field. Indeed, with respect to Darboux coordinates, we can represent γ by the map $t \mapsto (q^1(t), \dots, q^n(t), p_1(t), \dots, p_n(t))$ such that

$$\dot q^i(t) = (\partial_{p_i} H)(\gamma(t)), \tag{2.15}$$

$$\dot p_i(t) = -(\partial_{q^i} H)(\gamma(t)). \tag{2.16}$$

More generally, we define with $\Phi_t^H : \mathcal{M} \to \mathcal{M}$ the flow for time t with respect to the Hamiltonian vector field X_H, which sends $x \in M$ to $\gamma(t)$, where γ satisfies $\gamma(0) = x$ and $\dot\gamma(t) = (X_H)_{\gamma(t)} \in T_{\gamma(t)}\mathcal{M}$.

The triple $(\mathcal{M}, \omega, H)$ is called a *Hamiltonian system.*

Let $\{(q^i, p_i)\}_{1 \le i \le n}$ denote Darboux coordinates. Let $\Phi : \mathcal{M} \to \mathcal{M}$ be a diffeomorphism. Then the Hamilton equations (2.15)–(2.15) hold also with respect to the coordinates $\{(\tilde q^i, \tilde p_i)\}_{1 \le i \le n}$ associated to $\Phi : \mathcal{M} \to \mathcal{M}$ if and only if

$$\Phi^* \omega = \omega.$$

In this case, Φ is called a *symplectomorphism* and the transition functions between the coordinate charts $\{(q^i, p_i)\}_{1 \le i \le n}$ and $\{(\tilde q^i, \tilde p_i)\}_{1 \le i \le n}$ are called *canonical transformations.*

To connect with quantum mechanics, we will use our geometric reformulation of Hamiltonian systems to define the notion of a *Poisson bracket.*

Definition 2.22. *A Poisson bracket is a binary operation $\{\cdot, \cdot\} : C^\infty(\mathcal{M}) \times C^\infty(\mathcal{M}) \to C^\infty(\mathcal{M})$ defined as follows:*

$$\{f, g\} := \omega(X_f, X_g) = \omega((df)^\sharp, X_g) = df(X_g) = X_g(f).$$

The Poisson bracket with $\{f, g\}$ can be interpreted as measuring the rate of change of f with respect to the Hamiltonian flow corresponding to g. In particular, along $\gamma : I \to \mathcal{M}$ an orbit of X_H, we have that

$$\frac{d}{dt} f(\gamma(t)) = \{f, H\}(\gamma(t)).$$

With respect to Darboux coordinates, we have that

$$\{f, g\} = \sum_{i=1}^{n} \partial_{q^i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q^i} g.$$

In particular,

$$\{q^i, p_j\} = \delta^i_j.$$

Let $\rho : \mathbb{R}_t \times \mathcal{M} \to [0, \infty)$. Suppose that ρ stays constant along the Hamiltonian flow Φ^H_t. Then

$$0 = \frac{d}{dt}\rho(t, \gamma(t)) = (\partial_t \rho)(t, \gamma(t)) + \{\rho(t, \cdot), H\}(\gamma(t)).$$

Rearranging the above terms gives the *Liouville equation*:

$$(\partial_t \rho)(t, \cdot) = -\{\rho(t, \cdot), H\}. \tag{2.17}$$

The Liouville equation plays an important role in classical statistical mechanics.

We state without proof the key properties of Poisson brackets:

Proposition 2.25. *Let $(\mathcal{M}, \omega)$ be a symplectic manifold and let $f, g, h \in C^\infty(\mathcal{M})$. Then:*

(i) *(bilinearity) $\{f, g\}$ is linear in f and g over $\mathbb{R}$,*

(ii) *(antisymmetry) $\{f, g\} = -\{g, f\}$,*

(iii) *(Jacobi identity) $\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\} = 0$,*

(iv) $X_{\{f,g\}} = -[X_f, X_g]$.

In particular, the vector space $C^\infty(\mathcal{M})$ is a Lie algebra with respect to the Poisson bracket.

As we will see later in the quantum mechanics part of this course, position q^i and momentum p_i will become linear operators and the role of the Poisson bracket will be taken on by the commutator of linear operators, which also satisfies properties (i)–(iii) above.

The symplectic formulation has the advantage that *Liouville's theorem* reduces to a sequence of algebraic identities.

Theorem 2.26 (Liouville's Theorem)**.** *Let $(\mathcal{M}, \omega, H)$ be a Hamiltonian system. Then ω^n defines a volume form on $\mathcal{M}$,*[12] *which is preserved under the Hamiltonian flow, i.e.*

$$L_{X_H}(\omega^n) = 0,$$

with L_{X_H} the Lie derivative with respect to X_H and $\omega^n = \overbrace{\omega \wedge \ldots \wedge \omega}^{n}$. With respect to Darboux coordinates, we have that

$$\frac{1}{n!}\omega^n = dq^1 \wedge dp_1 \wedge dq^2 \wedge dp_2 \wedge \ldots \wedge dq^n \wedge dp_n.$$

Proof. We will apply *Cartan's "magic" formula*: for any $X \in \mathcal{T}(\mathcal{M})$

$$L_{X_H} = d \circ \iota_{X_H} + \iota_{X_H} \circ d,$$

[12] EXERCISE: Convince yourself of this, once you have become familiar with volume forms on general manifolds. Note that this implies in particular that any symplectic manifold is *orientable*.

where d is the exterior derivative and ι_Y, with $Y \in \mathcal{T}(\mathcal{M})$, denotes an interior product: recall, for $r \geq 1$ and $Y \in \mathcal{T}(\mathcal{M})$:

$$\iota_Y : \Omega^r(\mathcal{M}) \to \Omega^{r-1}(\mathcal{M}),$$
$$(\iota_Y \beta)(X_1, \ldots, X_{r-1}) = \beta(Y, X_1, \ldots, X_{r-1}).$$

Note that $d\omega^k = 0$ for all $k \geq 0$, so

$$L_{X_H}\omega^n = d(\iota_{X_H}\omega^n).$$

We have that $\iota_{X_H}\omega^n = \omega(X_H, \cdot) \wedge \omega^{n-1} = dH \wedge \omega^{n-1}$. Taking the exterior derivative d then gives zero.

It follows by an easy induction argument we can locally express in Darboux coordinates, $\frac{1}{n!}\omega^n = dp^1 \wedge dq^1 \wedge \ldots \wedge dp^n \wedge dq^n$. □

Consider the Hamiltonian system $(\mathbb{R}^{2n}, \sum_{i=1}^n dq^i \wedge dp_i, H)$ where Darboux coordinates are global and let $\Omega \subset \mathbb{R}^{2n}$ be bounded. Let $\Omega_t = \Phi_t^H(\Omega) \subset \mathbb{R}^{2n}$ denote the image of Ω under the Hamiltonian flow. Then Liouville's theorem, together with:

$$\left.\frac{d}{dt}\right|_{t=s} (\Phi_t^H)^*(\omega^n) = \left.\frac{d}{dt}\right|_{t=0} (\Phi_{t+s}^H)^*(\omega^n) = (\Phi_s^H)^* \left(\left.\frac{d}{dt}\right|_{t=0} (\Phi_t^H)^*(\omega^n) \right) = (\Phi_s^H)^* \left(L_X \omega^n\right)$$

implies that

$$\left.\frac{d}{dt}\right|_{t=s} \text{Volume}(\Omega_t) = \left.\frac{d}{dt}\right|_{t=s} \int_{\Omega_t} dq^1 dp_1 \ldots dq^n dp_n = \left.\frac{d}{dt}\right|_{t=s} \int_{\Omega} (\Phi_t^H)^* \left(\frac{\omega^n}{n!}\right)$$
$$= \int_{\Omega} \Phi_s^* \left(L_{X_H} \left(\frac{\omega^n}{n!} \right) \right) = 0,$$

so the volume of Ω_t is equal to the volume of Ω for all t (for which the flow is well-defined). Hence, the phase space volume is preserved under the Hamiltonian flow. This observation is one of the key results of the Hamiltonian approach!

Let $\rho : \mathbb{R}_t \times \mathcal{M} \to [0, \infty)$ be a non-negative solution to the Liouville equation, such that $\int_{\mathcal{M}} \rho(0, \cdot) \frac{\omega^n}{n!} = 1$, i.e. we can think of $\rho(0, \cdot)$ as a probability density at time $t = 0$. It follows by Liouville's theorem, that the probability of encountering a system in the phase space region $\Omega_t := \Phi_t^H(\Omega)$, $\int_{\Omega_t} \rho(t, \cdot) \frac{\omega^n}{n!}$, is invariant under time, i.e.

$$\frac{d}{dt} \int_{\Omega} \rho(t, \Phi_t^H(\cdot))(\Phi_t^H)^* \left(\frac{\omega^n}{n!} \right) = 0.$$

In particular,

$$\int_{\mathcal{M}} \rho(t, \cdot) \frac{\omega^n}{n!} = \int_{\mathcal{M}} \rho(0, \Phi_{-t}^H(\cdot))(\Phi_{-t}^H)^* \left(\frac{\omega^n}{n!} \right) = \int_{\mathcal{M}} \rho(0, \cdot) \frac{\omega^n}{n!} = 1.$$

Similarly, we can define the corresponding *Gibbs entropy* as follows:

$$S(t) = -\int_{\mathcal{M}} \rho(t, \cdot) \log \rho(t, \cdot) \, \frac{\omega^n}{n!}.$$

Since ρ and ω^n are preserved under the Hamiltonian flow, S is constant in time.[13]

Let $O \in C^\infty(\mathcal{M})$ represent a measurable quantity (for example, the total kinetic energy). Then we can define the expectation value of O at time t as follows:

$$\langle O \rangle(t) = \int_{\mathcal{M}} O\rho(t,\cdot)\frac{\omega^n}{n!}.$$

We will the following consequence version of "integrating by parts". Let $\alpha \in \Omega^m(\mathcal{N})$, $\mathcal{N}$ an m-manifold(with boundary), $f \in C^\infty(\mathcal{N})$ and $X \in \mathcal{T}(\mathcal{N})$. Assume that α vanishes on the boundary of $\mathcal{N}$ (if non-empty). By Cartan's magic formula and Stokes' theorem:

$$\int_{\mathcal{N}} X(f)\alpha = \int_{\mathcal{N}} L_X(f\alpha) - fL_X\alpha = \int_{\mathcal{N}} d\iota_X(f\alpha) + \overbrace{\iota_X(d\alpha)}^{=0} - fL_X\alpha = -\int_{\partial\mathcal{N}} f\iota_X\alpha - \int_{\mathcal{N}} fL_X\alpha.$$

By Liouville's equation and Liouville's theorem, and after integrating by parts in the above sense, assuming ρ decays suitably at infinity or the boundary of $\mathcal{M}$, we have that

$$\begin{aligned}
&\frac{d\langle O\rangle}{dt}(t) = \int_{\mathcal{M}} O\partial_t\rho(t,\cdot)\frac{\omega^n}{n!} = -\int_{\mathcal{M}} O\{\rho(t,\cdot),H\}\frac{\omega^n}{n!} \\
&= -\int_{\mathcal{M}} OX_H(\rho(t,\cdot))\frac{\omega^n}{n!} = \int_{\mathcal{M}} \rho(t,\cdot)L_{X_H}(O\frac{\omega^n}{n!}) = \int_{\mathcal{M}} \rho(t,\cdot)X_H(O)\frac{\omega^n}{n!} = \int_{\mathcal{M}} \{O,H\}\rho(t,\cdot)\frac{\omega^n}{n!} \\
&= \langle\{O,H\}\rangle(t).
\end{aligned}$$

We will encounter similar equations of motions when we study expectation values of observables in quantum mechanics.

As a corollary of Liouville's theorem, can obtain a remarkable result with far-reaching consequences, which really only uses the volume-preserving property of the Hamiltonian flow.

Corollary 2.27 (Poincaré's recurrence theorem)**.** *Let $(\mathcal{M},\omega,H)$ be a Hamiltonian system and let $\mathcal{K} \subset \mathcal{M}$ be a compact subset, such that $\Phi_t^H(\mathcal{K}) = \mathcal{K}$ for all $t \in \mathbb{R}$ for which $\Phi_t^H(\mathcal{K})$ is well-defined.*

Consider an arbitrary $x_0 \in \mathcal{K}$. Then for any neighbourhood U of x_0 and any $T > 0$, there exists a time $t_* > T$ such that the orbit corresponding to the Hamiltonian flow $t \mapsto x(t)$ with $x(0) = x_0$ satisfies $x(t_*) \in U$.

Proof. We first appeal to the following standard result: the flow corresponding to vector field on a compact manifold is global, i.e. defined for all times $t \in \mathbb{R}$. Since $\mathcal{K}$ is compact, $\Phi_t^H|_{\mathcal{K}}$ is defined for all $t \in \mathbb{R}$.

Now note that the volume of $\mathcal{K}$ is bounded with respect to the volume form ω^n. Indeed, consider around each point $x \in \mathcal{M}$ an open neighbourhood covered by Darboux coordinates with a bounded range. Then the volume of each neighbourhood is bounded with respect to ω^n. Since $\mathcal{K}$ is compact and is covered by a finite number of finite-volume Darboux neighbourhoods, it admits a finite subcover, so its volume is finite.

Let $U_t = \Phi_t^H(U)$. Suppose

$$U_t \cap U_s = \emptyset$$

[13] You may have encountered *increasing* entropies in statistical mechanics, which indicate a preferred time direction. Since the Liouville equation is invariant under time reflections, there is no preferred time direction. However, if we partition $\mathcal{M}$ into cells of finite volume and replace the probability density ρ with an appropriately averaged function over these cells, denoted $\overline{\rho}$, then the Gibbs entropy with respect to $\overline{\rho}$, denoted S_{cg}, the course-grained Gibbs entropy, can be shown to satisfy $S_{\text{cg}} \geq S$, using Jensen's inequality and the convexity of the function $x \mapsto x\log x$. So S_{cg} can increase in time. The picture to have in mind here is that a ball in phase space transported via Hamiltonian flow will maintain its volume, but need not maintain its shape, so it can in principle develop very thin filaments and cover more and more cells, which leads to an increase in course-grained entropy due to the finite resolution that we are applying to study the system.

for all $t, s \in \mathbb{R}$ with $t - s > T$. In particular, this means that for any sequence of times $\{t_i\}_{i\in\mathbb{N}}$ such that $t_{i+1} - t_i > T$ for all $i \in \mathbb{N}$, we have that $U_{t_i} \cap U_{t_j} = \emptyset$ for all $i, j \in \mathbb{N}$ with $i \neq j$.

By Liouville's theorem, the volume of each U_{t_i} equals the volume of U. Since the sets are disjoint and contained in $\mathcal{K}$, we therefore have that

$$\infty > \text{Volume}(\mathcal{K}) \geq \text{Volume}\left(\bigcup_{i\in\mathbb{N}} U_{t_i}\right) = \sum_{i\in\mathbb{N}} \text{Volume}(U_{t_i}) = \sum_{i\in\mathbb{N}} \text{Volume}(U) = \infty,$$

which is a contradiction.

There must therefore exist $s, t \in \mathbb{R}$ with $t - s > T$, such that $U_t \cap U_s \neq \emptyset$. Denote $t_* = t - s$, then

$$\Phi^H_{-s}(U_t \cap U_s) = U_{t_*} \cap U \neq \emptyset. \qquad \square$$

Poincaré's theorem says that for Hamiltonian systems restricted to a compact subset of phase space and for any initial configuration, the system will get arbitrarily close to its initial configuration after large enough time. A trivial example would be the harmonic oscillator. A counter-intuitive example would be the following: consider a box with a barrier containing only gas particles on one side of the barrier. Now remove the barrier and the gas particles will start filling the whole box. By Poincaré's recurrence theorem[14], however, all the gas particles will, after some time, move to one side of the barrier again. A resolution of this paradox is that Poincaré's recurrence theorem does not give an upper bound estimate for the time t_*. Therefore, t_* could in principle be much longer than the time scale for which the model is relevant (e.g. much greater than the age of the universe). We illustrate this below.

2.7.1 Poincaré vs box of gas (*non-examinable*)

Let $\Lambda = [0, L]^3$ describe a cubic box with sides of length L. Consider the Hamiltonian $H : \mathcal{M} =: (\Lambda \times \mathbb{R}^3)^N \to \mathbb{R}$, with $H(q, p) = \frac{|p|^2}{2m} + V(q)$, with bounded $V : \Lambda^N \to \mathbb{R}$, describing N particles of mass m interacting via conservative forces. We will assume perfectly reflecting walls.[15]

Let $\Lambda_{\frac{1}{2}} = [0, \frac{L}{2}] \times [0, L]^2$ describe exactly half of the box. Assume that $q(0) \in \Lambda_{\frac{1}{2}}$, i.e. the particles are initially all situated in one halve of the box. Let $E = H(q(0), p(0))$ be the total initial energy of all the particles and let $\Sigma_E = \{(q, p) \in \mathcal{M} \,|\, H(q, p) = E\}$. Since the total energy is conserved under Hamiltonian flow, $(q, p)(t) \in \Sigma_E$ for all $t \in \mathbb{R}$. Now let

$$A = \{(q, p) \in (\Lambda_{\frac{1}{2}} \times \mathbb{R}^3) \cap \Sigma_E\}.$$

Then A describes all points in phase space corresponding to all particles being contained in $\Lambda_{\frac{1}{2}}$ with total energy E. Let $D \subseteq \Sigma_E$. Consider the ratio of the volume of D to the volume of Σ_E, measured with respect the the restricted volume form on the $2n - 1$-dimensional manifold Σ_E:[16]

$$\mu_E(D) = \frac{\text{Vol}_E(D)}{\text{Vol}_E(\Sigma_E)}.$$

In the special case where $V \equiv 0$, the case of free particles, the phase space volume ratio corresponding to a restriction to $\Lambda_{\frac{1}{2}}$ is unaffected by the restriction to Σ_E, so we obtain $\mu_E(A) = 2^{-N}$,

[14] When the position space is bounded we also have boundedness in momentum space, assuming boundedness of the potential energy and conservation of energy. We can therefore restrict to a compact subset of phase space.

[15] This ensures that the momentum is reflected in the planes spanned by the sides of the cube, when the particle reaches the sides. Note that reflections are volume preserving maps, so Poincaré's recurrence theorem remains applicable.

[16] After you have defined measures in the mathematics part of the courses, you can note that $\mu_E : \mathcal{B}(\Sigma_E) \to \mathbb{R}$ defines a measure, with $\mathcal{B}(\Sigma_E)$ the Borel sigma algebra of Σ_E.

since that is the ratio of the volume of $\Lambda^N_{\frac{1}{2}}$ to Λ^N. More generally, note that:

$$\Sigma_E = \left\{(q,p) \in \mathcal{M} \,|\, |p| = \sqrt{2m(E - V(q))}\right\}.$$

Let $d\sigma_{\mathbb{S}^{3N-1}}$ be the volume form corresponding to integration over the unit round $3N-1$-sphere $\mathbb{S}^{3N-1}$. Let c_{3N-1} denote the area of $\mathbb{S}^{3N-1}$. After changing to spherical coordinates, we therefore obtain:

$$\mathrm{Vol}_E(D) = \int_D |p|_{\Sigma_E}|^{3N-1} d\sigma_{\mathbb{S}^{3N-1}} dq = (2m)^{\frac{3N-1}{2}} c_{3N-1} \int_{\{q \in \Lambda^N \,|\, (q,p) \in D\}} (E - V(q))^{\frac{3N-1}{2}}\, dq$$

and we can moreover write:

$$\mathrm{Vol}_E(\Sigma_E) = \mathrm{Vol}_E(A) + (2m)^{\frac{3N-1}{2}} c_{3N-1} \int_{\{q \notin \Lambda^N_{\frac{1}{2}}\}} |E - V(q)|^{\frac{3N-1}{2}}\, dq.$$

Hence, we can express

$$\mu_E(A) = \frac{1}{1 + \frac{\int_{\{q \notin \Lambda^N_{\frac{1}{2}}\}} |E-V(q)|^{\frac{3N-1}{2}}\, dq}{\int_{\{q \in \Lambda^N_{\frac{1}{2}}\}} |E-V(q)|^{\frac{3N-1}{2}}\, dq}}$$

and estimate:

$$\mu_E(A) \leq \frac{1}{1 + (2^N - 1)\frac{\min_{q \notin \Lambda_{\frac{1}{2}}} |E-V(q)|^{\frac{3N-1}{2}}}{\max_{q \in \Lambda_{\frac{1}{2}}} |E-V(q)|^{\frac{3N-1}{2}}}}.$$

If we assume that $E > \delta^{-1} \max_{q \in \Lambda} |V(q)|$ for some $\delta < 1$, which can be interpreted as the statement that most of the energy of the particles is kinetic, then:

$$\mu_E(A) \leq \frac{1}{1 + (2^N - 1)\left(\frac{1+\delta}{1-\delta}\right)^{\frac{3N-1}{2}}} = (2(1 + O(\delta))^{-N}(1 + NO(\delta)).$$

We will work with $\mu_E(A) \leq 2^{-N}$ for the sake of simplicity, as it will not change the main conclusion.

By the Poincaré recurrence theorem Corollary 2.27, we know that for all T we can find a t_* such that $(q,p)(t_*) \in A$.[17] Our goal is to try to say something about the size of t_*.

Let τ denote a fixed time scale, which we take to be the shortest time scale at which the configurations of particles can meaningfully change. This can, for example, be taken at the order of an average collision time, in the case of non-trivial potentials. As we will see, the choice of τ will not matter for our conclusions.

Consider the Hamiltonian flow corresponding to time steps of size τ:
$F := \Phi^H_\tau : \mathcal{M} \to \mathcal{M}$. In other words, we imagine taking snapshots of the system at small fixed time intervals. Let

$$n_{\mathrm{rec}}(q,p) = \min\{n \in \mathbb{N}_1 \,|\, F^n(q,p) \in A\},$$

i.e. $n_{\mathrm{rec}}(q,p)$ is the minimum number iterations of F (greater than 0) required for (q,p) initially in A to get back to A. If the minimum is taken over the empty set, we set $n_{\mathrm{rec}}(q,p) = \infty$. By the Poincaré recurrence theorem, the set $A^* = \{(q,p) \in A \,|\, n_{\mathrm{rec}}(q,p) = \infty\}$ must have zero volume (Exercise: Convince yourself of this.) Denote with $t_{\mathrm{rec},}(q,p) := n_{\mathrm{rec}}(q,p)\tau$ the recurrence time for an initial configuration (q,p) (in time-steps τ).

[17] Note that Σ_E is a compact subset of phase space!

We define the *mean recurrence time* $t_{\text{mean},A}$ as the average of $t_{\text{rec,A}}$ over all $(q,p) \in A$:

$$t_{\text{mean}} := \frac{1}{\mu_E(A)} \int_A t_{\text{rec}}\, d\mu_E,$$

where $d\mu_E$ denotes the induced volume form on Σ_E described above. We can interpret t_{mean} as the expectation value of the recurrence time, averaged over all possible initial configurations (ensembles) in $\Lambda_{\frac{1}{2}}$.

Now we appeal to *Kac's Lemma*, which states that (for general measure-preserving maps F):

$$\int_A n_{\text{rec}}\, d\mu_E = \mu_E(\Sigma_E) - \mu_E(A^*).$$

Since $\mu_E(A^*) = 0$, we therefore have that

$$t_{\text{mean}} = \frac{1}{\mu(A)} \int_A n_{\text{rec}}\, d\mu_E = \frac{1}{\mu(A)} \tau \geq 2^N \tau.$$

Note that this is the average recurrence time over all possible configurations in A. It does not yet tell us if this time is also *typical* for an initial configuration in A. Indeed, it could for example be that for most of phase space the recurrence time is much smaller than t_{mean}, but for a small region of phase space it is much, much larger than t_{mean}, so that on average we obtain t_{mean}.

We will now assume further that our Hamiltonian system is *ergodic*, which means that it reaches all elements of phase space Σ_E under the Hamiltonian flow. This can be shown to imply (by *Birkhoff's ergodic theorem*) that for any integrable function $O : \Sigma_E \to \mathbb{R}$ and for *almost every* $(q,p) \in A$:[18]

$$\lim_{n\to\infty} \frac{1}{n-1} \sum_{k=0}^{n} O(F^k(q,p)) = \lim_{T\to\infty} \frac{1}{T} \int_0^T O(\Phi_t^H(q,p))\, dt = \frac{1}{\mu_E(\Sigma_E)} \int_{\Sigma_E} O\, d\mu_E,$$

i.e. the average of any observable in time eventually equals the average over Σ_E (this is closely related to the statement that "all microstates are equiprobable after a sufficiently long time").[19]

Let us take $O = t_{\text{rec}}$. Then, we conclude that for almost every $(q,p) \in A$:

$$\lim_{n\to\infty} \frac{\tau}{n} \sum_{k=0}^{n-1} n_{\text{rec}}(F^k(q,p)) = t_{\text{mean}}.$$

The above identity can be interpreted as the statement that the time average of the number of time steps required to get back to A for almost every (q,p) in A is equal to t_{mean} as time goes to infinity.

If we take the cube to have sides of $1\,m$ and we use that $N \approx 10^{25}$ (for $1\,m^3$ of ideal gas at room temperature), we get that $t_{\text{mean}} \approx 2^{10^{25}} \tau$. This is such a large number, that calling it astronomical is not giving it enough justice. It is so large, that it does not really matter what time scale τ we pick. For boxes of gas, the Poincaré theorem is therefore not relevant for describing dynamics at relevant time scales.

[18] "Almost every" here has a precise mathematical meaning: in the complement of a set of *zero measure* (with respect to μ_E).

[19] You may think of this as a motivation for the study of *microcanonical ensembles* in statistical mechanics. Indeed, let $\rho_0 = \rho(0,\cdot)$ be the initial data for a probability density that solves the Liouville equation. Then $\rho(t,x) = \rho_0(\Phi_{-t}(x))$ since ρ is constant on integral curves of X_H, so by Birkhoff's ergodic theorem applied with Φ_{-t}^H instead of Φ_t^H, we obtain: $\lim_{T\to\infty} \frac{1}{T} \int_0^T \rho(t,x)\, dt = \lim_{T\to\infty} \frac{1}{T} \int_0^T \rho_0(\Phi_{-t}(x))\, dt = \frac{1}{\mu_E(\Sigma_E)} \int_{\Sigma_E} \rho_0\, d\mu_E$. Since $\int_{\Sigma_E} \rho_0\, d\mu_E = 1$, the time-average of $\rho(t,x)$ therefore converges to the constant $\frac{1}{\mu_E(\Sigma_E)}$ at almost every x. The probability of finding the system in any "microstate" (small, finite volume of phase space) therefore becomes equal when considering the time average. Proving the ergodicity of a system can however be highly non-trivial.

2.8 Lagrangian mechanics on manifolds

In this section, we will deal with Lagrangian mechanics involving constraints. We will present how to deal with constraints using the language of manifolds and a bit of geometry, and we will show how this relates to Lagrange multipliers.

Consider a C^2 Lagrangian $L : \mathbb{R}_t \times \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$. Let $n \in \mathbb{N}$, $n < N$ and let

$$\mathcal{M}_t = \{\mathbf{x} \in \mathbb{R}^N \,|\, G(t, \mathbf{x}) = 0\},$$

with $G : \mathbb{R}_t \times \mathbb{R}^N \to \mathbb{R}^{N-n}$ a smooth function. If for all $t \in \mathbb{R}$, $D_{\mathbf{x}}G(t, \mathbf{x})$ (derivative matrix with respect to spatial derivatives) is surjective, i.e. it has maximum rank $N - n$ at each $(t, \mathbf{x})$ with $\mathbf{x} \in \mathcal{M}_t$, then each $\mathcal{M}_t$ is an n-dimensional *embedded submanifold* of $\mathbb{R}^N$ for each t as a corollary of the Rank Theorem (which itself is related to the Inverse Function Theorem). This means that the inclusion maps $\iota_t : \mathcal{M}_t \to \mathbb{R}^N$ are smooth embeddings. Equivalently, the vectors $\{\nabla_{\mathbf{x}} G_A(t, \mathbf{x})\}_{1 \leq A \leq N-n}$ are non-zero and linearly independent for each $t \in \mathbb{R}$ and $\mathbf{x} \in \mathcal{M}_t$. The gradients $\{\nabla_{\mathbf{x}} G_A\}_{1 \leq A \leq N-n}$ are normal to $\mathcal{M}_t$. We refer to G as a *constraint function*.

It can moreover be shown that the level sets $\mathcal{M}_t$ and $\mathcal{M}_s$ are diffeomorphic for every $t, s \in \mathbb{R}$.[20]

We will refer to the family of level sets $\{\mathcal{M}_t\}_{t \in \mathbb{R}}$ satisfying the above conditions as a *holonomic constraints*. We say a system satisfies *non-holonomic constraints* if the constraints cannot be described by holonomic constraints (for example G could also depend on a velocity variable).

The above discussion motivates the consideration of Lagrangians on manifolds $\mathbb{R} \times \mathcal{M}$:

$$\mathcal{L} : T(\mathbb{R} \times \mathcal{M}) \to \mathbb{R},$$

with $T(\mathbb{R} \times \mathcal{M}) = \bigcup_{p \in \mathcal{M}} \{p\} \times T_p\mathcal{M}$ the tangent bundle of $\mathbb{R} \times \mathcal{M}$ (a disjoint union of tangent spaces).

Defining $\mathcal{L}$ as a map on $T(\mathbb{R} \times \mathcal{M})$ has the further advantage that its manifestly coordinate invariant, so it directly incorporates the desired transformation properties when we change coordinates. We will assume for the sake of simplicity that $\mathcal{L}$ is smooth. Note that the corresponding derivative map $d\mathcal{L}$ is a 1-form on the tangent bundle $T(\mathbb{R} \times \mathcal{M})$, i.e. $d\mathcal{L} \in \Omega^1(T(\mathbb{R} \times \mathcal{M}))$.

Given a smooth coordinate chart (U, ψ) with $U \subseteq \mathbb{R} \times \mathcal{M}$, we can express: $\psi = (q^0 = t, q^1, \ldots, q^n) : U \to \mathbb{R}^{n+1}$. Recall that for $q \in U$, we can express any $v_q \in T_q(\mathbb{R} \times \mathcal{M})$ as follows:

$$v_q = v_q^\mu (\partial_{q^\mu})_q,$$

with $(\partial_{q^\mu})_q$ the basis of coordinate vectors at p associated to the chart ψ. Recall also that for $\pi : T(\mathbb{R} \times \mathcal{M}) \to \mathbb{R} \times \mathcal{M}$, with $\pi(q, v_q) = q$ (the projection map), the pair $(\pi^{-1}(U), \phi)$ defines a coordinate chart if $\phi : \pi^{-1}(U) \to \mathbb{R}^{2(n+1)}$,

$$\phi(q, v_q) = (\psi(q), v_q^0, \ldots, v_q^n).$$

The *coordinate representation of* $\mathcal{L}$ is the map $\widehat{\mathcal{L}} : \mathbb{R}^{2(n+1)} \supseteq \phi(\pi^{-1}(U)) \to \mathbb{R}$ defined by

$$\widehat{\mathcal{L}}(q^0, \ldots, q^n, v^0, \ldots, v^n) = \mathcal{L}(\phi^{-1}(q^0, \ldots, q^n, v^0, \ldots, v^n)).$$

We can also express on $TU = \pi^{-1}(U)$:

$$d\mathcal{L} = \partial_{q^\mu} \mathcal{L} dq^\mu + \partial_{v^\mu} \mathcal{L} dv^\mu,$$

with q^μ and v^μ the components of the chart ϕ on TU.

[20] EXERCISE: Show this. Hint: consider the flow associated to the following curves. Let $x \in \mathcal{M}_t$ and consider curves satisfying $\gamma_x : I \to \mathbb{R}^N$ and $\gamma_x(0) = x$ and $G(t + \tau, \gamma_x(\tau)) = 0$.

Suppose $\tilde{\psi}$ is another coordinate chart on U and $\tilde{\phi}$ is the corresponding coordinate chart on $\pi^{-1}(U)$. Let $\widetilde{\mathcal{L}} : \tilde{\phi}(\pi^{-1}(U)) \to \mathbb{R}$ denote the corresponding coordinate representation of $\mathcal{L}$. Then we can express:

$$\widetilde{\mathcal{L}}(\tilde{q}^0, \dots, \tilde{q}^n, \tilde{v}^0, \dots, \tilde{v}^n) = \mathcal{L}(\tilde{\phi}^{-1}(\tilde{q}^0, \dots, \tilde{q}^n, \tilde{v}^0, \dots, \tilde{v}^n)) = \widehat{\mathcal{L}}((\phi \circ \tilde{\phi}^{-1})(\tilde{q}^0, \dots, \tilde{q}^n, \tilde{v}^0, \dots, \tilde{v}^n)) \tag{2.18}$$

Denote the corresponding transition function with $\Phi = \tilde{\psi} \circ \psi^{-1} : \mathbb{R}^{n+1} \supseteq \psi(U) \to \tilde{\psi}(U) \subseteq \mathbb{R}^{n+1}$. Then

$$(\phi \circ \tilde{\phi}^{-1})(\tilde{q}^0, \dots, \tilde{q}^n, \tilde{v}^0, \dots, \tilde{v}^n) = \begin{pmatrix} \Phi^{-1}(\tilde{q}^0, \dots, \tilde{q}^n), D\Phi^{-1}(\tilde{q}^0, \dots, \tilde{q}^n) \begin{pmatrix} \tilde{v}^0 \\ \vdots \\ \tilde{v}^n \end{pmatrix} \end{pmatrix}.$$

Theorem 2.28. *Let $\mathcal{M}$ be a smooth n-dimensional manifold and consider the coordinate chart (U, ψ), with $\psi = (t = q^0, q^1, \dots, q^n) : U \to \mathbb{R}^{n+1}$. We consider spacetime paths $\gamma \in C^1([a,b]; U)$, such that $\gamma^0(t) = t$. Let $\mathcal{L} \in C^\infty(T(\mathbb{R} \times \mathcal{M}))$. Consider the action $\mathcal{S}(\gamma) = \int_a^b \mathcal{L}(\gamma(t), \dot{\gamma}(t))\, dt$. Then γ is an extremal of $\mathcal{S}$ if and only if*

$$\frac{\partial \mathcal{L}}{\partial q^i}(\gamma(t), \dot{\gamma}(t)) - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v^i} \circ (\gamma, \dot{\gamma})\right)(t) = 0. \tag{2.19}$$

for all $1 \leq i \leq n$ and $t \in [a,b]$.

Proof. We consider the coordinate representation $\mathcal{L} \circ \phi^{-1}$ on $\phi(\pi^{-1}(U))$, with $\phi : \pi^{-1}(U) \to \mathbb{R}^{2(n+1)}$, $\phi(q, v_q) = (q^0, \dots, q^n, v_q^0, \dots, v_q^n)$, where $v_q = v_q^\mu \partial_{q^\mu}|_q$ and apply the argument from the $\mathcal{M} = \mathbb{R}^n$ setting, using that the consideration of spacetime paths γ with $\gamma^0(t) = t$ is equivalent to considering paths in space (with no time component). □

The above proposition, together with (2.18) implies in particular Proposition 2.17.

Recall that the constraint manifolds $\mathcal{M}_t \cong \mathcal{M}_0 =: \mathcal{M}$ are embedded submanifolds. Denote with $\Phi_t : \mathcal{M} \to \mathcal{M}_t$ a diffeomorphism between $\mathcal{M}$ and $\mathcal{M}_t$. Consider the following map:
$u : \mathbb{R}_t \times \mathcal{M} \to \mathbb{R}_t \times \mathbb{R}^N$, such that $u(t, \mathbf{x}) = (t, \iota(\Phi_t(\mathbf{x})))$, with $\iota : \mathcal{M}_t \to \mathbb{R}^N$ the inclusion map. Then u defines a smooth embedding of $\mathbb{R} \times \mathcal{M}$ into $\mathbb{R} \times \mathbb{R}^N$, i.e. u is an immersion (the derivative map du_p is injective for all $p \in \mathbb{R} \times \mathcal{M}$) and u is a homeomorphism to its image. Furthermore, $u(t, \mathbf{x}) \in \mathcal{M}_t$ for all $t \in \mathbb{R}$.

We can relate $\mathcal{L}$ to a different Lagrangian $L : \mathbb{R}_t \times \mathbb{R}^N \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$, such that for $(q, v) \in T(\mathbb{R}_t \times \mathcal{M})$:

$$\mathcal{L}(q, v) = L(u_*(q, v)).$$

where

$$\begin{aligned} u_* &: T(\mathbb{R}_t \times \mathcal{M}) \to T(\mathbb{R}_t \times \mathbb{R}^N) \cong \mathbb{R}_t \times \mathbb{R}^N \times \mathbb{R}_t \times \mathbb{R}^N, \\ u_*(q, v_q) &= (u(q), du_q(v_q)), \\ du_q(v_q) &= \partial_{q^\mu} u^A(q) v_q^\mu (\partial_{x^A})_{u(q)} \cong (\partial_{q^\mu} u^0(q) v_q^\mu, \partial_{q^\mu} u^1(q) v_q^\mu, \dots, \partial_{q^\mu} u^N(q) v_q^\mu), \end{aligned}$$

with respect to a coordinate chart $\{q^\mu\}_{0 \leq \mu \leq n}$ on $U \subseteq \mathbb{R}_t \times \mathcal{M}$ and Cartesian coordinates $\{x^A\}_{0 \leq A \leq N}$ on $\mathbb{R}_t \times \mathbb{R}^N$.

If $L(x, v) = \frac{1}{2} m |\mathbf{v}|^2 - U(\mathbf{x})$, then

$$\mathcal{L}(q, v) = \frac{1}{2} \sum_{A=1}^{N} m (du_q(v))_A (du_q(v))_A - U(u^1(q), \dots, u^N(q)).$$

Define $g_t \in \Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M})$ (section of $(0,2)$ tensor bundle) by $(g_t)_{\mathbf{q}}(v_1, v_2) = \sum_{A=1}^{N} (du_{(t,\mathbf{q})}(1, v_1))_A (du_{(t,\mathbf{q})}(1, v_2))_A$. Then g_t is a positive-definite symmetric (0,2)-tensor field and hence $(\mathcal{M}, g_t)$ can be viewed as a 1-parameter family of Riemannian manifolds.

Hence, we obtain in this case

$$\mathcal{L}((t,\mathbf{q}),(1,v)) = \frac{1}{2} m (g_t)_{\mathbf{q}}(v,v) - U(u^1(t,\mathbf{q}), \dots, u^N(t,\mathbf{q})).$$

Now, we return to the study of Lagrangians on $\mathbb{R}_t \times \mathbb{R}^N \times \mathbb{R} \times \mathbb{R}^N$ with constraints $G(t,\mathbf{x}) = 0$. By applying the chain rule, the equations (2.19) imply the following equations for L: with respect to a coordinate chart $\{q^\mu\}_{0 \le \mu \le n}$ on $U \subseteq \mathbb{R}_t \times \mathcal{M}$ and Cartesian coordinates $\{x^\alpha, w^\alpha\}_{0 \le \alpha \le N}$ on $T(\mathbb{R}_t \times \mathbb{R}^N)$, we have that the Euler–Lagrange equations with respect $\mathcal{L}$ for a path $\gamma \in C^1(U)$ give

$$\sum_{A=1}^{N} \left[\frac{\partial L}{\partial x^A}(u(\gamma(t)), \frac{d}{dt}(u \circ \gamma)(t)) - \frac{d}{dt}\left(\frac{\partial L}{\partial w^A}(u(\gamma(t)), \frac{d}{dt}(u \circ \gamma)(t) \right) \right] \partial_{q^i} u^A(\gamma(t)) = 0 \quad \forall \quad \forall 1 \le i \le n,$$

where in the above summation $1 \le A \le N$.

Note that we used here that $v \mapsto du_q(v)$ is a linear map, so

$$\partial_{v^i}(L(u(q), du_q(v)) = (\partial_{w^A} L)(u(q), du_q(v)) \partial_{v^i} du_q^A(v) = (\partial_{w^A} L)(u(q), du_q(v)) \partial_{q^i} u^A(q)$$

and hence:

$$\begin{aligned} \frac{d}{dt}\left[\partial_{v^i}(L\left(u(\gamma(t)), \frac{d}{dt}(u \circ \gamma)(t) \right) \right] = \frac{d}{dt}\left[(\partial_{w^A} L)\left(u(\gamma(t)), \frac{d}{dt}(u \circ \gamma)(t) \right) \right] \partial_{q^i} u^A(\gamma(t)) \\ + (\partial_{w^A} L)\left(u(\gamma(t)), \frac{d}{dt}(u \circ \gamma)(t) \right) \partial_{q^j} \partial_{q^i} u^A(\gamma(t)) \dot{\gamma}^j(t) \end{aligned}$$

Furthermore, we used that

$$\begin{aligned} \partial_{q^i}(L(u(q), du_q(v)) &= (\partial_{x^A} L)(u(q), du_q(v)) \partial_{q^i} u^A(q) + (\partial_{w^A} L)(u(q), du_q(v)) \partial_{q^i} du_q^A(v) \\ &= (\partial_{x^A} L)(u(q), du_q(v)) \partial_{q^i} u^A(q) + (\partial_{w^A} L)(u(q), du_q(v))(\partial_{q^i} \partial_{q^j} u^A(q)) v^j. \end{aligned}$$

This implies that the term in square brackets can be interpreted as the components of a vector in $\mathbb{R}^N$ that lies in the kernel of the matrix with components $A_{iA} = \partial_{q^i} u^A(\gamma(t))$. Since this kernel is spanned by $\{\nabla G^1(u(q), du_q(v)), \dots, \nabla G^{N-n}(u(q), du_q(v))\}$, which are orthogonal to $\mathcal{M}$, we can express the Euler–Lagrange equations as follows:

$$\left[\frac{\partial L}{\partial x^A}(t, \mathbf{x}(t), 1, \dot{\mathbf{x}}(t)) - \frac{d}{dt}\left(\frac{\partial L}{\partial w^A}(t, \mathbf{x}(t), 1, \dot{\mathbf{x}}(t)) \right) \right] = \sum_{j=1}^{N-n} \lambda^j(t) \partial_{x^A} G^j(t, \mathbf{x}(t)) \quad \forall\, 1 \le A \le N.$$

The functions λ^j are called *Lagrange multipliers*. Note that these can be interpreted as N equations for N unknowns $q^1(t), \dots, q^n(t), \lambda^1(t), \dots, \lambda^{N-n}(t)$. The above equations are equivalent to the standard Euler–Lagrange equations on $\mathbb{R}^N$ for the modified Lagrangian $\mathscr{L} : \mathbb{R} \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^{N-n} \to \mathbb{R}$:

$$\mathscr{L}(t, \mathbf{x}, \mathbf{v}, \lambda^1, \dots, \lambda^{N-n}) = L(t, \mathbf{x}, 1, \mathbf{v}) - \sum_{j=1}^{N-n} \lambda^j G^j(t, \mathbf{x})$$

coupled to the constraint equation $G(t, \mathbf{x}(t)) = 0$ (or equivalently, $\frac{\partial \mathscr{L}}{\partial \lambda^j}(t, \mathbf{x}(t)) = 0$). We then consider the paths $t \mapsto (\mathbf{x}(t), \lambda^1(t), \dots, \lambda^{N-n}(t))$, which solve the Euler–Lagrange equations with respect to $\mathbf{x}$ and $\mathbf{v}$ and $G(t, \mathbf{x}(t)) = 0$. In other words, the system of equations to consider is now:

$$\frac{\partial \mathscr{L}}{\partial x^A}(t, \mathbf{x}(t), \dot{\mathbf{x}}(t), \lambda(t)) - \frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial v^A}(t, \mathbf{x}(t), \dot{\mathbf{x}}(t), \lambda(t)) \right) = 0, \quad A \in \{1, \dots, N\},$$

$$\frac{\partial \mathscr{L}}{\partial \lambda^A}(t, \mathbf{x}(t), \dot{\mathbf{x}}(t), \lambda(t)) = 0, \quad A \in \{1, \dots, N-n\}.$$

We can interpret $\lambda^j G^j(t, \mathbf{x})$ as as kind of (time-dependent) potential corresponding to a *constraint force*, which takes the following form along a path:

$$\mathbf{F}_{\text{constraint}}(t, \mathbf{x}(t)) = -\lambda^j(t) \nabla_{\mathbf{x}} G^j(t, \mathbf{x}(t))$$

which points in the normal direction to the submanifold determined by G and can be interpreted as keeping the particle constrained to the surface.

We see therefore that the effect of a particles moving in a non-trivial geometry can be interpreted as an additional force in a larger ambient space, and vice versa! Having seen this correspondence between forces and geometry will perhaps remove some of the magic when we pass to the geometric setting of Einstein's theory of gravity and represent idealized, free-falling observers by geodesics in a spacetime manifold.[21]

Example 2.12. *Consider two particles with masses m_1 and m_2, attached to each other via an inextendible, massless string of length l over a massless pulley and initially at rest with negative z-coordinates. You may assume the radius of the pulley is R and that there is no friction between the rope and the pulley. We assume that constant gravitational forces $\mathbf{F}_i = -m_i g \hat{\mathbf{z}}$ act on the particles. Such a set-up is called an* Atwood machine*; see Figure 3.*

We would like to determine the motion of each particle. As the forces acting on the particles point all point in the $\hat{\mathbf{z}}$-direction, the total dimension of the problem is 2, with the coordinates (z_1, z_2) denoting the positions of the particles along the $\hat{\mathbf{z}}$-axis with the centre of the pulley sitting at $z = 0$. Note that the potential is given by $U(z_1, z_2) = mgz_1 + mgz_2$.

Since the length of the rope is constant, we have the following constraint: $z_2 = c - z_1$, with c a constant equal to $-\ell + \pi R$, the part of the string not touching the pulley. The coordinate on our 1-dimensional submanifold $\{z_1 + z_2 - c = 0\}$ is therefore $q = z^1$. The Lagrangian on $\{z_1 + z_2 - c = 0\}$ can then be expressed as follows:

$$\mathcal{L}(q, v) = \frac{1}{2} m_1 v^2 + \frac{1}{2} m_2 v^2 - m_1 g q - m_2 g (c - q).$$

The Euler Lagrange equation is:

$$(m_1 + m_2)\ddot{q} = -(m_1 - m_2)g$$

and hence, plugging in the initial conditions gives

$$q(t) = z_1(0) - \frac{1}{2} \frac{m_1 - m_2}{m_1 + m_2} g t^2$$

for t suitably small, so that $q(t) \leq 0$ (and we do not have to deal with the string detaching from the pulley). Finally, we have that $z_2(t) = c - q(t)$.

Example 2.13. *Consider a free particle in $\mathbb{R}^3$ of mass $m = 1$ that is constrained to a unit round 2-sphere $\mathbb{S}^2$. With respect to spherical coordinates (θ, φ), the induced Riemannian metric $g \in \Gamma(T^*\mathbb{S}^2 \otimes T^*\mathbb{S}^2)$ can be expressed as:*

$$g = d\theta^2 + \sin^2\theta d\varphi^2 = d\theta \otimes d\theta + \sin^2\theta d\varphi \otimes d\varphi.$$

The Lagrangian $\mathcal{L} : T\mathbb{S}^2 \to \mathbb{R}$ then takes the following expression with respect to a local coordinate chart q^i on $\mathbb{S}^2$:

$$\mathcal{L}(q, v) = \frac{1}{2} g_{ij}(q) v^i v^j.$$

[21] Though the actual equations of motion in Einstein's theory of gravity remain rather magical.

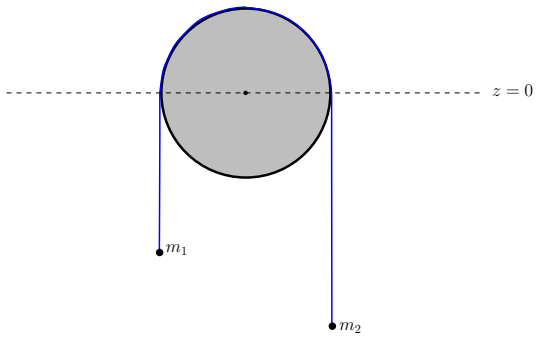


Figure 3: The Atwood machine.

The Euler–Lagrange equations give the following equation for paths $\gamma : I \to \mathbb{S}^2$ *in space*

$$\frac{1}{2}\partial_i g_{jk}(\gamma(t))\dot{\gamma}^j(t)\dot{\gamma}^k(t) - \frac{d}{dt}(g_{ij}(\gamma(t))\dot{\gamma}^j(t)) = 0.$$

By the chain rule, the above equation is equivalent to

$$0 = g_{ij}\ddot{\gamma}^j + \partial_k g_{ij}\dot{\gamma}^j\dot{\gamma}^k - \frac{1}{2}\partial_i g_{jk}\dot{\gamma}^j\dot{\gamma}^k = g_{ij}(\ddot{\gamma}^j + \Gamma^j_{mk}\dot{\gamma}^m\dot{\gamma}^k),$$

with

$$\Gamma^j_{mk} = \frac{1}{2}(g^{-1})^{jn}(\partial_m g_{kn} + \partial_k g_{mn} - \partial_n g_{mk}).$$

The numbers Γ^j_{mk} *are called the* Christoffel symbols. *They are intimately connected to* g *and we will later see them when we discuss natural notions of derivatives on manifolds equipped with metrics. We will see that these Euler–Lagrange equations are precisely the equations of an* affinely parametrized geodesic.

3 Lagrangian field theory

We have seen that Newton's equations, a system of ODE with solutions corresponding to paths

$$\gamma : [a, b] \to \mathbb{R}^{3N},$$

representing dynamical *particles*, can be obtained by extremizing an action of the form

$$\mathcal{S}(\gamma) = \int_a^b \mathcal{L}(t, \gamma(t), \dot{\gamma}(t))\, dt.$$

We will see that the equations of motion of *fields*, like the electric and magnetic fields, which we will interpret as tensor fields on $\mathbb{R}^{n+1}$ and on more general spacetime manifolds, can also be obtained by extremizing an action with an appropriate Lagrangian. In this case, the equations are PDE (partial differential equations) instead of ODE. Since fields are maps with as their domain spacetime regions, rather than time intervals, we need to make the following schematic form of the action precise:

$$\mathcal{S}(\text{field}) = \int_{\mathbb{R}^{n+1}} \mathcal{L}(\text{spacetime point, field, derivative of field}) \,(\text{volume form}).$$

To be able to define this action cleanly and to be able to easily generalize to the setting of manifolds, we take a detour and introduce some geometric concepts. In particular, it is convenient to have a suitable coordinate-independent notion of the *derivative of a tensor field*.

3.1 Preliminaries: metric, connections and volume forms

For $(r, s) \in (\mathbb{N}_0 \times \mathbb{N}_0) \backslash \{(0, 0)\}$, an (r, s)-tensor fields is an element of the following space of sections:

$$\mathcal{T}^{(r,s)}(\mathcal{M}) := \Gamma(T^{(r,s)}(\mathcal{M})) = \Gamma(\underbrace{T\mathcal{M} \otimes \ldots \otimes T\mathcal{M}}_{r \text{ times}} \otimes \underbrace{T^*\mathcal{M} \otimes \ldots \otimes T^*\mathcal{M}}_{s \text{ times}})$$

We moreover denote $T_x^{(r,s)}\mathcal{M} = \underbrace{T_x\mathcal{M} \otimes \ldots T_x\mathcal{M}}_{r \text{ times}} \otimes \underbrace{T_x^*\mathcal{M} \otimes \ldots T_x^*\mathcal{M}}_{s \text{ times}}$.

We will define $\mathcal{T}^{(0,0)}(\mathcal{M}) := C^\infty(\mathcal{M})$. In a slight abuse of notation, we will identify the space of (r, s)-tensor fields with the following space:

$$\mathcal{T}^{(r,s)}(\mathcal{M}) = \left\{ T : \underbrace{\Omega^1(\mathcal{M}) \times \ldots \Omega^1(\mathcal{M})}_{r \text{ times}} \times \underbrace{\mathcal{T}(\mathcal{M}) \times \ldots \times \mathcal{T}(\mathcal{M})}_{s \text{ times}} \to C^\infty(\mathcal{M}) \,|\, T \text{ multilinear over } C^\infty(\mathcal{M}) \right\}.$$

There is a natural identification between these two spaces. In one direction, this identification goes as follows. To any

$$T \in \Gamma(T^{(r,s)}(\mathcal{M})),$$

we associate $\widetilde{T} : (\Omega^1(\mathcal{M}))^r \times (\mathcal{T}(\mathcal{M}))^s$ defined as follows:

$$\widetilde{T}(\omega_1, \ldots, \omega_r, X_1, \ldots, X_s)(x) := T_x((\omega_1)_x, \ldots, (\omega_r)_x, (X_1)_x, \ldots, (X_s)_x)$$

for $\omega_i \in \Omega^1(\mathcal{M})$ and $X_i \in \mathcal{T}(\mathcal{M})$. This identification is explored further in the problem sheets.

Definition 3.1. *A (pseudo-Riemannian)* metric *(tensor field) is a* $(0, 2)$*-tensor field* $g \in \mathcal{T}^{(0,2)}(\mathcal{M})$ *that satisfies the additional two properties:*

- *(symmetry)* $g(X,Y) = g(Y,X)$ *for all* $X, Y \in \mathcal{T}(\mathcal{M})$,
- *(non-degeneracy) if* $g(X,Y) = 0$ *for all* $Y \in \mathcal{T}(\mathcal{M})$, *then* $X = 0$.

We say g *is a* Riemannian metric *(tensor field) if moreover*

- *(positive-definiteness)* $g(X,X) \geq 0$ *with equality if and only if* $X = 0$.

We call the pair $(\mathcal{M}, g)$ *a pseudo-Riemannian manifold.*

To define a Lorentzian metric (tensor field) we first make the following observation. The following map is bilinear and non-degenerate:

$$g_x : T_x\mathcal{M} \times T_x\mathcal{M} \to \mathbb{R},$$
$$g_x(v,w) = g(V,W)(x), \quad V_x = v, W_x = w.$$

Pick a basis $\{e_\mu\}$ of $T_x\mathcal{M}$. Let G be the matrix with coefficients $G_{\mu\nu} = g_x(e_\mu, e_\nu)$. Then, by the symmetry and non-degeneracy of g, G is invertible and symmetric. This means we can diagonalize G and that its eigenvalues are non-zero.

We define the *signature* (n_+, n_-) of g_x as follows: n_+ is the total number of positive eigenvalues and n_- is the total number of negative eigenvalues of G. This definition only makes sense if the signature is a quantity that is invariant under a choice of basis. That is to say, if we define $\tilde{G}$ as the matrix with coefficients $\tilde{G}_{\mu\nu} = g_x(\tilde{e}_\mu, \tilde{e}_\nu)$, where $\{\tilde{e}_\mu\}$ is a different basis, then we need to show that n_+ and n_- of $\tilde{G}$ are the same as n_+ and n_- of G. This follows from *Sylvester's Law of Inertia* (Exercise).[22]

For a Riemannian metric g, we have that $n_- = 0$ for all $x \in \mathcal{M}$.

Definition 3.2. *A metric* g *is* Lorentzian *if* $n_- = 1$ *for all* $x \in \mathcal{M}$. *The pair* $(\mathcal{M}, g)$ *is called a* Lorentzian manifold.

Every manifold admits a Riemannian metric, but not every manifold $\mathcal{M}$ admits a Lorentzian metric. For example, the sphere $\mathbb{S}^4$ does not admit a Lorentzian metric.[23]

[Exercise: Manifolds of the form $\mathcal{M} = \mathbb{R} \times \Sigma$ always admit a Lorentzian metric. *Hint*: You may use that Σ always admits a Riemannian metric.]

Definition 3.3. *A vector* $v \in T_x\mathcal{M}$ *is:*

- timelike *if* $g_x(v,v) < 0$,
- spacelike *if* $g_x(v,v) > 0$,
- null *if* $g_x(v,v) = 0$.

A vector field $X \in \mathcal{T}(\mathcal{M})$ *is timelike/spacelike/null if* X_x *is timelike/spacelike/null for all* $x \in \mathcal{M}$. *A smooth path* $\gamma : [a,b] \to \mathcal{M}$ *is timelike/spacelike/null if* $\dot{\gamma}(\tau)$ *is timelike/spacelike/null for all* $\tau \in [a,b]$.

Definition 3.4. *A spacetime is a triple* $(\mathcal{M}, g, [T])$, *where* $(\mathcal{M}, g)$ *is a Lorentzian manifold and* $[T]$ *is a* time orientation, *which is defined as follows:* $T \in \mathcal{T}(\mathcal{M})$ *is timelike and*

$$[T] := \{X \in \mathcal{T}(\mathcal{M}) \,|\, g(X,T) < 0\}.$$

[22] Sylvester's Law of Inertia: if A is an $n \times n$ symmetric matrix with n_+ positive eigenvalues and n_- negative eigenvalues, then, for any $n \times n$ matrix B, the symmetric matrix BAB^T will also have n_+ positive eigenvalues and n_- negative eigenvalues.

[23] By the so-called Hairy Ball Theorem, the sphere $\mathbb{S}^4$ does not admit a vector field that is non-vanishing everywhere. If we could equip $\mathbb{S}^4$ with a Lorentzian metric, then once can show that there must exist a vector field that does not vanish everywhere, which is a contradiction. Conversely, if a manifold admits a non-vanishing vector field X, we can construct the following Lorentzian metric: $g = -2\frac{X^\flat \otimes X^\flat}{g(X,X)} + h$, with h a Riemannian metric on $\mathcal{M}$ and where $X^\flat$ is a 1-form dual to X that we define below.

3.1.1 Musical isomorphisms: "raising and lowering indices"

On a manifold equipped with a metric $(\mathcal{M}, g)$, we can use the metric g to identify vector fields and 1-forms.

Let $X \in \mathcal{T}(\mathcal{M})$. Then we define

$$X^\flat := g(X, \cdot) \in \Omega^1(\mathcal{M}).$$

The symbol $\flat$ is called "flat" and is inspired by musical notation. Indeed $g(X, \cdot) : \mathcal{T}(\mathcal{M}) \to C^\infty(\mathcal{M})$ is multilinear over $C^\infty(\mathcal{M})$ so it defines a 1-form. Observe thatWe therefore have the existence of a map:

$$\flat : \mathcal{T}(\mathcal{M}) \to \Omega^1(\mathcal{M})$$

such that $\flat(X) = X^\flat$. Note that the associated maps $\flat_x : T_x\mathcal{M} \to T_x^*\mathcal{M}$, $\flat_x(v) = g_x(v, \cdot)$, with $x \in \mathcal{M}$, are linear maps between two vector spaces of equal dimension. Furthermore, by the non-degeneracy of g, $\flat_x$ must be injective. By the rank–nullity theorem of linear algebra, we therefore have that:

$$\dim \ker \flat_x + \dim \operatorname{ran} \flat_x = \dim T_x\mathcal{M}$$

and hence $\dim \operatorname{ran} \flat_x = \dim T_x^*\mathcal{M}$, so $\flat_x$ must also be surjective. It therefore has a well-defined inverse $\sharp_x : T_x^*\mathcal{M} \to T_x\mathcal{M}$ at each $x \in \mathcal{M}$ and we can make sense of the associated map:

$$\sharp : \Omega^1(\mathcal{M}) \to \mathcal{T}(\mathcal{M}).$$

For $\omega \in \Omega^1(\mathcal{M})$, we denote $\omega^\sharp := \sharp(\omega)$. The maps $\flat$ and $\sharp$ are called musical isomorphisms.

It is instructive to investigate the above maps with respect to a coordinate chart $(U, \{x^\mu\})$. We can then express:

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu =: g_{\mu\nu} dx^\mu dx^\nu,$$

with $g_{\mu\nu} \in C^\infty(U)$. Let $X = X^\alpha \partial_{x^\alpha} \in \mathcal{T}(U)$ and $Y = Y^\alpha \partial_{x^\alpha} \in \mathcal{T}(U)$. Then we can express in U:

$$X^\flat(Y) = g_{\mu\nu} X^\mu Y^\nu.$$

hence, $X^\flat = (g_{\nu\mu} X^\nu) dx^\mu = (g_{\mu\nu} X^\nu) dx^\mu$. In particular,

$$\left(\frac{\partial}{\partial x^\mu} \right)^\flat = g_{\mu\beta} dx^\beta.$$

We will denote the components of $X^\flat$ with an index in the subscript, i.e.

$$X_\mu := g_{\mu\nu} X^\nu.$$

For this reason, it is said that the map $\flat$, "lowers indices".

We now define $\mathfrak{g} \in \mathcal{T}^{(2,0)}(\mathcal{M})$ in the following way: for all $\omega, \theta \in \Omega^1(\mathcal{M})$,

$$\mathfrak{g}(\omega, \theta) := g(\omega^\sharp, \theta^\sharp).$$

We can express in U:

$$\mathfrak{g} = \mathfrak{g}^{\mu\nu} \partial_{x^\mu} \otimes \partial_{x^\nu} =: \mathfrak{g}^{\mu\nu} \partial_{x^\mu} \partial_{x^\nu}.$$

By the definition of $\mathfrak{g}$, we have that:

$$g_{\mu\nu} = g(\partial_{x^\mu}, \partial_{x^\nu}) = \mathfrak{g}((\partial_{x^\mu})^\flat, (\partial_{x^\nu})^\flat) = g_{\mu\alpha} g_{\nu\beta} \mathfrak{g}(dx^\alpha, dx^\beta) = g_{\mu\alpha} g_{\nu\beta} \mathfrak{g}^{\alpha\beta}$$

and hence

$$g_{\nu\beta} \mathfrak{g}^{\alpha\beta} = \delta^\alpha_\nu.$$

This means that the matrix with components $\mathfrak{g}^{\mu\nu}$ is the inverse of the matrix with components $g_{\mu\nu}$. We will therefore use the notation:

$$g^{-1} := \mathfrak{g}.$$

We can use g^{-1} to "raise indices". Let $\omega, \theta \in \Omega^1(\mathcal{M})$. We have that

$$\omega^\sharp(\theta) = g^{-1}(\omega, \theta) = (g^{-1})^{\mu\nu}\omega_\mu\theta_\nu.$$

Define $\omega^\mu := (g^{-1})^{\mu\nu}\omega_\nu$.

Let $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$. With respect to the coordinate chart $(U, \{x^\mu\})$, we can write:

$$T = T^{\mu_1\dots\mu_r}{}_{\nu_1\dots\nu_s} \frac{\partial}{\partial x^{\mu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\mu_r}} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_s}.$$

We can now create an $(r-1, s+1)$ tensor field in U with the components

$$T^{\mu_1\dots\mu_{k-1}}{}_{\mu_k}{}^{\mu_{k+1}\dots\mu_r}{}_{\nu_1\dots\nu_s} := g_{\mu_k\alpha}T^{\mu_1\dots\mu_{k-1}\alpha\mu_{k+1}\dots\mu_r}{}_{\nu_1\dots\nu_s}. \tag{3.1}$$

Similarly, we can create a $(r+1, s-1)$ tensor field in U with the components:

$$T^{\mu_1\dots\mu_r}{}_{\nu_1\dots\nu_{k-1}}{}^{\nu_k}{}_{\nu_{k+1}\dots\nu_s} := (g^{-1})^{\nu_k\alpha}T^{\mu_1\dots\mu_r}{}_{\nu_1\dots\nu_{k-1}\alpha\nu_{k+1}\dots\nu_s}. \tag{3.2}$$

Lemma 3.1.

(i *Let $T \in \mathcal{T}^{(r,r)}(\mathcal{M})$. Then there exists a unique function $\operatorname{tr} T \in C^\infty(\mathcal{M})$, such that $\operatorname{tr} T|_U = T^{\mu_1\dots\mu_r}{}_{\mu_1\dots\mu_r}$ with respect to any coordinate chart $(U, \{x^\mu\})$.*

(ii *Let $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$. Then there exists a unique $C^{(i,j)}T \in \mathcal{T}^{(r-1,s-1)}(\mathcal{M})$, such that with respect to any coordinate chart $(U, \{x^\mu\})$*

$$\begin{aligned} &C^{(i,j)}T|_U \\ &:= \sum_{\mu_i=0}^{n} T^{\mu_1\dots\mu_r}{}_{\nu_1\dots\nu_{j-1}\mu_i\nu_j\dots\nu_s} \partial_{x^{\mu_1}} \otimes \dots \partial_{x^{\mu_i-1}} \otimes \partial_{x^{\mu_i+1}} \dots \partial_{x^{\mu_r}} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_{j-1}} \otimes dx^{\nu_{j+1}} \otimes \dots \otimes dx^{\nu_s}. \end{aligned} \tag{3.3}$$

We refer to $C^{(i,j)}T$ as a contraction of T in the ith upper index and jth lower index.

Proof. "(i)": We will only prove the case $r = 1$ as the general r cases follows entirely analogously. Consider two coordinate charts (U, ϕ) and (V, ψ), where $\phi = (x^0 \dots, x^n)$ and $\psi = (\tilde{x}^0 \dots, \tilde{x}^n)$. Then $d\tilde{x}^\mu \in \Omega^1(U)$ and $\frac{\partial}{\partial \tilde{x}^\mu} \in \mathcal{T}(\mathcal{M})$ can be expressed as follows:

$$\begin{aligned} d\tilde{x}^\mu &= \frac{\partial \tilde{x}^\mu}{\partial x^\nu} dx^\nu, \\ \frac{\partial}{\partial \tilde{x}^\mu} &= \frac{\partial x^\nu}{\partial \tilde{x}^\mu} \frac{\partial}{\partial x^\nu}, \end{aligned}$$

where, recall, $\frac{\partial \tilde{x}^\mu}{\partial x^\nu} \in C^\infty(U)$ is defined as:

$$\frac{\partial \tilde{x}^\mu}{\partial x^\nu} := \frac{\partial(\tilde{x}^\mu \circ \phi^{-1})}{\partial x^\nu} \circ \phi,$$

and similarly for $\tilde{x}$ and x reversed, where ϕ is replaced with ψ. In particular, note that:

$$\frac{\partial \tilde{x}^\mu}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial \tilde{x}^\nu} = \delta^\mu_\nu.$$

The components $T_\mu^{\ \nu}$ with respect to a coordinate chart $(U, \{x^\mu\})$ and $\tilde{T}_\mu^{\ \nu}$ with respect to a coordinate chart $(U, \{\tilde{x}^\mu\})$ are therefore related in the following way:

$$T_\mu^{\ \nu} dx^\mu \otimes \frac{\partial}{\partial x^\nu} = T = \tilde{T}_\alpha^{\ \beta} d\tilde{x}^\alpha \otimes \frac{\partial}{\partial \tilde{x}^\beta} = \tilde{T}_\alpha^{\ \beta} \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} \frac{\partial x^\nu}{\partial \tilde{x}^\beta} dx^\mu \otimes \frac{\partial}{\partial x^\nu}.$$

Hence,

$$T_\mu^{\ \nu} = \tilde{T}_\alpha^{\ \beta} \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} \frac{\partial x^\nu}{\partial \tilde{x}^\beta}.$$

Taking the sum, we therefore obtain:

$$T_\mu^{\ \mu} = \tilde{T}_\alpha^{\ \beta} \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} \frac{\partial x^\mu}{\partial \tilde{x}^\beta} = \tilde{T}_\alpha^{\ \beta} \delta_\beta^\alpha = \tilde{T}_\alpha^{\ \alpha},$$

so the functions $T_\mu^{\ \mu}$ and $\tilde{T}_\alpha^{\ \alpha}$ agree.

"(ii)": Exercise. □

Let $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$ and $S \in \mathcal{T}^{(r',s')}(\mathcal{M})$. Then we define $S \otimes T \in \mathcal{T}^{(r+r',s+s')}(\mathcal{M})$ as follows: for all $\omega_i \in \Omega^1(\mathcal{M})$, $1 \le i \le r + r'$, $X_j \in \mathcal{T}(\mathcal{M})$, $1 \le j \le s + s'$:

$$(S \otimes T)(\omega_1, \ldots, \omega_{r+r'}, X_1, \ldots, X_{s+s'}) := S(\omega_1, \ldots, \omega'_r, X_1, \ldots, X'_s) T(\omega_{r+1}, \ldots, \omega_{r+r'}, X_{s+s'}, \ldots, X_{s+s'}).$$

Exercise: Show that the functions on the left-hand side of (3.1) are the components of the tensor field $C^{(k,2)}(g \otimes T) \in \mathcal{T}^{(r-1,s+1)}(\mathcal{M})$ restricted to U and functions on the left-hand side of (3.2) are the components of the tensor field $C^{(2,k)}(g^{-1} \otimes T) \in \mathcal{T}^{(r+1,s-1)}(\mathcal{M})$ restricted to U. We can therefore uniquely extend the lowering and raising of indices of tensor fields to the whole $\mathcal{M}$.

3.1.2 The Levi–Civita connection

A priori, there is no way to compare vectors in different tangent spaces on a general manifold $\mathcal{M}$. The Levi–Civita connection is a construction that provides a natural way to *transport* vectors from one tangent space to the other in the setting of manifolds equipped with a metric. For the sake of convenience, we will restrict ourselves to smooth tensor fields.

Definition 3.5. *An* affine connection *is a bilinear (over $\mathbb{R}$) map $\nabla : \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \to \mathcal{T}(\mathcal{M})$ with $(X, Y) \mapsto \nabla_X Y$, such that for all $X, Y \in \mathcal{T}(\mathcal{M})$ and $f \in C^\infty(\mathcal{M})$*

1. *(linearity over $C^\infty(\mathcal{M})$) $\nabla_{fX} Y = f \nabla_X Y$,*
2. *(Leibniz rule) $\nabla_X (fY) = X(f) Y + f \nabla_X Y$.*

A Levi-Civita connection *with respect to a metric g is an affine connection that satisfies additionally: for all $X, Y, Z \in \mathcal{T}(\mathcal{M})$:*

3. *(metric-preserving) $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$,*
4. *(torsion-free) $\nabla_X Y - \nabla_Y X = [X, Y]$.*

We define the corresponding *covariant derivative* as the map:

$$\nabla : \mathcal{T}(\mathcal{M}) \to \mathcal{T}^{(1,1)}(\mathcal{M}),$$
$$(\nabla Y)(X, \omega) := \omega(\nabla_X Y).$$

Note that ∇Y is a well-defined tensor field because it is bilinear over $C^\infty(\mathcal{M})$.

[Exercise: Verify this by considering $(\nabla Y)(fX + Z, h\omega + \alpha)$ for $f, h \in C^\infty(\mathcal{M})$, $X, Y, Z \in \mathcal{T}(\mathcal{M})$ and $\omega, \alpha \in \Omega^1(\mathcal{M})$.]

Theorem 3.2. *Let $(\mathcal{M}, g)$ be a pseudo-Riemannian manifold. Then a Levi–Civita connection with respect to g, must satisfy: for all $X, Y, Z \in \mathcal{T}(\mathcal{M})$.*

$$g(\nabla_X Y, Z) = \frac{1}{2}\left[X(g(Y,Z)) + Y(g(Z,X)) - Z(g(Y,X)) + g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X)\right]. \tag{3.4}$$

Furthermore, given a metric g, the corresponding Levi–Civita connection is unique.

Proof. We will first prove uniqueness. Let $X, Y, Z \in \mathcal{T}(M)$. By the metric preserving property and symmetry of g, we have that

$$X(g(Y,Z))+Y(g(Z,X))-Z(g(Y,X)) = g(\nabla_X Y+\nabla_Y X, Z)+g(\nabla_X Z-\nabla_Z X, Y)+g(\nabla_Y Z-\nabla_Z Y, X).$$

Applying the torsion free condition then gives

$$X(g(Y,Z)) + Y(g(Z,X)) - Z(g(Y,X)) = 2g(\nabla_X Y, Z) - g([X,Y],Z) + g([X,Z],Y) + g([Y,Z],X).$$

Rearranging terms then gives

$$g(\nabla_X Y, Z) = \frac{1}{2}\left(X(g(Y,Z)) + Y(g(Z,X)) - Z(g(Y,X)) + g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X)\right).$$

Suppose there exists another Levi-Civita connection $\widetilde{\nabla}$, then by the linearity of g

$$g(\nabla_X Y - \widetilde{\nabla}_X Y, Z) = 0$$

for all $X, Y, Z \in \mathcal{T}(\mathcal{M})$. By the non-degeneracy of g, this must mean that $\nabla_X Y - \widetilde{\nabla}_X Y = 0$ for all $X, Y \in \mathcal{T}(\mathcal{M})$.

We can similarly use (3.4) to prove the existence of ∇. Indeed, we have that $(\nabla_X Y)^\flat \in \Omega^1(\mathcal{M})$ can be defined as follows: let $X, Y, Z \in \mathcal{T}(M)$, then

$$(\nabla_X Y)^\flat(Z) := \frac{1}{2}\left(X(g(Y,Z)) + Y(g(Z,X)) - Z(g(Y,X)) + g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X)\right).$$

Indeed, $(\nabla_X Y)^\flat$ is a well-defined 1-form by linearity of the right-hand side above in Z. We leave it as an Exercise to verify that $(X,Y) \mapsto (\nabla_X Y) := \sharp((\nabla_X Y)^\flat)$ defines a Levi-Civita connection. □

We can express the Levi–Civita connection as follows with respect to a coordinate chart $\{x^\mu\}$:

$$(\nabla_{\partial_\mu}\partial_\nu)^\sigma = \frac{1}{2}(g^{-1})^{\sigma\rho}\left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}\right) =: \Gamma^\sigma_{\mu\nu}.$$

The expressions $\Gamma^\sigma_{\mu\nu}$ are called the Christoffel symbols. Hence,

$$\begin{aligned}(\nabla_X Y)^\sigma &= (\nabla_{X^\mu\partial_\mu}(Y^\nu\partial_\nu))^\sigma = X^\mu\partial_\mu Y^\sigma + \Gamma^\sigma_{\mu\nu}X^\mu Y^\nu,\\ (\nabla Y)_\mu{}^\sigma &= \partial_\mu Y^\sigma + \Gamma^\sigma_{\mu\nu}Y^\nu.\end{aligned}$$

We would like to extend ∇ to act on more general tensor fields: $\nabla : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r,s+1)}(\mathcal{M})$.[24] First, we let $\nabla f := df$ for $f \in C^\infty(\mathcal{M}) = T^{(0,0)}(\mathcal{M})$. This implies that $C^\infty(\mathcal{M}) \ni \nabla_X f = df(X) = X(f)$.

[24] Sometimes, ∇f is taken to mean $(df)^\sharp \in \mathcal{T}(\mathcal{M})$, the gradient of f. We will only use ∇f to denote df in these lecture notes.

Next, we consider $\omega \in \Omega^1(\mathcal{M}) = \mathcal{T}^{(0,1)}(\mathcal{M})$ and define

$$(\nabla\omega)(X,Y) := X(\omega(Y)) - \omega(\nabla_X Y)$$

for all $X, Y \in \mathcal{T}(\mathcal{M})$. Note that linearity over $C^\infty(\mathcal{M})$ in the first argument follows immediately, and linearity in the second argument follows from:

$$(\nabla\omega)(X,fY) = \nabla_X(\omega(fY)) - \omega(\nabla_X(fY)) = f(\nabla\omega)(X,Y) + (\nabla_X f)\omega(Y) - (\nabla_X f)\omega(Y) = f(\nabla\omega)(X,Y).$$

This definition ensures that ∇ is compatible with tensor contraction. With respect to a coordinate chart $(U, \{x^\mu\})$, we obtain for all $0 \leq \nu \leq n+1$

$$(\nabla(\omega_\rho Y^\rho))_\nu = (\nabla\omega)_{\nu\rho} Y^\rho + \omega_\rho (\nabla Y)^\rho{}_\nu.$$

We can therefore also express:

$$(\nabla\omega)_{\nu\mu} = \partial_\nu \omega_\mu - \Gamma^\rho_{\nu\mu}\omega_\rho.$$

We will use the following notational conventions:

$$\nabla_\nu \omega_\mu := (\nabla\omega)_{\nu\mu},$$
$$\nabla_\nu Y^\mu := (\nabla Y)_\nu{}^\mu$$

We now extend ∇ as a map on general tensor fields: $\nabla : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r,s+1)}(\mathcal{M})$ by making it compatible with tensor contraction.

For $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$, we define:

$$\begin{aligned}(\nabla T)(X,\omega_1,\ldots,\omega_r,Y_1,\ldots,Y_s) := (\nabla_X T)(\omega_1,\ldots,\omega_r,Y_1,\ldots,Y_s) := X(T(\omega_1,\ldots,\omega_r,Y_1,\ldots,Y_s))\\ - T(\nabla_X\omega_1,\ldots,\omega_r,Y_1,\ldots,Y_s) - \ldots - T(\nabla_X\omega_1,\ldots,\nabla_X\omega_r,Y_1,\ldots,Y_s)\\ - T(\omega_1,\ldots,\omega_r,\nabla_X Y_1,\ldots,Y_s) - \ldots - T(\omega_1,\ldots,\omega_r,Y_1,\ldots,\nabla_X Y_s).\end{aligned}$$

A tensor field T takes the following form with respect to a coordinate chat $(U, \{x^\mu\})$:

$$T = T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s}\partial_{x^{\nu_1}} \otimes \ldots \otimes \partial_{x^{\nu_r}} \otimes dx^{\mu_1} \otimes dx^{\mu_s},$$

with $T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} \in C^\infty(\mathcal{M})$ smooth functions. In the notation, we do not put the upper and lower indices directly above each other, so that we can lower and raise them using the musical isomorphisms.

We then obtain:

$$\begin{aligned}\nabla_\mu T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} = \partial_\mu T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} - \Gamma^\rho_{\mu\mu_1} T^{\nu_1\ldots\nu_r}{}_{\rho\ldots\mu_s} - \Gamma^\rho_{\mu\mu_2} T^{\nu_1\ldots\nu_r}{}_{\mu_1\rho\mu_3\ldots\mu_s} - \ldots - \Gamma^\rho_{\mu\mu_s} T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_{s-1}\rho}\\ + \Gamma^{\nu_1}_{\mu\rho} T^{\rho\nu_2\ldots\nu_r}{}_{\mu_1\ldots\mu_s} + \Gamma^{\nu_2}_{\mu\rho} T^{\nu_1\rho\nu_3\ldots\nu_r}{}_{\mu_1\ldots\mu_s} + \ldots + \Gamma^{\nu_r}_{\mu\rho} T^{\nu_1\ldots\nu_{r-1}\rho}{}_{\mu_1\ldots\mu_s}.\end{aligned}$$

By construction, we have in particular that $\nabla g = 0$.

When we want to specify to which metric g the covariant derivative ∇ is associated, we will use the notation ∇^g.

Definition 3.6. *The divergence of a vector field $X \in \mathcal{T}(\mathcal{M})$ is defined as follows:*

$$\operatorname{div} X := \operatorname{tr} \nabla X.$$

3.1.3 The natural volume form and Hodge dual

In order to derive the Euler–Lagrange equations, we will first need to establish a divergence theorem involving ∇. To make sense of that, we first need a natural way of integrating on $(\mathcal{M}, g)$. We will need assume that the $(n+1)$-dimensional manifold $\mathcal{M}$ is *orientable*. This means that there exists a $\omega \in \Omega^{n+1}(\mathcal{M})$ that is non-vanishing everywhere, which is called a volume form. The triple $(\mathcal{M}, g, \omega)$ is then called a *pseudo-Riemannian, oriented manifold*.

A coordinate chart $(U, \{x^\mu\})$ is said to be *positively oriented* if $\omega = f dx^0 \wedge \ldots \wedge dx^n$ in U, with f a positive function on U. We say a (local) basis of vector fields $(E_0, \ldots, E_n)$ is *positively oriented* if $\omega(E_0, \ldots, E_n) > 0$.

Note that the domain U of a coordinate chart $\{x^\mu\}$ is always an orientable manifold, since $dx^0 \wedge dx^1 \wedge \ldots \wedge dx^n$ is a well-defined volume form on U (but it may not necessary be extendible as a volume form to $\mathcal{M} \setminus U$). Therefore, any manifold is *locally* orientable, which is what we really need to be able to derive the Euler–Lagrange equations, which are local.

Lemma 3.3. *Let $(\mathcal{M}, g)$ be an $(n+1)$-dimensional orientable, pseudo-Riemannian manifold. Then there exists a unique volume form $\epsilon \in \Omega^{n+1}(\mathcal{M})$, which satisfies the property that*

$$\epsilon(E_0, \ldots, E_n) = 1$$

for any positively oriented, local basis of orthonormal vector fields $E_0, \ldots, E_n$.

The volume form ϵ, also denoted $\epsilon[g]$, is called a natural volume form[25] *and we will refer to the triple $(\mathcal{M}, g, \epsilon)$ as a* naturally oriented pseudo-Riemannian manifold.

With respect to any positively oriented coordinate chart $(U, \{x^\mu\})$, ϵ can be expressed as follows on U:

$$\epsilon = \sqrt{|\det g|} dx^0 \wedge \ldots \wedge dx^n,$$

with $\det g : U \to \mathbb{R}$ denoting the determinant of the $(n+1) \times (n+1)$ matrix with coefficients the functions $g(\partial_{x^\mu}, \partial_{x^\nu})$.

Proof. Exercise. □

Definition 3.7. *Let $r, n \in \mathbb{N}$ and $r \leq n+1$. Let $(\mathcal{M}, g, \epsilon)$ be an $(n+1)$-dimensional, naturally oriented pseudo-Riemannian manifold. The* Hodge dual *is a map*

$$\star : \Omega^r(\mathcal{M}) \to \Omega^{n+1-r}(\mathcal{M}),$$

with $(\star\omega) := \star(\omega)$, that is linear over $C^\infty(\mathcal{M})$, i.e. for all $\omega, \alpha \in \Omega^r(\mathcal{M})$ and $f \in C^\infty(\mathcal{M})$

$$\star(f\omega + \alpha) = f \star \omega + \star\alpha,$$

and $\star(1) = \epsilon$ with $1 \in C^\infty(\mathcal{M})$ and for $\omega = \omega^1 \wedge \ldots \wedge \omega^r \in \Omega^r(\mathcal{M})$ with $\omega^i \in \Omega^1(\mathcal{M})$ for $1 \leq i \leq r$,

$$(\star\omega)(X_1, \ldots, X_{n+1-r}) = \epsilon(X_1, \ldots, X_{n+1-r}, \omega_1^\sharp, \ldots, \omega_r^\sharp).$$[26]

With respect to a coordinate chart $(U, \{x^\mu\})$, we can express for $\omega \in \Omega^r(U)$ with $r \geq 1$:

$$(\star\omega)_{\mu_1 \ldots \mu_{n+1-r}} = \frac{1}{r!} \epsilon_{\mu_1 \ldots \mu_{n+1}} \omega^{\mu_{n+1-(r-1)} \cdots \mu_{n+1}},$$

where the $r!$ factor arises from:

$$\omega = \omega_{\mu_1 \ldots \mu_r} dx^{\mu_1} \otimes \ldots \otimes dx^{\mu_r} = \frac{1}{r!} \omega_{\mu_1 \ldots \mu_r} dx^{\mu_1} \wedge \ldots \wedge dx^{\mu_r}.$$

[25] It is "natural" because for any diffeomorphism $\Phi : \mathcal{M} \to \mathcal{M}$, $\Phi^*(\epsilon[g]) = \epsilon[\Phi^* g]$.

[26] Exercise: Convince yourself that for $\omega \in \Omega^r(\mathcal{M})$, $\star\omega$ is globally well-defined on $\mathcal{M}$.

Lemma 3.4. *Let $r, n \in \mathbb{N}$ and $r \leq n+1$. Let $(\mathcal{M}, g, \epsilon)$ be an $(n+1)$-dimensional, naturally oriented pseudo-Riemannian manifold.*

(i) *The following identity holds:*

$$\epsilon_{\mu_1 \dots \mu_{n+1}} \epsilon^{\nu_1 \dots \nu_r \mu_{r+1} \dots \mu_{n+1}} = \operatorname{sign}(\det g) r!(n+1-r)! \delta^{\nu_1}_{[\mu_1} \dots \delta^{\nu_r}_{\mu_r]},$$

where square brackets denote the antisymmetrization of the indices, i.e.

$$A_{[\mu_1 \dots \mu_r]} = \frac{1}{r!} \sum_{\sigma \in S(\{\mu_1, \dots, \mu_r\})} \operatorname{sign}(\sigma) A_{\sigma(\mu_1) \dots, \sigma(\mu_r)},$$

with $S(\{\mu_1, \dots, \mu_r\})$ the set of all permutations of $\{\mu_1, \dots, \mu_r\}$.

(ii) *Let $\omega \in \Omega^r(\mathcal{M})$, then*

$$\star(\star\omega) = \operatorname{sign}(\det g)(-1)^{r(n+1-r)}\omega.$$

(iii) *Let $\omega \in \Omega^r(\mathcal{M})$, then with respect to an arbitrary coordinate chart:*

$$\star(d \star \omega)_{\mu_1 \dots \mu_{r-1}} = \operatorname{sign}(\det g)(-1)^{r(n+1-r)} \nabla^{\mu_r} \omega_{\mu_1 \dots \mu_r}.$$

Proof. Problem sheet. □

Lemma 3.5. $\nabla \epsilon = 0$

Proof. By (i) of Lemma 3.4, we have that in the domain of a coordinate chart:

$$\nabla_\nu(\epsilon_{\mu_0 \dots \mu_n} \epsilon^{\mu_0 \cdots \mu_n}) = \nabla_\nu(\operatorname{sign}(\det g)(n+1)!) = 0.$$

Furthermore, by $\nabla g^{-1} = 0$ and the compatibility of ∇ with contraction, we can express

$$\nabla_\nu(\epsilon_{\mu_0 \dots \mu_n} \epsilon^{\mu_0 \cdots \mu_n}) = (\nabla_\nu \epsilon_{\mu_0 \dots \mu_n}) \epsilon^{\mu_0 \cdots \mu_n} + \epsilon_{\mu_0 \dots \mu_n}(\nabla_\nu \epsilon^{\mu_0 \cdots \mu_n}) = 2\epsilon^{\mu_0 \cdots \mu_n} \nabla_\nu \epsilon_{\mu_0 \dots \mu_n}.$$

Finally, note that by the antisymmetry properties of ϵ

$$\epsilon^{\mu_0 \cdots \mu_n} \nabla_\nu \epsilon_{\mu_0 \dots \mu_n} = (n+1)! \epsilon^{0 \dots n} (\nabla \epsilon)_{\nu 0 \dots n}.$$

Since $\epsilon^{0 \dots n} \neq 0$, we must have that $(\nabla \epsilon)_{\nu 0 \dots n}$ vanishes for all ν, so, by the antisymmetry properties of ϵ, we must have that $\nabla_\nu \epsilon_{\mu_0 \dots \mu_n} = 0$ for and hence $\nabla \epsilon = 0$. □

Proposition 3.6 (Divergence theorem)**.** *Let $(\mathcal{M}, g, \epsilon)$ be an $(n+1)$-dimensional, naturally oriented, pseudo-Riemannian manifold (with boundary) and let $X \in \mathcal{T}(\mathcal{M})$. Then*

$$\int_{\mathcal{M}} \operatorname{div} X \, \epsilon = (-1)^n \int_{\partial \mathcal{M}} \iota^*(\star X^\flat),$$

where $\iota : \partial \mathcal{M} \to \mathcal{M}$ is the inclusion map.

Proof. Consider $X^\flat \in \Omega^1(\mathcal{M})$. Then $\star X^\flat \in \Omega^n(\mathcal{M})$. By (iii) and (ii) of Lemma 3.4, we have that in the domain U of a chart $(U, \{x^\mu\})$

$$\operatorname{div} X \, \epsilon = (\nabla_\mu X^\mu)\epsilon = \star(\nabla^\mu X_\mu) = \star(\operatorname{sign}(\det g)(-1)^n \star (d \star X^\flat)) = (-1)^n d \star X^\flat.$$

By Stokes' theorem, we have that

$$\int_{\mathcal{M}} (-1)^n d \star X^\flat = (-1)^n \int_{\partial \mathcal{M}} \iota^*(\star X^\flat).$$ □

3.2 Lagrangians and field manifolds

Let $(\mathcal{M}, g)$ be an $(n+1)$-dimensional, orientable, pseudo-Riemannian manifold(with boundary).

Definition 3.8. *The* field manifold $\mathcal{N}$ *is the following union:*

$$\mathcal{N} = \bigcup_{x \in \mathcal{M}} \{x\} \times T_x^{(0,r)}\mathcal{M} \times T_x^{(0,r+1)}\mathcal{M}$$

with projection map $\pi : \mathcal{N} \to \mathcal{M}$, $\pi(x, q, v) = x$ *for* $q \in T_x^{(0,r)}\mathcal{M}$ *and* $v \in T_x^{(0,r+1)}\mathcal{M}$.

Analogously to the construction of the tangent and cotangent bundle, it can be shown that given a coordinate chart (U, ψ), $\psi = (x^0, \dots, x^n)$ on $\mathcal{M}$, the pair $(\pi^{-1}(U), \varphi)$ defines a coordinate chart on $\mathcal{N}$, if we take $\varphi(x, q, v) = (\{x^\mu\}, \{q_{\mu_1 \dots \mu_r}\}, \{v_{\nu_1 \dots \nu_{r+1}}\})$, where

$$q = q_{\mu_1 \dots \mu_r}(dx^{\mu_1} \otimes \dots \otimes dx^{\mu_r})_x \in T_x^{(0,r)}\mathcal{M},$$
$$v = v_{\nu_1 \dots \nu_{r+1}}(dx^{\nu_1} \otimes dx^{\nu_2} \otimes \dots \otimes dx^{\nu_{r+1}})_x \in T_x^{(0,r+1)}\mathcal{M}.$$

In this way $\mathcal{N}$ can be equipped with the structure of a manifold, i.e. it is a (second-countable, Hausdorff, locally Euclidean) topological space equipped with a differentiable structure. We will moreover apply the convention $T_x^{(0,0)}\mathcal{M} := \mathbb{R}$.

Definition 3.9. *The* (field) Lagrangian[27] *is a smooth map:*

$$\mathcal{L} : \mathcal{N} \to \mathbb{R}.$$

A field *is an element* $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M})$.

Note that $d\mathcal{L} \in \Omega^1(\mathcal{N})$ takes the following form on $\pi^{-1}(U)$:

$$d\mathcal{L}|_{\pi^{-1}(U)} = \frac{\partial \mathcal{L}}{\partial x^\mu} dx^\mu + \frac{\partial \mathcal{L}}{\partial q_{\mu_1 \dots \mu_r}} dq_{\mu_1 \dots \mu_r} + \frac{\partial \mathcal{L}}{\partial v_{\nu_1 \dots \nu_{r+1}}} dv_{\nu_1 \dots \nu_{r+1}}.$$

Consider a field $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M})$. Note that $\nabla\phi \in \mathcal{T}^{(0,r+1)}(\mathcal{M})$. We will denote $(\phi, \nabla\phi)$ the elements of $\Gamma(\mathcal{N})$ (sections of the bundle $\mathcal{N}$) defined as follows: $(\phi, \nabla\phi)(x) := (x, \phi_x, (\nabla\phi)_x) \in \mathcal{N}$.

Definition 3.10. *Let* $\mathcal{M}$ *be compact. The action* $\mathcal{S} : \mathcal{T}^{(0,r)}(\mathcal{M}) \to \mathbb{R}$ *is the following integral:*

$$\mathcal{S}(\phi) = \int_{\mathcal{M}} \mathcal{L} \circ (\phi, \nabla\phi)\, \epsilon.$$

Note that $\mathcal{L} \circ (\phi, \nabla\phi) : \mathcal{M} \to \mathbb{R}$ is smooth. Exercise: Convince yourself that the above notions can easily be extended to the setting where the fields are general (r, s)-tensor fields, rather than just $(0, s)$-tensor fields.

Just like in the case of Lagrangians with domain $T(\mathbb{R} \times \mathcal{M})$, we can extremize the action and obtain Euler–Lagrange equations.

Definition 3.11. *Let* $U \subset V \subseteq \mathcal{M}$ *be open, with* V *compact. We say* $\phi \in \mathcal{T}^{(0,r)}(U)$ *is an extremal of the action* $\mathcal{S} : \mathcal{T}^{(0,r)}(U) \to \mathbb{R}$ *if for any* $h \in \mathcal{T}^{(0,r)}(\mathcal{M})$, *such that* $h|_{\partial U} = 0$ *if* $\partial U \neq \emptyset$,

$$\frac{d}{ds}\Big|_{s=0} \mathcal{S}(\phi + sh) = 0.$$

[27] Sometimes $\mathcal{L}$ is referred to as a *Lagrangian density* in the literature and the form $\mathcal{L} \cdot \epsilon \in \Omega^{(n+1)}(\mathcal{M})$ is called the corresponding *Lagrangian*.

Proposition 3.7. *Consider a coordinate chart $(U, \{x^\mu\})$ on $\mathcal{M}$, such that $U \subseteq V$, with V compact. Then $\phi \in \mathcal{T}^{(0,r)}(U)$ is an extremal of the action $\mathcal{S} : \mathcal{T}^{(0,r)}(U) \to \mathbb{R}$ if and only if ϕ satisfies the following equations on U: for all $0 \leq \nu_i \leq n$,*

$$\frac{\partial \mathcal{L}}{\partial q_{\nu_1 \dots \nu_r}} \circ (\phi, \nabla\phi) - \nabla_\nu P^{\nu\nu_1\dots\nu_r} = 0, \tag{3.5}$$

where $P \in \mathcal{T}^{(r+1,0)}(U)$, with

$$P = \left[\frac{\partial \mathcal{L}}{\partial v_{\nu\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi)\right] \partial_{x^\nu} \otimes \partial_{x^{\nu_1}} \otimes \dots \partial_{x^{\nu_r}}.$$

The equations (3.12) are called Euler–Lagrange (field) equations.

Proof. Consider

$$\mathcal{L} : \pi^{-1}(U) \to \mathbb{R},$$

with $\pi : \mathcal{N} \to \mathcal{M}$. Let $h \in \mathcal{T}^{(0,r)}r(U)$, such that $h|_{\partial U} = 0$. Then we apply the chain rule to obtain:

$$\begin{aligned}
\frac{d}{ds}|_{s=0}\mathcal{S}(\phi + sh) &= \int_U \frac{d}{ds}|_{s=0}\mathcal{L} \circ (\phi + sh, \nabla\phi + s\nabla h)\epsilon \\
&= \sum_{\nu_1,\dots,\nu_r=0}^{n} \int_U \frac{\partial \mathcal{L}}{\partial q_{\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi) h_{\nu_1\dots\nu_r}\epsilon + \sum_{\nu=0}^{n} \frac{\partial \mathcal{L}}{\partial v_{\nu\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi)\nabla_\nu h_{\nu_1\dots\mu_r}\epsilon \\
&= \sum_{\nu_1,\dots,\nu_r=0}^{n} \int_U \left[\frac{\partial \mathcal{L}}{\partial q_{\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi) - \sum_{\nu=0}^{n} \nabla_\nu P^{\nu\nu_1\dots\nu_r}\right] h_{\nu_1\dots\mu_r}\,\epsilon,
\end{aligned} \tag{3.6}$$

where we arrived at the final inequality as follows: we consider the tensor field $P \in \mathcal{T}^{(r+1,0)}(U)$ and define $J = J^\nu \partial_{x^\nu} \in \mathcal{T}(U)$ via the following tensor contraction:

$$J^\nu := P^{\nu\nu_1\dots\nu_r} h_{\nu_1\dots\mu_r}.$$

In this way, we can apply the rules for covariant differentiation of tensor fields to obtain:

$$P^{\nu\nu_1\dots\nu_r}\nabla_\nu h_{\nu_1\dots\mu_r} = \operatorname{div} J - (\nabla_\nu P^{\nu\nu_1\dots\nu_r}) h_{\nu_1\dots\mu_r}.$$

Then we use that

$$\int_U \operatorname{div} J\epsilon = 0$$

by Proposition 3.6 and the fact that J vanishes on ∂U by $h|_{\partial U} = 0$.

Since h was arbitrary, we arrive at (3.12) by taking h to have only one non-vanishing component and applying Lemma 2.15. □

We will also refer to (3.12) as the *Euler–Lagrange equations* corresponding to the Lagrangian $\mathcal{L}$.

Example 3.1 (Klein–Gordon equation)**.** *Let $\mathcal{M} = \mathbb{R}^{n+1}$ and $r = 0$. Let $\{x^\mu\}$ be Cartesian coordinates. Consider $m = m_{\mu\nu}dx^\mu \otimes dx^\nu$, with $m_{00} = -1$, $m_{0i} = 0$ and $m_{ij} = \delta_{ij}$. The Lorentzian metric m is known as the Minkowski metric and $(\mathcal{M}, m)$ is called the Minkowski spacetime. Consider the field manifold:*

$$\mathcal{N} = \bigcup_{x\in\mathcal{M}} \{x\} \times \mathbb{R} \times T^*_x\mathcal{M}.$$

Consider the Lagrangian:

$$\mathcal{L} : \mathcal{N} \to \mathbb{R}, \quad \text{with}$$
$$\mathcal{L}(x, q, v) = \frac{1}{2} m_x^{-1}(v, v) + \frac{\mathfrak{m}^2}{2} q^2,$$

with $\mathfrak{m} \in \mathbb{R}$. *Then* $\mathcal{L} \circ (\phi, \nabla\phi) = \frac{1}{2} m^{-1}(\nabla\phi, \nabla\phi) + \frac{\mathfrak{m}^2}{2}\phi^2$.

We have that

$$\frac{\partial \mathcal{L}}{\partial q} \circ (\phi, \nabla\phi) = \mathfrak{m}^2 \phi,$$
$$P^\mu = \frac{\partial \mathcal{L}}{\partial v_\mu} \circ (\phi, \nabla\phi) = \nabla^\mu \phi,$$
$$\nabla_\mu P^\mu = \nabla_\mu \nabla^\mu \phi.$$

The Euler–Lagrange equations are equivalent to:

$$\nabla_\mu \nabla^\mu \phi = \mathfrak{m}^2 \phi,$$

We denote $\Box_g \phi := \nabla_\mu \nabla^\mu \phi$, *with* ∇ *the Levi-Civita connection corresponding to* g.

Note that in Cartesian coordinates, $\Gamma^\mu_{\nu\rho} = 0$, *so* $\Box_m \phi = -\partial^2_{x^0}\phi + \sum_{i=1}^n \partial^2_{x^i}\phi$ *so with* $t = x^0$

$$-\partial_t^2 \phi + \sum_{i=1}^n \partial_i^2 \phi = \mathfrak{m}^2 \phi,$$

which is the Klein–Gordon equation with mass $\mathfrak{m}$. *The* $\mathfrak{m} = 0$ *case is called the (massless) wave equation.*

Example 3.2 (Maxwell equations in vacuum)**.** *Let* $\mathcal{M} = \mathbb{R}^{3+1}$ *and* $r = 1$ *and let* $\{x^\mu\}$ *be Cartesian coordinates. Consider the Minkowski spacetime* $(\mathcal{M}, m)$ *and the field manifold:*

$$\mathcal{N} = \bigcup_{x \in \mathcal{M}} \{x\} \times T^*_x \mathcal{M} \times T^{(0,2)}_x \mathcal{M}.$$

Denote with m *the Minkowski metric on* $\mathbb{R}^{3+1}$. *We define:*

$$\mathcal{L} : \mathcal{N} \to \mathbb{R}, \quad \text{with}$$
$$\mathcal{L}(x, q, v) = \frac{1}{4}(m^{-1})^{\alpha\gamma}_x (m^{-1})^{\beta\delta}_x (v_{\alpha\beta} - v_{\beta\alpha})(v_{\gamma\delta} - v_{\delta\gamma}).$$

Consider the field $A \in \Omega^1(\mathcal{M})$ *with covariant derivative* $\nabla A \in \mathcal{T}^{(0,2)}(\mathcal{M})$.

Then

$$\mathcal{L} \circ (A, \nabla A) = \frac{1}{4}(m^{-1})^{\alpha\gamma}(m^{-1})^{\beta\delta}(\nabla_\alpha A_\beta - \nabla_\beta A_\alpha)(\nabla_\gamma A_\delta - \nabla_\delta A_\gamma).$$

We have that

$$\frac{\partial \mathcal{L}}{\partial q_\mu} \circ (A, \nabla A) = 0,$$
$$P^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial v_{\mu\nu}} \circ (A, \nabla A) = (\nabla^\mu A^\nu - \nabla^\mu A^\nu),$$
$$\nabla_\mu P^{\mu\nu} = \nabla_\mu (\nabla^\mu A^\nu - \nabla^\mu A^\nu) = \nabla_\mu F^{\mu\nu},$$

with $F = dA$. Indeed,

$$F = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu = (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \otimes dx^\nu = (\nabla_\mu A_\nu - \nabla_\nu A_\mu) dx^\mu \otimes dx^\nu.$$

Using F, the Lagrangian takes the form

$$\mathcal{L} \circ (A, \nabla A) = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

By (iii) *of Lemma 3.4, we can put the Euler–Lagrange equations in the following form:*

$$\begin{aligned} dF &= 0, \\ d \star F &= 0. \end{aligned}$$

Note that F has $\binom{4}{2} = 6$ independent components when expanded in a coordinate basis. Let us denote $\mathbf{E} = (F_{01}, F_{02}, F_{03})^T$ and $\mathbf{B} = (-F_{23}, -F_{31}, -F_{12})^T$, i.e. $E_i = F_{0i}$ and $B_i = -\frac{1}{2}\epsilon_{ijk} F^{jk}$. Then the equation $dF = 0$ implies that

$$\partial_{[\mu} F_{\nu\rho]} = 0.$$

Using moreover the antisymmetry of $F_{\mu\nu}$ in its indices, we obtain: for all $0 \leq \mu, \nu, \rho \leq n$

$$\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0$$

Note that this gives four independent equations, with $(\mu, \nu, \rho) = \{(0,1,2), (0,1,3), (0,2,3), (1,2,3)\}$. One can verify that the first three choices result in the equation:

$$\partial_t \mathbf{B} + \nabla \times \mathbf{E} = 0. \tag{3.7}$$

The final choice results in

$$\operatorname{div} \mathbf{B} = 0. \tag{3.8}$$

The equation $\nabla_\nu F^{\mu\nu} = 0$ gives four equations. The case $\mu = 0$ results in

$$\operatorname{div} \mathbf{E} = 0 \tag{3.9}$$

and $1 \leq \mu \leq 3$ give

$$\partial_t \mathbf{E} - \nabla \times \mathbf{B} = 0. \tag{3.10}$$

The equations (3.7)–(3.10) are the Maxwell equations of electromagnetism in vacuum and F is called the Faraday tensor*.*

Example 3.3 (Minimal surface equation)**.** *Let $\mathcal{M} = \Omega \subseteq \mathbb{R}^2$ be compact, $r = 0$ and let e be the Euclidean metric on $\mathbb{R}^2$. Consider the field manifold:*

$$\mathcal{N} = \bigcup_{x \in \mathcal{M}} \{x\} \times \mathbb{R} \times T_x^* \mathcal{M}.$$

Consider

$$\begin{aligned} \mathcal{L} &: \mathcal{N} \to \mathbb{R}, \\ \mathcal{L}(x, q, v) &= \sqrt{1 + e_x^{-1}(v, v)}. \end{aligned}$$

Then, for $u \in C^\infty(\Omega)$, $\nabla u \in \Omega^1(\Omega)$,

$$\mathcal{L} \circ (u, \nabla u) = \sqrt{1 + |\nabla u|^2}.$$

We therefore have that

$$\frac{\partial \mathcal{L}}{\partial q} \circ (u, \nabla u) = 0,$$
$$P^i = \frac{\partial \mathcal{L}}{\partial v_i} \circ (u, \nabla u),$$
$$\nabla_i P^i = \nabla_i \left(\frac{\nabla^i u}{\sqrt{1 + |\nabla u|^2}} \right).$$

The Euler–Lagrange equations in the interior of Ω *are therefore:*

$$\nabla_i \left(\frac{\nabla^i u}{\sqrt{1 + |\nabla u|^2}} \right) = 0. \tag{3.11}$$

The graph $S = \{(x, u(x)) \in \mathbb{R}^3 \,|\, x \in \Omega\}$ *is a 2-surface in* $\mathbb{R}^3$ *with boundary* $u(\partial\Omega) \subset \mathbb{R}^3$ *and the action* $\mathcal{S}(u)$ *gives the area of the surface. Extremals of the action can therefore be associated to surfaces with boundary* $u(\partial\Omega)$ *that extremize the area locally. It can be shown that the extremals are local minimizers of the action. The equation* (3.11) *is therefore called the* minimal surface equation. *Note that the above discussion does not tell us anything about the existence of minimal surfaces.*

3.3 Noether's theorem

We will now investigate the relation between symmetries of actions and Lagrangians with the existence of conservation laws. In this section, we will slightly generalize the notion of a field and a field Lagrangian, introduced in §3.2, in order to incorporate vector-valued fields.

Let V be some fixed, m-dimensional vector space. We define the field manifold as follows:

$$\mathcal{N} = \bigcup_{x \in \mathcal{M}} \{x\} \times (T_x^{(0,r)}\mathcal{M} \otimes V) \times (T_x^{(0,r+1)}\mathcal{M} \otimes V)$$

with projection map $\pi : \mathcal{N} \to \mathcal{M}$, $\pi(x, q, v) = x$ for $q \in T_x^{(0,r)}\mathcal{M} \otimes V$ and $v \in T_x^{(0,r+1)}\mathcal{M} \otimes V$. Given a coordinate chart (U, ψ), $\psi = (x^0, \ldots, x^n)$ on $\mathcal{M}$ and a choice of basis $\{e_A\}_{1 \le A \le m}$ on V, the pair $(\pi^{-1}(U), \phi)$ defines a coordinate chart on $\mathcal{N}$, if we take $\phi(x, q, v) = (\{x^\mu\}, \{q^A_{\mu_1 \ldots \mu_r}\}, \{v^A_{\nu_1 \ldots \nu_{r+1}}\})$, with

$$q = q^A_{\mu_1 \ldots \mu_r} (dx^{\mu_1} \otimes \ldots \otimes dx^{\mu_r})_x \otimes e_A \in T_x^{(0,r)}\mathcal{M} \otimes V,$$
$$v = v^A_{\nu_1 \ldots \nu_{r+1}} (dx^{\nu_1} \otimes dx^{\nu_2} \otimes \ldots \wedge dx^{\nu_{r+1}})_x \otimes e_A \in T_x^{(0,r+1)}\mathcal{M} \otimes V,$$

where we sum over all double indices.

In this way $\mathcal{N}$ takes on the structure of a manifold.

Definition 3.12. *The* field Lagrangian *is a smooth map of the form:*

$$\mathcal{L} : \mathcal{N} \to \mathbb{R}.$$

A field *is an element* $\phi \in \Gamma(T^{(0,r)}\mathcal{M} \otimes V)$*, with*

$$T^{(0,r)}\mathcal{M} \otimes V := \bigcup_{x \in \mathcal{M}} \{x\} \times (T_x^{(0,r)}\mathcal{M} \otimes V).$$

An element of $\Gamma(T^{(0,r)}\mathcal{M} \otimes V)$ *is also called a* vector-valued $(0, r)$-tensor field. *The above definition can straightforwardly be extended to include* vector-valued (r, s)-tensor fields.

When $r = 0$, $\Gamma(T^{(0,0)}\mathcal{M} \otimes V) = C^\infty(\mathcal{M}; V)$, which is the space of smooth functions with values in V. We will shorten the notation by writing $\mathcal{T}^{(r,s)}(\mathcal{M}; V)$ for the space of vector valued (r, s)-tensor fields.

Given $T \in \mathcal{T}^{(r,s)}(\mathcal{M}; V)$, we define $T^A \in \mathcal{T}^{(r,s)}(\mathcal{M})$ with $A \in \{1, \dots, m\}$ via:

$$T^A = T^{\mu_1 \dots \mu_r A}{}_{\nu_1 \dots \nu_s} \partial_{x^{\mu_1}} \otimes \dots \otimes \partial_{x^{\mu_r}} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_s}.$$

Then $T = T^A \otimes e_A$.

Given $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M}; V)$, we define $(\phi, \nabla\phi) \in \Gamma(\mathcal{N})$, by $(\phi, \nabla\phi)(x) = (x, \phi_x, (\nabla\phi)_x)$.

Note that the Euler–Lagrange generalize in this setting to:

$$\frac{\partial \mathcal{L}}{\partial q^A_{\nu_1 \dots \nu_r}} \circ (\phi, \nabla\phi) - \nabla_\nu P^{\nu\nu_1 \dots \nu_r A} = 0, \tag{3.12}$$

with $P \in \mathcal{T}^{(r+1,0)}(\mathcal{M}; V)$ satisfying $P^{\nu\nu_1 \dots \nu_r A} = \frac{\partial \mathcal{L}}{\partial v^A_{\nu\nu_1 \dots \nu_r}} \circ (\phi, \nabla\phi)$. Exercise: Show this.

Definition 3.13. One-parameter group actions Ψ on $\mathcal{T}^{(0,r)}(\mathcal{M}; V)$ *are defined as follows:*

$$\begin{aligned} \Psi : \mathbb{R} \times \mathcal{T}^{(0,r)}(\mathcal{M}; V) &\to \mathcal{T}^{(0,r)}(\mathcal{M}; V), \\ \Psi(s, \cdot) =: \Psi_s : \mathcal{T}^{(0,r}(\mathcal{M}; V) &\to \mathcal{T}^{(0,r)}(\mathcal{M}; V) \quad (s \in \mathbb{R}), \\ \Psi_0 &= \mathrm{id}_{\mathcal{T}^{(0,r)}(\mathcal{M}; V)}, \\ \Psi_{s+t} &= \Psi_s \circ \Psi_t \quad \text{for all } s, t \in \mathbb{R}. \end{aligned}$$

We will be interested in one-parameter group actions for which the following map is well-defined:[28]

$$\begin{aligned} \frac{d}{ds}|_{s=0}\Psi_s : \mathcal{T}^{(0,r)}(\mathcal{M}; V) &\to \mathcal{T}^{(0,r)}(\mathcal{M}; V), \\ \left(\frac{d}{ds}|_{s=0}\Psi_s(\phi)\right)_x &:= \lim_{s \to 0} \frac{(\Psi_s(\phi))_x - \phi_x}{s}. \end{aligned}$$

If $\frac{d}{ds}\Big|_{s=s_0} \Psi_s(\phi)$ is well-defined, we say the one-parameter group action on $\mathcal{T}^{(0,r)}(\mathcal{M}; V)$ is differentiable at $s = 0$.

Example 3.4 (External transformations)**.** *Let $\psi_s : \mathcal{M} \to \mathcal{M}$ be a one-parameter group of diffeomorphisms (a global flow on $\mathcal{M}$). Then the pullbacks*

$$\psi_s^* : \mathcal{T}^{(0,r)}(\mathcal{M}; V) \to \mathcal{T}^{(0,r)}(\mathcal{M}; V),$$

with $\psi_s^*(\phi)_x := (\psi_s^* \phi^A)_x \otimes e_A$ also form a 1-parameter group action on $\mathcal{T}^{(0,r)}(\mathcal{M}; V)$, which is differentiable at $s = 0$, with

$$\frac{d}{ds}\Big|_{s=0} (\psi_s^* \phi)_x = (L_X \phi^A)_x \otimes e_A =: (L_X \phi)_x,$$

where $X \in \mathcal{T}(\mathcal{M})$, such that $X(f) = \frac{d}{ds}|_{s=0} f(\psi_s)$ for all $f \in C^\infty(\mathcal{M})$.

Group actions of the form $\Psi_s = \psi_s^*$ are sometimes called external transformations.

[28] In expression below, we take the limit with respect to any choice of norm on $\wedge^r T^*_x\mathcal{M} \otimes V$, as the norms on finite dimensional vector spaces are all equivalent.

Example 3.5 (Internal transformations)**.** *Let $A : \mathbb{R} \times V \to V$ be a one-parameter group action, with $A_s = A(s, \cdot) \in GL(V)$, such that $\frac{d}{ds}\Big|_{s=s_0} A_s : V \to V$ is a well-defined linear map. Let*

$$\Psi : \mathbb{R} \times \mathcal{T}^{(0,r)}(\mathcal{M}; V) \to \mathcal{T}^{(0,r)}(\mathcal{M}; V)$$
$$(\Psi_s(\phi))_x = \phi^A_x \otimes A_s(e_A).$$

Then Ψ defines a one-parameter group action on $\mathcal{T}^{(0,r)}(\mathcal{M}; V)$. Furthermore,

$$\frac{d}{ds}\Big|_{s=0} (\Psi_s(\phi))_x = \phi^A_x \otimes \left(\frac{d}{ds}\Big|_{s=0} A_s\right)(e_A) \in \mathcal{T}^{(0,r)}(\mathcal{M}; V).$$

These type of 1-parameter group actions, which do <u>not</u> arise from diffeomorphisms in spacetime, are sometimes called internal transformations*.*

Definition 3.14. *Let $\Psi : \mathbb{R} \times \mathcal{T}^{(0,r)}(\mathcal{M}; V) \to \mathcal{T}^{(0,r)}(\mathcal{M}; V)$ be a one-parameter group action. We say Ψ is a* continuous symmetry transformation *of the action $\mathcal{S}$ if for all $s \in \mathbb{R}$ and all $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M}; V)$:*

$$\mathcal{S}(\Psi_s(\phi)) = \mathcal{S}(\phi).$$

Suppose that Ψ is a continuous symmetry transformation of $\mathcal{S}$ such that $\frac{d}{ds}|_{s=0}\Psi_s : \mathcal{T}^{(0,r)}(\mathcal{M}; V) \to \mathcal{T}^{(0,r)}(\mathcal{M}; V)$ is well-defined, and $\mathcal{L}$ denotes the corresponding Lagrangian. Then for all $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M}; V)$:

$$\int_{\mathcal{M}} \left[\frac{d}{ds}\Big|_{s=0} \mathcal{L} \circ (\Psi_s(\phi), \nabla\Psi_s(\phi))\right] \epsilon = 0. \tag{3.13}$$

The above statement is *implied*[29] by the following statement: There exists a $\mathbb{F}[\phi] \in \mathcal{T}(\mathcal{M})$ with $(i_{\partial\mathcal{M}})^*(\star\mathbb{F}[\phi]^\flat) = 0$ if $\partial\mathcal{M} \neq \emptyset$ such that:

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L} \circ (\Psi_s(\phi), \nabla\Psi_s(\phi)) = \operatorname{div} \mathbb{F}[\phi]. \tag{3.14}$$

Indeed, this follows by integrating both sides of (3.14) over $\mathcal{M}$ with respect to the natural volume form ϵ and applying the divergence theorem.

Motivated by (3.14), we will define an *infinitesimal symmetry* as an analogue of (3.14) for transformations of the form $\Psi_s(\phi) = \phi + s\frac{d}{ds}\big|_{s=0}\Psi_s(\phi)$. That is to say, we will ignore the terms of "order $o(s)$" (little o-notation) that would appear formally in an expansion of a general $\Psi_s(\phi)$ around $s = 0$, if we were to equip $\mathcal{T}^{(0,r)}(\mathcal{M}; V)$ with a norm:

$$\Psi_s(\phi) = \phi + s\frac{d}{ds}\Big|_{s=0} \Psi_s(\phi) + o(s).$$

Definition 3.15. *Let $\Psi : \mathbb{R} \times \mathcal{T}^{(0,r)}(\mathcal{M}; V) \to \mathcal{T}^{(0,r)}(\mathcal{M}; V)$ be a one-parameter group action that is differentiable at $s = 0$. We say Ψ is an* infinitesimal symmetry tranformation *of $\mathcal{S}$ if for all $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M}; V)$, there exists an $\mathbb{F}[\phi] \in \mathcal{T}(\mathcal{M})$ such that*

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L} \circ \left(\phi + s\frac{d}{ds}\Big|_{s=0} \Psi_s(\phi), \nabla\phi + s\nabla\frac{d}{ds}\Big|_{s=0} \Psi_s(\phi)\right) = \operatorname{div} \mathbb{F}[\phi].$$

[29] Note that if Ψ is a symmetry information, the existence of a vector field $\mathbb{F}[\phi]$ satisfying (3.14) and $\mathbb{F}[\phi]|_{\partial\mathcal{M}} = 0$ is not guaranteed! Denote $f = \frac{d}{ds}\big|_{s=0} \mathcal{L} \circ (\Psi_s(\phi), \nabla\Psi_s(\phi))$. Then we are looking for solutions to $\operatorname{div}\mathbb{F} = f$ with $\int_{\mathcal{M}} f\epsilon = 0$ and $i^*_{\partial M} \star\mathbb{F}^\flat = 0$. If $(\mathcal{M}, g)$ is Riemannian then we can restrict to $\mathbb{F}^\flat = dh$ for some function h. Then one can show (Exercise) that $\Delta_g h = 0$ and $n_{\partial\mathcal{M}}(h) = 0$ (Neumann boundary condition), with $n_{\partial\mathcal{M}}$ a normal vector field to $\partial\mathcal{M}$ (if non-empty). This boundary value problem can be solved (for suitably nice boundary $\partial\mathcal{M}$) because the corresponding PDE is *elliptic*. For Lorentzian manifolds, the above argument no longer works, since equation becomes $\Box_g h = f$, which is *hyperbolic* and does not have a well-posed boundary value problem in general.

Theorem 3.8 (Noether's theorem for fields)**.** *Let Ψ be an infinitesimal symmetry of $\mathcal{S}$ with corresponding $\mathbb{F}[\phi] \in \mathcal{T}(\mathcal{M})$. Let $\phi \in \mathcal{T}^{(0,r)}(\mathcal{M}; V)$ and let $(U, \{x^\mu\}$ be a coordinate chart. Define $\mathbb{J}[\phi] \in \mathcal{T}(U)$ as follows:*

$$\mathbb{J}^\nu[\phi] = \left(\frac{\partial \mathcal{L}}{\partial v^B_{\nu\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi)\right) \left(\frac{d}{ds}\Big|_{s=0} \Psi_s(\phi)\right)^B_{\nu_1\dots\nu_r} - \mathbb{F}^\nu[\phi].$$

Then if ϕ a solution to the Euler–Lagrange equations on U, we have that:

$$\operatorname{div} \mathbb{J}[\phi] = 0.$$

We refer to $\mathbb{J}[\phi]$ as a conserved current.

Proof. Denote $h = \frac{d}{ds}\Big|_{s=0} \Psi_s(\phi)$ and consider the field $\phi + sh$. By using the infinitesimal symmetry property and applying the chain rule, we obtain in U:[30]

$$\begin{aligned}
\operatorname{div} \mathbb{F}[\phi] = \frac{d}{ds}\Big|_{s=0} \mathcal{L} \circ (\phi + sh, \nabla(\phi + sh)) &= \frac{\partial \mathcal{L}}{\partial q^B_{\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi) h^B_{\nu_1\dots\nu_r} + \frac{\partial \mathcal{L}}{\partial v^B_{\nu\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi) \nabla_\nu h^B_{\nu_1\dots\nu_r} \\
&= \left[\frac{\partial \mathcal{L}}{\partial q^B_{\nu_1\dots\nu_r}} \circ (\phi, \nabla\phi) - \nabla_\nu P^{\nu\nu_1\dots\nu_r B}\right] h^B_{\nu_1\dots\nu_r} \\
&\quad + \nabla_\nu \left(P^{\nu\nu_1\dots\nu_r B} h^B_{\nu_1\dots\nu_r}\right),
\end{aligned}$$

where the last term indicates components of the covariant derivative of a contraction of the P and h tensor fields, and not of the exterior derivative of the corresponding component functions.

If ϕ is a solution to the Euler–Lagrange equations in U, then term inside the square brackets vanishes and we conclude that

$$\nabla_\nu \left(P^{\nu\nu_1\dots\nu_r B} h^B_{\nu_1\dots\mu_r} - \mathbb{F}^\nu[\phi]\right) = 0. \qquad \square$$

Example 3.6. *Recall Example 3.1: let $\mathcal{M} = \mathbb{R}^{n+1}$ and $r = 0$. Then $\mathcal{N} \cong \mathbb{R}^{n+1} \times \mathbb{R} \times \mathbb{R}^{n+1}$. Let $\{x^\mu\}$ be Cartesian coordinates and m the Minkowski metric. Consider*

$$\mathcal{L} : \mathcal{N} \to \mathbb{R}, \quad \text{with}$$
$$\mathcal{L}(x, q, v) = \frac{1}{2} m_x^{-1}(v, v) + \frac{\mathfrak{m}^2}{2} q^2,$$

with $\mathfrak{m} \in \mathbb{R}$. Then $\mathcal{L} \circ (\phi, \nabla\phi) = \frac{1}{2} m^{-1}(\nabla\phi, \nabla\phi) + \frac{\mathfrak{m}^2}{2}\phi^2$.

We consider the one-parameter group of diffeomorphisms $\psi : \mathbb{R} \times \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ corresponding to spacetime translations in an arbitrary direction $b \in \mathbb{R}^{n+1}$. In Cartesian coordinates, this takes the form:

$$\psi_s(x^0, \dots, x^n) = (x^0 + sb^0, \dots, x^n + sb^n).$$

We will b also as a (constant) vector field on $\mathbb{R}^{n+1}$.

Then $\frac{d}{ds}|_{s=0}(\psi_s^* \phi) = L_b \phi$ and

$$\nabla \frac{d}{ds}|_{s=0}(\psi_s^* \phi) = \nabla L_b \phi = L_b \nabla \phi.$$[31]

So for $h = \frac{d}{ds}|_{s=0}(\psi_s^* \phi)$:

[30] To avoid ambiguity: $\nabla_\nu \left(P^{\nu\nu_1\dots\nu_r B} h^B_{\nu_1\dots\nu_r}\right)$ indicates the divergence of the vector field whose components are $P^{\nu\nu_1\dots\nu_r B} h^B_{\nu_1\dots\nu_r}$ and *not* the sum over ν of ordinary derivatives of the product of functions $P^{\nu\nu_1\dots\nu_r B} h^B_{\nu_1\dots\nu_r}$.

[31] Exercise: Show that $[L_X, \nabla] = [L_X, d] = 0$ when acting on $C^\infty(\mathcal{M})$.

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L} \circ (\phi + sh, \nabla\phi + s\nabla h) = m^{-1}(\nabla\phi, \nabla h) + \mathfrak{m}^2 \phi h$$
$$= m^{-1}(\nabla\phi, L_b \nabla\phi) + \mathfrak{m}^2 \phi L_b \phi$$
$$= L_b(\mathcal{L} \circ (\phi, \nabla\phi)) - \frac{1}{2} L_b(m^{-1})(\nabla\phi, \nabla\phi).$$

and, since $L_b(m^{-1}) = \frac{d}{ds}|_{s=0} \psi_s^* m^{-1} = 0$, *because* $\psi_s^* m^{-1} = m^{-1}$, *i.e.* ψ_s *is an isometry of the Minkowski spacetime, we therefore have that*

$$\frac{d}{ds}|_{s=0} \mathcal{L} \circ (\phi + sh, \nabla\phi + s\nabla h) = b^\mu \nabla_\mu (\mathcal{L} \circ (\phi, \nabla\phi)) = \nabla_\mu (b^\mu \mathcal{L} \circ (\phi, \nabla\phi)).$$

We therefore have that Ψ *defined by* $\Psi_s = \psi_s^*$ *is an infinitesimal symmetry with* $\mathbb{F}^\mu[\phi] = b^\mu \mathcal{L} \circ (\phi, \nabla\phi)$.

By Noether's theorem, the conserved current $\mathbb{J}[\phi]$ *takes the form with respect to Cartesian coordinates:*

$$\mathbb{J}^\mu[\phi] = \frac{\partial \mathcal{L}}{\partial v_\mu} \circ (\phi, \nabla\phi) b^\nu \nabla_\nu \phi - \mathbb{F}^\mu[\phi] = b^\nu \nabla^\mu \phi \nabla_\nu \phi - b^\mu \frac{1}{2} \left((m^{-1})^{\alpha\beta} \nabla_\alpha \phi \nabla_\beta \phi + \mathfrak{m}^2 \phi^2 \right),$$

with ϕ *a solution to the Euler–Lagrange equations.*

Since b *was arbitrary and constant, we can write* $\mathbb{J}_\mu[\phi] = b^\nu \mathbb{T}_{\mu\nu}[\phi]$, *with*

$$\mathbb{T}_{\mu\nu}[\phi] := \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m_{\mu\nu} \left((m^{-1})^{\alpha\beta} \nabla_\alpha \phi \nabla_\beta \phi + \mathfrak{m}^2 \phi^2 \right).$$

Then $\nabla^\mu \mathbb{T}_{\mu\nu}[\phi] = 0$. *The tensor field*

$$\mathbb{T}[\phi] = \mathbb{T}_{\mu\nu}[\phi] dx^\mu \otimes dx^\nu$$

is called the stress-energy tensor *or* energy-momentum tensor. *The key property of the stress-energy tensor is that it is* divergence-free *(i.e.* $\nabla^\mu \mathbb{T}_{\mu\nu}[\phi] = 0$*).*

We refer to

$$E[\phi](t) = \int_{\mathbb{R}^n} \mathbb{T}^{00}[\phi]|_{x^0 = t} \, d\mathbf{x}$$

as the energy of ϕ *at time* t *and*

$$P^i[\phi](t) = \int_{\mathbb{R}^n} \mathbb{T}^{0i}[\phi]|_{x^0 = t} \, d\mathbf{x}$$

as the i*-th component of the linear momentum of* ϕ *at time* t.

Example 3.7. *We consider the 1-parameter group of diffeomorphisms* $\psi_s : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ *corresponding to Lorentz transformations. In Cartesian coordinates, these must take the form:*

$$\psi_s^\mu(x) = \Lambda_\nu^\mu(s) x^\nu,$$

where $\Lambda_\nu^\mu(s)$ *are the components of a 1-parameter group of proper, ortochronous Lorentz transformations* $\Lambda(s) \in SO^+(n;1)$.[32] *It will be useful to characterize* $O(n;1)$ *as follows* $O(n;1) = \{\Lambda \in GL(\mathbb{R}^{n+1}) \,|\, \Lambda^T \eta \Lambda = \eta\}$, *with* η *the diagonal matrix with entries* $-1, +1, \ldots, +1$.[33] *Then* $\Lambda \in SO^+(n;1)$ *if additionally* $\det \Lambda = 1$ *and* $\Lambda_{00} > 0$.

[32] Exercise: Explain why the Lorentz transformations have to be proper and ortochronous.
[33] Exercise: Show that is characterization is equivalent to the definition of $O(n;1)$ from §2.2.

We will write $\Lambda(s)x$ for $\psi_s(x)$. Note that

$$\frac{d}{ds}\Big|_{s=0}\psi_s^\mu(x) = \frac{d}{ds}|_{s=0}(\Lambda^\mu_\nu(s)x^\nu) = A^\mu_\nu x^\nu,$$

with $A \in \mathfrak{so}(n;1) = \{A \in \mathrm{Mat}(n+1;\mathbb{R}) \,|\, \eta A^T \eta = -A\}$,[34] *where $\mathfrak{so}(n;1)$ is the Lie algebra corresponding to $O(n;1)$, $SO(n;1)$ and $SO^+(n;1)$. In fact, for any $A \in \mathfrak{so}(n;1)$, the matrix exponential $\Lambda(s) = \exp(sA)$ defines a 1-parameter group of proper, ortochronous Lorentz transformations such that $\frac{d}{ds}|_{s=0}\Lambda(s) = A$.*

Then, for $X = A^\mu_\nu x^\nu \partial_{x^\mu} \in \mathcal{T}(\mathcal{M})$:

$$\frac{d}{ds}\Big|_{s=0}\psi_s^*\phi = L_X\phi = X(\phi)$$

and

$$\nabla\frac{d}{ds}|_{s=0}(\psi_s^*\phi) = \nabla L_X\phi = L_X\nabla\phi,$$

so just like in the previous example, we obtain for $h = \frac{d}{ds}|_{s=0}(\psi_s^*\phi)$:

$$\frac{d}{ds}|_{s=0}\mathcal{L}\circ(\phi+sh,\nabla\phi+s\nabla h) = L_X(\mathcal{L}\circ(\phi,\nabla\phi)) - \frac{1}{2}L_X(m^{-1})(\nabla\phi,\nabla\phi).$$

In this case, we use the expression for the pullback in coordinates:

$$(\psi_s^* m)_{\mu\nu} = \frac{\partial\psi_s^\alpha}{\partial x^\mu}\frac{\partial\psi_s^\beta}{\partial x^\nu}m_{\alpha\beta} = \Lambda(s)^\alpha_\mu m_{\alpha\beta}\Lambda(s)^\beta_\nu = (\Lambda^T\eta\Lambda)_{\mu\nu} = m_{\mu\nu},$$

so $L_X m = 0$ and hence also $L_X m^{-1} = 0$, and we conclude that

$$\begin{aligned}\frac{d}{ds}|_{s=0}\mathcal{L}\circ(\phi+sh,\nabla\phi+s\nabla h) &= L_X(\mathcal{L}\circ(\phi,\nabla\phi))\\ &= A^\mu_\nu x^\nu\partial_\mu((\mathcal{L}\circ(\phi,\nabla\phi)))\\ &= x^\nu\partial_\mu(A^\mu_\nu(\mathcal{L}\circ(\phi,\nabla\phi))),\end{aligned}$$

Using that $\mathrm{tr}(A) = -\mathrm{tr}(\eta A\eta) = -\mathrm{tr}\,A$, we have that $\mathrm{tr}\,A = 0$, so we can further rewrite:

$$x^\nu\partial_\mu(A^\mu_\nu(\mathcal{L}\circ(\phi,\nabla\phi)))(x) = \partial_\mu(A^\mu_\nu x^\nu(\mathcal{L}\circ(\phi,\nabla\phi))).$$

Hence,

$$\mathbb{F}^\mu[\phi](x) = A^\mu_\nu x^\nu(\mathcal{L}\circ(\phi,\nabla\phi))(x)$$

and in analogy with the previous example,

$$\begin{aligned}\mathbb{J}^\mu[\phi](x) &= \frac{\partial\mathcal{L}}{\partial v_\mu}(\phi(x),\nabla\phi(x))A^\alpha_\nu x^\nu\partial_\alpha\phi(x) - \mathbb{F}^\mu[\phi](x)\\ &= A^\nu_\beta x^\beta\mathbb{T}^\mu{}_\nu[\phi](x).\end{aligned}$$

Since the dimension of $\mathfrak{so}(n;1)$ is $\frac{1}{2}(n+1)n$, there are $\frac{1}{2}n(n+1)$ linearly independent conserved currents associated to $SO(n;1)$. For example, we can label:

$$({}^{\alpha\beta}\mathbb{J})^\mu[\phi](x) := x^\alpha\mathbb{T}^{\mu\beta}[\phi](x) - x^\beta\mathbb{T}^{\mu\alpha}[\phi](x),$$

[34] By definition of Lorentz transformations $\frac{d}{ds}|_{s=0}(\Lambda^T(s)\eta\Lambda(s)) = 0$. Since taking the transpose of a matrix is a linear operation, we have that for $A = \frac{d}{ds}|_{s=0}\Lambda(s))$, $A^T\eta + \eta A = 0$. Equivalently, since $\eta = \eta^{-1}$, $\eta A^T\eta = -A$.

which are antisymmetric in α, β and therefore result in $\frac{1}{2}(n+1)n$ linearly independent currents.

The following $\frac{1}{2}n(n-1)$ quantities can be associated to the rotation subgroup $SO(n) \subseteq SO(n;1)$:

$$Q^{ij}[\phi](t) = \int_{\mathbb{R}^3} ({}^{ij}\mathbb{J})^0[\phi]|_{x^0=t}\, d\mathbf{x} = \int_{\mathbb{R}^3} (x^i \mathbb{T}^{0j}[\phi] - x^j \mathbb{T}^{0i}[\phi])|_{x^0=t}\, d\mathbf{x}$$

and can be interpreted as the angular momentum of ϕ at time t. The following n quantities are associated to Lorentz boosts:

$$Q^{0i}[\phi](t) = \int_{\mathbb{R}^3} ({}^{0i}\mathbb{J})^0[\phi]|_{x^0=t}\, d\mathbf{x} = \int_{\mathbb{R}^3} (x^0 T^{0i}[\phi] - x^i T^{00}[\phi])|_{x^0=t}\, d\mathbf{x}.$$

Proposition 3.9. *Suppose $\phi(t,\mathbf{x})$ and $\nabla\phi(t,\mathbf{x})$ go to zero sufficiently fast as $|\mathbf{x}| \to \infty$. Then $E[\phi](t)$, $P_i[\phi](t)$, $Q^{ij}[\phi](t)$ and $Q^{0i}[\phi](t)$ are conserved in time.*

Proof. EXERCISE. □

Example 3.8. *Let $\mathcal{M} = \mathbb{R}^{n+1}$, $r = 0$ and $V = \mathbb{R}^2$. We can identify $\mathbb{R}^2$ with $\mathbb{C}$ by letting $q^1 = \Re(q)$ and $q^2 = \Im(q)$, for $q \in \mathbb{C}$. Consider the Lagrangian:*

$$\mathcal{L} : \mathcal{N} \to \mathbb{R},$$
$$\mathcal{L}(x,q,v) = \frac{1}{2} m_x^{-1}(v^1, v^1) + \frac{1}{2} m_x^{-1}(v^2, v^2) + \frac{\mathfrak{m}^2}{2}|q|^2.$$

Let $\phi \in C^\infty(\mathbb{R}^{n+1};\mathbb{C})$ be a complex-valued field. Then, we consider the following one-parameter group action: $\Psi_s : C^\infty(\mathbb{R}^{n+1};\mathbb{C}) \to C^\infty(\mathbb{R}^{n+1};\mathbb{C})$, with

$$\Psi_s(\phi)(x) = e^{is}\phi(x).$$

Then

$$\frac{d}{ds}\Big|_{s=0} \Psi_s(\phi) = i\phi,$$
$$\nabla \frac{d}{ds}\Big|_{s=0} \Psi_s(\phi) = i\nabla\phi,$$

so for $h = \frac{d}{ds}\Big|_{s=0} \Psi_s(\phi) = i\phi = -\phi^2 + i\phi^1$,

$$\frac{d}{ds}|_{s=0} \mathcal{L} \circ (\phi + sh, \nabla\phi + s\nabla h) = m^{-1}(\nabla\phi^1, \nabla h^1) + m^{-1}(\nabla\phi^2, \nabla h^2) + \mathfrak{m}^2 \Re(\phi\overline{h}) = 0,$$

so $\mathbb{F}[\phi] \equiv 0$.

The corresponding Noether current $\mathbb{J}[\phi]$ is given by:

$$\mathbb{J}^\mu[\phi] = \frac{\partial \mathcal{L}}{\partial v_\mu^B} \circ (\phi, \nabla\phi) \left(\frac{d}{ds}|_{s=0} \Psi_s(\phi) \right)^B = \Re(\nabla^\mu\phi)\Re(i\phi) + \Im(\nabla^\mu\phi)\Im(i\phi) = \Im(\overline{\phi}\nabla^\mu\phi).$$

3.4 The Einstein–Hilbert action

We will now show that we can also derive the *Einstein equations of general relativity*, by extremizing an appropriate action. We start by considering the action corresponding to a field $\phi \in \Omega^r(\mathcal{M})$ with $(\mathcal{M}, g)$ an orientable $n+1$-dimensional Lorentzian manifold. Rather than keeping g fixed, we will

be considering one-parameter families of Lorentzian metrics on a fixed manifold $\mathcal{M}$.[35] Consider the action:

$$\mathcal{S}_{\text{matter}}(\phi, g) = \int_{\mathcal{M}} \mathcal{L}_{\text{matter}} \circ (\phi, \nabla^g \phi, g)\, \epsilon[g],$$

with $\epsilon[g]$ the natural volume form associated to g, ∇^g the covariant derivative associated to the Levi–Civita connection corresponding to g and

$$\begin{aligned} &\mathcal{L}_{\text{matter}} : \mathcal{N} \to \mathbb{R}, \\ &\mathcal{N} := \bigcup_{x \in \mathcal{M}} \{x\} \times (\wedge^r T_x^* \mathcal{M}) \times (T_x^* \mathcal{M} \otimes \wedge_x^r T^* \mathcal{M}) \times \text{Lor}_x, \end{aligned}$$

with Lor_x the set of symmetric, non-degenerate elements of $T_x^*\mathcal{M} \otimes T_x^*\mathcal{M}$ with signature $(-+\ldots+)$.

We will assume that $\mathcal{L}_{\text{matter}}(x, \phi_x, (\nabla^{g_1}\phi)_x, k) = \mathcal{L}_{\text{matter}}(x, q, (\nabla^{g_2}\phi)_x, k)$ for any two Lorentzian metrics g_1 and g_2, $\phi \in \Omega^r(\mathcal{M})$, $x \in \mathcal{M}$ and $k \in \text{Lor}_x$! All the Lagrangians we have encountered so far satisfy this property.

For example, we could consider a Klein–Gordon Lagrangian, which satisfies:

$$\mathcal{L}_{\text{matter}}(x, q, v, k) = \frac{1}{2} k^{-1}(v, v) + \frac{\mathfrak{m}^2}{2} q^2,$$

with $k^{-1} \in T_x\mathcal{M} \otimes T_x\mathcal{M}$ the unique tensor associated to $k \in \text{Lor}_x$ via musical isomorphisms induced by k. In this case $\nabla^{g_1}\phi = d\phi = \nabla^{g_2}\phi$ for any two Lorentzian metrics g_1, g_2.

The *equations of motion* of ϕ are the Euler–Lagrange equations corresponding to $\mathcal{S}_{\text{matter}}$, where g is held fixed. Since they depend on on the metric g, they can be thought of describing how the behaviour of the field on $(\mathcal{M}, g)$ is affected by the curvature of $(\mathcal{M}, g)$ ("spacetime tells matter how to move"). In §3.2, we showed that they can be derived by considering $\frac{d}{ds}|_{s=0}\mathcal{S}_{\text{matter}}(\phi + sh, g)$, with h vanishing at the boundary $\partial\mathcal{M}$.

The Einstein equations describe instead how the metric g is affected by the presence of the field ϕ ("matter tells spacetime how to curve"). We will derive them by defining the following *total action*:

$$\mathcal{S}(\phi, g) = -\frac{1}{16\pi} \int_{\mathcal{M}} R[g]\, \epsilon[g] + \mathcal{S}_{\text{matter}}(\phi, g),$$

where $R[g]$ is the Ricci scalar corresponding to the metric g and considering $\frac{d}{ds}|_{s=0}\mathcal{S}(\phi, g + sh)$ with h compactly supported in the interior of the manifold(with boundary)). The part of the action

$$\mathcal{S}_{\text{EH}}(g) := \frac{1}{16\pi} \int_{\mathcal{M}} R[g]\, \epsilon[g]$$

is called the *Einstein–Hilbert action*.

Remark 3.1. *The Ricci scalar R is a smooth function on $(\mathcal{M}, g)$ that is invariant under isometries by Proposition B.9 in the appendix. It is not the unique function on $\mathcal{M}$ with that property. We could have, for example, also considered contractions of the Riemann tensor $R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta}$ or the Ricci tensor $R_{\mu\nu}R^{\mu\nu}$ or functions thereof.*

We will restrict to a compact subset $\mathcal{K} \subset \mathcal{M}$. Let g be a Lorentzian metric and consider the one-parameter family of tensor fields $g^s = g + sh \in \mathcal{T}^{(0,2)}(\mathcal{K})$, with $h \in \mathcal{T}^{(0,2)}(\mathcal{K})$ symmetric and $s \in I \subset \mathbb{R}$, with $|s|$ suitably small (depending on g) such that g^s is a Lorentzian metric. If $\mathcal{K}$ has a

[35] It is at this point good to point out that not every manifold admits a Lorentzian metric $\mathcal{M}$, let alone a one-parameter family of Lorentzian metrics. If $\mathcal{M}$ is covered by a single coordinate chart $\{x^\mu\}$, however, we can express $g = g_{\alpha\beta}dx^\alpha \otimes dx^\beta$ and there is a bijection between the set of Lorentzian metrics and the set of smooth, symmetric, non-degenerate, matrix-valued functions, which can trivially be constructed.

boundary, then we will also assume that h is supported in the interior of $\mathcal{K}$ so that $h_{\mu\nu}$ and $\partial_\rho h_{\mu\nu}$ vanish at the boundary $\partial\mathcal{K}$ with respect to an arbitrary coordinate chart.

Since we have that, schematically,

$$\begin{aligned}\text{Riem} &= \partial\Gamma + \Gamma\Gamma,\\ \Gamma &= g^{-1}\partial g\end{aligned}$$

in the domain of a coordinate chart, we can see that $R[g]$ does not just depend on g and its partial derivatives, but it also depends on **second-order derivatives of** g. In view of our derivation of the Euler–Lagrange equations for general Lagrangians (with covariant derivatives replaced by partial derivatives), we might expect to obtain *third-order* equations of motion for the components of g (as the Euler–Lagrange equations corresponding to Lagrangians depending on the field and its first-order derivatives resulted in second-order equations). We will see however, that the equations of motion are in fact second order.

Lemma 3.10. *With respect to an arbitrary coordinate chart $(U, \{x^\mu\})$, we can express*

$$\frac{d}{ds}\Big|_{s=0} \epsilon[g^s] = \frac{1}{2}(g^{-1})^{\mu\nu} h_{\mu\nu}\, \epsilon[g].$$

Proof. We have that with respect an arbitrary coordinate chart $(U, \{x^\mu\})$:

$$\epsilon[g^s] = \sqrt{-\det g^s}\, dx^0 \wedge \ldots \wedge dx^n.$$

From the chain rule, it follows that

$$\frac{d}{ds}\Big|_{s=0} \sqrt{-\det g^s} = -\frac{1}{2}(-\det g)^{-\frac{1}{2}} \frac{d}{ds}\Big|_{s=0} \det g^s.$$

We use the following fact from linear algebra. For any symmetric $(n+1)\times(n+1)$ matrix Y, we have that:[36]

$$\det e^Y = e^{\operatorname{tr} Y}.$$

This follows immediately by using that a symmetric matrix Y can be written as $Y = B^{-1}DB$, where D is diagonal, and $e^Y = e^{B^{-1}DB} = B^{-1}e^D B$. In fact the identity holds also for general matrices, but we will not need that fact.

Let A, X be a symmetric matrices and consider $A_s := A + sX = A(\mathbf{1} + sA^{-1}X)$. We obtain

$$A_s = A e^{sA^{-1}X}(\mathbf{1} + B_s),$$

where $B_0 = 0$ and $\frac{d}{ds}|_{s=0} B_s = 0$.

Hence,

$$\det A_s = \det A \det(e^{sA^{-1}X}) \det(I + B_s).$$

We therefore have that

$$\frac{d}{ds}|_{s=0} \det A_s = \det A \frac{d}{ds}|_{s=0} \det(e^{sA^{-1}X}) = \frac{d}{ds}|_{s=0} e^{s \operatorname{tr}(A^{-1}X)} = \operatorname{tr}(A^{-1}X) \det A.$$

Hence, $\frac{d}{ds}|_{s=0} \det g^s = (g^{-1})^{\mu\nu} h_{\mu\nu} \det g$ and we conclude that

$$\frac{d}{ds}\Big|_{s=0} \sqrt{-\det g^s} = \frac{1}{2}\sqrt{-\det g}(g^{-1})^{\mu\nu} h_{\mu\nu}. \qquad \square$$

[36] The matrix exponent e^A of a general matrix A is given by $e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$.

Lemma 3.11. (i) *Denote with $\Gamma^{\rho}_{\mu\nu}[g^s]$ the Christoffel symbols with respect to g^s. Then with respect to an arbitrary coordinate chart*

$$\frac{d}{ds}\Big|_{s=0}\Gamma^{\rho}_{\mu\nu}[g^s] = \frac{1}{2}(g^{-1})^{\rho\sigma}(\nabla_\nu h_{\mu\sigma} + \nabla_\mu h_{\nu\sigma} - \nabla_\sigma h_{\mu\nu}),$$

where ∇ denotes the Levi–Civita covariant derivative with respect to g.

(ii) *For $R_{\mu\nu}[g^s]$ denoting the components of the Ricci tensors corresponding to the metrics g^s, we can express with respect to an arbitrary coordinate chart*

$$\frac{d}{ds}\Big|_{s=0}R[g^s] = -R_{\mu\nu}[g]h^{\mu\nu} + \nabla_\rho J^\rho,$$

with

$$J^\rho = \nabla^\mu h^\rho{}_\mu - \nabla^\rho h^\mu{}_\mu.$$

Proof. Recall that with respect to normal coordinates around a point $p \in \mathcal{M}$ corresponding to g: $g_{\mu\nu}(p) = m_{\mu\nu}$, $\Gamma^{\rho}_{\mu\nu} = 0$ and $\partial_\rho g_{\mu\nu}(p) = 0$. Hence,

$$\begin{aligned}\Gamma^{\rho}_{\mu\nu}[g^s](p) &= \frac{s}{2}(g^{-1})^{\rho\sigma}(p)(\partial_\nu h_{\mu\sigma} + \partial_\mu h_{\nu\sigma} - \partial_\sigma h_{\mu\nu})(p) + o(s),\\ &= \frac{s}{2}(g^{-1})^{\rho\sigma}(p)(\nabla_\nu h_{\mu\sigma} + \nabla_\mu h_{\nu\sigma} - \nabla_\sigma h_{\mu\nu})(p) + o(s),\end{aligned}$$

with $o(s)$ denoting terms that vanish in the limit $s \to 0$ and ∇ the covariant derivative with respect to g. We obtain

$$\frac{d}{ds}\Big|_{s=0}\Gamma^{\rho}_{\mu\nu}[g^s](p) = \frac{1}{2}(g^{-1})^{\rho\sigma}(\nabla_\nu h_{\mu\sigma} + \nabla_\mu h_{\nu\sigma} - \nabla_\sigma h_{\mu\nu})(p).$$

While the above expression was derived with respect to normal coordinates around p, it remains valid under a change of coordinates. Hence, we can define a tensor field $C \in \mathcal{T}^{(1,2)}(\mathcal{M})$ globally, with $C^\rho{}_{\mu\nu}$ equal to the right-hand side above in *any* coordinate chart. We conclude (i).

To obtain (ii), we turn again to normal coordinates around $p \in \mathcal{M}$ with respect to the metric g, in which the components of the Riemann tensor with respect to g^s, $R^\sigma{}_{\mu\rho\nu}[g^s]$, satisfy

$$R^\sigma{}_{\mu\rho\nu}[g^s](p) = \partial_\rho\Gamma^\sigma_{\nu\mu}[g^s](p) - \partial_\nu\Gamma^\sigma_{\rho\mu}[g^s](p) + o(s).$$

and

$$R_{\mu\nu}[g^s](p) = \partial_\rho\Gamma^\rho_{\nu\mu}[g^s](p) - \partial_\nu\Gamma^\rho_{\rho\mu}[g^s](p) + o(s).$$

Hence,

$$R[g^s](p) = ((g^s)^{-1})^{\mu\nu}(p)R_{\mu\nu}[g^s](p) + o(s).$$

We therefore have that

$$\begin{aligned}\frac{d}{ds}\Big|_{s=0}R[g^s](p) &= R_{\mu\nu}[g](p)\frac{d}{ds}\Big|_{s=0}((g^s)^{-1})^{\mu\nu}(p) + (g^{-1})^{\mu\nu}\partial_\rho\frac{d}{ds}\Big|_{s=0}\Gamma^\rho_{\nu\mu}[g^s](p) - (g^{-1})^{\mu\nu}\partial_\nu\frac{d}{ds}\Big|_{s=0}\Gamma^\rho_{\rho\mu}[g^s](p)\\ &= R_{\mu\nu}[g](p)\frac{d}{ds}\Big|_{s=0}((g^s)^{-1})^{\mu\nu}(p) + \frac{1}{2}\nabla^\rho(\nabla^\mu h_{\mu\rho} + \nabla^\mu h_{\mu\rho} - \nabla_\rho h^\mu_\mu)(p)\\ &\quad - \frac{1}{2}\nabla^\mu(\nabla_\mu h^\rho_\rho + \nabla^\rho h_{\mu\rho} - \nabla^\rho h_{\rho\mu})(p)\\ &= R_{\mu\nu}[g](p)\frac{d}{ds}\Big|_{s=0}((g^s)^{-1})^{\mu\nu}(p) + \nabla^\rho(\nabla^\mu h_{\mu\rho} - \nabla_\rho h^\mu_\mu).\end{aligned}$$

Since $((g^s)^{-1})^{\mu\nu} g^s_{\nu\alpha} = \delta^\mu_\alpha$, we have that

$$g_{\nu\alpha} \frac{d}{ds}\Big|_{s=0} ((g^s)^{-1})^{\mu\nu} = -(g^{-1})^{\mu\nu} \frac{d}{ds}\Big|_{s=0} g^s_{\nu\alpha} = -(g^{-1})^{\mu\nu} h_{\nu\alpha}$$

From this it follows that

$$\frac{d}{ds}\Big|_{s=0} ((g^s)^{-1})^{\mu\nu} = -(g^{-1})^{\mu\alpha}(g^{-1})^{\nu\beta} h_{\alpha\beta} = -h^{\mu\nu}. \tag{3.15}$$

We conclude that

$$\frac{d}{ds}|_{s=0} R[g^s](p) = -R_{\mu\nu}[g] h^{\mu\nu}(p) + (\nabla_\rho J^\rho)(p),$$

with respect to normal coordinates, with

$$J^\rho = \nabla^\mu h^\rho_\mu - \nabla^\rho h^\mu_\mu.$$

As above, we can conclude that in fact the right-hand side of $\frac{d}{ds}|_{s=0} R[g^s](p)$ takes the same form also with respect to arbitrary coordinates around p.

Note that the term $\nabla_\rho J^\rho$ involves second-order derivatives of h, because it is the divergence of a vector field, there will be no need to integrate it by parts, since it will vanish after integration. □

Proposition 3.12. *Assume that* $\mathcal{L}_{\text{matter}}(x, \phi_x, (\nabla^{g_1}\phi)_x, k) = \mathcal{L}_{\text{matter}}(x, \phi_x, (\nabla^{g_2}\phi)_x, k)$ *for all Lorentzian metrics* g_1, g_2, $\phi \in \Omega^r(\mathcal{M})$, $x \in \mathcal{M}$ *and* $k \in \text{Lor}_x$.

A pair $(\phi, g) \in \Omega^r(\mathcal{K}) \times \mathcal{T}^{(0,2)}(\mathcal{K})$, *with* g *a Lorentzian metric, is an extremum of* $\mathcal{S}(g, \phi)$ *restricted to a compact subset* $\mathcal{K} \subseteq \mathcal{M}$ *implies that in the interior of* $\mathcal{K}$*:*

$$G_{\mu\nu}[g] = 8\pi \mathbb{T}_{\mu\nu}[\phi, g], \tag{3.16}$$

$$\nabla_\mu P^{\mu\mu_1\dots\mu_r} - \frac{\partial \mathcal{L}_{\text{matter}}}{\partial q_{\mu_1\dots\mu_r}} \circ (\phi, \nabla^g\phi, g) = 0. \tag{3.17}$$

with $P \in \mathcal{T}^{(r+1,0)}(\mathcal{M})$ *satisfying* $P^{\mu\mu_1\dots\mu_r} = \frac{\partial \mathcal{L}_{\text{matter}}}{\partial v_{\mu\mu_1\dots\mu_r}} \circ (\phi, \nabla^g\phi, g)$ *and*

$$\mathbb{T}_{\mu\nu}[\phi, g] := 2\left(\frac{\partial \mathcal{L}_{\text{matter}}}{\partial (k^{-1})^{\mu\nu}} \circ (\phi, \nabla^g\phi, g) - \frac{1}{2} g_{\mu\nu} \mathcal{L}_{\text{matter}} \circ (\phi, \nabla^g\phi, g)\right). \tag{3.18}$$

The equations (3.16) *are called* the Einstein equations of general relativity. *By the contracted Bianchi identity (Corollary B.11), we must have that*

$$\nabla^\mu \mathbb{T}_{\mu\nu}[\phi, g] = 0,$$

if ϕ *and* g *solve* (3.16) *and* (3.17).

We will refer to the corresponding tensor field $\mathbb{T}[\phi, g]$ *as the* Hilbert energy-momentum tensor *to contrast it with the energy-momentum tensor obtained from Noether's Theorem, which we will call the* Noether energy momentum-tensor.

Proof. By Lemma 3.10 and (ii) of Lemma 3.11, we have that

$$-\frac{d}{ds}\Big|_{s=0} \mathcal{S}_{\text{EH}}(g^s) = \frac{1}{16\pi} \int_{\mathcal{K}} (R_{\mu\nu}[g] - \frac{1}{2} g_{\mu\nu}) h^{\mu\nu}\, \epsilon[g] = \frac{1}{16\pi} \int_{\mathcal{K}} G_{\mu\nu}[g] h^{\mu\nu}\, \epsilon[g],$$

where we used the divergence theorem together with $J|_{\partial\mathcal{K}} = 0$ to conclude that the integral of $\nabla_\rho J^\rho$ vanishes.

We will now investigate $\mathcal{S}_{\text{matter}}$. By the chain rule, we have that

$$\frac{d}{ds}\Big|_{s=0}(\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^{g^s}\phi,g^s)\epsilon[g^s]) = \frac{d}{ds}\Big|_{s=0}\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g^s)\,\epsilon[g]+\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g^s)\frac{d}{ds}\Big|_{s=0}\epsilon[g^s].$$

By applying (3.15), we therefore have that

$$\frac{d}{ds}\Big|_{s=0}\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g^s) = -\frac{\partial\mathcal{L}_{\text{matter}}}{\partial(k^{-1})^{\mu\nu}}\circ(\phi,\nabla^g\phi,g)h^{\mu\nu},$$

where $h^{\mu\nu}$ are the components of the (2,0)-tensor field dual to h, defined via the musical isomorphisms with respect to g.

Together with Lemma 3.10, it follows therefore that

$$\frac{d}{ds}\Big|_{s=0}(\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g^s)\epsilon[g^s]) = -\left(\frac{\partial\mathcal{L}_{\text{matter}}}{\partial(k^{-1})^{\mu\nu}}(\phi,\nabla^g\phi,g) - \frac{1}{2}g_{\mu\nu}\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g)\right)h^{\mu\nu}\,\epsilon[g].$$

Hence, $\frac{d}{ds}\Big|_{s=0}(\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g^s)\epsilon[g^s]) = -\frac{1}{2}\mathbb{T}_{\mu\nu}[\phi,g]h^{\mu\nu}$, with

$$\mathbb{T}_{\mu\nu}[\phi,g] = 2\left(\frac{\partial\mathcal{L}_{\text{matter}}}{\partial(k^{-1})^{\mu\nu}}\circ(\phi,\nabla^g\phi,g) - \frac{1}{2}g_{\mu\nu}\mathcal{L}_{\text{matter}}\circ(\phi,\nabla^g\phi,g)\right).$$

We then conclude that:

$$\frac{d}{ds}\Big|_{s=0}\mathcal{S}(\phi,g^s) = \frac{1}{16\pi}\int_{\mathcal{K}}(-8\pi\mathbb{T}_{\mu\nu}[\phi,g]+G_{\mu\nu}[g])h^{\mu\nu}\,\epsilon[g].$$

Since h was arbitrary, we can apply Lemma 2.15 to the components $h^{\mu\nu}$ to conclude that (3.16) holds in the interior of $\mathcal{K}$.

We conclude (3.17) by instead considering $\frac{d}{ds}|_{s=0}\mathcal{S}_{\text{matter}}(\phi+s\psi,g)$, with $\psi\in\Omega^r(\mathcal{K})$, such that $\psi|_{\partial\mathcal{K}}\equiv 0$, which results in the Euler–Lagrange equations, as discussed in the previous chapters.

From the contracted Bianchi identity Corollary B.11 together with (3.16), it follows immediately that $\nabla^\mu\mathbb{T}_{\mu\nu}[\phi,g]=0$. □

Remark 3.2. *The Hilbert energy-momentum tensor $\mathbb{T}$ is automatically symmetric, since the tensor G is symmetric. This $\mathbb{T}$ does not need to coincide with the Noether energy momentum tensor. As we have seen in the problem sheets, the canonical energy-momentum tensor need not be symmetric. One can show that:*

$$\mathbb{T}^{\text{Hilbert}}_{\mu\nu} = \mathbb{T}^{\text{Noether}}_{\mu\nu} + \nabla_\lambda\mathbb{B}^\lambda{}_{\mu\nu},$$

for an appropriate choice of (1,2) tensor $\mathbb{B}$.[37]

The definition (3.18) therefore provides a convenient way of constructing symmetric energy-momentum tensors, even when one is only interested in the Euler–Lagrange equations with a fixed g. It moreover singles out a unique of energy-momentum tensor that is relevant when studying the interactions of the field theory with gravity.

Remark 3.3. *One can also study the gravitational properties of matter with equations of motions that do not necessarily arise from an action principle, by considering (3.16) with a given symmetric (0,2) tensor field $\mathbb{T}$ such that $\nabla_\mu\mathbb{T}^{\mu\nu}=0$. For example, when studying the gravitational properties of perfect fluids (Einstein–Euler equations).*

[37] This is called the Belinfante–Rosenfeld modification.

By redefining $\mathcal{S}_{\mathrm{EH}}(g) := \frac{1}{16\pi}\int_{\mathcal{M}}(R[g]-2\Lambda)\,\epsilon[g]$ with $\Lambda \in \mathbb{R}$ a constant, we obtain instead the equations:

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi \mathbb{T}_{\mu\nu}[\phi, g]. \tag{3.19}$$

EXERCISE: Show this.

The constant Λ is called a *cosmological constant* and it plays an important role when gravitational properties at a cosmological scale, where $\Lambda > 0$ (and very small) must be included to account for experimental observations at cosmological scales. By putting the $\Lambda g_{\mu\nu}$ on the right-hand side of the equation with $8\pi\mathbb{T}_{\mu\nu}$, it can be interpreted as part of the energy-momentum tensor. It is therefore often referred to as *dark energy*.

In four spacetime dimensions, the Einstein equations with cosmological constant are the most general equations with a symmetric, divergence-free (0,2)-tensor on the left-hand side:

Theorem 3.13 (Lovelock's Theorem)**.** *Let $(\mathcal{M}, g)$ be a four-dimensional Lorentzian manifold. Let $H \in \mathcal{T}^{(0,2)}(\mathcal{M})$ be symmetric and divergence free ($\nabla^\mu H_{\mu\nu} = 0$), such for any $p \in \mathcal{M}$ and with respect to any coordinate chart with domain containing p, $H_{\mu\nu}(p)$ can be expressed as a function of $g_{\mu\nu}(p)$, $\partial_\rho g_{\mu\nu}(p)$ and $\partial_\sigma\partial_\rho g_{\mu\nu}(p)$. Then, there exist $\alpha, \beta \in \mathbb{R}$, such that*

$$H_{\mu\nu} = \alpha G_{\mu\nu} + \beta g_{\mu\nu}.$$

4 Gauge theories

4.1 Gauge transformations in electromagnetism

We consider the electromagnetic Lagrangian from Example 3.2, which corresponds to the Maxwell equations of electromagnetism in vacuum, but we replace $(\mathbb{R}^{n+1}, m)$ by a general Lorentzian manifold $(\mathcal{M}, g)$:

$$\begin{aligned}\mathcal{L} : \mathcal{N} &\to \mathbb{R}, \quad \text{with} \\ \mathcal{N} &= \bigcup_{x\in\mathcal{M}} \{x\} \times \wedge^1 T_x^*\mathcal{M} \times (T_x^*\mathcal{M} \otimes \wedge^1 T_x^*\mathcal{M}), \\ \mathcal{L}(x,q,v) &= -\frac{1}{4}(g^{-1})_x^{\alpha\mu}(g^{-1})_x^{\beta\nu}(v_{\mu\nu} - v_{\nu\mu})(v_{\alpha\beta} - v_{\beta\alpha}).\end{aligned}$$

Recall that for $A \in \Omega^1(\mathcal{M})$,

$$\mathcal{L} \circ (A, \nabla A) = \frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

with $F = dA \in \Omega^2(\mathcal{M})$.

Now, consider the map $\Psi_f : \Omega^1(\mathcal{M}) \to \Omega^1(\mathcal{M})$, $\Psi_f(A) = A - df$, where $f \in C^\infty(\mathcal{M})$. Since $d\Psi_f(A) = dA = F$, it follows immediately that:

$$\mathcal{L} \circ (\Psi_f(A), \nabla\Psi_f(A)) = \mathcal{L} \circ (A, \nabla A).^{38}$$

The maps Ψ_f are called a *gauge transformations*. Below, we will show that they can be associated to a Lie group (a *gauge group*) by adding another field to the electromagnetic Lagrangian. We consider (ϕ, A), with $\phi \in C^\infty(\mathcal{M}; \mathbb{C})$ and $A \in \Omega^1(\mathcal{M})$. Fix $f \in C^\infty(\mathcal{M})$. Consider the maps:

$$\begin{aligned}\Phi_f : C^\infty(\mathcal{M}; \mathbb{C}) \times \Omega^1(\mathcal{M}) &\to C^\infty(\mathcal{M}; \mathbb{C}) \times \Omega^1(\mathcal{M}), \\ \Phi_f(\phi, A)(x) &= (e^{i\mathfrak{e}f(x)}\phi(x), A(x) - df(x)),\end{aligned}$$

with $\mathfrak{e} \in \mathbb{R}$ arbitrary. In slight abuse of notation, we will also denote $\Phi_f(\phi)(x) = e^{i\mathfrak{e}f(x)}\phi(x)$ and $\Phi_f(A) = A - df$.

We wish to combine the electromagnetic Lagrangian with the Lagrangian from Example 3.8 with $\mathfrak{m} = 0$, such that the resulting Lagrangian is invariant under Φ_f ("gauge invariant"). For this purpose, we introduce the following operator:

$$\begin{aligned}d^A &: C^\infty(\mathcal{M}; \mathbb{C}) \to \Omega^1(\mathcal{M}; \mathbb{C}) \\ d^A\phi &= d\phi + i\mathfrak{e}\phi A,\end{aligned}$$

with $\mathfrak{e} \in \mathbb{R}$.

Proposition 4.1. *Consider the Lagrangian:*

$$\begin{aligned}\mathcal{L} : \mathcal{N} &\to \mathbb{R}, \quad \text{with} \\ \mathcal{N} &= \bigcup_{x\in\mathcal{M}} \{x\} \times \mathbb{C} \times (T_x^*\mathcal{M} \otimes \mathbb{C}) \times \wedge^1 T_x^*\mathcal{M} \times (T_x^*\mathcal{M} \otimes \wedge^1 T_x^*\mathcal{M}) \\ \mathcal{L}(x,q,v,p,w) &= \underbrace{\frac{1}{2}(g^{-1})_x^{\mu\nu} v_\mu \overline{v}_\nu}_{\text{"scalar field part"}} + \underbrace{\frac{1}{4}(g^{-1})_x^{\alpha\mu}(g^{-1})_x^{\beta\nu}(w_{\mu\nu} - w_{\nu\mu})(w_{\alpha\beta} - w_{\beta\alpha})}_{\text{"electromagnetic part"}}.\end{aligned}$$

Then

$$\mathcal{L} \circ (\phi, A, d^A\phi, \nabla A) = \mathcal{L} \circ (\Phi_f(\phi, A), d^{\Phi_f(A)}\Phi_f(\phi), \nabla\Phi_f(A)).$$

[38] Exercise: Determine the corresponding Noether currents.

Proof. We will show that $d^A\Phi_f(\phi)(x) = e^{i\mathfrak{e}f(x)}d^A\phi(x)$.

$$\begin{aligned} d_\mu^{\Phi_f(A)}\Phi_f(\phi) &= d_\mu^A(e^{i\mathfrak{e}f}\phi) = \partial_\mu(e^{i\mathfrak{e}f}\phi) + i\mathfrak{e}\Phi_f(A)_\mu e^{if}\phi \\ &= e^{i\mathfrak{e}f}\partial_\mu\phi + i\mathfrak{e}\partial_\mu f e^{i\mathfrak{e}f}\phi + i\mathfrak{e}(A_\mu - \partial_\mu f)e^{i\mathfrak{e}f}\phi \\ &= e^{i\mathfrak{e}f}(\partial_\mu\phi + i\mathfrak{e}\phi A_\mu) \\ &= e^{i\mathfrak{e}f}d_\mu^A\phi. \end{aligned}$$

Hence,

$$(g^{-1})^{\mu\nu}d_\mu^{\Phi_f(A)}\Phi_f(\phi)\overline{d_\nu^{\Phi_f(A)}\Phi_f(\phi)} = (g^{-1})^{\mu\nu}d^A\phi_\mu\overline{d^A\phi_\nu}.$$

We already showed that $d\Phi_f(A) = dA = F$, so the electromagnetic part of the Lagrangian is also invariant under Φ_f. □

We can interpret the map $\Phi_f : C^\infty(\mathcal{M};\mathbb{C}) \to C^\infty(\mathcal{M};\mathbb{C})$, $\Phi_f(\phi)$, as a group action corresponding to the Lie group $U(1)$ at each $x \in \mathbb{R}^{n+1}$. Indeed, if we identify $\mathbb{R}^2$ with $\mathbb{C}$, we can express

$$U(1) = \{z \in \mathbb{C} \,|\, |z| = 1\},$$

so $z = e^\theta$, where $\theta \in i\mathbb{R} \cong \mathfrak{u}(1)$. Consider the map

$$\begin{aligned} \Phi : C^\infty(\mathcal{M};U(1)) \times C^\infty(\mathbb{R}^{n+1};\mathbb{C}) &\to \mathbb{C}^\infty(\mathbb{R}^{n+1};\mathbb{C}), \\ \Phi(g,\phi)(x) &= g(x)\phi(x). \end{aligned}$$

Note that $\Phi(gh,\phi)(x) = \Phi(g,\Phi_h(\phi))(x)$ and $\Phi(e,\phi) = \phi$, so Φ is a left group action.

We have that $\Phi(g,\phi) = \Phi_f(\phi)$, for $f \in C^\infty(\mathcal{M})$ such that $e^{i\mathfrak{e}f} = g$.

4.2 Gauge transformations for Lie groups

The above characterization of gauge transformations as group actions corresponding to a Lie group provides a path towards generalizing gauge transformations in the context of more general Lie groups.

In this section, we will assume that $\mathcal{M}$ is an $n+1$ dimensional manifold that is covered by a single coordinate chart. For example, we could consider $\mathcal{M} = \mathbb{R}^{n+1}$ in Cartesian coordinates. Equivalently, we could restrict to $U \subset \mathcal{M}$ covered by a single chart. This is implicitly done in most physics texts. It is possible to introduce all the notions below more generally outside the domain of a single coordinate chart, but this requires more advanced geometric notions like *principal G-bundles* and *adjoint bundles*.

Definition 4.1. *Let G be a Lie group of dimension m and V a finite-dimensional vector space. Then a* representation *of G is a smooth group homomorphism*[39]*:*

$$\Pi : G \to GL(V).$$

The derivative map of Π at the identity $\pi : \mathfrak{g} \cong T_eG \to \mathfrak{gl}(V)$ then defines a representation *of the associated Lie algebra $\mathfrak{g}$, which a Lie algebra homomorphism, i.e. a linear map:*

$$\pi : \mathfrak{g} \to \mathfrak{gl}(V),$$

such that $\pi([X,Y]) = \rho(X)\rho(Y) - \rho(Y)\rho(X)$ for all $X, Y \in \mathfrak{g}$.

[39]This means: $\Pi(gh) = \Pi(g)\Pi(h)$ for all $g, h \in G$.

We can express:

$$\pi(X) = \frac{d}{ds}|_{s=0}\Pi(\exp(sX)),$$

where $\exp : \mathfrak{gl}(V) \to GL(V)$ is the (matrix) exponential map.

Let $\phi \in C^\infty(\mathcal{M}; V)$. Then

$$\begin{aligned}\Phi : C^\infty(\mathcal{M}; G) \times C^\infty(\mathcal{M}; V) &\to C^\infty(\mathcal{M}; V),\\ \Phi(g, \phi)(x) &= \Pi(g(x))\phi(x)\end{aligned}$$

is a left-group action. Denote $\Phi_g = \Phi(g, \cdot) : C^\infty(\mathcal{M}; V) \to C^\infty(\mathcal{M}; V)$.

Definition 4.2. *Let $\mathcal{M}$ be a manifold, G a Lie group, V a vector space and $\mathfrak{g}$ the associated Lie algebra. Let $A \in \Omega^1(\mathcal{M}; \mathfrak{g})$ and $\phi \in C^\infty(\mathcal{M}; V)$.*

Then the covariant derivative associated to (G, Π, A) on $(\mathcal{M}, g)$ *is the following map:*

$$\begin{aligned}d^A : C^\infty(\mathcal{M}; V) &\to \Omega^1(\mathcal{M}; V),\\ d^A\phi &= d\phi + \pi(A)\phi,\end{aligned}$$

where $\pi(A) \in \Omega^1(\mathcal{M}; \mathfrak{gl}(V)) = \Gamma(\mathfrak{gl}(V) \otimes T^*\mathcal{M})$ defined as follows: $\pi(A) = \pi(e_i) \otimes A^i$, with e_i, $i = 1, \ldots, \dim G$ a basis of $\mathfrak{g}$.

An important class of Lie groups are matrix Lie groups. They are the main Lie groups that play a role in physics.

Definition 4.3. *A matrix Lie group G is a subgroup of $GL(m)$ ($m \times m$ invertible matrices) or $GL(m; \mathbb{C})$ ($m \times m$ invertible matrices with complex entries) such that G is topologically closed in $GL(m)$.*

Example 4.1. *Let G be a matrix Lie group such that $G \subset GL(V)$, for some finite dimensional vector space V. Then the fundamental representation is defined as $\Pi : G \to GL(V)$ with $\Pi(g) = g$ and the corresponding Lie algebra representation is $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$, with $\pi(X) = X$. The corresponding covariant derivative then takes the form:*

$$d^A\phi = d\phi + A\phi = d\phi + e_i(\phi) \otimes A^i.$$

Example 4.2. *Let G be a Lie group. Consider the conjugation maps $\Psi^g : G \to G$, $\Psi^g(h) = ghg^{-1}$ and their derivative maps at the identity $d\Psi^g_e : T_eG \to T_eG$. Then $\mathrm{Ad} : g \mapsto d\Psi^g_e$ defines a Lie group representation with $V = T_eG \cong \mathfrak{g}$. We denote $\mathrm{Ad}_g = \mathrm{Ad}(g)$. Furthermore, if G is a matrix Lie group, then $\mathrm{Ad}(g)(X) = gXg^{-1}$. (*Exercise*)*

*It can be shown (*Exercise*) that the adjoint representation of the corresponding Lie algebra $\mathfrak{g}$ then satisfies: for all $X, Y \in \mathfrak{g}$*

$$\begin{aligned}\mathrm{ad} : \mathfrak{g} &\to \mathfrak{gl}(\mathfrak{g}),\\ \mathrm{ad}(X)(Y) &= [X, Y].\end{aligned}$$

One can directly verify that ad *is a Lie algebra homomorphism by applying the Jacobi identity:*

$$\mathrm{ad}([X,Y])(Z) = [[X,Y],Z] = -[[Y,Z],X] - [[Z,X],Y] = [X,[Y,Z]] - [Y,[X,Z]] = \mathrm{ad}(Y)\mathrm{ad}(X)(Z) - \mathrm{ad}(X)\mathrm{ad}(Y)(Z).$$

The corresponding covariant derivative of $\phi \in C^\infty(\mathcal{M}; \mathfrak{g})$ is given by:

$$d^A\phi = d\phi + \mathrm{ad}(A)\phi = d\phi + [A, \phi] := d\phi + [e_i, \phi] \otimes A^i.$$

In order to define a gauge transformation for a pair (ϕ, A) we still need to find a transformation $\Phi_g(A)$ of $A \in \Omega^1(\mathcal{M}; \mathfrak{g})$, with $g \in C^\infty(\mathcal{M}; G)$, that guarantees that

$$d^{\Phi_g(A)}(\Phi_g\phi)(x) = \Pi(g(x))(d^A\phi)(x) := \Pi(g)(x)\left(\partial_\mu\phi(x) + \pi(A_\mu(x))\phi(x)\right) \otimes dx^\mu_x,$$

which is analogous to what we saw for $G = U(1)$.

Proposition 4.2. *Let G be a compact and connected matrix Lie group. Let $g \in C^\infty(\mathcal{M}; G)$. The map*

$$\Phi_g : \Omega^1(\mathcal{M}; \mathfrak{g}) \to \Omega^1(\mathcal{M}; \mathfrak{g}),$$
$$\Phi_g(A)(x) = \mathrm{Ad}_{g(x)}(A(x)) - \partial_\mu g(x) g^{-1}(x) \otimes dx^\mu = g(x) e_i g^{-1}(x) \otimes A^i - \partial_\mu g(x) g^{-1}(x) \otimes dx^\mu$$

is well-defined and for all $\phi \in C^\infty(\mathcal{M}; V)$

$$d^{\Phi_g(A)}(\Phi_g(\phi))(x) = \Pi(g(x))(d^A\phi)(x).$$

We refer to Φ_g as a (local) gauge transformation.

Remark 4.1. *Note that the transformation in Proposition 4.2 only makes sense in the domain of a coordinate chart and for G a matrix Lie group. Recall that on the abelian group $G = U(1)$, we had $\Phi_g(A) = A - df$ which does make sense globally.*

We first prove the following lemma:

Lemma 4.3. *Consider (G, Π), with G a connected and compact Lie group and Π a Lie group representation. Let $\mathfrak{g}$ be the corresponding Lie algebra.*

(i) *If $X \in \mathfrak{g}$ and $g \in G$, then*
$$\pi(X) = \Pi(g^{-1})\pi(\mathrm{Ad}_g(X))\Pi(g).$$

(ii) *Let G be a matrix Lie group and $\gamma : I \to G$ a smooth curve. Then*
$$\frac{d\gamma}{ds}(s)\gamma(s)^{-1} \in \mathfrak{g}$$
and
$$\frac{d}{ds}\left(\Pi(\gamma)\right)(s)\Pi(\gamma(s)^{-1}) = \pi\left(\frac{d\gamma}{ds}(s)\gamma(s)^{-1}\right).$$

(iii) *Let $g \in C^\infty(\mathcal{M}; G)$, with G a matrix Lie group. Then for all $0 \le \mu \le n+1$ and $x \in \mathcal{M}$, $\partial_\mu g(x) g(x)^{-1} \in \mathfrak{g}$ and*
$$\partial_\mu(\Pi \circ g)(x)(\Pi(g(x)))^{-1} = \pi(\partial_\mu g(x) g(x)^{-1}).$$

Proof. Let $g \in G$. Note that there always exists a $\theta \in \mathfrak{g}$, such that $g = \exp(\theta)$, if $\exp : \mathfrak{g} \to G$ is surjective. A sufficient condition for surjectivity turns out out to be connectedness and compactness for G. Then $\Pi(g) = \exp(\pi(\theta))$ and $\Pi(g^{-1}) = \exp(-\pi(\theta))$, so

$$\Pi(g^{-1})\pi(\mathrm{Ad}_g(X))\Pi(g) = \exp(-\pi(\theta))\pi(\mathrm{Ad}_{\exp(\theta)}(X))\exp(\pi(\theta)).$$

We consider $f : I \to \mathfrak{g}$, with $I \subset \mathbb{R}$ a neighbourhood of 0 and

$$f(s) = \exp(-\pi(s\theta))\pi(\mathrm{Ad}_{\exp(s\theta)}(X))\exp(\pi(s\theta)).$$

Then $f(0) = \pi(X)$. We will show that $f'(s) = 0$ to conclude (i).

Since $\mathrm{Ad}_{gh}(X) = \mathrm{Ad}_g(\mathrm{Ad}_h(X))$ for all $g, h \in G$, we can write

$$\frac{d}{ds}|_{s=s_0}\mathrm{Ad}_{\exp(s\theta)}(X) = \frac{d}{ds}|_{s=0}(\mathrm{Ad}_{\exp(s\theta)}(\mathrm{Ad}_{\exp(s_0\theta)}(X)) = \mathrm{ad}_\theta(\mathrm{Ad}_{\exp(s_0\theta)}(X)) = [\theta, \mathrm{Ad}_{\exp(s_0\theta)}(X)].$$

We have that by the product rule and linearity of π:

$$\begin{aligned} f'(s_0) &= \exp(-s_0\pi(\theta))\Big(- \pi(\theta)\pi(\mathrm{Ad}_{\exp(s_0\theta)}(X)) + \pi(\mathrm{Ad}_{\exp(s_0\theta)}(X))\pi(\theta) \\ &\quad + \pi([\theta, \mathrm{Ad}_{\exp(s_0\theta)}(X)])\Big)\exp(s_0\pi(\theta)) \\ &= \exp(-s_0\pi(\theta)\Big(- [\pi(\theta), \pi(\mathrm{Ad}_{\exp(s_0\theta)}(X))] + \pi([\theta, \mathrm{Ad}_{\exp(s_0\theta)}(X)])\Big) \\ &= 0. \end{aligned}$$

This concludes (i).

We now turn to (ii). Let $s_0 \in I$. Define $\rho : I \to G$ as follows: $\rho(s) = \gamma(t + s_0)\gamma(s_0)^{-1}$. Then $\rho(0) = e$, so $\rho'(0) \in \mathfrak{g}$. Since we can express

$$\rho'(0) = \gamma'(s_0)\gamma(s_0)^{-1},$$

we have that $\gamma'(s_0)\gamma(s_0)^{-1} \in \mathfrak{g}$ for all $s_0 \in I$. Furthermore, we can express:

$$\frac{d}{ds}|_{s=s_0}\left(\Pi(\gamma)\right)(s)\Pi(\gamma^{-1}(s_0)) = \frac{d}{ds}|_{s=s_0}\left(\Pi(\rho(s)\gamma(s_0))\right)(s_0)\Pi(\gamma^{-1}(s_0)) = (\Pi \circ \rho)'(0).$$

We also have that

$$\pi\left(\frac{d\gamma}{ds}(s_0)\gamma^{-1}(s_0)\right) = \pi(\rho'(0)).$$

By definition of the Lie algebra representation,

$$\pi(\rho'(0)) = \frac{d}{ds}|_{s=0}\Pi(\exp(t\rho'(0))) = \frac{d}{ds}|_{s=0}\Pi(\rho)(s) = (\Pi \circ \rho)'(0).$$

This gives the desired identity.

We obtain (iii) by letting $\gamma(s) = g(\delta(s))$, with $\delta : I \to \mathcal{M}$, $\delta(s_0) = x$ and $\delta'(s_0) = \partial_\mu|_x$. Then $\gamma'(s_0) = \partial_\mu g(x)$ and applying (ii) at $s = s_0$. □

Proof of Proposition 4.2. By Lemma 4.3, the map Φ_g is well-defined. We denote with $\Pi(g)$ the map $x \mapsto \Pi(g(x))$. We can write locally, in the domain of a chart:

$$\begin{aligned} d^{\Phi_g(A)}\Pi(g)(\phi(x)) &= d(\Pi(g)\phi)(x) + \pi\left(\mathrm{Ad}_{g(x)}(A_\mu) - \partial_\mu g(x)g^{-1}(x)\right)\Pi(g)\phi(x) \otimes dx^\mu \\ &\overset{\text{(i) of Lemma 4.3}}{=} \Pi(g(x))\left(\partial_\mu\phi(x) + \pi(A_\mu(x))\phi(x)\right) \otimes dx^\mu + \partial_\mu\Pi(g)(x)\phi(x) \otimes dx^\mu \\ &\quad - \pi(\partial_\mu g(x)g^{-1}(x))\Pi(g)\phi(x) \otimes dx^\mu \\ &\overset{\text{(iii) of Lemma 4.3}}{=} \Pi(g(x))\left(\partial_\mu\phi(x) + \pi(A_\mu(x))\phi(x)\right) \otimes dx^\mu + \partial_\mu\Pi(g)(x)\phi(x) \otimes dx^\mu - \partial_\mu\Pi(g)(x)\phi(x) \otimes dx^\mu \\ &= \Pi(g)(x)\left(\partial_\mu\phi(x) + \pi(A_\mu(x))\phi(x)\right) \otimes dx^\mu = \Pi(g(x))(d^A\phi(x)). \end{aligned}$$

To obtain the third identity, we moreover used that $\partial_\mu g(x) = \frac{d}{ds}(g \circ \gamma)(0)$, where $\gamma(0) = x$ and $\gamma'(0) = \partial_\mu$, and then applied (ii) of Lemma 4.3 to the curve $s \mapsto (g \circ \gamma)(s)$. □

Definition 4.4. *Let G be a connected and compact matrix Lie group, V a finite-dimensional vector space and Π the corresponding Lie group representation. Let $g \in C^\infty(\mathcal{M}; G)$. Then a (local) gauge transformation corresponding to (G, V, Π) is defined as follows:*

$$\begin{aligned} \Phi_g &: C^\infty(\mathcal{M}; V) \times \Omega^1(\mathcal{M}; \mathfrak{g}) \to C^\infty(\mathcal{M}; V) \times \Omega^1(\mathcal{M}; \mathfrak{g}), \\ \Phi_g(\phi, A)(x) &= \Big(\Pi(g(x))\phi(x), \mathrm{Ad}_{g(x)}(A(x)) - \partial_\mu g(x)g(x)^{-1} \otimes dx^\mu_x\Big). \end{aligned}$$

4.3 Yang–Mills theory

In this section, we will construct gauge invariant Lagrangians corresponding to connected, compact matrix Lie groups G. As in the previous section, we will assume that $\mathcal{M}$ is covered by a single coordinate chart.

We will first extend the covariant derivative d^A to $\Omega^r(\mathcal{M};\mathfrak{g})$.

Definition 4.5. *Let $\theta \in \Omega^s(\mathcal{M};\mathfrak{g})$ and $\omega \in \Omega^r(\mathcal{M};\mathfrak{g})$. Then we define the wedge product of $\mathfrak{g}$-valued differential forms as follows:*

$$[\theta \wedge \omega] := [e_i, e_j] \otimes \theta^i \wedge \omega^j \in \Omega^{r+s}(\mathcal{M};\mathfrak{g}).$$

Definition 4.6. *Let $A \in \Omega^1(\mathcal{M};\mathfrak{g})$ and $\omega \in \Omega^r(\mathcal{M};\mathfrak{g})$ with $r \geq 1$. Then we define the covariant derivative d^A associated to the adjoint representation as follows:*

$$\begin{aligned} d^A &: \Omega^r(\mathcal{M};\mathfrak{g}) \to \Omega^{r+1}(\mathcal{M};\mathfrak{g}), \\ d^A\omega &= d\omega + [A \wedge \omega], \end{aligned}$$

With respect to a coordinate chart, we have that for $\omega \in \Omega^r(\mathcal{M};\mathfrak{g})$:

$$\begin{aligned} (d^A\omega)_{\mu\mu_1\ldots\mu_r} &= (r+1)\nabla_{[\mu}\omega_{\mu_1\ldots\mu_r]} + [e_i, e_j](A^i \wedge \omega^j)_{\mu\mu_1\ldots\mu_r} \\ &= (r+1)\nabla_{[\mu}\omega_{\mu_1\ldots\mu_r]} + \frac{(r+1)!}{r!}[e_i, e_j]A^i_{[\mu}\omega^j_{\mu_1\ldots\mu_r]} \\ &= (r+1)\Big(\nabla_{[\mu}\omega_{\mu_1\ldots\mu_r]} + [A_{[\mu}, \omega_{\mu_1\ldots\mu_r]}]\Big) \end{aligned}$$

Definition 4.7. *Let $\mathcal{M}$ be a manifold and let G be a Lie group with Lie algebra $\mathfrak{g}$. Let d^A denote the covariant derivative associated to the adjoint representation. Then the* Yang–Mills field strength *$F \in \Omega^2(\mathcal{M};\mathfrak{g})$ corresponding to G is defined as follows:*

$$F = dA + \frac{1}{2}[A \wedge A],$$

or, locally,

$$F = (\partial_\mu A_\nu - \partial_\nu A_\mu) \otimes dx^\mu \otimes dx^\nu + [A_\mu, A_\nu] \otimes dx^\mu \otimes dx^\nu.$$

In contrast with partial derivatives, the covariant derivatives d^A associated to (G, Π, A) do not commute. We show below that the Yang–Mills field strength corresponds to their commutator.

Proposition 4.4. *Let $\omega \in \Omega^r(\mathcal{M};\mathfrak{g})$. Then*

$$d^A d^A \omega = [F \wedge \omega].$$

In particular, with respect to any coordinate chart:

$$d^A_\mu d^A_\nu \phi - d^A_\nu d^A_\mu \phi = (d^A d^A \phi)_{\mu\nu} = [F_{\mu\nu}, \phi].$$

Proof. We have that

$$\begin{aligned} d^A(d^A\omega) = d^A\left(d\omega + [A \wedge \omega]\right) &= d^2\omega + [dA \wedge \omega] - [A \wedge d\omega] + [A \wedge d\omega] + \overbrace{[A \wedge [A \wedge \omega]]}^{=\frac{1}{2}[[A\wedge A]\wedge\omega]} \\ &= \left[\left(dA + \frac{1}{2}[A \wedge A]\right) \wedge \omega\right] = [F \wedge \omega], \end{aligned}$$

where we used that by the Jacobi identity:

$$\begin{aligned}[A \wedge [A \wedge \omega]] &= [e_i, [e_j, e_k]] \otimes A^i \wedge A^j \wedge \omega^k \\ &= -[e_j, [e_k, e_i]] \otimes A^i \wedge A^j \wedge \omega^k - [e_k, [e_i, e_j]] \otimes A^i \wedge A^j \wedge \omega^k \\ &= -[e_j, [e_i, e_k]] \otimes A^j \wedge A^i \wedge \omega^k + [[e_i, e_j], e_k] \otimes A^i \wedge A^j \wedge \omega^k \\ &= -[A \wedge [A \wedge \omega]] + [[A \wedge A] \wedge \omega]. \end{aligned}$$

□

Proposition 4.5. *Let $\Phi_g : \Omega^1(\mathcal{M}; \mathfrak{g}) \to \Omega^1(\mathcal{M}; \mathfrak{g})$ be the gauge transformation from Proposition 4.2. Write*

$$F^{\Phi_g(A)} = d\Phi_g(A) + \frac{1}{2}[\Phi_g(A) \wedge \Phi_g(A)].$$

Then, locally,

$$F^{\Phi_g(A)} = g(x)F_{\mu\nu}(x)g^{-1} \otimes dx^\mu \otimes dx^\nu = g(x)e_i(x)g^{-1}(x) \otimes F^i(x).$$

Proof. We will use that

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

Let $g \in C^\infty(\mathcal{M}; G)$. Observe that $0 = \partial_\mu(gg^{-1}) = (\partial_\mu g)g^{-1} + g\partial_\mu g^{-1}$, so $\partial_\mu g^{-1} = -g^{-1}(\partial_\mu g)g^{-1}$, from which it follows that

$$\begin{aligned}(F^{\Phi_g(A)})^i_{\mu\nu} &= \Big(\partial_\mu(gA_\nu g^{-1} - (\partial_\nu g)g^{-1}) - \partial_\nu(gA_\mu g^{-1} - (\partial_\mu g)g^{-1}\Big) \\ &\quad + [gA_\mu g^{-1} - (\partial_\mu g)g^{-1}, gA_\nu g^{-1} - (\partial_\nu g)g^{-1}] \\ &= g(\partial_\mu A_\nu - \partial_\nu A_\mu)g^{-1} + g[A_\mu, A_\nu]g^{-1} + (\partial_\mu g)A_\nu g^{-1} - (\partial_\nu g)A_\mu g^{-1} \\ &\quad - (gA_\nu g^{-1})(\partial_\mu g)g^{-1} + (gA_\mu g^{-1})(\partial_\nu g)g^{-1} \\ &\quad - \partial_\nu g\partial_\mu g^{-1} + \partial_\mu g\partial_\nu g^{-1} \\ &\quad - gA_\mu g^{-1}(\partial_\nu g)g^{-1} - (\partial_\mu g)A_\nu g^{-1} \\ &\quad + gA_\nu g^{-1}(\partial_\mu g)g^{-1} + (\partial_\nu g)A_\mu g^{-1} \\ &\quad + \underbrace{(\partial_\mu g)g^{-1}(\partial_\nu g)g^{-1} - (\partial_\nu g)g^{-1}(\partial_\mu g)g^{-1}}_{=-(\partial_\mu g)\partial_\nu g^{-1} + \partial_\nu g\partial_\mu g^{-1}} \\ &= gF_{\mu\nu}g^{-1}. \end{aligned}$$

□

In order to construct a gauge invariant Lagrangian, we still need a way of combining components of F to obtain a real-valued function.

Definition 4.8. *The Killing form $B : \mathfrak{g} \times \mathfrak{g} \to \mathbb{R}$ is a symmetric bilinear form defined as follows:*

$$B(X, Y) = \mathrm{tr}(\mathrm{ad}\,(X)\mathrm{ad}\,(Y)).$$

Remark 4.2. *A simply connected Lie group G is* semisimple *if and only if the Killing form κ is non-degenerate. The Killing form of a compact, semisimple Lie group is negative definite.*

Lemma 4.6. *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. Let $X, Y \in \mathfrak{g}$ and $g \in G$. Then*

$$B(gXg^{-1}, gYg^{-1}) = B(X, Y).$$

Proof. By (i) of Lemma 4.3, we have that

$$\mathrm{ad}(gXg^{-1}) = \mathrm{Ad}(g)\mathrm{ad}(X)\mathrm{Ad}(g^{-1}).$$

Then, by the cyclic property of the trace,

$$\begin{aligned} B(gXg^{-1}, gYg^{-1}) &= \operatorname{tr}(\operatorname{ad}(gXg^{-1})\operatorname{ad}(gYg^{-1})) = \operatorname{tr}(\operatorname{Ad}(g)\operatorname{ad}(X)\operatorname{Ad}(g^{-1})\operatorname{Ad}(g)\operatorname{ad}(Y)\operatorname{Ad}(g^{-1})) \\ &= \operatorname{tr}(\operatorname{ad}(X)\operatorname{ad}(Y)) = B(X,Y). \end{aligned}$$

□

Now we define the Yang–Mills Lagragian.

Definition 4.9. *Let G be a semisimple and compact matrix Lie group. Then the Yang–Mills Lagrangian is defined as follows:*

$$\begin{aligned} \mathcal{L}_{YM} &: \mathcal{N}_{YM} \to \mathbb{R}, \\ \mathcal{N}_{YM} &= \bigcup_{x \in \mathcal{M}} \{x\} \times T_x^*\mathcal{M} \otimes \mathfrak{g} \times \wedge^2(T_x^*\mathcal{M}) \otimes \mathfrak{g}, \\ \mathcal{L}_{YM}(x,q,v) &= \frac{1}{4\mathfrak{e}^2} B\Big(v_{\alpha\beta} - v_{\beta\alpha} + [q_\alpha, q_\beta], v_{\mu\nu} - v_{\nu\mu} + [q_\mu, q_\nu]\Big)(g^{-1})_x^{\alpha\mu}(g^{-1})_x^{\beta\nu}. \end{aligned}$$

The fields are $A \in \Omega^1(\mathcal{M}; \mathfrak{g})$ and the corresponding Yang–Mills action *is then given by:*

$$\mathcal{S}_{YM}(A) = \frac{1}{4\mathfrak{e}^2} \int_{\mathcal{M}} B(e_i, e_j) \otimes (F^{\mu\nu})^i (F_{\mu\nu})^j \epsilon,$$

with F the Yang–Mills field strength corresponding to A.

Corollary 4.7. *The Yang–Mills Lagrangian is gauge invariant:*

$$\mathcal{L}_{YM} \circ (\Phi_g(A), \nabla\Phi_g(A)) = \mathcal{L}_{YM} \circ (A, \nabla A).$$

Proof. Combine Proposition 4.5 and Lemma 4.6. □

Remark 4.3. *We can get rid of the factor $\mathfrak{e}^{-2}$ by rescaling $A = \mathfrak{e}\hat{A}$, changing the covariant derivative to $d^{\hat{A}} = d + \mathfrak{e}\pi(\hat{A})\phi$ and the transformation of $\hat{A}$ to:*

$$\Phi_g(\hat{A})(x) = \operatorname{Ad}_{g(x)}(\hat{A}) - \mathfrak{e}^{-1}\partial_\mu g(x) g(x)^{-1} \otimes dx^\mu.$$

We leave it to the interested reader to derive the corresponding Euler–Lagrange equations, or *Yang–Mills equations*:

$$\nabla_\mu F^{\mu\nu} + [A_\mu, F^{\mu\nu}] = 0.$$

You will need the following associativity property of the Killing form: $B([X,Y],Z) = B(X,[Y,Z])$ for all $X, Y, Z \in \mathfrak{g}$ (Exercise).

F also satisfies (Exercise)

$$d^A F = 0.$$

In components, the above equation is equivalent to

$$(\partial_\mu F_{\nu\rho} + [A_\mu, F_{\nu\rho}]) + (\partial_\nu F_{\rho\mu} + [A_\nu, F_{\rho\mu}]) + (\partial_\rho F_{\mu\nu} + [A_\rho, F_{\mu\nu}]) = 0$$

and is called the *Bianchi identity*. We will encounter the analogue of this identity when we study the curvature properties of $(\mathcal{M}, g)$.

Remark 4.4. *The standard model revolves around the Yang–Mills Lagrangian with $G = U(1) \times SU(2) \times SU(3)$ where the $U(1) \times SU(2)$ part describes the electroweak interaction (electromagnetism unified with weak nuclear interaction, the latter which is responsible for radioactive decay) and $SU(3)$ describes the strong nuclear interaction, which is responsible for binding quarks and neutrons and protons together in atomic nuclei).*

5 Quantum mechanics

5.1 The kinematic postulates of quantum mechanics

Recall the definition of a Hilbert space.

Definition 5.1. *A Hilbert space* $(\mathbf{H}, \langle\ ,\ \rangle)$ *is a vector space* $\mathbf{H}$ *equipped with an inner product* $\langle\ ,\ \rangle$*, which is a* sesquilinear form *on* $\mathbf{H}$*, i.e.* $\langle\ ,\ \rangle : \mathbf{H} \times \mathbf{H} \to \mathbb{C}$ *satisfies*

$$\langle x, \lambda y + z\rangle = \lambda\langle x, y\rangle + \langle x, z\rangle \quad \forall x, y, z \in \mathbf{H},\ \forall \lambda \in \mathbb{C},$$
$$\langle \lambda x + z, y\rangle = \overline{\lambda}\langle x, y\rangle + \langle x, z\rangle \quad \forall x, y, z \in \mathbf{H},\ \forall \lambda \in \mathbb{C},$$

that is moreover symmetric

$$\langle x, y\rangle = \overline{\langle y, x\rangle} \quad \forall x, y \in \mathbf{H}$$

and positive-definite:

$$\langle x, x\rangle > 0 \quad \forall x \in \mathbf{H} \setminus \{0\}.$$

Furthermore, $\mathbf{H}$ *is complete with respect to the corresponding norm defined as follows:* $|x| = \sqrt{\langle x, x\rangle}$*. We will moreover assume all Hilbert spaces are separable.*[40]

We will consider linear maps on $\mathbf{H}$ (defined on dense subspaces).

Definition 5.2.

- *A* linear operator *is a linear map*

$$A : \mathcal{D}(A) \to \mathbf{H},$$

with $\mathcal{D}(A) \subseteq \mathbf{H}$ *a linear subspace. We refer to* A *as an* densely defined operator *if* $\mathcal{D}(A)$ *is dense.*

- *We say* A *is a* bounded linear operator *if*

$$||A|| := \sup_{\substack{x \in \mathcal{D}(A) \\ |x|=1}} |Ax| < \infty.$$

A linear operator that does not satisfy A *is said to be* unbounded*.*

- *The (Hermitian) adjoint operator* A^* *corresponding to an densely defined operator* A *is defined as follows: let*

$$\mathcal{D}(A^*) = \{\psi \in \mathbf{H} \,|\, \phi \mapsto \langle \psi, A\phi\rangle \text{ is a bounded linear functional on } \mathcal{D}(A)\}.$$

Then $A^* : \mathcal{D}(A^*) \to \mathbf{H}$ *is a linear map defined as follows:* $A^*\psi = \chi$*, where* $\chi \in \mathbf{H}$ *is uniquely determined by* $\langle \chi, \phi\rangle = \langle \psi, A\phi\rangle$ *for all* $\phi \in \mathcal{D}(A)$ *by the Riesz Representation Theorem.*

Densely defined bounded linear operators admit a unique extension $A : \mathbf{H} \to \mathbf{H}$ by the Bounded Linear Transformation (BLT) Theorem. Furthermore, $A : \mathbf{H} \to \mathbf{H}$ is bounded if and only if A is (sequentially) continuous: $A\psi_n \to A\psi$ for all sequences $\{\psi_n\}$ such that $\psi_n \to \psi$. We denote with $B(\mathbf{H})$ the space of bounded linear operators on $\mathbf{H}$.

Exercise: If A is a densely defined bounded linear operator, then $A^* : \mathbf{H} \to \mathbf{H}$ is well-defined.

In general, we do not know how big the set $\mathcal{D}(A^*)$ is. We certainly have that $0 \in \mathcal{D}(A^*)$, but the set need not be dense in $\mathbf{H}$.

Recall also:

[40] It can be shown, using Zorn's Lemma, that every Hilbert space admits an orthonormal basis. Exercise: If $\mathbf{H}$ is separable, show that every orthonormal basis must be countable. In practice, one can often construct a countable orthonormal basis explicitly (without appealing to Zorn's Lemma).

Definition 5.3. *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a linear operator. A number $\lambda \in \mathbb{C}$ belongs to $\rho(A)$, the* resolvent set *of A, if there exists a $R \in B(\mathbf{H})$ such that:*

1. *For all $\psi \in \mathbf{H}$, $R\psi \in \mathcal{D}(A)$ and $(A - \lambda \mathbf{1})R\psi = \psi$,*
2. *For all $\psi \in \mathcal{D}(A)$, $R(A - \lambda \mathbf{1})\psi = \psi$.*

The operator R is referred to as a resolvent operator *of A and is denoted by $(A - \lambda I)^{-1}$. The complement $\sigma(A) := \mathbb{C} \setminus \rho(A)$ is called the* spectrum *of A.*

In these lectures, we will can mostly restrict our consideration to symmetric or self-adjoint operators, which are defined as follows:

Definition 5.4. *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a linear operator.*

- *A is* symmetric *if*
$$\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle \quad \forall \psi, \phi \in \mathcal{D}(A).$$
- *Let A be densely defined. Then A is* self-adjoint *if $\mathcal{D}(A) = \mathcal{D}(A^*)$ and $A = A^*$.*
- *Let A be densely defined. Then A is* closeable, *if $\overline{\mathcal{D}(A^*)} = \mathbf{H}$. The* closure *of A is defined as $\overline{A} := (A^*)^*$.*
- *Let A be densely defined and symmetric, so it is closeable by Proposition 5.1. Then A is* essentially self-adjoint *if $\overline{A}$ is self-adjoint.*
- *Let A be densely defined. Then A is* closed *if the graph $\mathcal{G}(A) = \{(\psi, \phi) \in \mathcal{D}(A) \times \mathbf{H} \ , \ \phi = A\psi\}$ is closed in $\mathbf{H} \times \mathbf{H}$.*

Will see for densely defined symmetric operators $\mathcal{D}(A) \subseteq \mathcal{D}(\overline{A})$. In practice, we are often given a symmetric operator on a dense domain and we consider the self-adjoint operator $\overline{A}$. In the physics literature, symmetry and self-adjointness are often mixed up and a symmetric/self-adjoint operator is sometimes called a "Hermitian operator".

The following properties of symmetric operators will be important for identifying probability within quantum mechanics.

Proposition 5.1. *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a symmetric operator. Then:*

(i) *$\langle \psi, A\psi \rangle \in \mathbb{R}$ for all $\psi \in \mathcal{D}(A)$. Furthermore, for all $m \in \mathbb{N}_1$, $\langle \psi, A^m\psi \rangle \in \mathbb{R}$ if $A^k\psi \in \mathcal{D}(A)$ for all $0 \leq k \leq m - 1$.*

(ii) *Let $\lambda \in \mathbb{C}$ be an eigenvalue for A, i.e. there exists a $\psi \in \mathcal{D}(A)$ such that $A\psi = \lambda\psi$. Then $\lambda \in \mathbb{R}$.*

(iii) *A is self-adjoint if and only if $\sigma(A) \subseteq \mathbb{R}$.*

(iv) *If A is densely defined, then $\mathcal{D}(A) \subseteq \mathcal{D}(A^*)$ and $A = A^*$ on $\mathcal{D}(A)$, so A is closeable. Furthermore, $\mathcal{D}(A) \subseteq \mathcal{D}(\overline{A})$.*

(v) *If A is densely defined, but not necessarily symmetric, then A^* is a closed operator. In particular, if A is closeable, then $\overline{A}$ is a closed operator.*

Proof. "(i)": By the symmetry property of inner products
$$\langle \psi, A\psi \rangle = \overline{\langle A\psi, \psi \rangle}.$$
By the symmetry of A:
$$\overline{\langle A\psi, \psi \rangle} = \overline{\langle \psi, A\psi \rangle}.$$

Hence, $\langle \psi, A\psi \rangle \in \mathbb{R}$.

Let $m \geq 1$ and $A^k \psi \in \mathcal{D}(A)$ for all $0 \leq k \leq m-1$. Suppose m is even, then

$$\langle \psi, A^m \psi \rangle = \langle A^{\frac{m}{2}} \psi, A^{\frac{m}{2}} \psi \rangle = |A^{\frac{m}{2}} \psi|^2 \in \mathbb{R}.$$

Suppose m is odd. Then

$$\langle \psi, A^m \psi \rangle = \langle A^{\frac{m-1}{2}} \psi, A A^{\frac{m-1}{2}} \psi \rangle \in \mathbb{R},$$

by the fact that $\langle \phi, A\phi \rangle \in \mathbb{R}$ for all $\phi \in \mathcal{D}(A)$.

"(ii)": By (i) and sesquilinearity of the inner product:

$$\lambda |\psi|^2 = \langle \psi, \lambda \psi \rangle = \langle \psi, A\psi \rangle \in \mathbb{R}. \qquad \square$$

"(iii)": We will appeal to the following useful claim that we will state without proof, but that is featured in the problem sheets.

Claim 5.2. *A densely defined symmetric operator $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ is self-adjoint if and only if* $\operatorname{ran}(A - \lambda \mathbf{1}) = \mathbf{H}$ *for all* $\lambda \in \mathbb{C} \setminus \mathbb{R}$.[41]

Suppose $\sigma(A) \subseteq \mathbb{R}$. Then $\lambda \in \mathbb{C} \setminus \mathbb{R}$ implies that $\lambda \in \rho(A)$, so the resolvent $(A - \lambda \mathbf{1})^{-1} : \mathbf{H} \to \mathcal{D}(A)$ is well-defined and hence $\operatorname{ran}(A - \lambda \mathbf{1}) = \mathbf{H}$, so A must be self-adjoint by the above claim.

Now suppose that A is self-adjoint. By the above claim we must have that $\operatorname{ran}(A - \lambda \mathbf{1}) = \mathbf{H}$ for all $\lambda \in \mathbb{C} \setminus \mathbb{R}$. In order for $(A - \lambda \mathbf{1})^{-1} : \mathbf{H} \to \mathcal{D}(A)$ to be a well-defined bounded linear operator or all $\lambda \in \mathbb{C} \setminus \mathbb{R}$ and hence $\sigma(A) \subseteq \mathbb{R}$, we therefore only need to show that $\ker(A - \lambda \mathbf{1}) = \{0\}$ and $||(A - \lambda \mathbf{1})|| < \infty$. Both of these will follow from the following important inequality that is valid in general for symmetric operators: for all $\psi \in \mathcal{D}(A)$:

$$\begin{aligned} |(A - \lambda \mathbf{1})\psi|^2 = \langle (A - \lambda \mathbf{1})\psi, (A - \lambda \mathbf{1})\psi \rangle &= |A\psi|^2 + |\lambda|^2 |\psi|^2 - \lambda \langle \psi, A\psi \rangle - \overline{\lambda} \langle A\psi, \psi \rangle \\ &\overset{A \text{ symmetric}}{=} |A\psi|^2 + |\lambda|^2 |\psi|^2 - (\overline{\lambda} + \lambda) \langle A\psi, \psi \rangle \geq |A\psi|^2 + |\lambda|^2 |\psi|^2 - 2|\Re \lambda| |\langle \psi, A\psi \rangle| \\ &\overset{\text{Cauchy–Schwarz}}{\geq} |A\psi|^2 + |\lambda|^2 |\psi|^2 - 2|\Re \lambda| |\psi| |A\psi| \\ &\overset{2xy \leq x^2 + y^2}{\geq} |\Im \lambda|^2 |\psi|^2. \end{aligned}$$

Hence, if $\psi \neq 0$, then $(A - \lambda \mathbf{1})\psi \neq 0$, so $\ker(A - \lambda \mathbf{1}) = \{0\}$. Furthermore, by taking $\psi = (A - \lambda \mathbf{1})\psi$, we obtain:

$$|(A - \lambda \mathbf{1})\psi| \leq \frac{1}{|\Im \lambda|^2} |\psi|^2,$$

so $||(A - \lambda \mathbf{1})|| \leq \frac{1}{|\Im \lambda|} < \infty$.

"(iv)": Let $\psi, \phi \in \mathcal{D}(A)$. Then $\langle A\psi, \phi \rangle = \langle \psi, A\phi \rangle$ by the symmetry of A. By Cauchy–Schwarz, we therefore have that

$$|\langle \psi, A\phi \rangle| = |\langle A\psi, \phi \rangle| \leq |A\psi| |\phi|.$$

Hence, $\phi \mapsto \langle \psi, A\phi \rangle$ defines a bounded functional and therefore $\psi \in \mathcal{D}(A^*)$ and $A^* \phi = A\phi$. Since $\mathcal{D}(A) \subseteq \mathcal{D}(A^*)$ and $\mathcal{D}(A)$ is dense in $\mathbf{H}$, $\mathcal{D}(A^*)$ must also be dense in $\mathbf{H}$. Therefore $\overline{A} = (A^*)^*$ is well-defined, and by the above $\mathcal{D}(A^*) \subseteq \mathcal{D}(\overline{A})$.

"(v)": Let $\{\psi_n\}$ be a sequence in $\mathcal{D}(A^*)$, such that $\psi_n \to \psi$, $A^* \psi_n \to \phi$ in $\mathbf{H}$ as $n \to \infty$. To conclude that A^* is closed, we need to show that $\psi \in \mathcal{D}(A^*)$ and $A^* \psi = \phi$.

We have that for all $\chi \in \mathcal{D}(A)$,

$$\langle A^* \psi_n, \chi \rangle = \langle \psi_n, A\chi \rangle$$

[41] Exercise: Show that if there exists a $\lambda_0 \in \mathbb{C} \setminus \mathbb{R}$ such that $\operatorname{ran}(A - \lambda_0 \mathbf{1}) = \mathbf{H}$, then $\operatorname{ran}(A - \lambda \mathbf{1}) = \mathbf{H}$ for all $\lambda \in \mathbb{C} \setminus \mathbb{R}$. Hint: Write $A - \lambda \mathbf{1} = (A - \lambda_0 \mathbf{1})(\mathbf{1} + (\lambda_0 - \lambda)(A - \lambda_0 \mathbf{1})^{-1})$ and then invert both sides.

an hence

$$\langle \phi, \chi \rangle = \lim_{n\to\infty} \langle A^* \psi_n, \chi \rangle = \lim_{n\to\infty} \langle \psi_n, A\chi \rangle = \langle \psi, A\chi \rangle.$$

From this it follows that the functional $\chi \mapsto \langle \psi, A\chi \rangle$ is bounded on $\mathcal{D}(A)$, so $\psi \in \mathcal{D}(A^*)$ and moreover $A^*\psi = \phi$. Since $\overline{A} = (A^*)^*$, we can apply the above argument to A^* instead of A to conclude that $\overline{A}$ is also closed.

EXERCISE: Let $A : \mathcal{D}(A) \to \mathbf{H}$ be densely defined and assume that A^* is an extension of A, i.e. $\mathcal{D}(A) \subseteq \mathcal{D}(A^*)$ and $A = A^*$ on $\mathcal{D}(A)$. Then A must be symmetric.

Symmetric bounded operators are self-adjoint. In general, (essential) self-adjointness implies symmetry, but the converse is not always true.

We will now list the main kinematic (i.e. non-dynamic) postulates of quantum mechanics and introduce some additional mathematical concepts along the way.

Postulate 1. *A* (quantum) state of a physical system *is the following equivalence class of unit norm elements of a Hilbert space* $(\mathbf{H}, \langle\ ,\ \rangle)$*:* let $\psi \in \mathbf{H}$ with $|\psi| = 1$, then the corresponding (quantum) state is the set:

$$[\psi] := \{\phi \in \mathbf{H} \,|\, \phi = c\psi,\ c \in \mathbb{C},\ |c| = 1\}.$$

In practice, one often refers to unit norm elements $\psi \in \mathbf{H}$ as states, but, as we will see when we define the measurement postulate, any ϕ such that $\phi = c\psi$ with $|c| = 1$ produces the same measurement results. Recall that in classical mechanics, a state of a 1-particle system in $\mathbb{R}^n$ is a point in phase space $\mathbb{R}^{2n}$ (a "classical state").

Remark 5.1. *The above definition of a state is also called a "pure state" to distinguish it from a more general notion of state called a "mixed state", which we will define later.*

In the physics literature, you will often encounter the "Dirac bra-ket" notation, where $|\psi\rangle$ (a "ket") denotes $\psi \in \mathbf{H}$. The notation $\langle\phi|$ (a "bra") is used for a bounded linear functional $\phi \in \mathbf{H}^*$ and the notation $\langle\phi|\psi\rangle$ means $\phi(\psi)$. This notation is also used for the inner product $\langle \phi, \psi \rangle$, with ϕ identified with its dual element in $\mathbf{H}$. Furthermore, $|\psi\rangle\langle\phi|$ denotes a linear operator A on $\mathbf{H}$, with $A\chi = \phi(\chi)\psi$ for any $\chi \in \mathbf{H}$.

Postulate 2. *To each "appropriate" function $f : \mathbb{R}^{2n} \to \mathbb{R}$ on the classical phase space $\mathbb{R}^{2n}$ (a "classical observable"), we can associate a self-adjoint operator $\hat{f} : \mathbf{H} \supseteq \mathcal{D}(\hat{f}) \to \mathbf{H}$ (a "quantum observable"). More generally, observables are represented by self-adjoint operators.*

In Hamiltonian mechanics, *observables* are represented by real-valued functions on the phase space $\mathbb{R}^{2n}$ (for example, describing N particles in $\mathbb{R}^3$ in classical mechanics, with $n = 3N$). In quantum mechanics, observables are instead represented by self-adjoint operators on Hilbert spaces.

As we will later see, for $\mathbf{H} = L^2(\mathbb{R}^n)$ the coordinates (x^i, p_i) on phase space correspond to operators $\hat{x}^i\psi = x^i\psi$ and $\hat{p}_i\psi = -i\hbar\frac{\partial}{\partial x^i}\psi$, when restricted to appropriate domains. These are both unbounded operators.

Remark 5.2. *Taking products of real-valued functions $f \cdot g$, which is commutative, corresponds to considering composition of operators $\hat{f}\hat{g}$, which is non-commutative, i.e. we could have $\hat{f}\hat{g} \neq \hat{g}\hat{f}$. There is therefore an* ordering ambiguity *when mapping real-valued functions to operators on Hilbert spaces. A particular choice of ordering is called a* quantization scheme.

The proposition below motivates why cannot restrict to operators with $\mathcal{D}(A) = \mathbf{H}$ if we want to consider unbounded operators like position and momentum.

Proposition 5.3. *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a densely defined symmetric operator. If $\mathcal{D}(A) = \mathbf{H}$, then A must be bounded.*

Proof. By Proposition 5.1 (iv), $A = A^*$ on $\mathbf{H} = \mathcal{D}(A) \subseteq \mathcal{D}(A^*)$ by Proposition 5.1 (iv). By Proposition 5.1 (v), we moreover have that $A = A^*$ is closed. Now we can appeal to the Closed Graph Theorem, which states that *closed operators with closed domains must be bounded.* □

Postulate 3. [Born Rule] *Consider a quantum observable $A : \mathcal{D}(A) \to \mathbf{H}$. For a physical system in a state represented by $\psi \in \mathcal{D}(A)$, the result of the measurement of A is governed by a* probability measure *μ_ψ^A and the corresponding expected value (or "expectation value") $\int_{\mathbb{R}} x \, d\mu_\psi^A(x) = \int_{\mathbb{R}} \mathrm{id}_{\mathbb{R}} \, d\mu_\psi^A$ is given by*

$$\langle A \rangle_\psi := \langle \psi, A\psi \rangle.$$

And more generally, when $\psi \in \mathcal{D}(A^k)$ for $1 \leq k \leq m$, the m-th moment with respect to μ_ψ^A, $\int_{\mathbb{R}} x^m \, d\mu_\psi^A(x)$, is given by: $\langle A^m \rangle_\psi = \langle \psi, A^m \psi \rangle$.

Suppose we prepare a large ensemble of independent physical systems to lie in the same state ψ and perform measurements of an observable A (like position, momentum or energy) on each system. By Postulate 3, the measurement values should approach a probability distribution, determined by ψ and A, with expectation value $\langle A \rangle_\psi$. Hence measurements are allowed to behave in a probabilistic, *non-deterministic* manner.

The above postulate motivates the consideration of equivalence classes of vectors in $\mathbf{H}$ in Postulate 1, as for $\psi_1, \psi_2 \in \mathbf{H}$ with $\psi_2 = c\psi_1$, $|c| = 1$,

$$\langle A \rangle_{\psi_2} = \langle \psi_2, A\psi_2 \rangle = \langle c\psi_1, cA\psi_1 \rangle = \overline{c}c \langle \psi_1, A\psi_1 \rangle = \langle A \rangle_{\psi_1},$$

so ψ_1 and ψ_2 are indistinguishable by measurements.

Example 5.1. *In the case $\mathbf{H} = L^2(\mathbb{R})$ the measurement of position x has the following expectation value:*

$$\langle \hat{x} \rangle_\psi = \langle \psi, \hat{x}\psi \rangle_{L^2(\mathbb{R})} = \int_{\mathbb{R}} x |\psi|^2(x) \, dx.$$

Hence, $|\psi|^2$ defines a probability density function and a suitable probability measure is given by:

$$\mu_\psi^{\hat{x}}(\Omega) = \int_\Omega |\psi|^2 \, dx,$$

for $\Omega \in \mathcal{B}_{\mathbb{R}}$, the Borel σ-algebra of $\mathbb{R}$.

Example 5.2. *Let $A : \mathcal{D}(A) \to \mathbf{H}$ be a self-adjoint operator with unit eigenvector $\psi \in \mathcal{D}(A)$, such that $A\psi = \lambda\psi$. We have that $A^m\psi = \lambda^m\psi$, so $A^m\psi \in \mathcal{D}(A)$ for all $m \in \mathbb{N}_0$. Then*

$$\langle A^m \rangle_\psi = \langle \psi, A^m\psi \rangle = \lambda^m |\psi|^2 = \lambda^m.$$

Note that the Dirac measure $\delta_\lambda : \mathcal{B}_{\mathbb{R}} \to \mathbb{R}$ satisfies also

$$\int_{\mathbb{R}} \mathrm{id}_{\mathbb{R}}^m \, d\delta_\lambda = \int_{\mathbb{R}} x^m d\delta_\lambda(x) = x^m|_{x=\lambda} = \lambda^m.$$

Hence, in this case $\mu_\psi^A = \delta_\lambda$. It can even be shown that δ_λ is the unique *probability measure with expectation value $\langle A^m \rangle_\psi = \lambda^m$, but this lies outside the scope of this course.*

Example 5.3. *Let $A : \mathcal{D}(A) \to \mathbf{H}$ be a self-adjoint operator, such that $\mathbf{H}$ has an orthonormal basis of eigenvectors $\{e_i\}_{i\in I}$ of A with distinct eigenvalues λ_i. Let $\psi \in \mathcal{D}(A)$, such that $A^m\psi \in \mathcal{D}(A)$ for all $m \in \mathbb{N}_0$. We can express $\psi = \sum_{i\in I} a_i e_i$, with $a_i \in \mathbb{C}$. The probability measure $\mu_\psi^A = \sum_{i\in I} |a_i|^2 \delta_{\lambda_i}$ results in the expectation value $\langle A\rangle_\psi$, since*

$$\langle \psi, A^m\psi\rangle = \left\langle \sum_{j\in I} a_j e_j, \sum_{i\in I} a_i \lambda_i^m e_i \right\rangle = \sum_{i\in I} |a_i|^2 \lambda_i^m.$$

Measurements of the observable corresponding to A can therefore only take values in a discrete subset of $\mathbb{R}$.

More generally, the *Spectral Theorem* provides a canonical way of assigning a probability measure μ_ψ^A on $\mathcal{B}_\mathbb{R}$ to any self-adjoint operator A with m-th moments agreeing with $\langle A^m\rangle_\psi$.

To understand the formulation of the Spectral Theorem, we first define the notion of a *projection-valued measure* or PVM.

Definition 5.5. *Let $\mathcal{B}_\mathbb{R}$ denote the Borel σ-algebra on $\mathbb{R}$, i.e. the smallest σ-algebra in $\mathbb{R}$ generated by the open sets in $\mathbb{R}$. A* projection-valued measure *or PVM is a map $P : \mathcal{B}_\mathbb{R} \to B(\mathbf{H})$ satisfying the following properties:*

1. *For all $\Omega \in \mathcal{B}_\mathbb{R}$, $P(\Omega)^* = P(\Omega)$.*
2. *For all $\Omega \in \mathcal{B}_\mathbb{R}$, $P(\Omega)P(\Omega) = P(\Omega)$.*
3. *$P(\mathbb{R}) = \mathbf{1}$.*
4. *For all countable collections $\{\Omega\}_{i\in\mathbb{N}}$ of pairwise disjoint sets in $\mathcal{B}_\mathbb{R}$ and for all $\psi \in \mathbf{H}$,*

$$P\left(\bigcup_{i\in\mathbb{N}} \Omega_i\right)\psi = \sum_{i\in\mathbb{N}} P(\Omega_i)\psi.$$

Note that the properties 1. and 2. imply that each $P(\Omega)$ is an orthogonal projection operator, whereas 3. and 4. are reminiscent of the defining properties of a measure.[42]

One can make sense of integration with respect to a PVM P. That is to say, let $f : \mathbb{R} \to \mathbb{C}$ be measurable (with respect to $\mathcal{B}_\mathbb{R}$ and $\mathcal{B}_\mathbb{C}$), then it is possible to make sense of

$$\int_\mathbb{R} f\, dP$$

on an appropriate domain.

In particular, if f is simple, $f = \sum_{n=1}^N c_n \mathbf{1}_{\Omega_n}$, with $c_n \in \mathbb{C}$ and $\{\Omega_n\}$ pairwise disjoint sets in $\mathcal{B}_\mathbb{R}$, then

$$\int_\mathbb{R} f\, dP := \sum_{n=1}^N c_n P(\Omega_n)$$

is a bounded operator on $\mathbf{H}$.

Definition 5.6. *For any PVM P and unit vector $\psi \in \mathbf{H}$, define a probability measure $\mu_\psi : \mathcal{B}_\mathbb{R} \to [0,1]$ as follows:*[43]

$$\mu_\psi(\Omega) = \langle \psi, P(\Omega)\psi\rangle.$$

[42] Exercise: Show that $P(\emptyset) = 0$ (the zero operator).

[43] Exercise: Show that μ_ψ is indeed a well-defined $\mathbb{R}$-valued probability measure on $\mathcal{B}_\mathbb{R}$.

We will see below that for measurable $f : \mathbb{R} \to \mathbb{C}$ and ψ in an appropriate subspace of $\mathbf{H}$:

$$\left\langle \psi, \left(\int_{\mathbb{R}} f \, dP \right) \psi \right\rangle = \int_{\mathbb{R}} f \, d\mu_\psi. \tag{5.1}$$

EXERCISE: Prove the above identity for simple f and $\psi \in \mathbf{H}$.

Theorem 5.4 (Spectral Theorem)**.** *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a self-adjoint operator. Then there is a unique PVM $P^A : \mathcal{B}_\mathbb{R} \to B(\mathbf{H})$, such that*

$$A = \int_{\mathbb{R}} \iota_\mathbb{R} \, dP^A,$$

with $\iota_\mathbb{R} : \mathbb{R} \to \mathbb{C}$ the inclusion map, $\iota_\mathbb{R}(x) = x + i0$. Furthermore, $P^A(\Omega) = 0$ if $\Omega \subseteq \mathbb{R} \setminus \sigma(A)$, so the integral can be restricted to $\sigma(A)$.

In the literature, one sometimes encounters the alternative notations:

$$\int_{\mathbb{R}} \lambda \, P^A(d\lambda) \quad \text{and} \int_{\mathbb{R}} \lambda \, dP^A(\lambda).$$

We denote with μ_ψ^A the measure μ_ψ on $\mathcal{B}_\mathbb{R}$ associated to P^A.

Theorem 5.5 (Functional Calculus)**.** *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a self-adjoint operator and let $f : \mathbb{R} \to \mathbb{C}$ be measurable. Then the following operator is well-defined*

$$f(A) := \int_{\mathbb{R}} f \, dP^A$$

on the dense subspace $\mathcal{D}(f(A)) = \{\psi \in \mathbf{H} \,|\, \int_\mathbb{R} |f|^2 \, d\mu_\psi^A < \infty\} \subseteq \mathbf{H}$. In particular, we can characterize $\mathcal{D}(A) = \{\psi \in \mathbf{H} \,|\, \int_\mathbb{R} |\lambda|^2 \, d\mu_\psi^A(\lambda) < \infty\}$.

Furthermore,

(i) *If $f : \mathbb{R} \to \mathbb{R}$, then $f(A)$ is self-adjoint.*

(ii) *If $f \in L^\infty(\mathbb{R}; \mathbb{C})$, then $f(A) \in B(\mathbf{H})$.*

(iii) *For all $\psi \in \mathcal{D}(f(A))$, $\langle \psi, f(A)\psi \rangle = \int_\mathbb{R} f d\mu_\psi^A$.*

(iv) *If $g : \mathbb{R} \to \mathbb{C}$ is measurable, then $f(A)g(A) = (fg)(A)$ and $\mathcal{D}(f(A)g(A)) = \mathcal{D}((fg)(A))$.*

Note that for all $m \in \mathbb{N}_0$ and ψ satisfying $\int_\mathbb{R} \lambda^{2m} \, d\mu_\psi^A(\lambda) < \infty$

$$\langle \psi, A^m \psi \rangle = \left\langle \psi, \left(\int_{\mathbb{R}} \lambda^m dP^A(\lambda) \right) \psi \right\rangle = \int_{\mathbb{R}} \lambda^m d\mu_\psi^A(\lambda). \tag{5.2}$$

EXERCISE: Show that $\psi \in \mathcal{D}(A^k)$ for all $0 \leq k \leq m$.

Remark 5.3. *In the case of bounded operators, it can be shown that μ_ψ^A is the unique probability density satisfying (5.2).*

5.2 Position and momentum operators

In this section, we define the position and momentum operators $\hat{x}^i$ and $\hat{p}_i$ with $1 \leq i \leq n$ on the Hilbert space $L^2(\mathbb{R}^n)$.

Proposition 5.6. *Let $V : \mathbb{R}^n \to \mathbb{R}$ be a measurable function. Let $\hat{V} : L^2(\mathbb{R}^n) \supseteq \mathcal{D}(\hat{V}) \to L^2(\mathbb{R}^n)$ be defined as follows:*

$$\begin{aligned} \mathcal{D}(\hat{V}) &:= \{\psi \in L^2(\mathbb{R}^n) \,|\, V \cdot \psi \in L^2(\mathbb{R}^n)\}, \\ (\hat{V}\psi)(x) &:= V(x)\psi(x). \end{aligned}$$

Then $\mathcal{D}(\hat{V})$ is dense in $L^2(\mathbb{R}^n)$ and $\hat{V}$ is a self-adjoint linear operator.

Proof. Let $E_m = \{x \in \mathbb{R}^n \,|\, |V(x)| \leq m\}$, so that $\cup_{m\in\mathbb{N}_0} E_m = \mathbb{R}^n$ and E_m are elements of the Borel σ-algebra $\mathcal{B}_{\mathbb{R}^n}$. By construction, we have that for any $\psi \in L^2(\mathbb{R}^n)$, $\mathbf{1}_{E_m}\psi \in \mathcal{D}(\hat{V})$.

Hence, we can apply the Dominated Convergence Theorem to conclude that $\mathbf{1}_{E_m}\psi \to \psi$ with respect to the L^2-norm, so $\mathcal{D}(\hat{V})$ must be dense.

It is clear that $\hat{V}$ is symmetric with respect to the L^2-inner product. By Proposition 5.1(iv), $\hat{V}^*$ must therefore be an extension of $\hat{V}$. Now suppose that $\psi \in \mathcal{D}(\hat{V}^*)$, i.e.

$$\phi \mapsto \int_{\mathbb{R}^n} \overline{\psi(x)} V(x)\phi(x)\, dx$$

defines a bounded linear functional on $\mathcal{D}(\hat{V})$ and

$$\int_{\mathbb{R}^n} \overline{\psi(x)} V(x)\phi(x)\, dx = \int_{\mathbb{R}^n} \overline{(\hat{V}^*\psi)(x)}\phi(x)\, dx.$$

or equivalently,

$$\int_{\mathbb{R}^n} (\overline{\psi(x)} V(x) - \overline{(\hat{V}^*\psi)(x)})\phi(x)\, dx = 0.$$

Now take $\chi = (\psi V - \hat{V}^*(\psi))$. Then by the above identity, $\langle \chi, \phi\rangle = 0$ for all $\phi \in \mathcal{D}(\hat{V})$ and by $\overline{\mathcal{D}(\hat{V})} = L^2(\mathbb{R}^n)$, $(\mathcal{D}(\hat{V}))^\perp = \{0\}$, so $\chi = 0$ (as an element of $L^2(\mathbb{R}^n)$) , which in turn implies that $V\psi = \hat{V}^*\psi$ in $L^2(\mathbb{R}^n)$ and hence, $\psi \in \mathcal{D}(\hat{V})$. We infer that $\mathcal{D}(\hat{V}^*) \subseteq \mathcal{D}(\hat{V})$, from which $\mathcal{D}(\hat{V}^*) = \mathcal{D}(\hat{V})$ follows by the fact that $\hat{V}^*$ is an extension of $\hat{V}$. □

Corollary 5.7. *The position operators $\hat{x}^i$, with $1 \leq i \leq n$, are self-adjoint on the domains $\mathcal{D}(\hat{x}^i) = \{\psi \in L^2(\mathbb{R}^n) \,|\, x^i\psi \in L^2(\mathbb{R}^n)\}$.*

We interpret the mommentum operator as a multiplication operator in Fourier space.

Proposition 5.8.

(i) *For each $1 \leq i \leq n$, define $\hat{p}_i : L^2(\mathbb{R}^n_x) \supseteq \mathcal{D}(\hat{p}_i) \to L^2(\mathbb{R}^n_x)$ as follows: let $\mathcal{F} : L^2_x(\mathbb{R}^n) \to L^2_k(\mathbb{R}^n)$ denote the Fourier transform[44] and let*

$$\begin{aligned} \mathcal{D}(\hat{p}_i) &= \{\psi \in L^2(\mathbb{R}^n_x) \,|\, k_i\mathcal{F}(\psi) \in L^2(\mathbb{R}^n_k)\}, \\ \hat{p}_i\psi &= \mathcal{F}^{-1}(\hbar\hat{k}_i\mathcal{F}(\psi)), \end{aligned}$$

with $\hbar$ the reduced Planck constant. Then $\mathcal{D}(\hat{p}_i)$ is dense and $\hat{p}_i$ is self-adjoint.

[44] For $\psi \in \mathcal{S}(\mathbb{R}^n)$ (a Schwartz function), define $\mathcal{F}(\psi)(k) := (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{-ik\cdot x}\psi(x)\, dx$. Then $\mathcal{F}(\psi) \in \mathcal{S}(\mathbb{R}^n)$. By Plancherel's theorem, $||\mathcal{F}\psi||_{L^2(\mathbb{R}^n)} = ||\psi||_{L^2(\mathbb{R}^n)}$ for all $\psi \in \mathcal{S}(\mathbb{R}^n)$. Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$, $\mathcal{F}$ admits a unique extension as a bounded linear operator with $||\mathcal{F}|| = 1$, which in fact is also unitary.

(ii) *We can alternatively characterize:*

$$\mathcal{D}(\hat{p}_i) := \{\psi \in L^2(\mathbb{R}^n_x) \,|\, \partial_{x_i}\psi \in L^2(\mathbb{R}^n_x)\},$$
$$\hat{p}_i\psi := -i\hbar\partial_{x^i}\psi$$

with $\partial_{x^i}\psi$ denoting a distributional derivative.

Proof. "(i)": By Proposition 5.6, the multiplication operators $\hat{k}_i\psi := k_i\psi$ are self-adjoint on $\mathcal{F}(\mathcal{D}(\hat{p}_i))$. By the unitarity of $\mathcal{F}$, $\hat{p}_j = \hbar\mathcal{F}^{-1}\hat{k}_i\mathcal{F}$ must also be self-adjoint on $\mathcal{D}(\hat{p}_i)$.

"(ii)": We will only sketch the proof. It relies on the fact that for all $\chi \in C^\infty_c(\mathbb{R}^n)$, the distribution derivative $\partial_{x^j}\psi \in L^2(\mathbb{R}^n)$ of $\psi \in L^2(\mathbb{R}^n)$ satisfies by Plancherel's theorem:

$$\langle \mathcal{F}(\chi), \mathcal{F}(\partial_{x^j}\psi)\rangle = \langle \chi, \partial_{x^j}\psi\rangle = -\langle \partial_{x^j}\chi, \psi\rangle = -\langle i\hat{k}_j\mathcal{F}(\chi), \mathcal{F}(\psi)\rangle = \langle \mathcal{F}(\chi), i\hat{k}_j\mathcal{F}(\psi)\rangle.$$

Since χ was arbirary, it can be shown that $\mathcal{F}(\partial_{x^j}\psi)(k) = ik_j\mathcal{F}(\psi)(k)$ for almost every k. By the bijectivity of $\mathcal{F}$, it therefore holds that $\partial_{x^j}\psi = \mathcal{F}^{-1}(ik_j\mathcal{F}(\psi))$ almost everywhere. □

We will encounter moreover the Laplacian Δ, which by analogous arguments, can also shown to be self-adjoint on appropriate domains. We leave the proof as an Exercise.

Proposition 5.9. *Define $\Delta : L^2(\mathbb{R}^n) \supset \mathcal{D}(\Delta) \to L^2(\mathbb{R}^n)$ as follows:*

$$\mathcal{D}(\Delta) := \{\psi \in L^2(\mathbb{R}^n_x) \,|\, |k|^2\mathcal{F}(\psi) \in L^2(\mathbb{R}^n_k)\},$$
$$\Delta\psi := -\mathcal{F}^{-1}(|k|^2\mathcal{F}(\psi)).$$

Then $\mathcal{D}(\Delta)$ is dense and Δ is self-adjoint. We have that $\mathcal{D}(\Delta) = W^{2,2}(\mathbb{R}^n) = H^2(\mathbb{R}^n)$ and $\Delta\psi$ as defined above agrees with $\sum_{i=1}^n \partial^2_{x^i}\psi$ defined in the distributional sense.[45]

We can associate to a Hamiltonian

$$H(x,p) = \sum_{i=1}^n \frac{p_i^2}{2m} + V(x)$$

the operator

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta + \hat{V}.$$

Operators of the form $\hat{H}$ are also called *Schrödinger operators*.

By the above properties, we have the following inequality for $\psi \in \mathcal{D}(\Delta)$:

$$\langle \psi, -\frac{\hbar^2}{2m}\Delta\psi\rangle = \langle \mathcal{F}^{-1}(\mathcal{F}(\psi)), \frac{\hbar^2}{2m}\mathcal{F}^{-1}(|k|^2\mathcal{F}(\psi))\rangle = \langle \mathcal{F}(\psi), \frac{\hbar^2}{2m}|k|^2\mathcal{F}(\psi)\rangle \geq 0.$$

One can show that $\hat{H}$ is self-adjoint under suitable assumptions on V.

Theorem 5.10. *Let $V : \mathbb{R}^n \to \mathbb{R}$ be a measurable function such that we can decompose $V = V_1 + V_2$, with $V_1 \in L^p(\mathbb{R}^n)$, where $p > \frac{n}{2}$ and $p \geq 2$, and with $V_2 \in L^\infty(\mathbb{R}^n)$. Then $\hat{H} : \mathcal{D}(\Delta) \to L^2(\mathbb{R}^n)$ is self-adjoint and bounded below, i.e. there exists a $c \in \mathbb{R}$ such that*

$$\langle \psi, \hat{H}\psi\rangle \geq c|\psi|^2$$

for all $\psi \in \mathcal{D}(\Delta) = H^2(\mathbb{R}^n)$.

[45] More generally, $W^{s,p}(\mathbb{R}^n)$ with $s \in \mathbb{R}$ and $p \in (1,\infty)$ denote the so-called *Sobolev spaces* and they consist of functions $\psi \in L^p(\mathbb{R}^n)$ satisfying $\mathcal{F}^{-1}((1+|k|^2)^{\frac{s}{2}}\mathcal{F}(\psi)) \in L^p(\mathbb{R}^n)$. For $s \in \mathbb{N}_0$, we can alternatively characterize: $W^{s,p}(\mathbb{R}^n) = \left\{\psi \in L^p(\mathbb{R}^n) \,|\, \sum_{\alpha\in\mathbb{N}_0^n,\, |\alpha|\leq s} \int_{\mathbb{R}^n} |\partial^{\alpha_1}_{x^1}\dots\partial^{\alpha_n}_{x^n}\psi|^p\, dx < \infty\right\}$.

We will leave the proof of Theorem 5.10 for later.

Remark 5.4. *Theorem 5.10 illustrates an important difference between classical and quantum mechanics. Suppose V is unbounded, e.g. $V(x) = -\frac{Q^2}{|x|}$ in the case of the Hamiltonian corresponding to an electron orbiting a proton: the hydrogen atom. Then in classical mechanics, the energy (kinetic+potential) can be made arbitrarily negative by considering as initial data $p = 0$ and x close to 0.*

In contrast, the energy in quantum mechanics is bounded below by the ground state energy

$$E_{GS} = \inf_{\psi \in H^2(\mathbb{R}^3)} \frac{\langle \psi, \hat{H}\psi \rangle}{\langle \psi, \psi \rangle} \geq c.$$

*This lower bound property is called "stability of the first kind". Indeed, we can split $V(x) = V_1(x) + V_2(x)$, with $V_1(x) = -\frac{Q^2}{|x|}\mathbf{1}_{|x|\leq 1}$ and $V_2(x) = -\frac{Q^2}{|x|}\mathbf{1}_{|x|>1}$. Then (*Exercise*) V satisfies the assumptions in Theorem 5.10.*

Stability of the second kind concerns a multiparticle physical system consisting of N electrons and K nuclei (at fixed positions) and requires the existence of constant $C \geq 0$ that is independent of N, K and the positions of the nuclei, such that $E_{GS} \geq -C(N + K)$: the size of the ground state energy should not grow more than linearly with the number of particles.[46] *Stability of the second kind was proved by Dyson–Lenard in the 60s and a more conceptual proof was provided by Lieb–Thirring in the 70s (using crucially the* fermionic *nature of electrons). These results have since also been generalized to relativistic settings.*

This difference between the classical and quantum cases is sometimes attributed to the fact that in quantum mechanics, one cannot localize both position and momentum (that is to say, one cannot choose both p and x close to 0), which is known as the uncertainty principle. *As we will later see, it is more accurate to say that the kinetic part of $\langle \psi, \hat{H}\psi \rangle$ controls the potential part via a* Sobolev inequality*, so the potential part cannot be much bigger than the kinetic part, which is not the case in the classical mechanics setting. In contrast with the uncertainty principle (see below), Sobolev inequalities do not solely rely on the Hilbert space structure of $\mathbf{H}$, but they also rely on the fact that $\mathbf{H} = L^2(\mathbb{R}^n)$.*

5.3 The dynamic postulates of quantum mechanics

Recall that

$$[\hat{x}^i, \hat{p}_j] = i\hbar\delta^i_j.$$

This resembles very much the identity

$$\{x^i, p_j\} = \delta^i_j$$

where $\{\cdot, \cdot\} : C^\infty(\mathbb{R}^{2n}) \times C^\infty(\mathbb{R}^{2n}) \to C^\infty(\mathbb{R}^{2n})$ is the *Poisson bracket* which we saw in the earlier part of the course:

$$\{f, g\} = \sum_{j=1}^{n} \partial_{x^j} f \partial_{p_j} g - \partial_{p_j} f \partial_{x^j} g.$$

If $t \mapsto (x, p)(t)$ is a solution to Hamilton's equations of classical mechanics, then we also saw that

$$\frac{df}{dt}(x(t), p(t)) = \{f, H\}(x(t), p(t)).$$

[46] For large N (order of 10^{23}), the energy of matter should be an *extensive* thermodynamic quantity in order to be compatible with the behaviour of ordinary matter in the thermodynamic limit. For example, suppose that $-E_{GS} \sim N^\alpha$, with $\alpha > 1$. Then pouring water from a big glass consisting of $2N$ particles to two small glasses such that each contain N particles should result in a change of energy proportional to $(2N)^\alpha - 2N^\alpha = (2^\alpha - 2)N^\alpha$, which will be very large for large N.

If we want expectation values of observables to evolve in time like classical observables, the correspondence

$$\{\cdot,\cdot\} \longleftrightarrow \frac{1}{i\hbar}[\cdot,\cdot]$$

suggests

$$\frac{d\langle\hat{f}\rangle_{\psi(t)}}{dt} = \frac{1}{i\hbar}\langle[\hat{f},\hat{H}]\rangle_{\psi(t)}.$$

When passing from classical to quantum mechanics, we can take $\hat{H}$ to be as in Theorem 5.10. However, we can consider also more general self-adjoint operators with no classical analogue. If we take the viewpoint that ψ is time dependent and the observables are time-dependent, this is guaranteed by the following postulate:

Postulate 4. *Let $\hat{H} : \mathcal{D}(\hat{H}) \to \mathbf{H}$ be self-adjoint. A 1-parameter family $\psi(t) \in \mathcal{D}(\hat{H})$ with $t \in I \subseteq \mathbb{R}$ describes the time evolution (of a representation of) a quantum state with respect to $\hat{H}$ if it satisfies the equation:*

$$\frac{d\psi}{dt}(t) = \frac{1}{i\hbar}\hat{H}\psi(t). \tag{5.3}$$

The equation (5.13) is called the *Schrödinger equation.*

Indeed, from Postulate 4, (sesqui)linearity and self-adjointness of $\hat{H}$, it follows that for appropriate $\psi \in \mathbf{H}$

$$\frac{d\langle\hat{f}\rangle_{\psi}}{dt}(t) = \frac{d}{dt}\langle\psi(t),\hat{f}\psi(t)\rangle = \langle\frac{1}{i\hbar}\hat{H}\psi(t),\hat{f}\psi(t)\rangle + \langle\psi,\frac{1}{i\hbar}\hat{f}\hat{H}\psi(t)\rangle = \frac{1}{i\hbar}\langle\psi(t),[\hat{f},\hat{H}]\psi(t)\rangle = \frac{1}{i\hbar}\langle[\hat{f},\hat{H}]\rangle_{\psi(t)}.$$

EXERCISE: Determine the conditions on the domain of ψ that guarantee that the above computation makes sense.

In particular, for $\hat{f} = \mathbf{1}$, this implies that

$$\frac{d}{dt}|\psi(t)|^2 = \frac{1}{i\hbar}\langle[\mathbf{1},\hat{H}]\rangle_{\psi(t)} = 0.$$

So if $\psi(0)$ is a unit vector, $\psi(t)$ will stay a unit vector for all times t. The time evolution is therefore norm preserving and $\psi(t)$ still represents a quantum state in accordance with Postulate 1.

One can alternatively make the following postlulate

Postulate 4'. *Let $\hat{H} : \mathcal{D}(\hat{H}) \to \mathbf{H}$ be self-adjoint. The time evolution of a 1-parameter family observables $\hat{f}(t)$ is governed by the equation:*

$$\frac{d\hat{f}}{dt}(t) = \frac{1}{i\hbar}[\hat{f}(t),\hat{H}]. \tag{5.4}$$

Postulate 4' also immediately implies that $\frac{d\langle\hat{f}\rangle_{\psi}}{dt} = \frac{1}{i\hbar}\langle[\hat{f},\hat{H}]\rangle_{\psi}$. Postulate 4 is called the *Schrödinger picture* and Postulate 4' is called the *Heisenberg picture.*

Let $\psi \in \mathcal{D}(\hat{p}_j\hat{x}_i) \cap \mathcal{D}(\hat{x}_i\hat{p}_j)$ for all $1 \leq i,j \leq n$,

$$[\hat{x}^i,\hat{p}_j]\psi = i\hbar\delta^i_j\psi.$$

Suppose that instead of defining $\hat{x}_i$ and $\hat{p}_i$ as above, we started instead with self-adjoint operators $A_i : \mathbf{H} \supseteq \mathcal{D}(A_i) \to \mathbf{H}$, $B_i : \mathbf{H} \supseteq \mathcal{D}(B_i) \to \mathbf{H}$, $i = 1,\ldots,n$, such that for all $1 \leq i,j \leq n$:

$$[A_i,A_j] = 0,$$

$$[B_i, B_j] = 0,$$
$$[A_i, B_j] = i\hbar\delta_{ij}\mathbf{1}.$$

The operators A_i and B_i are our candidates for position and momentum operators. We first perform an orthogonal decomposition: $\mathbf{H} = \bigoplus_l V_l$, with V_l the smallest closed subspaces that are invariant under the 1-parameter unitary groups e^{itA_i} and e^{itB_i}, $t \in \mathbb{R}$.

Then it turns out that, under additional technical assumptions on the commutativity properties of e^{itA_i} and e^{isB_i} for $t, s \in \mathbb{R}$ (the "Weyl relations"), there exists a unitary operator $U_l : V_l \to L^2(\mathbb{R}^n)$ for each l such that $U_l A_i U_l^{-1} = \hat{x}^i$ and $U_l B_i U_l^{-1} = \hat{p}_i$. This result is known as the *Stone–von Neumann theorem*. It may be interpreted as the statement that, up to conjugation with unitary operators, we may take $\mathbf{H} = L^2(\mathbb{R}^n)$ and represent the functions x_i and p_i by the operators $\hat{x}_i$ and $\hat{p}_i$.

We will show that the Schrödinger equation can be associated to *1-parameter unitary groups*.

Definition 5.7. *A* 1-parameter unitary group *is a family of unitary operators $U(t) : \mathbf{H} \to \mathbf{H}$ with $t \in \mathbb{R}$ such that*

- $U(0) = \mathbf{1}$,
- $U(s)U(t) = U(s+t)$ *for all* $s, t \in \mathbb{R}$.

A 1-parameter unitary group is strongly continuous *if moreover*

$$\lim_{t\to 0} |U(t)\psi - \psi| = 0$$

for all $\psi \in \mathbf{H}$.

We can associate to a strongly continuous 1-parameter unitary group a kind of "derivative" at $t = 0$:

Definition 5.8. *If $U(\cdot)$ is a strongly continuous 1-parameter unitary group on $\mathbf{H}$, then the* infinitesimal generator *of $U(\cdot)$ is the linear operator $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$, with*

$$A\psi := \lim_{t\to 0} \frac{1}{i}\frac{U(t)\psi - \psi}{t},$$
$$\mathcal{D}(A) := \left\{\psi \in \mathbf{H} \,|\, \lim_{t\to 0} \frac{1}{i}\frac{U(t)\psi - \psi}{t} \text{ exists in } \mathbf{H}\right\}.$$

We will now see that

$$U(t) = e^{itA}$$

define a strongly continuous 1-parameter unitary group if A is self-adjoint. In the discussion of the Spectral Theorem, we saw that we can apply functions on $\mathbb{R}$ to self-adjoint operators as follows ("functional calculus"):

$$f(A) = \int_{\sigma(A)} f(\lambda)\, dP^A(\lambda)$$

to obtain an operator that is well-defined on the domain $\mathcal{D}(f(A)) = \{\psi \in \mathbf{H} \,|\, \int_{\mathbb{R}} |f|^2\, d\mu_\psi^A\} \subseteq \mathbf{H}$ for $f : \mathbb{R} \to \mathbb{R}$ a bounded measurable function. Furthermore, for $f, g : \mathbb{R} \to \mathbb{R}$ bounded and measurable, it is possible to show that $(fg)(A) = f(A)g(A)$.

Since $\mathbb{R} \ni \lambda \mapsto e^{it\lambda}$ is bounded for $t \in \mathbb{R}$, e^{itA} is well defined.

Proposition 5.11. *Suppose $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ is self-adjoint. Then $U(t) = e^{itA}$ defines a strongly continuous 1-parameter unitary group. Furthermore, for all $\psi \in \mathbf{H}$, the limit*

$$\lim_{t\to 0} \frac{1}{i}\frac{U(t)\psi - \psi}{t}$$

is well-defined if and only if $\psi \in \mathcal{D}(A)$ and the limit is equal to $A\psi$.

Proof. Denote $f_t(\lambda) = e^{it\lambda}$. Since $f_t(\lambda)\overline{f_t(\lambda)} = 1$, we have by the multiplicative property of the functional calculus that $f_t(A)f_t(A)^* = f_t(A)^* f_t(A) = \mathbf{1}$, so $f_t(A)$ is unitary (and bounded). It is therefore well-defined on the domain $\mathbf{H}$.

Since $f_0(A) = \mathbf{1}$ and $f_{t+s} = f_t f_s$, $U(t) = f_t(A)$ defines a 1-parameter unitary group. To establish strong continuity, note that

$$|U(t)\psi - \psi|^2 = \langle U(t)\psi - \psi, U(t)\psi - \psi\rangle = \langle (U^*(t) - \mathbf{1})(U(t) - \mathbf{1})\psi, \psi\rangle = \int_{\sigma(A)} |e^{it\lambda} - 1|^2 \, d\mu^A_\psi(\lambda).$$

By the Dominated Convergence the integral on the RHS vanishes as $t \to 0$, so $U(\cdot)$ is strongly continuous.

Now, let $\psi \in \mathcal{D}(A)$. Then

$$\left| \frac{1}{i}\frac{U(t)\psi - \psi}{t} - A\psi \right| = \int_{\sigma(A)} \left| \frac{1}{i}\frac{e^{it\lambda} - 1}{t} - \lambda \right|^2 \, d\mu^A_\psi(\lambda). \tag{5.5}$$

We will show that RHS vanishes as $t \to 0$ by Dominated Convergence. First, note by the fundamental theorem of calculus that

$$\left|e^{it\lambda} - 1\right| = \left| \int_0^\lambda \frac{d}{d\lambda}(e^{it\lambda}) \, d\lambda \right| \leq |t||\lambda|.$$

so

$$\left| \frac{1}{i}\frac{e^{it\lambda} - 1}{t} - \lambda \right|^2 \leq \lambda^2.$$

Since

$$\int_{\sigma(A)} \lambda^2 \, d\mu^A_\psi(\lambda) = \langle A\psi, A\psi\rangle < \infty$$

the function $\lambda \mapsto \lambda^2$ is integrable with respect to $d\mu^A_\psi(\lambda)$. This means that we can apply once again the Dominated Convergence Theorem to conclude that the RHS of (5.5) goes to zero as $t \to 0$.

Now suppose $\psi, \phi \in \mathcal{D}(B)$, with B the infinitesimal generator of $U(\cdot)$. Then

$$\begin{aligned}
\langle \phi, B\psi\rangle &= \lim_{t\to 0} \left\langle \phi, \frac{1}{i}\frac{U(t)\psi - \psi}{t} \right\rangle \\
&= \lim_{t\to 0} \left\langle -\frac{1}{i}\frac{U^*(t)\phi - \phi}{t}, \psi \right\rangle \\
&= \lim_{t\to 0} \left\langle \frac{1}{i}\frac{U(-t)\phi - \phi}{-t}, \psi \right\rangle \\
&= \langle B\phi, \psi\rangle.
\end{aligned}$$

Note that by the above $\mathcal{D}(A) \subseteq \mathcal{D}(B)$, so B is a symmetric operator and an extension of A.

Exercise: Symmetric operators B that are extensions of self-adjoint operators A must satisfy $\mathcal{D}(A) = \mathcal{D}(B)$ and $B = A$. □

Corollary 5.12. *Let $\hat{H} : L^2(\mathbb{R}^n) \supseteq \mathcal{D}(\hat{H}) \to L^2(\mathbb{R}^n)$ be a self-adjoint Hamiltonian operator. Then the initial value problem for the Schrödinger equation*

$$\begin{aligned}
\partial_t \psi(t, \cdot) &= \frac{1}{i\hbar}\hat{H}\psi(t, \cdot), \\
\psi(0, \cdot) &= \psi_0 \in \mathcal{D}(\hat{H})
\end{aligned}$$

admits a unique (weak) solution $\psi \in C^0(\mathbb{R}; \mathcal{D}(\hat{H})) \cap C^1(\mathbb{R}; L^2(\mathbb{R}^n))$, with $\psi(t, x) = (e^{\frac{1}{i\hbar}t\hat{H}}\psi_0)(x)$ almost everywhere.

Proof. Let $\mathbf{H} = L^2(\mathbb{R}^n)$ (the proof works for general Hilbert spaces $\mathbf{H}$). By Proposition 5.11, $\psi(t,\cdot) = U(t)\psi_0 = e^{-\frac{i}{\hbar}t\hat{H}}\psi_0$ is well-defined with infinitesimal generator $-\frac{1}{\hbar}\hat{H}$ and since $\hat{H}(U(t)\psi_0) = U(t)(\hat{H}\psi_0) \in \mathbf{H}$, we have that $U(t)\psi_0 \in \mathcal{D}(\hat{H})$. By the strong continuity of $U(\cdot)$, we moreover have that the map $t \mapsto \psi(t,\cdot) \in \mathcal{D}(\hat{H})$ is continuous.

Furthermore, the following limit property holds with respect to the $\mathbf{H}$-norm:

$$\lim_{h\to 0} \frac{\psi(t+h,\cdot) - \psi(t,\cdot) - h\frac{1}{i\hbar}(\hat{H}\psi(t,\cdot))}{h} = i \lim_{h\to 0} \frac{(U(h)U(t)\psi_0) - (U(t)\psi_0) - t\frac{1}{i\hbar}(\hat{H}U(t)\psi_0)}{ih} = 0,$$

so $\partial_t\psi(t,\cdot) = \frac{1}{i\hbar}(\hat{H}U(t)\psi_0) = \frac{1}{i\hbar}(\hat{H}\psi(t,\cdot)$ almost everywhere. We apply the strong continuity of $U(\cdot)$ again to obtain:

$$\lim_{h\to 0}((\partial_t\psi)(t+h,\cdot) - \partial_t\psi(t,\cdot)) = \frac{1}{i\hbar}\lim_{h\to 0}(\hat{H}U(t+h)\psi_0 - \hat{H}U(t)\psi_0) = \frac{1}{i\hbar}\lim_{h\to 0}(U(h)(\hat{H}U(t)\psi_0) - \hat{H}U(t)\psi_0) = 0,$$

so $t \mapsto \partial_t\psi(t,\cdot) \in \mathbf{H}$ is also continuous. □

It turns out that every strongly continuous unitary group produces a self-adjoint operator.

Theorem 5.13 (Stone's theorem)**.** *The infinitesimal generator A of a strongly continuous unitary group $U(\cdot)$ is a unique, well-defined self-adjoint operator with $\mathcal{D}(A) \subseteq \mathbf{H}$ dense and $U(t) = e^{itA}$.*

By the above theorem, we could have also arrived at the Schrödinger equation by demanding that time evolution behaves according to a strongly continuous 1-parameter unitary group. This would, however, not tell us that the self-adjoint infinitesimal generator should be the operator corresponding to the Hamiltonian of classical mechanics.

Postulate 4." *A 1-parameter family $\psi(t) \in \mathbf{H}$ with $t \in \mathbb{R}$ describes the time evolution of (a representation of) a quantum state if it satisfies the equation:*

$$\psi(t) = U(t)\psi_0. \tag{5.6}$$

for some $\psi_0 \in \mathbf{H}$ and $U(\cdot)$ a strongly continuous 1-parameter unitary group on $\mathbf{H}$. We define the Hamiltonian $\hat{H}$ by $\hat{H} := -\frac{1}{\hbar}A$, with $A : \mathcal{D}(A) \to \mathbf{H}$ the infinitesimal generator of $U(\cdot)$.

The final postulate is also known as the postulate of "wave function collapse".

Postulate 5. *Let $A : \mathcal{D}(A) \to \mathbf{H}$ be a self-adjoint operator representing an observable. Consider a physical system in a state represented by $\psi \in \mathcal{D}(A) \setminus \{0\}$ and suppose that a measurement of A produces an outcome contained in $\sigma(A) \supseteq \Omega \in \mathcal{B}_{\mathbb{R}}$. Then, immediately after the measurement the state will change ("collapse") to the state represented by:*

$$\psi_{A,\Omega} := \frac{P^A(\Omega)\psi}{|P^A(\Omega)\psi|},$$

with $P^A : \mathcal{B}_{\mathbb{R}} \to B(\mathbf{H})$ the PVM associated to A.[47]

Exercise: Explain why $P^A(\Omega)\psi \neq 0$ by appealing to Postulate 3.

By combining Postulates 3 and 5, we have that the new probability measure associated to A directly after a measurement whose value sits in Ω will be $\mu^A_{\psi_{A,\Omega}}$.

[47] Stated in this precise way, "wave function collapse" is also known as the *von Neumann–Lüders postulate.*

In particular, if $\lambda \in \sigma(A)$ is an eigenvalue, then $P^A(\{\lambda\})$ will correspond to a projection to the eigenspace of λ and the associated probability measure will be a Dirac measure.

In the case of the position operator, $\hat{x}$, which has a purely continuous spectrum (EXERCISE), we have that

$$(P^{\hat{x}}(\Omega)\psi)(x) = \mathbf{1}_\Omega(x)\psi(x).$$

The more precise the position measurement, the smaller the set Ω and the more the probability measure associated to $\psi_{\hat{x},\Omega}$ is localized.

Postulate 5 implies that if an observable is measured, then the probability measure corresponding to a second measurement must be updated to take into account the first measurement. In particular, the probability of the second measurement taking on values in $\mathbb{R} \setminus \Omega$ is now:

$$\mu^A_{\psi_{A,\Omega}}(\mathbb{R} \setminus \Omega) = \frac{1}{|P^A(\Omega)\psi|^2}\langle P^A(\Omega)\psi, P^A(\mathbb{R} \setminus \Omega)P^A(\Omega)\psi\rangle = 0,$$

using that by the defining properties of PVMs:

$$P(\Omega)\psi = P(\mathbb{R})P(\Omega)\psi = P(\Omega \cup (\mathbb{R}\setminus\Omega))P(\Omega)\psi = P(\Omega)P(\Omega)\psi + P(\mathbb{R}\setminus\Omega)P(\Omega)\psi = P(\Omega)\psi + P(\mathbb{R}\setminus\Omega)P(\Omega)\psi,$$

so $P(\mathbb{R} \setminus \Omega)P(\Omega)\psi = 0$.

Subsequent measurements of the same observable therefore have the effect of localizing the probability measure. The same need not be true if our subsequent measure is of a *different* observable due to the uncertainty principle, which we will discuss later.

The interpretation of Postulate 5 and its role in the time evolution of the state is the source of a lot of ongoing debate, often of a philosophical nature.

5.4 The uncertainty principle

Definition 5.9. *Let $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ be a symmetric operator and let $\psi \in \mathcal{D}(A)$ with $|\psi| = 1$. Then we define the standard deviation corresponding to A and ψ as follows:*

$$\Delta_\psi A := \sqrt{\langle A\psi - \langle A\rangle_\psi \psi, A\psi - \langle A\rangle_\psi \psi\rangle}.$$

We will also refer to A as the uncertainty *of A in a state represented by ψ. If $\psi, A\psi \in \mathcal{D}(A)$, we can use the symmetry of A to express:*

$$\Delta_\psi A = \sqrt{\langle (A - \langle A\rangle_\psi \mathbf{1})^2\rangle_\psi}.$$

EXERCISE: Show that if A is self-adjoint and $\psi, A\psi \in \mathcal{D}(A)$ then $\Delta_\psi A$ agrees with the standard deviation with respect to the probability measure μ^A_ψ, i.e. $\Delta_\psi A = \sqrt{\int_{\mathbb{R}} (x - \langle A\rangle_\psi)^2 \, d\mu^A_\psi}$.

The goal of this section is to establish the following inequality, called the *Heisenberg uncertainty principle*:

$$\Delta_\psi \hat{x} \cdot \Delta_\psi \hat{p} \geq \frac{\hbar}{2},$$

for ψ with $|\psi| = 1$ in an appropriate subset of $\mathbf{H}$. This inequality tells us that if we choose ψ to be supported in a small interval so that $\Delta_\psi \hat{x}$ is very small, then $\Delta_\psi \hat{p}$ must necessarily be large, and similarly, the other way around: of we choose ψ so that $\Delta_\psi \hat{p}$ is very small, then $\Delta_\psi \hat{x}$ must be very large.

Another way of thinking about this is the following: suppose ψ is localized around a point $x_0 \in \mathbb{R}$, then the Fourier transform $\mathcal{F}(\psi)$ cannot also be localized around a point $k_0 \in \mathbb{R}$.

By Postulate 5, the uncertainty principle implies that we cannot simultanously measure position and momentum. Indeed, if we measure position (with small measurement error), then ψ will

collapse to a state localized around the measurement value $x_0 \in \mathbb{R}$. If we then measure the momentum, it can no longer be localized around $x_0 \in \mathbb{R}$. It is fundamentally impossible to measure position and momentum precisely. The reason that this does not create issues on macroscopic scales is that $\hbar$ is *very* small in macroscopic units, i.e.

$$\hbar = 1.055 \times 10^{-34} J \cdot s.$$

To prove the uncertainty principle, we first introduce some notation for the domains of products of operators.

Definition 5.10. *Let A, B be linear operators on $\mathbf{H}$ with domains $\mathcal{D}(A)$ and $\mathcal{D}(A)$, respecetively. We define AB to be the operator with domain*

$$\mathcal{D}(AB) = \{\psi \in \mathcal{D}(B) \,|\, B\psi \in \mathcal{D}(A)\},$$

such that $AB\psi = A(B\psi)$.

Theorem 5.14. *Let A, B be symmetric operators and $\psi \in \mathcal{D}(AB) \cap \mathcal{D}(BA)$, such that $|\psi| = 1$. Then*

$$\Delta_\psi A \Delta_\psi B \geq \frac{1}{2} |\langle [A, B] \rangle_\psi|$$

Proof. Define $A' := A - \langle \psi, A\psi \rangle \mathbf{1}$ on $\mathcal{D}(A)$ and $B' := B - \langle \psi, B\psi \rangle \mathbf{1}$ on $\mathcal{D}(B)$. It can easily be shown that A' and B' are also symmetric.

Observe that $(\Delta_\psi A)^2 = \langle A'\psi, A'\psi \rangle$ and $(\Delta_\psi B)^2 = \langle B'\psi, B'\psi \rangle$.

Now, apply Cauchy–Schwarz to obtain:

$$\langle A'\psi, A'\psi \rangle \langle B'\psi, B'\psi \rangle \geq |\langle A'\psi, B'\psi \rangle|^2 \geq |\Im \langle A'\psi, B'\psi \rangle|^2 = \left| \frac{1}{2} \left(\langle A'\psi, B'\psi \rangle - \langle B'\psi, A'\psi \rangle \right) \right|^2$$

By symmetry of A' and B' and $\psi \in \mathcal{D}(A'B') \cap \mathcal{D}(B'A')$:

$$(\Delta_\psi A)^2 (\Delta_\psi B)^2 = \langle A'\psi, A'\psi \rangle \langle B'\psi, B'\psi \rangle \geq \frac{1}{4} |\langle \psi, [A', B']\psi \rangle|^2 = \frac{1}{4} |\langle \psi, [A, B]\psi \rangle|^2 . \qquad \square.$$

Corollary 5.15 (Heisenberg Uncertainty Principle)**.** *Let A, B be symmetric operators such that $[A, B] = i\hbar \mathbf{1}$ and $\psi \in \mathcal{D}(AB) \cap \mathcal{D}(BA)$, with $|\psi| = 1$. Then*

$$\Delta_\psi A \Delta_\psi B \geq \frac{\hbar}{2}.$$

In particular, for all $\psi \in L^2(\mathbb{R})$ such that $\psi \in \mathcal{D}(\hat{x}\hat{p}) \cap \mathcal{D}(\hat{p}\hat{x})$ and $|\psi| = 1$,

$$\Delta_\psi \hat{x} \Delta_\psi \hat{p} \geq \frac{\hbar}{2}.$$

If we are not careful with domain considerations, we can produce violations of the uncertainty principle.

Let $\mathbf{H} = L^2([-1, 1])$. Let $A\psi(x) = x\psi(x)$ be the analogue of the position operator $\hat{x}$. Since A is now bounded, we have that $\mathcal{D}(A) = L^2([-1, 1])$ and A is self-adjoint.

Let $B = -i\hbar \frac{d}{dx}$ be the analogue of the momentum operator $\hat{p}$. We will take $\mathcal{D}(B) = \{\psi \in C^1([-1, 1]) \,|\, \psi(-1) = \psi(1)\}$. It can be shown that $\mathcal{D}(B)$ is dense in $L^2[-1, 1]$. As in the case of the usual position and momentum operators, we have that for all $\psi \in \mathcal{D}(B)$

$$[A, B]\psi = i\hbar\psi.$$

Note that for $\phi, \psi \in \mathcal{D}(B)$, we can integrate by parts to obtain

$$\langle \phi, B\psi \rangle = -i\hbar \int_{-1}^{1} \overline{\phi(x)} \frac{d\psi}{dx}(x)\, dx = -i\hbar(\overline{\phi(1)}\psi(1) - \overline{\phi(-1)}\psi(-1)) + i\hbar \int_{-1}^{1} \overline{\frac{d\phi}{dx}(x)}\psi(x)\, dx = \langle B\phi, \psi \rangle,$$

where we applied the boundary conditions in the definition of $\mathcal{D}(B)$. Hence, B is symmetric. To conclude that B is essentially self-adjoint, we will apply the following lemma:

Lemma 5.16. *Let $B : \mathbf{H} \supseteq \mathcal{D}(B) \to \mathbf{H}$ be a densely defined symmetric operator with a countable orthonormal basis of eigenvectors. Then B is essentially self-adjoint.*

Proof. Essential self-adjointness will follow if we can show that the ranges $\operatorname{ran}(B - i\mathbf{1})$ and $\operatorname{ran}(B + i\mathbf{1})$ are dense subspaces of $\mathbf{H}$ (see the problem sheet). Let $\{e_j\}$ denote the orthonormal basis of eigenvectors of B. Then $(B \pm i\mathbf{1})e_j = (\lambda_j \pm i)e_j$. Since $\lambda_j \in \mathbb{R}$, $\operatorname{ran}(B \pm i\mathbf{1})$ contains any non-zero multiple of e_j for all j. Hence, it must be dense in $\mathbf{H}$. □

Observe now that

$$\psi_j(x) = \frac{1}{\sqrt{2}} e^{i\pi j x}$$

define a orthonormal basis of eigenvectors for B, so B must be essentially self-adjoint. Denote with $\overline{B}$ its closure, which must be self-adjoint. Note now that

$$(\Delta_{\psi_j} \overline{B})^2 = \langle B\psi_j - \langle \psi_j, B\psi_j \rangle \psi, B\psi_j - \langle \psi_j, B\psi_j \rangle \rangle = \langle \lambda_j \psi_j - \lambda_j \psi_j, \lambda_j \psi_j - \lambda_j \psi_j \rangle = 0.$$

Since $\psi_j \in \mathcal{D}(A)$, $\Delta_{\psi_j} A$ is also well-defined, and therefore

$$\Delta_{\psi_j} A \Delta_{\psi_j} \overline{B} = 0.$$

Why is this not in contradiction with the uncertainty principle? It is because $A\psi_j \notin \mathcal{D}(B)$ as it does not satisfy the required boundary conditions. Hence $\psi_j \notin \mathcal{D}(BA)$, which is required for the uncertainty principle to hold.

In the case of the position and momentum operators on $L^2(\mathbb{R}^n)$, a stronger version of the uncertainty property holds, which we will state without proof.

Theorem 5.17. *Let $\hat{x}^j$ and $\hat{p}_j$ denote the position and momentum operators on $L^2(\mathbb{R}^n)$. If $\psi \in \mathcal{D}(\hat{x}_j) \cap \mathcal{D}(\hat{p}_j)$ and $|\psi| = 1$, then,*

$$\Delta_\psi \hat{x}_j \Delta_\psi \hat{p}_j \geq \frac{\hbar}{2}.$$

5.5 The self-adjointness of Schrödinger operators

We will now consider the following Hamiltonian on $\mathbf{H} = L^2(\mathbb{R}^n)$:

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta + \hat{V},$$

with measurable $V : \mathbb{R}^n \to \mathbb{R}$. We refer to these operators as Schrödinger operators.

Since both $-\frac{\hbar^2}{2m}\Delta$ and $\hat{V}$ are self-adjoint on $\mathcal{D}(\Delta) \cap \mathcal{D}(\hat{V})$, you may think that establishing self-adjointness of the sum is straightforward. In particular, if $\hat{V}$ is very singular, (essential) self-adjointness may fail on natural domains.

Definition 5.11. *If $A : \mathbf{H} \supseteq \mathcal{D}(A) \to \mathbf{H}$ and $B : \mathbf{H} \supseteq \mathcal{D}(B) \to \mathbf{H}$ are linear operators on $\mathbf{H}$, then we take $A + B$ to be the operator with domain $A + B := \mathcal{D}(A) \cap \mathcal{D}(B)$ and $(A + B)\psi = A\psi + B\psi$.*[48]

[48] Exercise: Show that $\mathcal{D}(A) \cap \mathcal{D}(B) \subseteq \mathbf{H}$ is a subspace.

Theorem 5.18 (Kato–Rellich)**.** *Let A and B be self-adjoint linear operators on $\mathbf{H}$ with $\mathcal{D}(A) \subseteq \mathcal{D}(B)$. Suppose that there exist $a \in [0,1)$, $b \in (0,\infty)$, such that for all $\psi \in \mathcal{D}(A)$*

$$|B\psi| \leq a|A\psi| + b|\psi|.$$

Then $A+B$ is self-adjoint on $\mathcal{D}(A)$. Furthermore, if $A \geq 0$, i.e. $\langle A\phi, \phi\rangle \geq 0$ for all $\phi \in \mathcal{D}(A)$, then

$$\sigma(A+B) \subseteq \left[-\frac{b}{1-a}, \infty\right).$$

Proof. Consider the operator $A + i\mu\mathbf{1}$, with $\mu \in \mathbb{R}_+$ to be determined. Then, for any $\psi \in \mathcal{D}(A)$,

$$(A+B+i\mu\mathbf{1})\psi = (B(A+i\mu\mathbf{1})^{-1} + \mathbf{1})(A+i\mu\mathbf{1})\psi, \tag{5.7}$$

where we used that $(A+i\mu\mathbf{1})^{-1}$ is a well-defined bounded linear operator (the resolvent) and $(A+i\mu\mathbf{1})^{-1}\phi \in \mathcal{D}(A)$ for all $\phi \in \mathbf{H}$, since $-i\mu \notin \sigma(A) \subseteq \mathbb{R}$. Now, note that for all $\psi \in \mathbf{H}$:

$$\begin{aligned}|\psi|^2 &= |A(A+i\mu\mathbf{1})^{-1}\psi + i\mu(A+i\mu\mathbf{1})^{-1}\psi|^2 \\ &= |A(A+i\mu\mathbf{1})^{-1}\psi|^2 + \mu^2|(A+i\mu\mathbf{1})^{-1}\psi|^2 + \langle A(A+i\mu\mathbf{1})^{-1}\psi, i\mu(A+i\mu\mathbf{1})^{-1}\psi\rangle \\ &\quad + \langle i\mu(A+i\mu\mathbf{1})^{-1}\psi, A(A+i\mu\mathbf{1})^{-1}\psi\rangle.\end{aligned}$$

By self-adjointness of A and sequilinearity of the inner product, the last two terms cancel, so

$$\mu^2|(A+i\mu\mathbf{1})^{-1}\psi|^2 + |A(A+i\mu\mathbf{1})^{-1}\psi|^2 = |\psi|^2,$$

from which it follows that $||(A+i\mu\mathbf{1})^{-1}|| \leq \frac{1}{\mu}$ and $||A(A+i\mu\mathbf{1})^{-1}|| \leq 1$.

By the invoking the inequality assumption on B, we therefore have that for all $\psi \in \mathbf{H}$

$$\begin{aligned}|B(A+i\mu\mathbf{1})^{-1}\psi| &\leq a|A(A+i\mu\mathbf{1})^{-1}\psi| + b|(A+i\mu\mathbf{1})^{-1}\psi| \\ &\leq \left(a + \frac{b}{\mu}\right)|\psi|.\end{aligned}$$

Now we take μ sufficiently large such that $a + \frac{b}{\mu} < 1$. Then for $C := B(A+i\mu\mathbf{1})^{-1}$, we have that $||C|| < 1$. It can then easily be shown that

$$(\mathbf{1}+C)^{-1} = \sum_{n\in\mathbb{N}_0} (-1)^n C^n$$

is a well-defined bounded linear operator, so by (5.7):

$$(A+B+i\mu\mathbf{1})^{-1} = (A+i\mu\mathbf{1})^{-1}(C+\mathbf{1})^{-1} : \mathbf{H} \to \mathcal{D}(A)$$

is a well-defined bounded linear operator and in particular $(A+B+i\mu\mathbf{1}) : \mathcal{D}(A) \to \mathbf{H}$ is surjective.

Repeating the above with μ replaced by $-\mu$ also tells use that $A+B-i\mu\mathbf{1} : \mathcal{D}(A) \to \mathbf{H}$ is surjective. Since $\mathrm{ran}(A+B\pm i\mu\mathbf{1}) = \mathbf{H}$, we must, by the above observation, therefore have that $A+B$ is self-adjoint.

Now suppose A is non-negative. Note that then $\sigma(A) \subseteq [0,\infty)$ (Exercise). We will consider $A+\lambda\mathbf{1}$, with $\lambda \in (0,\infty)$. We proceed as above with $i\mu$ replaced by λ to obtain:

$$\begin{aligned}|\psi|^2 = |A(A+\lambda\mathbf{1})^{-1}\psi|^2 + \lambda^2|(A+\lambda\mathbf{1})^{-1}\psi|^2 + \langle A(A+\lambda\mathbf{1})^{-1}\psi, \lambda(A+\lambda\mathbf{1})^{-1}\psi\rangle \\ + \langle \lambda(A+\lambda\mathbf{1})^{-1}\psi, A(A+\lambda\mathbf{1})^{-1}\psi\rangle,\end{aligned} \tag{5.8}$$

where we note that $(A+\lambda\mathbf{1})^{-1}$ is a well-defined bounded linear operator with $(A+\lambda\mathbf{1})^{-1}\phi \in \mathcal{D}(A)$ for all $\phi \in \mathbf{H}$, since $-\lambda \notin \sigma(A) \subseteq [0,\infty)$.

By the fact that $\langle A\phi, \phi\rangle \geq 0$, we moreover have that the last two terms on the RHS of (5.8) are non-negative, so that

$$|\psi|^2 \geq |A(A+\lambda\mathbf{1})^{-1}\psi|^2 + \lambda^2|(A+\lambda\mathbf{1})^{-1}\psi|^2.$$

Hence, $||(A+\lambda\mathbf{1})^{-1}|| \leq \frac{1}{\lambda}$. Proceeding as above with λ replacing $i\mu$, we obtain

$$|B(A+\lambda\mathbf{1})^{-1}\psi| \leq \left(a + \frac{b}{\lambda}\right)|\psi|$$

We have that $a + \frac{b}{\lambda} < 1$ if and only if $\lambda > \frac{b}{1-a}$. Applying (5.7) with $i\mu$ replaced by λ and $C = B(A+\lambda\mathbf{1})^{-1}$ gives that

$$(A+B+\lambda\mathbf{1})^{-1} = (A+\lambda\mathbf{1})^{-1}(C+\mathbf{1})^{-1} : \mathbf{H} \to \mathcal{D}(A)$$

is a well-defined bounded linear operator, so $-\lambda \in \rho(A+B)$ and $\sigma(A+B) \subseteq [-\frac{b}{1-a}, \infty)$. □

To be able to satisfy the inequality assumption in the Kato–Rellich Theorem, we need the following important inequalities on L^p-norms, which we will state without proof:

Lemma 5.19.

(i) (Hölder inequality) *Let $p, q \in [1,\infty]$ and $r \in [1,\infty)$, with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$ and let $f \in L^p(\mathbb{R}^n)$, $g \in L^q(\mathbb{R}^n)$. Then $fg \in L^r(\mathbb{R}^n)$, with*

$$||fg||_{L^r(\mathbb{R}^n)} \leq |f||_{L^p(\mathbb{R}^n)}||g||_{L^q(\mathbb{R}^n)}.$$

(ii) (Hausdorff–Young inequality) *Let $p \in [1,2]$ and $q \in [2,\infty]$, such that $\frac{1}{q} + \frac{1}{p} = 1$. Then the Fourier transform is well-defined as the map $\mathcal{F} : L^p(\mathbb{R}^n) \to L^q(\mathbb{R}^n)$ and*

$$||\mathcal{F}(f)||_{L^q(\mathbb{R}^n)} \leq ||f||_{L^p(\mathbb{R}^n)}.$$

Recall the statement of Theorem 5.10:

Theorem (Theorem 5.10)**.** *Let $V : \mathbb{R}^n \to \mathbb{R}$ be a measurable function such that we can decompose $V = V_1 + V_2$, with $V_1 \in L^p(\mathbb{R}^n)$, where $p > \frac{n}{2}$ and $p \geq 2$, and with $V_2 \in L^\infty(\mathbb{R}^n)$. Then $\hat{H} : \mathcal{D}(\Delta) \to L^2(\mathbb{R}^n)$ is self-adjoint and bounded below, i.e. there exists a constant $c \in \mathbb{R}$ such that*

$$\langle \psi, \hat{H}\psi\rangle \geq c|\psi|^2$$

for all $\psi \in \mathcal{D}(\Delta) = H^2(\mathbb{R}^n)$.

We are now in a position to prove this theorem.

Proof of Theorem 5.10. We will take $A = -\frac{\hbar^2}{2m}\Delta$ and $B = \hat{V}$ and apply the Kato–Rellich Theorem. Exercise: Show that $A \geq 0$.

First, we need to show that $\mathcal{D}(\hat{V}) \supseteq \mathcal{D}(\Delta) = \{\psi \in L^2(\mathbb{R}^n_x) \,|\, |k|^2\mathcal{F}(\psi) \in L^2(\mathbb{R}^n_k)\}$.

Observe first that that there exists a constant $c_{n,p} > 0$ such that for $t \in \mathbb{R}_+$

$$\int_{\mathbb{R}^n} (1+t|k|^2)^{-p}\, dk = \int_0^\infty \int_{\mathbb{S}^{n-1}} (1+t|k|^2)^{-p}|k|^{n-1}\, d|k| d\sigma_{\mathbb{S}^{n-1}} < c_{n,p}^p t^{-\frac{n}{2}}$$

if $-2p+n-1<-1$ or $p>\frac{n}{2}$, in which case $(1+t|k|^2)^{-1}\in L^p(\mathbb{R}^n)$ and $||(1+t|k|^2)^{-1}||_{L^p(\mathbb{R}^n)}<c_{n,p}t^{-\frac{n}{2p}}$. By Hölder's inequality and Minkowski's inequality $||f+g||_{L^2(\mathbb{R}^n)}\leq||f||_{L^2(\mathbb{R}^n)}+||g||_{L^2(\mathbb{R}^n)}$:

$$\begin{aligned}||\mathcal{F}(\psi)||_{L^r(\mathbb{R}^n)}&=||(1+t|k|^2)^{-1}(1+t|k|^2)\mathcal{F}(\psi)||_{L^r(\mathbb{R}^n)}\\&\leq||(1+t|k|^2)^{-1}||_{L^p(\mathbb{R}^n)}||(1+t|k|^2)\mathcal{F}(\psi)||_{L^2(\mathbb{R}^n)}\\&\leq c_{n,p}t^{1-\frac{n}{2p}}(t^{-1}||\psi||_{L^2(\mathbb{R}^n)}+||\Delta\psi||_{L^2(\mathbb{R}^n)})\end{aligned}$$

for $\frac{1}{r}=\frac{1}{p}+\frac{1}{2}$.

By the Hausdorff–Young inequality (applied to $\mathcal{F}(\psi)$) we then obtain

$$||\psi||_{L^q(\mathbb{R}^n)}\leq||\mathcal{F}(\psi)||_{L^r(\mathbb{R}^n)}\leq c_{n,p}t^{1-\frac{n}{2p}}(t^{-1}||\psi||_{L^2(\mathbb{R}^n)}+||\Delta\psi||_{L^2(\mathbb{R}^n)})$$

for $\frac{1}{q}+\frac{1}{r}=1$, if $1\leq r\leq 2$, which requires $p\geq 2$.

Now, using that $\frac{1}{q}+\frac{1}{p}=\frac{1}{q}+\frac{1}{r}-\frac{1}{2}=\frac{1}{2}$, we can apply Hölder again to the product $V_1\psi$ to conclude that $V_1\psi\in L^2(\mathbb{R}^n)$ if $\psi\in\mathcal{D}(\Delta)$ and, since V_2 is bounded, we immediately have that $V_2\psi\in L^2(\mathbb{R}^n)$, so $\psi\in\mathcal{D}(\hat{V})$.

We now turn to the inequality assumed in Kato–Rellich. The same Hölder inequality as above gives

$$\begin{aligned}||V\psi||_{L^2(\mathbb{R}^n)}&\leq||V_1||_{L^p(\mathbb{R}^n)}||\psi||_{L^q(\mathbb{R}^n)}+||V_2||_{L^\infty(\mathbb{R}^n)}||\psi||_{L^2(\mathbb{R}^n)}\\&\leq c_{n,p}||V_1||_{L^p(\mathbb{R}^n)}t^{1-\frac{n}{2p}}(||\Delta\psi||_{L^2(\mathbb{R}^n)}+t^{-1}||\psi||_{L^2(\mathbb{R}^n)})+||V_2||_{L^\infty(\mathbb{R}^n)}||\psi||_{L^2(\mathbb{R}^n)}.\end{aligned}$$

Since $2p>n$, we can take t sufficiently small such that $a:=\frac{2m}{\hbar^2}c_{n,p}||V_1||_{L^p(\mathbb{R}^n)}t^{1-\frac{n}{2p}}<1$ so that the assumption in Kato–Rellich is satisfied with $A=-\frac{\hbar^2}{2m}\Delta$ and $B=\hat{V}$.

Note that smaller t results in a larger constant b in the lower bound in Kato–Rellich. □

The above theorem can be improved to include potentials which do not satisfy the assumptions in the theorem but are non-negative, i.e. $V\geq 0$. This includes for example, the harmonic oscillator potential $V(x)=x^2$. We state the corresponding theorem without proof.

Theorem 5.20. *Let $V:\mathbb{R}^n\to\mathbb{R}$ be a measurable function such that we can decompose $V=V_1+V_2+V_+$, with $V_1\in L^p(\mathbb{R}^n)$, where $p=2$ (if $n\leq 3$), $p>\frac{n}{2}$ (if $n=4$), $V_2\in L^\infty(\mathbb{R}^n)$, and with $V_+\in L^2_{\mathrm{loc}}(\mathbb{R}^n)$ and $V_+\geq 0$.*[49] *Then $\hat{H}:C_c^\infty(\mathbb{R}^n)\to L^2(\mathbb{R}^n)$ is essentially self-adjoint and there exists a $c\in\mathbb{R}$ such that*

$$\langle\psi,\hat{H}\psi\rangle\geq c|\psi|^2$$

for all $\psi\in C_c^\infty(\mathbb{R}^n)$.

Exercise: Show that in the case $n=1$, $V(x)=-x^2$, the following statement holds: for any $M>0$, there exists a $\psi_M\in C_c^\infty(\mathbb{R})$, such that $\langle\psi,\hat{H}\psi\rangle\leq -M$.

It can moreover be shown that $\hat{H}:C_c^\infty(\mathbb{R})\to L^2(\mathbb{R})$ is not even essentially self-adjoint if $V(x)=-x^4$.

[49] The space $L^2_{\mathrm{loc}}(\mathbb{R}^n)$ consists of measurable functions f such that $\chi_K f\in L^2(\mathbb{R}^n)$ for all compact $K\subset\mathbb{R}^n$.

5.6 The free Schrödinger equation

We consider now the Schrödinger equation corresponding to the Schrödinger operator $\hat{H} = -\frac{\hbar^2}{2m}\Delta$:

$$\begin{aligned} \partial_t \psi(t,\cdot) &= \frac{i\hbar}{2m}\Delta\psi(t,\cdot), \\ \psi(0,\cdot) &= \psi_0. \end{aligned} \tag{5.9}$$

We will analyse solutions to this equations by means of the Fourier transform in x. Note the transformation of $-i\partial_{x^i}$ to multiplication k_i under the Fourier transform, when acting on suitably regular functions.

In Fourier space, the Schrödinger equation therefore becomes

$$\partial_t \mathcal{F}(\psi)(t,\cdot) = -\frac{i\hbar|k|^2}{2m}\mathcal{F}(\psi)(t,\cdot).$$

This equation can easily be solved:

$$\mathcal{F}(\psi)(t,k) = \mathcal{F}(\psi_0)(k)e^{-\frac{i\hbar|k|^2}{2m}t}.$$

The above expression suggests that $\psi(t,x)$ can therefore be obtained via an inverse Fourier transform:

$$\psi(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k)e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)}\,dk.$$

In the proposition below, we show that the above expression indeed solves the initial value problem (5.9). For this, we observe that the exponential in the integrand solves the free Schrödinger equation:

$$\left(\partial_t - \frac{i\hbar}{2m}\Delta\right)e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)} = 0.$$

Theorem 5.21. *Let $\psi_0 \in L^2(\mathbb{R}^n)$. Then the corresponding solution to (5.9) (in a distributional sense) is given by:*

$$\psi(t,\cdot) = \mathcal{F}^{-1}\left(\mathcal{F}(\psi_0)e^{-i\frac{\hbar|k|^2}{2m}t}\right) \tag{5.10}$$

In particular, if ψ_0 is a Schwartz function, then

$$\psi(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k)e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)}\,dk. \tag{5.11}$$

Proof. Suppose that ψ_0 is a Schwartz function and ψ is given by (5.11). Note first that

$$\frac{\psi(t+h,x)-\psi(t,x)}{h} = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k)e^{ik\cdot x}\frac{e^{-i\frac{\hbar|k|^2}{2m}(t+h)} - e^{-i\frac{\hbar|k|^2}{2m}t}}{h}\,dk$$

Note that, by the mean-value theorem

$$\left|\frac{e^{-i\frac{\hbar|k|^2}{2m}(t+h)} - e^{-i\frac{\hbar|k|^2}{2m}t}}{h}\right| \leq \frac{\hbar}{2m}|k|^2.$$

Since $\mathcal{F}$ maps Schwartz functions to Schwartz functions, $\mathcal{F}(\psi_0)$ is Schwartz and $|k|^2|\mathcal{F}(\psi_0)|$ is integrable and we can apply the Dominated Convergence Theorem to conclude that:

$$\partial_t\psi(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}}\int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k)\partial_t e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)}\,dk$$

Similarly, using that for $h \in \mathbb{R}^n$ with $h^j = a\delta_r^j$

$$\left| \frac{e^{ik\cdot(x+h)} - e^{ik\cdot x}}{a} \right| \leq |k_r|.$$

We have that

$$\partial_{x^r}\psi(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k) \partial_{x^r} e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)}\, dk$$

and repeating this, we obtain

$$\partial_t\psi(t,x) - \frac{i\hbar}{2m}\Delta\psi(t,x) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \mathcal{F}(\psi_0)(k) \left(\partial_t - \frac{i\hbar}{2m}\Delta\right) e^{i\left(k\cdot x - \frac{\hbar|k|^2}{2m}t\right)}\, dk = 0.$$

Furthermore,

$$\mathcal{F}(\psi)(t,k) = \mathcal{F}(\psi_0)(k) e^{-\frac{i\hbar|k|^2}{2m}t}.$$

Now let $\psi_0 \in L^2(\mathbb{R}^n)$ and, by density of Schwartz functions in $L^2(\mathbb{R}^n)$, let $\psi_0^{(j)}$ be a sequence of Schwartz functions approaching ψ_0. Denote with $\psi^{(j)}$ the corresponding solutions to (5.9). Then

$$\psi^{(j)}(t,\cdot) := \mathcal{F}^{-1}(\mathcal{F}(\psi_0^{(j)}) e^{-\frac{i\hbar|k|^2}{2m}t})$$

are well–defined Schwartz functions. Since, $\mathcal{F}$ is well-defined and unitary on $L^2(\mathbb{R}^n)$, $\{\psi^{(j)}(t,\cdot\}$ form a Cauchy sequence in $L^2(\mathbb{R}^n)$ and admit a limit $\psi(t,\cdot)$ which satisfies $\mathcal{F}(\psi(t,\cdot)) = \mathcal{F}(\psi_0)e^{-\frac{i\hbar|k|^2}{2m}t}$.

It remains to show that for all $\chi \in C_c^\infty(\mathbb{R}_t \times \mathbb{R}_x^n)$

$$\int_{\mathbb{R}\times\mathbb{R}^n} \overline{\psi} \left(\partial_t\chi - \frac{i\hbar}{2m}\Delta\chi\right) dxdt = 0.$$

Exercise: Show that the above is equivalent to $(\partial_t - \frac{i\hbar}{2m}\Delta)\psi = 0$ if $\psi \in C^2(\mathbb{R}_t \times \mathbb{R}_x^n)$.

By Plancherel and integrating by parts in t:

$$\begin{aligned}
\int_{\mathbb{R}\times\mathbb{R}^n} \overline{\psi} \left(\partial_t\chi - \frac{i\hbar}{2m}\Delta\chi\right) dxdt &= \int_{\mathbb{R}\times\mathbb{R}^n} \overline{\mathcal{F}(\psi)(t,k)} \left(\partial_t\mathcal{F}(\chi) + \frac{i\hbar}{2m}|k|^2\mathcal{F}(\chi)\right) dkdt \\
&= \int_{\mathbb{R}\times\mathbb{R}^n} \overline{\mathcal{F}(\psi_0)(k)} e^{\frac{i\hbar|k|^2}{2m}t} \left(\partial_t\mathcal{F}(\chi) + \frac{i\hbar}{2m}|k|^2\mathcal{F}(\chi)\right) dkdt \\
&\overset{\text{int. by pt.}}{=} \int_{\mathbb{R}\times\mathbb{R}^n} \overline{\mathcal{F}(\psi_0)(k)} e^{\frac{i\hbar|k|^2}{2m}t} \left(-\frac{i\hbar|k|^2}{2m}\mathcal{F}(\chi) + \frac{i\hbar}{2m}|k|^2\mathcal{F}(\chi)\right) dkdt = 0.
\end{aligned}$$

□

In the proposition below, we will see that we can think of solutions ψ to (5.9) as convolution with $K_t : \mathbb{R}^n \to \mathbb{C}$, which is defined as follows:

$$K_t(x) := (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ik\cdot x} e^{-i\frac{\hbar|k|^2 t}{2m}}\, dk = \left(\frac{m}{2\pi i\hbar t}\right)^{\frac{n}{2}} e^{\frac{im|x|^2}{2\hbar t}}.$$

For this, we need the following lemma.

Lemma 5.22. *Let $\psi, \phi \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. Then $\phi * \psi \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$ and*

$$(2\pi)^{-\frac{n}{2}}\mathcal{F}(\phi * \psi) = \mathcal{F}(\phi)\mathcal{F}(\psi).$$

Proof. We can write

$$\begin{aligned}
\mathcal{F}(\phi * \psi)(k) &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n_x} \left(\int_{\mathbb{R}^n_y} \phi(x-y)\psi(y)\, dy \right) e^{-ik\cdot x}\, dx \\
&= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n_x} \left(\int_{\mathbb{R}^n_y} e^{-ik\cdot(x-y)}\phi(x-y)e^{-ik\cdot y}\psi(y)\, dy \right) dx \\
&\overset{\text{Fubini}}{=} (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n_z \times \mathbb{R}^n_y} e^{-ik\cdot z}\phi(z)e^{-ik\cdot y}\psi(y)\, dydz \\
&= (2\pi)^{\frac{n}{2}} \mathcal{F}(\phi)\mathcal{F}(\psi).
\end{aligned}$$

□

Theorem 5.23. *Suppose $\psi_0 \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. Then ψ as defined in (5.10) can be expressed as follows for all $t \neq 0$:*

$$\psi(t,\cdot) = K_t * \psi_0,$$

or

$$\psi(t,x) = \left(\frac{m}{2\pi i\hbar t}\right)^{\frac{n}{2}} \int_{\mathbb{R}^n} e^{\frac{im|x-y|^2}{2\hbar t}} \psi_0(y)\, dy.$$

Proof. Consider $B_j(0)$ the ball of radius j around $0 \in \mathbb{R}^n$. Then $1_{B_j(0)}K_t \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. By Lemma 5.22, we have that

$$\mathcal{F}(1_{B_j(0)}K_t * \psi_0) = (2\pi)^{\frac{n}{2}} \mathcal{F}(1_{B_j(0)}K_t)\mathcal{F}(\psi_0). \tag{5.12}$$

We would like to conclude that

$$(2\pi)^{\frac{n}{2}} \mathcal{F}(1_{B_j(0)}K_t)\mathcal{F}(\psi_0) \xrightarrow{L^2} e^{-i\frac{\hbar|k|^2 t}{2m}} \mathcal{F}(\psi_0)$$

with respect to the L^2 norm as $j \to \infty$. Since $\psi_0 \in L^2(\mathbb{R}^n)$ and one can show that 1) the above convergence holds pointwise (see definition of K_t) and 2) $\mathcal{F}(1_{B_j(0)}K_t)$ can be bounded uniformly by a constant independent of j(Exercise: prove 1) and 2)), we can apply the Dominated Convergence Theorem to conclude the above L^2-convergence.

Now, by Plancherel, and the identity (5.12), we must also have L^2-convergence of the corresponding inverse Fourier transforms:

$$(1_{B_j(0)}K_t) * \psi_0 \xrightarrow{L^2} \mathcal{F}^{-1}(e^{-i\frac{\hbar|k|^2 t}{2m}} \mathcal{F}(\psi_0))$$

As clearly, the left-hand side above also converges pointwise to $K_t * \psi_0$ in $L^2(\mathbb{R}^n)$, since $\psi_0 \in L^1(\mathbb{R}^n)$, and the pointwise limit must coincide with the L^2 limit, we can conclude that:

$$K_t * \psi_0 = \mathcal{F}^{-1}(e^{-i\frac{\hbar|k|^2 t}{2m}})\mathcal{F}(\psi_0)) \in L^2(\mathbb{R}^n).$$

Note that the right-hand side above is equal to ψ as defined in (5.10). □

Remark 5.5. *By the conservation of $||\psi(t,\cdot)||_{L^2(\mathbb{R}^n)}$ in t and the fact that we can estimate*

$$||\psi(t,\cdot)||_{L^\infty(\mathbb{R}^n)} \leq \left(\frac{|m|}{2\pi\hbar}\right)^{\frac{n}{2}} ||\psi_0||_{L^1(\mathbb{R}^n)} t^{-\frac{n}{2}},$$

we can see that ψ disperses *or spreads out as $t \to \infty$.*

Remark 5.6. *Note that the free Schrödinger equation satisfies the* infinite speed of propagation property*. Indeed, consider initial data ψ_0 with support in $B_R(0)$ for some $R > 0$. Let $x \in \mathbb{R}^n$, with $|x|$ arbitrarily large. Then*

$$\psi(t,x) = \Big(\frac{m}{2\pi i\hbar t}\Big)^{\frac{n}{2}} \int_{B_R(0)} e^{\frac{im|x-y|^2}{2\hbar t}} \psi_0(y)\, dy$$

will not be vanishing for general data ψ_0. This means that arbitrarily far away points in space are instantaneously affected by initially localized perturbations.

5.7 Multiple particles and mixed states

In quantum mechanics, N distinguishable particles[50] in $\mathbb{R}^3$ are represented by a state $\psi \in L^2(\mathbb{R}^{3N})$. We can make sense of statements about the position $x_j^{(k)}$ and momentum $p_j^{(k)}$ coordinates of the k-th particle by considering appropriate operators $\hat{x}_j^{(k)}$ and $\hat{p}_j^{(k)}$, $j = 1, 2, 3$, which only involve the coordinates (x_j, p_j) on the k-th copy of $\mathbb{R}^3$ in $\mathbb{R}^{3N}$.

Suppose we would like to associate to the k-th particle a unit vector $\psi^{(k)} \in L^2(\mathbb{R}^3)$. Then we encounter an ambiguity as there is no unique way of decomposing a vector $\psi \in L^2(\mathbb{R}^{3N})$ into N vectors in $L^2(\mathbb{R}^3)$. The notation of a state as introduced in Postulate 1 is therefore inadequate to describe a *subsystem* within a physical system.

Since there is no issue with associating quantum observables to classical observables that are restricted to the phase space describing a subset of particles, there is also no issue with describing the corresponding expectation values. Indeed, consider the Hilbert space $L^2(\mathbb{R}^6)$. The components x_j, $j = 1, 2, 3$, of the position vector $x \in \mathbb{R}^6$ describe the position of particle 1. The corresponding expectation value is then given by:

$$\langle \hat{x}_j \rangle_\psi = \langle \psi, \hat{x}_j \psi \rangle.$$

We will therefore try to redefine a "state" by *starting* from a family of expectation values. We start by considering expectation values associated to bounded linear operators.

Definition 5.12. *A linear map $\Phi : B(\mathbf{H}) \to \mathbb{C}$ is a* family of expectation values *if the following conditions hold:*

1. $\Phi(\mathbf{1}) = 1$,
2. $\Phi(A) \in \mathbb{R}$ *if A is self-adjoint,*
3. $\Phi(A) \geq 0$ *if A is self-adjoint and $A \geq 0$,*
4. *Let $\{A_n\}$ be a strongly converging sequence in $B(\mathbf{H})$, i.e. there exists a $A \in B(\mathbf{H})$ such that for all $\psi \in \mathbf{H}$, $|A_n\psi - A\psi| \to 0$ as $n \to \infty$, then $\Phi(A_n) \to \Phi(A)$ in $B(\mathbf{H})$.*

We will see below that families of expectation values are closely linked to so-called *density matrices*.

Definition 5.13. *A* trace $\operatorname{tr} A$ *of an operator $A \in B(\mathbf{H})$ with $A \geq 0$ is defined as follows: let $\{e_j\}$ be an orthonormal basis of $\mathbf{H}$*[51]*, then*

$$\operatorname{tr} A = \sum_{j=1}^{\infty} \langle e_j, A e_j \rangle.$$

[50] If the particle k and l are indistinguishable, then the classical observables should be invariant under $\mathbf{x}_k \in \mathbb{R}^3$ and $\mathbf{x}_l \in \mathbb{R}^3$.

[51] Exercise: Show that the trace is independent of the choice of orthonormal basis.

The operator $A \in B(\mathbf{H})$ is said to be trace class *if* $\operatorname{tr}\sqrt{AA^*} < \infty$.[52]

An operator $\rho \in B(\mathbf{H})$ is called a density matrix *if ρ is self-adjoint, $\rho \geq 0$ and* $\operatorname{tr}\rho = 1$.[53]

It can be shown (see Chapter VI of Reed–Simon Volume 1) that $\operatorname{tr}(AB) = \operatorname{tr}(BA) < \infty$ for A a trace class operator and $B \in B(\mathbf{H})$. Furthermore, it can be shown that self-adjoint trace class operators are compact, which implies that they admit a basis of eigenvectors.

We will state the following theorem without proof:

Theorem 5.24. *Let $\rho \in B(\mathbf{H})$ be a density matrix. Then the linear map $\Phi_\rho : B(\mathbf{H}) \to \mathbb{C}$,*

$$\Phi_\rho(A) = \operatorname{tr}(\rho A) = \operatorname{tr}(A\rho)$$

is a family of expectation values. Conversely, for any family of expectation values $\Phi : B(\mathbf{H}) \to \mathbb{C}$, there exists a unique density matrix ρ such that $\Phi = \Phi_\rho$.

Recall the physics notation $|\psi\rangle\langle\psi|$ for the orthogonal projection onto the span of a unit vector ψ, i.e. $|\psi\rangle\langle\psi|(\phi) = \langle\psi, \phi\rangle\psi$. The proposition below demonstrates that via this operator, we can include our previous notion of a state.

Proposition 5.25. *For any unit vector $\psi \in \mathbf{H}$, $\rho = |\psi\rangle\langle\psi|$ is a density matrix and for any $A \in B(\mathbf{H})$:*

$$\operatorname{tr}(|\psi\rangle\langle\psi|\, A) = \langle\psi, A\psi\rangle = \langle A\rangle_\psi.$$

Proof. Since $|\psi\rangle\langle\psi|$ is an orthogonal projection, it is bounded, self-adjoint and non-negative. We compute its trace by considering a countable orthonormal basis $\{e_j\}$ of $\mathbf{H}$ with $e_1 = \psi$. Note that $\rho(e_j) = \delta_{1j}e_1$, so

$$\operatorname{tr}\rho = \sum_{j=1}^{\infty}\langle e_j, \rho e_j\rangle = \sum_{j=1}^{\infty}\delta_{1j}\langle e_j, e_1\rangle = 1.$$

Hence, $\rho = |\psi\rangle\langle\psi|$ is a density matrix.

For any $A \in B(\mathbf{H})$, we moreover have that

$$\operatorname{tr}(|\psi\rangle\langle\psi|\, A) = \sum_{j=1}^{\infty}\langle e_j, \psi\rangle\langle\psi, Ae_j\rangle = \langle\psi, A\psi\rangle.$$

□

Note moreover that for $c \in \mathbb{C}$, with $|c| = 1$, $|c\psi\rangle\langle c\psi| = |\psi\rangle\langle\psi|$, so each element in the equivalence class $[\psi]$ (see Postulate 1) gives the same density matrix.

Definition 5.14. *We refer to a density matrix $\rho \in B(\mathbf{H})$ as a* pure state *if there exists a unit vector $\psi \in \mathbf{H}$ such that $\rho = |\psi\rangle\langle\psi|$. We refer to a density matrix $\rho \in B(\mathbf{H})$ as a* mixed state *if there exists no such unit vector $\psi \in \mathbf{H}$.*

We can identify (up to unitary maps) $L^2(\mathbb{R}^{3N})$ with the completion of $L^2(\mathbb{R}^3)\underbrace{\otimes \ldots \otimes}_{N} L^2(\mathbb{R}^3)$.

We denote the completion of the tensor product of two Hilbert spaces $\mathbf{H}_1$ and $\mathbf{H}_2$ by $\mathbf{H}_1\widehat{\otimes}\mathbf{H}_2$ and we take $\langle\cdot,\cdot\rangle_{\mathbf{H}}$ on $\mathbf{H} = \mathbf{H}_1\widehat{\otimes}\mathbf{H}_2$ to be the unique inner product satisfying: $\langle v_1 \otimes w_1, v_2 \otimes w_2\rangle_{\mathbf{H}} = \langle v_1, v_2\rangle_{\mathbf{H}_1}\langle w_1, w_2\rangle_{\mathbf{H}_2}$ for all $v_1, v_2 \in \mathbf{H}_1$ and $w_1, w_2 \in \mathbf{H}_2$.

Since our Hilbert spaces are allowed to be infinite dimensional, we need a more general construction of a "tensor product" to make sense of the above statement. There are multiple such

[52] Here $\sqrt{AA^*} = f(AA^*)$, with $f(z) = \sqrt{z}$, is defined via functional calculus applied to AA^*, using that AA^* is self-adjoint.

[53] For complex Hilbert spaces $\mathbf{H}$, any non-negative bounded operator must always be self-adjoint (Exercise).

constructions, which we will not perform here. Instead, we will state the following property which captures the essence of any such construction and is therefore called a *universal property*. Let V, W be vector spaces. A tensor product of V and W is a pair $(V \otimes W, \otimes)$, with $V \otimes W$ a vector space and $\otimes : V \times W \to V \otimes W$ a linear map satisfying the following property: for any vector space Z and *bilinear* map $B : V \times W \to Z$, there exists a unique *linear* map $\widetilde{B} : V \otimes W \to Z$ such that $\widetilde{B} \circ \otimes = B$. We denote $v \otimes w := \otimes(v, w)$ for $v \in V$ and $w \in W$.

The above definition is non-constructive and does not provide the *existence* of a pair $(V \otimes W, \otimes)$ satisfying the above property: for this a more explicit construction is required. Assuming the existence, however, it turns out that the all constructions are equivalent in the following sense: suppose that $(V \otimes W, \otimes)$ and $(V \tilde{\otimes} W, \tilde{\otimes})$ satisfy the universal property. Then there exists a linear isomorphism $\Psi : V \otimes W \to V \tilde{\otimes} W$ such that $\Psi \circ \otimes = \tilde{\otimes}$.

By the universal property, any bilinear map on $V \to W$ admits a unique extension (as a linear map) to $V \otimes W$. In other words, it is sufficient to define a linear map on $V \otimes W$ on elements of the form $v \otimes w$, with $v \in V$ and $w \in W$.

Let

$$\langle \cdot, \cdot \rangle : (\mathbf{H}_1 \otimes \mathbf{H}_2) \times (\mathbf{H}_1 \otimes \mathbf{H}_2) \to \mathbb{C}$$

be defined as follows on elements of the form $v \otimes w$: $\langle v_1 \otimes v_2, w_1 \otimes w_2 \rangle_{\mathbf{H}} = \langle v_1, w_2 \rangle_{\mathbf{H}_1} \langle w_1, w_2 \rangle_{\mathbf{H}_2}$ for all $v_1, w_1 \in \mathbf{H}$ and $v_2, w_2 \in \mathbf{H}_2$. Let $B_{v_1,v_2} : \mathbf{H}_1 \otimes \mathbf{H}_2 \to \mathbb{C}$ be defined as follows:

$$B_{v_1,v_2}(w_1, w_2) = \langle v_1, w_2 \rangle_{\mathbf{H}_1} \langle w_1, w_2 \rangle_{\mathbf{H}_2}.$$

Then B_{v_1,v_2} is a bilinear map and therefore extends uniquely to a linear map $\tilde{B}_{v_1,v_2}$ such that $\tilde{B}_{v_1,v_2}(w_1 \otimes w_2) = B_{v_1,v_2}(w_1, w_2)$.

Now consider the bilinear map $\Psi : \mathbf{H}_1 \times \mathbf{H}_2 \to L(\mathbf{H}_1 \otimes \mathbf{H}_2)$, with $L(\mathbf{H}_1 \otimes \mathbf{H}_2)$ the space of linear maps $\mathbf{H}_1 \otimes \mathbf{H}_2 \to \mathbb{C}$, such that $\Psi(v_1, v_2) = \overline{\widetilde{B}}_{v_1,v_2}$, with $\overline{\widetilde{B}}_{v_1,v_2}(w_1, w_2) := \overline{\widetilde{B}_{v_1,v_2}(w_1, w_2)}$. Then Ψ extends uniquely to a linear map $\widetilde{\Psi} : (\mathbf{H}_1 \otimes \mathbf{H}_2 \to L(\mathbf{H}_1 \otimes \mathbf{H}_2)$.

Now define $\langle T, S \rangle = \overline{(\widetilde{\Psi}(T))(S)}$. Then $\langle v_1 \otimes v_2, w_1 \otimes w_2 \rangle_{\mathbf{H}} = \langle v_1, w_2 \rangle_{\mathbf{H}_1} \langle w_1, w_2 \rangle_{\mathbf{H}_2}$. We leave it as an exercise to check that $\langle \cdot, \cdot \rangle$ is a well-defined inner product. Uniqueness follows from the uniqueness of $\widetilde{\Psi}$.

Any property we need about tensor products should therefore not depend on the precise construction, but rather follow from the universal property.

Proposition 5.26. *There exists a unique unitary map $U : L^2(\mathbb{R}^n) \widehat{\otimes} L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^{2n})$, such that for all $\psi, \phi \in L^2(\mathbb{R}^n)$*

$$U(\psi \otimes \phi)(x, y) = \psi(x)\phi(y).$$

Proof. Sketch of proof: Part 1: the existence and uniqueness of U restricted to $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^n)$ satisfying the desired property follows from the universal property of tensor products which says that to any bilinear map $B : L^2(\mathbb{R}^n) \times L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^{2n})$ corresponds a unique linear map U such that $U : L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^{2n})$ with $U(\psi \otimes \phi) = B(\psi, \phi)$.

Note that $|U(\psi \otimes \phi)| = |\psi| \cdot |\phi|$. Since any element of $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^n)$ can be written as a linear combination of a finite number of elements of the form $\psi_j \otimes \phi_j$, we can conclude that U is an isometry. It therefore admits a unique isometric extension to the completion $L^2(\mathbb{R}^n) \widehat{\otimes} L^2(\mathbb{R}^n)$.

Part 2: To conclude unitarity of U, it remains to show surjectivity, i.e. $\operatorname{ran} U = L^2(\mathbb{R}^n)$. This is done by showing that $(\operatorname{ran} U)^\perp = \{0\}$. We will not prove this part.

□

We now consider a more general version of Postulate 1, adapted to the notion of mixed states and to the tensor product property above:

Postulate 1'. *A* (quantum) state of a physical system *is a density matrix* $\rho \in B(\mathbf{H})$*, with* $(\mathbf{H}, \langle\ ,\ \rangle)$ *a Hilbert space. The (quantum) state of a physical system composed of two subsystems associated to the Hilbert spaces* $\mathbf{H}_1$ *and* $\mathbf{H}_2$ *is described by a density matrix* $\rho \in B(\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2)$*.*

If a pure state $\psi \in \mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2$ is of the form $\psi_1 \otimes \psi_2$, with $\psi_1 \in \mathbf{H}_1$ and $\psi_1 \in \mathbf{H}_2$, we says ψ is *non-entangled*. Otherwise, we says ψ is *entangled*.

In the case of a single particle with spin s, $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$ moving in $\mathbb{R}^3$, the Hilbert space is given by $\mathbf{H} = L^2(\mathbb{R}^3) \hat{\otimes} V_s$ with V_s a complex vector space of dimension $2s+1$. We can identify $L^2(\mathbb{R}^3) \hat{\otimes} V_s \cong L^2(\mathbb{R}^3; V_s)$.

Now suppose we consider 2 identical particles of spin s. In view of the above theorem, a natural guess for the corresponding Hilbert space would be $L^2(\mathbb{R}^3; V_s) \hat{\otimes} L^2(\mathbb{R}^3; V_s) \cong L^2(\mathbb{R}^6; V_s \otimes V_s)$. However, in view of the *Spin-statistics theorem*, which goes beyond the scope of quantum mechanics, the following holds:

- In the case of spin $s = l \in \mathbb{N}_0$ we say the particles are *bosons* and we have to consider $L^2(\mathbb{R}^6; \mathrm{Sym}(V_s \otimes V_s))$, with $\mathrm{Sym}(V_s \otimes V_s)$ the symmetric $(2,0)$-tensors on V_s. More generally for N identical bosons, the relevant Hilbert space is $L^2(\mathbb{R}^{3N}; \mathrm{Sym}(\overbrace{V_s \otimes \ldots \otimes V_s}^{N}))$.
- In the case of $s = l + \frac{1}{2}$, $l \in \mathbb{N}_0$, we say the particles are *fermions* and we have to consider $L^2(\mathbb{R}^6; \mathrm{Alt}(V_s \otimes V_s))$, with $\mathrm{Alt}(V_s \otimes V_s)$ the antisymmetric $(2,0)$-tensors on V_s. In the case of N identical fermions, the relevant Hilbert space is $L^2(\mathbb{R}^{3N}; \mathrm{Alt}(\overbrace{V_s \otimes \ldots \otimes V_s}^{N}))$.

Since $\mathrm{tr}(\rho A)$ is only defined for $A \in B(\mathbf{H})$, we have to be a little careful with the interpretation of $\mathrm{tr}(\rho A)$ as an expectation value, in the case of unbounded A.

We keep Postulate 2 as is. We change Postulate 3 as follows:

Postulate 3'. *Consider an observable* $\hat{f}$ *corresponding to a function* $f : \mathbb{R}^{2n} \to \mathbb{R}$*. For a physical system in a state* $\rho \in B(\mathbf{H})$*, the result of the measurement of* f *is governed by a* probability measure $\mu_\rho^{\hat{f}} : \mathcal{B}_\mathbb{R} \to [0,1]$ *that is defined as follows:*

$$\mu_\rho^{\hat{f}}(\Omega) := \mathrm{tr}(\rho P^{\hat{f}}(\Omega)),$$

with $P^{\hat{f}} : \mathcal{B}_\mathbb{R} \to B(\mathbf{H})$ *the PVM associated to* $\hat{f}$ *via the spectral theorem.*

Note that in view of Postulate 3', the expectation value associated to the probability measure is given by:

$$\langle \hat{f} \rangle_\rho = \int_\mathbb{R} \lambda \, d\mu_\rho^{\hat{f}}(\lambda).$$

Let $\{e_j\}_{j \in \mathbb{N}}$ be an orthonormal basis of ρ with corresponding eigenvalues α_j (which exists by compactness and self-adjointness of ρ). For $A \in B(\mathbf{H})$, we can then express:[54]

$$\mathrm{tr}(\rho A) = \mathrm{tr}(A\rho) = \sum_{j \in \mathbb{N}} \langle e_j, A\rho e_j \rangle = \sum_{j \in \mathbb{N}} \alpha_j \langle e_j, A e_j \rangle = \sum_{j \in \mathbb{N}} \int_{\sigma(A)} \lambda \, \alpha_j d\mu_{e_j}^A(\lambda) = \int_{\sigma(A)} \lambda \, d\mu_\rho^A(\lambda) = \langle A \rangle_\rho,$$

[54] For unbounded A, we could try to *define*:

$$\mathrm{tr}(\rho A) := \int_\mathbb{R} \lambda \, d\mu_\rho^A(\lambda),$$

provided the integral is appropriately convergent. We will omit a careful analysis of this.

where in the third inequality and fourth inequality we used that

$$\sum_{j\in\mathbb{N}} \alpha_j \mu^A_{e_j}(\Omega) = \sum_{j\in\mathbb{N}} \langle e_j, P^A(\Omega)\alpha_j e_j\rangle = \sum_{j\in\mathbb{N}} \langle e_j, P^A(\Omega)\rho e_j\rangle = \mu^A_\rho(\Omega)$$

and we used boundedness of $\sigma(A)$ to be able to exchange the integral and the infinite sum $\sum_{j\in\mathbb{N}}$.

The map $B(\mathbf{H}) \ni A \mapsto \langle A\rangle_\rho$ therefore satisfies the defining properties of an family of expectation values.

Recall that our original aim was to define states of N particles in such a way that we can uniquely split off the state corresponding to a single particle. With Postulate 1' and 3' this can be done by the proposition below. Given $A \in B(\mathbf{H}_1)$ and $C \in B(\mathbf{H}_2)$, we first denote with $A \otimes C \in B(\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2)$ the unique linear operator with

$$(A \otimes C)(\phi \otimes \psi) = A(\phi) \otimes C(\phi).$$

for all $\phi \in \mathbf{H}_1$ and $\psi \in \mathbf{H}_2$.

Proposition 5.27. *Let ρ be a density matrix on $\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2$. Then there exists a unique density matrix $\rho^{(1)}$ on $\mathbf{H}_1$ such that for all $A \in B(\mathbf{H}_1)$:*

$$\mathrm{tr}(\rho(A \otimes \mathbf{1})) = \mathrm{tr}(\rho^{(1)} A).$$

Proof. By Theorem 5.24, $\Phi = \mathrm{tr}(\rho((\cdot) \otimes \mathbf{1}))$ defines a family of expectation values on $B(\mathbf{H}_1)$,[55] so there must exist a unique $\rho^{(1)}$ such that $\Phi = \Phi_{\rho^{(1)}}$. □

Hence, to each state $\rho \in B(\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2)$ of a composite system, there exists a unique state $\rho^{(1)} \in B(\mathbf{H}_1)$ that describes the subsystem corresponding to $\mathbf{H}_1$ and its associated probability measure satisfies for self-adjoint A on $\mathbf{H}_1$: for all $\Omega \in \mathcal{B}_\mathbb{R}$

$$\mu^A_{\rho^{(1)}}(\Omega) = \mathrm{tr}(\rho^{(1)} P^A(\Omega)) = \mathrm{tr}(\rho P^A(\Omega) \otimes \mathbf{1}) = \mu^{A\otimes\mathbf{1}}_\rho(\Omega),$$

so it produces the correct expectation value for operators restricted to $\mathbf{H}_1$.

Proposition 5.28. *Let $\{f_k\}_{k\in\mathbb{N}_0}$ be an orthonormal basis for $\mathbf{H}_2$. Let ρ be a density matrix on $\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2$ and let $\rho^{(1)}$ be the density matrix on $\mathbf{H}_1$ satisfying $\mathrm{tr}(\rho(A \otimes \mathbf{1})) = \mathrm{tr}(\rho^{(1)} A)$ for al $A \in B(\mathbf{H}_1)$. Then for all $\phi, \psi \in \mathbf{H}_1$:*

$$\langle \phi, \rho^{(1)} \psi\rangle = \sum_{k\in\mathbb{N}_0} \langle \phi \otimes f_k, \rho(\psi \otimes f_k)\rangle$$

Proof. Let $\{e_j\}_{j\in\mathbb{N}_0}$ be an orthonormal basis of $\mathbf{H}_1$. We will use that $\{e_j \otimes f_k\}_{j,k\in\mathbb{N}_0}$ is then an orthonormal basis of $\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2$. We can therefore express:

$$\begin{aligned}
\langle \phi, \rho^{(1)} \psi\rangle = \mathrm{tr}(\rho^{(1)} |\psi\rangle\langle\phi|) &= \mathrm{tr}(\rho(|\psi\rangle\langle\phi| \otimes \mathbf{1})) \\
&= \sum_{j,k\in\mathbb{N}_0} \langle e_j \otimes f_k, \rho(|\psi\rangle\langle\phi| \otimes \mathbf{1})(e_j \otimes f_k)\rangle \\
&= \sum_{j,k\in\mathbb{N}_0} \langle e_j \otimes f_k, \rho(\langle\phi, e_j\rangle\psi \otimes f_k)\rangle \\
&= \sum_{k\in\mathbb{N}_0} \langle \sum_{j\in\mathbb{N}_0} \langle\phi, e_j\rangle e_j \otimes f_k, \rho(\psi \otimes f_k)\rangle = \sum_{k\in\mathbb{N}_0} \langle \phi \otimes f_k, \rho(\psi \otimes f_k)\rangle.
\end{aligned}$$

□

[55] Exercise: Verify this.

The above proposition can be interpret as the statement that $\rho^{(1)}$ can be obtained from ρ by performing a trace over $\mathbf{H}_2$.

EXERCISE: Suppose that $\rho \in B(\mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2)$ is a pure state representing a composite system and write $\rho = |\psi\rangle\langle\psi|$ for some $\psi \in \mathbf{H}_1 \widehat{\otimes} \mathbf{H}_2$. Then the state $\rho^{(1)}$ describing the subsystem associated to $\mathbf{H}_1$ is pure if and only if ρ is non-entangled (a pure state ρ is non-entangled if we can write $\psi = \psi_1 \otimes \psi_2$ for some $\psi_1 \in \mathbf{H}_1$ and $\psi_2 \in \mathbf{H}_2$). [Hint: Suppose that $\rho^{(1)}$ is pure. Then we can write $\rho^{(1)} = |\psi_1\rangle\langle\psi_1|$ for some $\psi_1 \in \mathbf{H}_1$. Let $\{e_j\}$ be an orthonormal basis of $\mathbf{H}_1$ with $e_1 = \psi$. Apply Proposition 5.28 to show that $\psi = \psi_1 \otimes \psi_2$ for some $\psi_2 \in \mathbf{H}$.]

Time-evolution takes the following form (ignoring issues of domain):

Postulate 4'. *The time evolution of a physical system in a state $\rho \in B(\mathbf{H})$ is governed by the equation:*

$$\frac{d\rho}{dt} = -\frac{1}{i\hbar}[\rho, \hat{H}] \tag{5.13}$$

and $\rho(t) = e^{-\frac{it}{\hbar}\hat{H}} \rho(0) e^{+\frac{it}{\hbar}\hat{H}}$.

Note the similarity of the above equation to the Liouville equation (2.17) from Hamiltonian mechanics:

$$(\partial_t \rho)(t, \cdot) = -\{\rho(t, \cdot), H\}.$$

Note also that in Hamiltonian mechanics, the probability of encountering particle 1 in an N-particle system at the 1-particle phase space point (x^1, p_1) is governed by the following probability density:

$$\rho^{(1)}(x^1, p_1) = \int_{\mathbb{R}^{6(N-1)}} \rho(x^1, p_1, x^2, p_2, \ldots, x^N, p_N) dx^2 dp^2 \ldots dx^N dp^N.$$

The above naturally leads us to an analogous, statistical interpretation of density matrices and mixed states. Suppose we prepare a (1-particle) system to lie in a state represented by $\psi \in \mathbf{H}$, $|\psi| = 1$, but there is a degree of uncertainty regarding exactly what state the system is in. Then we can describe the system by a density matrix, which encodes the probabilities that the system is in a set of different states. In other words, density matrices describe the state of a *statistical ensemble* of particles.

For example, let $\psi, \phi \in \mathbf{H}$ such that $|\psi| = |\phi| = 1$ and $\langle \psi, \phi \rangle = 0$, then

$$\rho = \frac{1}{3}|\psi\rangle\langle\psi| + \frac{2}{3}|\phi\rangle\langle\phi|$$

is a density matrix that can be interpreted as saying that the system is in state ψ with probability $\frac{1}{3}$ and in state ϕ with probability $\frac{2}{3}$.

Note that the notion of probability coming from the uncertainty about which state a system is in is fundamentally different from the notion of probability regarding measurement outcomes that is intrinsic to quantum mechanics and is present even if we know exactly what state a system is in.

Postulate 5'. *Let $\hat{f}$ be an observable corresponding to a function $f : \mathbb{R}^{2n} \to \mathbb{R}$. Consider a physical system in a state $\rho \in B(\mathbf{H})$. Suppose that a measurement of f produces an outcome contained in $\Omega \in \mathcal{B}_{\mathbb{R}}$, where $\Omega \subset \sigma(\hat{f})$. Then, immediately after the measurement, the state will change ("collapse") to the state*

$$\rho_{\hat{f},\Omega} = \frac{P^{\hat{f}}(\Omega) \rho P^{\hat{f}}(\Omega)}{Z},$$

with $Z := \mathrm{tr}(P^{\hat{f}}(\Omega) \rho P^{\hat{f}}(\Omega))$ and $P^{\hat{f}} : \mathcal{B}_{\mathbb{R}} \to B(\mathbf{H})$ the PVM associated to $\hat{f}$ via the spectral theorem.

Note that $P^{\hat{f}}(\Omega)\rho P^{\hat{f}}(\Omega)$ defines a positive operator with finite trace.

EXERCISE: Check that in the case $\rho = |\psi\rangle\langle\psi|$, Postulates 3'–5' imply Postulates 3–5, respectively.

Example 5.4. *Consider the finite dimensional 2-particle Hilbert space $\mathbf{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ with self-adjoint operator $A : \mathbb{C}^2 \to \mathbb{C}^2$ that has two linearly independent norm 1 eigenvectors $e_1 \in \mathbf{H}$ and $e_2 \in \mathbf{H}$ with distinct eigenvalues λ_1 and λ_2, respectively. Now consider the following element of $\mathbf{H}$ representing an entangled state:*

$$\psi = \frac{1}{\sqrt{2}}(e_1 \otimes e_2 - e_2 \otimes e_1).$$

By Proposition 5.28, the density matrix corresponding to particle 1 satisfies:

$$\begin{aligned}\langle e_i, \rho^{(1)} e_j\rangle = \sum_{k=1}^{2}\langle e_i \otimes e_k, \rho(e_j \otimes e_k)\rangle &= \sum_{k=1}^{2}\langle e_i \otimes e_k, |\psi\rangle\langle\psi|(e_j \otimes e_k)\rangle \\ &= \sum_{k=1}^{2}\langle \psi, e_j \otimes e_k\rangle\langle e_i \otimes e_k, \psi\rangle = \frac{1}{2}(\delta_{1j}\delta_{1i} + \delta_{2j}\delta_{2i}) = \frac{1}{2}\delta_{ij}.\end{aligned}$$

Hence, $\rho^{(1)}$ can be represented by the matrix $\frac{1}{2}\mathbf{1}$ and it represents a mixed state.

The operator $A \otimes \mathbf{1}$ is associated to measurements of the observable A of only particle 1. The corresponding PVM sends λ_1 to $|e_1\rangle\langle e_1| \otimes \mathbf{1}$ and λ_2 to $|e_2\rangle\langle e_2| \otimes \mathbf{1}$, leading to collapse of ψ to either $\psi_1 = e_1 \otimes e_2$ or $\psi_2 = -e_2 \otimes e_1$, respectively, each with probability $\frac{1}{2}$. Similarly, measurements of $\mathbf{1} \otimes A$ lead to either ψ_1 or ψ_2 with probability $\frac{1}{2}$. Since the collapsed states ψ_i are non-entangled, the corresponding density matrices $\rho^{(1)}$ and $\rho^{(2)}$ are pure and each correspond to either $|e_1\rangle\langle e_1|$ or $|e_2\rangle\langle e_2|$.

If the measurement outcome of $A \otimes \mathbf{1}$ is λ_1, then the probability of measuring λ_2 when considering $\mathbf{1} \otimes \mathbf{A}$ acting on the collapsed state is 1 and the probability of measuring λ_1 is 0. Hence, measuring A of particle 1 has an effect on the probability measure assigned to measurements of A of particle 2, even if the particles are very far away from each other. This illustrates that measurements carried out on two particles can be strongly correlated.

In the proposition below, we establish several different ways of checking whether a state is pure.

Proposition 5.29.

(i) *A density matrix $\rho \in B(\mathbf{H})$ corresponds to a pure state if and only if $\rho^2 = \rho$.*

(ii) *A density matrix $\rho \in B(\mathbf{H})$ corresponds to a pure state if and only if $\operatorname{tr}\rho^2 = 1$.*

(iii) *A density matrix $\rho \in B(\mathbf{H})$ corresponds to a pure state if and only if $S(\rho) = \operatorname{tr}(-\rho\log\rho) = 0$. The quantity $S(\rho)$ is called the* von Neumann entropy.

Proof. Suppose $\rho = |\psi\rangle\langle\psi|$, let $\{e_j\}$ be an orthonormal basis with $e_1 = \psi$. Then $\rho(e_j) = \delta_{1j}e_1$. Let $\phi \in \mathbf{H}$. Then we can write $\phi = \sum_{j=1}^{\infty}\phi^j e_j$, so

$$\rho^2(\phi) = \sum_{j=1}^{\infty}\phi^j\rho^2(e_j) = \phi^1 e_1 = \rho(\phi).$$

Hence $\rho^2 = \rho$.

This immediately implies that $\operatorname{tr}\rho^2 = \operatorname{tr}\rho = 1$ and $2\log\rho = \log\rho^2 = \log\rho$, so $\log\rho = 0$ and $S(\rho) = 0$.

Now suppose ρ is a density matrix. Since ρ is self-adjoint and compact, it has an orthonormal basis of eigenvectors $\{e_j\}$ with corresponding eigenvalues λ_j such that $\lambda_j \to 0$ as $j \to \infty$. Suppose that $\rho^2 = \rho$. We can then express:

$$\sum_{j\in\mathbb{N}_1} |\lambda_j|^2 = ||\rho||^2 = ||\rho^2||^2 = \sum_{j\in\mathbb{N}_1} |\lambda_j|^4.$$

This implies that

$$\sum_{j\in\mathbb{N}_1} |\lambda_j|^2(1 - |\lambda_j|^2) = 0,$$

so $\lambda_j = 1$ or $\lambda_j = 0$. Since $1 = \operatorname{tr}\rho = \sum_{j\in\mathbb{N}_1} \lambda_j$, there exists precisely one j such that $\lambda_j = 1$. Hence, $\rho = |e_j\rangle\langle e_j|$.

Now suppose $\operatorname{tr}\rho^2 = 1$. Then $\operatorname{tr}\rho^2 = \operatorname{tr}\rho$, so

$$\sum_{j=1}^{\infty} \lambda_j^2 = \sum_{j=1}^{\infty} \lambda_j.$$

Using that $\lambda_j \geq 0$, as $\rho \geq 0$, this implies that

$$\sum_{j=1}^{\infty} \lambda_j(1 - \lambda_j) = 0,$$

so, as above, there exists precisely one j such that $\lambda_j = 1$ and $\rho = |e_j\rangle\langle e_j|$.

Finally, suppose that $\operatorname{tr}(-\rho\log\rho) = 0$. Then

$$0 = \sum_{j=1}^{\infty} -\lambda_j \log\lambda_j.$$

Since all $\rho \geq 0$ and $\operatorname{tr}\rho = 1$, $0 \leq \lambda_j \leq 1$, so $-\lambda_j\log\lambda_j \geq 0$. By the vanishing of the above infinite sum, we must therefore have that $-\lambda_j\log\lambda_j = 0$ for all j, so either $\lambda_j = 1$ or $\lambda_j = 0$. By $\operatorname{tr}\rho = 1$, there is precisely one j such that $\lambda_j = 1$ and $\rho = |e_j\rangle\langle e_j|$. □

Since we can express $S(\rho) = \sum_{j\in\mathbb{N}_1} -\lambda_j\log\lambda_j$, with $0 \leq \lambda_j \leq 1$ the eigenvalues of ρ and $\sum_{j\in\mathbb{N}_1}\lambda_j = 1$, we see a strong resemblance to the Gibbs entropy from statistical mechanics: consider a classical system described by a discrete set of possible microstates, then the Gibbs entropy is given by

$$S_{\text{Gibbs}} = -k_B \sum_{j\in\mathbb{N}_1} p_j \log p_j,$$

with p_j the probability of the system being in the j-th microstate.

A Deriving Poincaré transformations

Here we provide some additional motivation for the definition of the Poincaré transformations given in the lecture notes. We will show that restriction on coordinate transformations in the definition of Poincaré transformations follows from three assumptions: 1) **homogeneity of spacetime**, 2) **constancy of the speed of light**, 3) **isotropy of spacetime**.

Recall that we defined:

Definition A.1. *Poincaré transformations are affine maps $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$, with $n \geq 1$, that preserve Lorentzian length, i.e.*

$$m(\Phi(x) - \Phi(y), \Phi(x) - \Phi(y)) = m(x - y, x - y) \quad \forall x, y \in \mathbb{R}^{n+1}.$$

We denote the set of Poincaré transformations in $\mathbb{R}^{n+1}$ by $P(n; 1)$.

First, we will rephrase the affine condition on diffeomorphisms $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ as a statement of *homogeneity*. We interpret the standard Cartesian coordinates $(t = x^0, \ldots, x^n)$ on $\mathbb{R}^{n+1}$ as the spacetime coordinates assigned by Observer 0 and the coordinates $\Phi = (t' = y^0, y^1, \ldots, y^n) : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ as the spacetime coordinates assigned by Observer 1, who is moving in $\mathbb{R}^{n+1}$ relative to Observer 0 along the path $t' \mapsto \Phi^{-1}(t', 0)$ (parametrized with respect to t'). We want the coordinate chart Φ to be an *inertial coordinate chart*, but we do not want to appeal to the validity of Newton's First Law to define what that means (this would lead to the set of Galilean coordinate transformations).

We still want inertial coordinates to be coordinates with respect to which equations governing physical models remain invariant. Rather than taking a particular choice of physical model (for example, the Maxwell equations coupled to appropriate matter equations), we will define inertial coordinates via a set of reasonable conditions, which capture the following statements: there are no preferred points in space and time, no preferred directions in space and all observers corresponding to inertial coordinates should agree on the speed of light.

We will first assume that spatial difference vectors and time intervals measured by Observer 1 are independent of what Observer 0 declares to be their coordinate origin: there is no preferred origin in spacetime. More precisely:

Condition 1 (Homogeneity of spacetime)**.** *Let $x, y, z \in \mathbb{R}^{n+1}$. Then $\Phi(x + z) - \Phi(y + z) = \Phi(x) - \Phi(y)$.*

Proposition A.1. *A diffeomorphism Φ satisfies Condition 1 if and only if Φ is affine.*

Proof. If Φ is affine, then we immediately obtain Condition 1. Now suppose Φ satisfies Condition 1. Then for any $x, y, v \in \mathbb{R}^{n+1}$, we have that

$$\Phi(x + hv) - \Phi(x) = \Phi(y + hv) - \Phi(y).$$

From this it follows that the directional derivatives $D_v\Phi$ satisfy:

$$(D_v\Phi)(y) = \lim_{h \to 0} \frac{\Phi(y + hv) - \Phi(y)}{h} = \lim_{h \to 0} \frac{\Phi(x + hv) - \Phi(h)}{h} = (D_v\Phi)(x).$$

Hence, the derivative of Φ satisfies $D\Phi(x) = D\Phi(0)$. This means that $D^2\Phi(x) = 0$, so Φ is affine. □

In order to derive preservation of Lorentzian length, we will restrict to observers that agree on the speed of light. More precisely,

Condition 2 (Constancy of the speed of light)**.** *Any particle travelling with a constant velocity vector and speed 1 (i.e. the speed of light with $c = 1$) according to Observer 0, will also travel with a constant velocity vector and speed 1 according to Observer 1.*

In order to exploit Condition 2, we first prove the following lemma:

Lemma A.2. *Let $b : \mathbb{R}^{n+1} \times \mathbb{R}^{n+1} \to \mathbb{R}$ be a symmetric non-degenerate bilinear form that can be represented by a matrix with 1 negative eigenvalue and n positive eigenvalues. Let $p : \mathbb{R}^{n+1} \times \mathbb{R}^{n+1} \to \mathbb{R}$ be a symmetric bilinear form satisfying: for all $v \in \mathbb{R}^{n+1}$, such that $b(v, v) = 0$, we must have that $p(v, v) = 0$. Then there exists a constant $\alpha_p \in \mathbb{R}$ such that for all $v, w \in \mathbb{R}^{n+1}$:*

$$p(v, w) = \alpha_p b(v, w).$$

If p has the same number of positive and negative eigenvalues as b, then $\alpha_p > 0$.

Proof. Without loss of generality, we can choose a basis $\{e_\mu\}_{0 \le \mu \le n}$ on $\mathbb{R}^{n+1}$, such that $b(e_0, e_i) = 0$, $b(e_0, e_0) = -1$ and $b(e_i, e_j) = \delta_{ij}$, for $1 \le i, j \le n$. Denote $V = \operatorname{span}\{e_0\}$ and $W = \operatorname{span}\{e_1, \dots, e_n\}$. Let $v \in V$ and $w \in W$. Then

$$b(v \pm w, v \pm w) = -(v^0)^2 + \sum_{i=1}^{n} (w^i)^2.$$

Hence, there exists a $\lambda \in \mathbb{R}_+$ such that $b(v \pm \lambda w, v \pm \lambda w) = 0$.

Note also that by symmetry and bilinearity of p:

$$p(v, \lambda w) = \frac{1}{4} \left(p(v + \lambda w, v + \lambda w) - p(v - \lambda w, v - \lambda w) \right).$$

We use that $b(v \pm \lambda w, v \pm \lambda w) = 0$ implies $p(v \pm \lambda w, v \pm \lambda w) = 0$ to obtain $p(v, \lambda w) = 0$. By bilinearity of p, we therefore have

$$p(v, w) = 0$$

for all $v \in V$ and $w \in W$.

Consider $w, w' \in W$ such that $b(w', w') = b(w, w)$. Let $v \in V$ such that $b(v \pm w, v \pm w) = 0$. We can then express:

$$b(v \pm w', v \pm w') = b(v, v) + b(w', w') = b(v, v) + b(w, w) = b(v \pm w, v \pm w) = 0.$$

We therefore also have that $0 = p(v \pm w', v \pm w') = p(v \pm w, v \pm w)$ and hence:

$$p(w', w') = \overbrace{p(v + w', v + w')}^{=0} - p(v, v) = \overbrace{p(v + w, v + w)}^{=0} - p(v, v) = p(w, w).$$

Furthermore, we obtain for $w, w' \in W$ satisfying $b(w, w') = 0$:

$$p(w, w') = \frac{1}{4} \left(p(w + w', w + w') - p(w - w', w - w') \right) = 0.$$

Hence, we have that $p(e_i, e_j) = p(e_1, e_1)\delta_{ij}$. Denote $\alpha_p = p(e_1, e_1)$.

We also obtain the following equality: let $v \in V$ and $w \in W$ such that $p(v + w, v + w) = 0$. Then

$$p(v, v) = p(v + w, v + w) - p(w, w) = -p(w, w) = -\alpha_p.$$

We conclude that $p(e_\mu, e_\nu) = \alpha_p b(e_\mu, e_\nu)$. If the diagonal matrix associated to p has the same number of positive and negative eigenvalues as the diagonal matrix associated to b, namely n and 1, respectively, then $\alpha_p > 0$. □

Proposition A.3. *Let $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a coordinate transformation satisfying Conditions 1 and 2. Then there exists a constant α_Φ such that for all $x, y \in \mathbb{R}^{n+1}$:*

$$m(\Phi(x) - \Phi(y), \Phi(x) - \Phi(y)) = \alpha_\Phi m(\Phi(x) - \Phi(y), \Phi(x) - \Phi(y))$$

Proof. By Condition 1 and Proposition A.1, Φ is affine. We will first assume that Φ is linear, i.e. $\Phi(0) = 0$. Let $\gamma : I \to \mathbb{R}^{n+1}$ describe a particle travelling at speed 1 with respect to coordinates $\{x^0, \dots, x^n\}$, parametrized by $s = x^0$. Let X denote the constant vector field satisfying $X = \dot{\gamma}$. Then the statement that γ represents a particle travelling at constant speed 1 implies:

$$m(X, X) = -1 + |\dot{\gamma}|^2 = 0.$$

By Condition 2, $\Phi(\gamma)(t')$ must also represent a particle travelling at speed 1 (with respect to the parameter $y^0 = t'$). In the coordinate chart Φ, the components of the tangent vector field to γ are given by $D_X\Phi(0) = \Phi(X)$, by using linearity of Φ. So we obtain:

$$0 = m(D\Phi(X), D\Phi(X)) = m(\Phi(X), \Phi(X)).$$

Hence, $b(\cdot, \cdot) := m(\cdot, \cdot)$ and $p(\cdot, \cdot) := m(\Phi(\cdot), \Phi(\cdot))$ satisfy the assumptions in Lemma A.2 and we can conclude that there exists a positive constant α_Φ such that:

$$m(\Phi(x), \Phi(y)) = \alpha_\Phi m(x, y).$$

The above applies moreover that for arbitrary $z \in \mathbb{R}^{n+1}$:

$$\begin{aligned} m((\Phi(x)+z)-(\Phi(y)+z), (\Phi(x)+z)-(\Phi(y)+z)) = m(\Phi(x)-\Phi(y), \Phi(y)-\Phi(y)) = m(\Phi(x-y), \Phi(x-y)) \\ = \alpha_\Phi m(x - y, x - y). \end{aligned}$$

Since any affine map $\widetilde{\Phi}$ can be decomposed as follows: $\widetilde{\Phi}(x) = \Phi(x)+z$ with Φ linear and $z = \widetilde{\Phi}(0)$, we conclude the statement in the proposition. □

In order to conclude that $\alpha_\Phi = 1$ and hence that Φ is a Poincaré transformation, we need to impose one more condition: *isotropy*. We will assume that spatial difference vectors and time intervals measured by Observer 1 are independent of how Observer 0 happens to be rotated: space has no preferred direction. More precisely:

Condition 3 (Isotropy)**.** *Let $x, y \in \mathbb{R}^{n+1}$ and $R \in SO(n)$. Then*

$$\Phi\left(\begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} x\right) - \Phi\left(\begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} y\right) = \Phi(x) - \Phi(y).$$

Proposition A.4. *Let $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a coordinate transformation satisfying Conditions 1, 2 and 3. Then $\Phi \in P(n; 1)$.*

Proof. We will show that we must have that $\alpha_\Phi = 1$ in Proposition A.3.

Note that the frame of coordinate vector fields $\{f^0, \dots, f^n\}$ associated to Observer 1 satisfies $f_\mu = \Phi_*(e_\mu)$, with e_μ the standard basis of coordinate vector fields on $\mathbb{R}^{n+1}$, or the frame of vector fields associated to Observer 1, and where Φ_* is the pushforward of Φ. Since e_μ are constant in spacetime and $D\Phi(x) = D\Phi(0)$ by the fact that Φ is affine, we have that $f_\mu = D\Phi(0)e_\mu$, and hence f_μ are also constant in spacetime. Note that the spatial origin of Observer 1, satisfies $\Phi^{-1}(t', 0) = (x^0(t', 0), x^1(t', 0), \dots, x^n(t', 0))$. Since x^i depend linearly on t', and since t' depends linearly on t, we must have that the origin of Observer 1 moves at constant velocity with respect to Observer 0.

By the above observations, there exist rotations $R, \widetilde{R} \in SO(n)$ and $b \in \mathbb{R}^{n+1}$, such that $\Psi := \begin{pmatrix} 1 & 0 \\ 0 & \widetilde{R} \end{pmatrix} \circ \Phi \circ \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \circ T_b$ describes an observer moving at constant velocity ve_n, $v > 0$ according to Observer 0 and satisfies $\Psi(0) = 0$. In other words, we can translate and rotate Observer 0 so that Observer 1 moves at a constant velocity along the x^n-axis in the direction of increasing x^n, and then rotate Observer 1 so that its frame of coordinate vector fields agrees with the frame of Observer 1. Hence, the constant α_Φ can only depend on $R, \widetilde{R}, b$ and the speed v of Observer 1 according to Observer 0. By appealing to Conditions 1 and 3 we moreover obtain:

$$\begin{aligned}
&m(\Psi(x) - \Psi(y), \Psi(x) - \Psi(y))) \\
&= m\left(\Phi \circ \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \circ T_b(x) - \Phi \circ \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \circ T_b(y), \Phi \circ \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \circ T_b(x) - \Phi \circ \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \circ T_b(y)\right) \\
&= m(\Phi(x) - \Phi(y), \Phi(x) - \Phi(y)),
\end{aligned}$$

so in fact α_Φ only depends on the speed v (the magnitude of the velocity) of Observer 1 according to Observer 0. We will write $\alpha_\phi = a_v$. Now consider another observer, Observer 2, who assigns coordinates to $\mathbb{R}^{n+1}$ according to a coordinate transformation $\widetilde{\Phi}$ satisfying Conditions 1, 2 and 3. Then $m(\widetilde{\Phi}(x) - \widetilde{\Phi}(y), \widetilde{\Phi}(x) - \widetilde{\Phi}(y)) = a_w m(x - y, x - y)$, with w the speed of Observers 2 and according to Observer 0. We also have that $m(\widetilde{\Phi}(x) - \widetilde{\Phi}(y), \widetilde{\Phi}(x) - \widetilde{\Phi}(y)) = a_{w_1} m(\Phi(x) - \Phi(y), \Phi(x) - \Phi(y))$, with w_1 the speed of Observer 2 according to Observer 1.

By combining the above identities, we obtain moreover:

$$a_w = a_{w_1} a_v. \tag{A.1}$$

Hence, $a_{w_1} = \frac{a_w}{a_v}$, so a_{w_1} can only depend on the speeds v and w. But the speed w_1 does not just depend on the speeds v and w, but also on the angle between the corresponding velocity vectors. This implies that the constants a_v, a_w, a_{w_1} must all be equal to some fixed constant $a \in \mathbb{R}$. Furthermore, by (A.1), we must have that $a = a^2$, which implies that $a = 1$. We can therefore conclude that $\alpha_\Phi = 1$. □

B Geometry

B.1 Geodesics and normal coordinates

Let $(\mathcal{M}, g)$ be a Riemannian or Lorentzian manifold and denote with ∇ the corresponding Levi–Civita connection.

We first look closer at the locality property of the covariant derivative.

Lemma B.1. *Let $p \in \mathcal{M}$ and $X, Y \in \mathcal{T}(\mathcal{M})$. Let U be an arbitrarily small neighbourhood containing p. Then $\nabla_X Y(p)$ depends only on $Y|_U$ and $X(p)$.*

Proof. Let $\phi \in C^\infty(\mathcal{M})$ be a smooth bump function such that $\operatorname{supp} \varphi \subset U$ and $\varphi(p) = 1$. Let $Y, \tilde{Y} \in \mathcal{T}(\mathcal{M})$ such that $Y|_U = \tilde{Y}|_U$. Denote $Z = Y - \tilde{Y}$. By linearity of ∇_X, it is sufficient to show that $\nabla_X Z(p) = 0$. Since Z vanishes on U, we have that $\varphi Z \equiv 0$. Hence, by the product rule for covariant derivatives:

$$0 = \nabla_X(\varphi Z)(p) = X(\varphi)(p) Z(p) + \varphi(p) \nabla_X Z(p) = 0 + 1 \cdot \nabla_X Z(p).$$

Now let $X, \tilde{X} \in \mathcal{T}(\mathcal{M})$ be such that $X(p) = \tilde{X}(p)$ and define $V = X - \tilde{X}$. We then need to show that for any $Y \in \mathcal{T}(\mathcal{M})$, $\nabla_V Y(p) = 0$. Let $p \in \mathcal{U}$, with $\mathcal{U}$ the domain of a coordinate chart $\{x^\mu\}$. Then we can express $V = V^\mu \partial_\mu$. Furthermore, we can write:

$$\nabla_V Y(p) = V^\mu(p) \nabla_{\partial_\mu} Y(p) = 0.$$

□

We will introduce the notion of the covariant derivative along curves corresponding to ∇. Let $\gamma : I \to \mathcal{M}$ be a smooth curve with $I \subset \mathbb{R}$ open. Then we define $\mathcal{T}(\gamma)$, the set of *vector fields along* $\gamma : I \to \mathcal{M}$, as follows:

$$\mathcal{T}(\gamma) = \{X : I \to T\mathcal{M} \mid X(t) \in T_{\gamma(t)}\mathcal{M} \; \forall t \in I\}.$$

Proposition B.2. *For each smooth curve $\gamma : I \to \mathcal{M}$, with $I \subset \mathbb{R}$ an open interval, ∇ determines a unique operator $D_t : \mathcal{T}(\gamma) \to \mathcal{T}(\gamma)$ satisfying the following properties:*

(i) *(Linearity) Let $X, Y \in \mathcal{T}(\gamma)$ and $\lambda \in \mathbb{R}$, then*

$$D_t(\lambda X + Y) = \lambda D_t X + D_t Y.$$

(ii) *(Product rule) Let $f \in C^\infty(I)$ and $X \in \mathcal{T}(\gamma)$. Then*

$$D_t(fX) = \dot{f}X + f D_t X.$$

(iii) *(Extensions) If $X \in \mathcal{T}(\gamma)$ is extendible to $\mathcal{T}(\mathcal{M})$, then for any extension $\widetilde{X} \in \mathcal{T}(\mathcal{M})$ of X,*

$$D_t X(t) = \nabla_{\dot{\gamma}(t)} \widetilde{X},$$

where $\nabla_{\dot{\gamma}(t)}$ is well-defined by Lemma B.1.

We refer to $D_t X$ as the covariant derivative of X along γ.

Proof. We first establish uniqueness. Let $t_0 \in I$. By Lemma B.1 and property (iii), $D_t X$ depends only on $X|_{(t_0-\epsilon, t_0+\epsilon)}$, with $\epsilon > 0$ arbitrarily small. Let $t_0 \subset U$, with U covered by a single coordinate chart $\{x^\mu\}$. We write $X(t) = X^\mu(t)\partial_\mu$. Since ∂_μ is extendible, we use all three properties of D_t to obtain:

$$\begin{aligned} D_t X(t_0) &= \dot{X}^\mu(t_0)\partial_\mu + X^\mu(t_0)\nabla_{\dot{\gamma}(t_0)}\partial_\mu \\ &= (\dot{X}^\nu(t_0) + \dot{X}^\mu(t_0)\dot{\gamma}(t_0)^\rho \Gamma^\nu_{\mu\rho}(\gamma(t_0))\partial_\nu. \end{aligned}$$

Hence, D_t must be unique if it exists. In fact, we can use the above expression to define D_t if $\gamma(I)$ lies in the domain of a single chart. If not, we can use it to define D_t locally and then conclude that the expressions agree when two charts overlap. □

Definition B.1. *We define the* acceleration *of curve $\gamma : \mathbb{R} \subseteq I \to \mathcal{M}$ as $D_t\dot{\gamma}$. We say γ is an affinely parametrized* geodesic *if $D_t\dot{\gamma} \equiv 0$, which we refer to as the geodesic equation.*

Proposition B.3. *Let $p \in \mathcal{M}$ and $X_p \in T_p\mathcal{M}$. Then there exists an open interval $0 \in I \subset \mathbb{R}$ and a smooth geodesic $\gamma : I \to \mathcal{M}$ such that*

$$\begin{aligned} \gamma(t_0) &= p, \\ \dot{\gamma}(t_0) &= X_p. \end{aligned}$$

Any two such geodesics agree on their common domain.

Proof. The equation $D_t\dot{\gamma} \equiv 0$ reduces to a second-order ODE in the domain of a chart around p. Apply existence and uniqueness for ODE (Picard–Lindelöf). □

Let $X_p \in T_p\mathcal{M}$. Then we denote with γ_{X_p} the corresponding geodesic with $\gamma_{X_p}(0) = p$ and $\dot{\gamma}_{X_p}(0) = X_p$.

Proposition B.4. *Let $\gamma : I \to \mathcal{M}$ be a geodesic. Then*

$$\frac{d}{dt} g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t)) = 0.$$

Proof.

$$\frac{d}{dt} g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t)) = \nabla_{\dot\gamma} g(\dot\gamma, \dot\gamma)(t) = 2g(D_t\dot\gamma, \dot\gamma)(t) = 0.$$

□

Let g be a Lorentzian metric. We say $X \in T_p\mathcal{M}$ is *spacelike* if $g_p(X, X) > 0$, *timelike* if $g_p(X, X) < 0$ and *null* if $g_p(X, X) = 0$. By the above proposition, the corresponding geodesics γ_X can similarly be classified as timelike, spacelike or null geodesics.

We similarly say a general curve $\gamma : I \to \mathcal{M}$ is spacelike, timelike or null if the tangent vector field $\dot\gamma$ satisfies $g(\dot\gamma, \dot\gamma) > 0$, $g(\dot\gamma, \dot\gamma) < 0$ and $g(\dot\gamma, \dot\gamma) = 0$ respectively.

Remark B.1. *Recall that we already saw that for submanifolds on $\mathbb{R}^n$ with the metric induces by the Euclidean metric on $\mathbb{R}^n$ the geodesic equation corresponds to the Euler–Lagrange equations of the action:*

$$\mathcal{S}(\gamma) = \int_I g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t)\, dt.$$

Geodesics allow use to define a natural coordinate chart in the neighbourhood of any point in $\mathcal{M}$ called *normal coordinates*. We first define an exponential map.

Definition B.2. *Let $p \in \mathcal{M}$ and $\mathcal{E}_p := \{X \in T_p\mathcal{M} \,|\, \gamma_X : I \to \mathcal{M}$ well-defined with $[0, 1] \subset I\}$. Then the (restricted)* exponential map $\mathrm{Exp}_p : \mathcal{E}_p \to \mathcal{M}$ *is defined as:*

$$\mathrm{Exp}_p(X) = \gamma_X(1).$$

We use the notation "Exp" so as not to confuse this map with the exponential map "exp" on Lie groups.

Lemma B.5. *The exponential map Exp_p satisfies the following properties:*

(i) *For any $X \in T_pM$, $\gamma_{cX} = \gamma_X(ct)$, with $c \geq 0$, $t \in \mathbb{R}$ such that either side is well-defined.*

(ii) *For each $X \in \mathcal{E}_p$ we can express $\gamma_X(t) = \mathrm{Exp}_p(tX)$ for all t such that either side is well-defined.*

(iii) *$\mathcal{E}_p$ is starshaped with respect to 0, i.e. if $X \in \mathcal{E}_p$ then $tX \in \mathcal{E}_p$ for all $t \in [0, 1]$. Furthermore, there exists a $r > 0$ such that $B_r(0) \subset \mathcal{E}_p$.*

(iv) *The map Exp_p is smooth.*

Proof. We will first prove (i). Without loss of generality, we will show that if $\gamma_X(ct)$ is well-defined, then $\gamma_{cX}(t)$ is also well-defined. Indeed, if we instead start with the assumption that $\gamma_{cX}(t)$ is well-defined, we repeat the argument with X replaced with cX and c replaced with $\frac{1}{c}$.

Suppose the domain of $\gamma = \gamma_X$ is I, with $0 \in I \subset \mathbb{R}$ open. Define $\tilde\gamma : c^{-1}I \to \mathcal{M}$ by $\tilde\gamma(t) = \gamma(ct)$. Then $\tilde\gamma(0) = p$ and

$$\dot{\tilde\gamma}(0) = c\dot\gamma(0) = cX.$$

It remains to show that $\tilde\gamma$ is a geodesic. Let $\widetilde{D}_t$ and D_t be the covariant derivatives along γ and $\tilde\gamma$, respectively. We compute in coordinates:

$$\begin{aligned}
\tilde{D}_t\dot{\tilde\gamma}^\nu(t) &= \ddot{\tilde\gamma}^\nu(t) + \dot{\tilde\gamma}^\mu(t)\dot{\tilde\gamma}^\rho(t)\Gamma^\nu_{\mu\rho}(\tilde\gamma(t)) \\
&= c^2(\ddot\gamma^\nu(ct) + \dot\gamma^\mu(t)\dot\gamma^\rho(t)\Gamma^\nu_{\mu\rho}(\gamma(ct)) \\
&= c^2 D_t\dot\gamma^\nu(ct) = 0
\end{aligned}$$

and conclude that $\tilde{\gamma} = \gamma_{cX}$ by the uniqueness property of geodesics.

The statement (ii) immediately follows since: $\mathrm{Exp}_p(tX) = \gamma_{tX}(1) = \gamma_X(t)$.

We turn to (iii). Let $X \in \mathcal{E}_p$. Then the domain of γ_X includes $[0,1]$ by definition. By (ii), we have that for $0 \leq t \leq 1$, $\mathrm{Exp}_p(tX) = \gamma_{tX}(1) = \gamma_X(t)$ is well-defined, so $tX \in \mathcal{E}_p$. From Picard–Lindelöf it follows moreover that there exists an $r > 0$ such that for all $X \in T_p\mathcal{M}$ with $|X| = 1$, γ_X is well defined in $[0, r]$. By (ii), this means that Exp is well-defined on $B_r(0) \subset \mathcal{E}_p$.

Finally, smoothness of Exp_p follows from smoothness of the metric g and the continuous dependence of the solution to the ODE on initial data, applied also to higher-order time derivatives of γ. □

We will use the map Exp_p to construct a particularly convenient coordinate chart.

Proposition B.6. *For any $p \in \mathcal{M}$, there exists a neighbourhood V of $0 \in T_p\mathcal{M}$ and a neighbourhood U of p in $\mathcal{M}$ such that*

$$\mathrm{Exp}_p : V \to U$$

is a diffeomorphism.

The set U is called a *normal neighbourhood of p*.

Proof. This follows from the inverse function theorem, once we establish invertibility of $(\mathrm{Exp}_p)_* : T_p\mathcal{M} \cong T_0(T_p\mathcal{M}) \to T_p\mathcal{M}$. Let $X \in T_p\mathcal{M}$ and let $\delta : \mathbb{R} \to T_p\mathcal{M}$ be defined as $\delta(t) = tX$. Then $X = \dot{\delta}(0) \in T_0(T_p\mathcal{M})$. Hence,

$$(\mathrm{Exp}_p)_*(X) = \frac{d}{dt}|_{t=0}(\mathrm{Exp} \circ \delta)(t) = \frac{d}{dt}|_{t=0}\mathrm{Exp}(tX) = \frac{d}{dt}|_{t=0}\gamma_X(t) = X.$$

So $(\mathrm{Exp}_p)_*$ is the identity map and is thus clearly invertible. □

Let $p \in T_p\mathcal{M}$ and let $\{e_\mu\}$ be an orthonormal basis on T_pM. Then $E : \mathbb{R}^n \to T_p\mathcal{M}$, $E(x^1, \ldots, x^n) = x^\mu e_\mu$ defines an isomorphism.

We refer to $\mathrm{Exp}(B_r(0))$ as the *geodesic ball* of radius r centred at p.

Definition B.3. *Let U be a normal neighbourhood of $p \in \mathcal{M}$. The map $\psi : U \to \mathbb{R}^n$, $\psi = E^{-1} \circ \mathrm{Exp}_p^{-1} : U \to \mathbb{R}^n$ defines a coordinate chart. The corresponding coordinates $x^\mu = \psi^\mu$ are called* normal coordinates.

Proposition B.7. *Normal coordinates satisfy the following properties:*

(i) *Let $X = X^\mu e_\mu \in T_p\mathcal{M}$. Then γ_X in normal coordinates is represented by:*

$$\gamma_V(t) = t(X^1, \ldots, X^n).$$

for all $t \in \mathbb{R}$ such that $\mathrm{Exp}_p(tX) \in U$.

(ii) *The coordinates of p are $(0, \ldots, 0)$.*

(iii) *The components of g at p are equal to the components in Cartesian coordinates of the Euclidean metric in the Riemannian case or the Minkowski metric in the Lorentzian case, i.e. $g_{\mu\nu} = \delta_{\mu\nu}$ (Riemannian) and $g_{\mu\nu} = m_{\mu\nu}$ (Lorentzian).*

(iv) *The derivatives of $g_{\mu\nu}$ vanish at p: $\partial_\rho g_{\mu\nu}(p) = 0$ with respect to normal coordinates. In particular, $\Gamma^\rho_{\mu\nu}(p) = 0$.*

Proof. Property (i) follows from the fact that $\psi(\gamma_X(t)) = \psi(\mathrm{Exp}_p(tX)) = E^{-1}(tX) = t(X^1, \ldots, X^n)$. Similarly, (ii) follows from $\psi(p) = \psi(\mathrm{Exp}_p(0)) = (0, \ldots, 0)$.

Consider now (iii). We have that $g_{\mu\nu}(p) = g_p(\partial_\mu|_p, \partial_\nu|_p)$, where $\partial_\mu|_p \in T_p\mathcal{M}$ is the basis vector field associated to the normal coordinates $\{x^\mu\}$. By construction, $\partial_\mu|_p = \dot{\gamma}_{e_\mu}(0) = e_\mu$. Hence, (iii) follows from the assumption that $\{e_\mu\}$ form an orthonormal basis of $T_p\mathcal{M}$ with respect to $g(p)$.

Finally, we will show (iv). We first show that the Christoffel symbols must vanish. This follows from the geodesic equation expressed in normal coordinates: let $X \in T_p\mathcal{M}$, then

$$0 = (D_t\dot{\gamma}_X)^\nu(0) = \ddot{\gamma}^\mu(0) + \Gamma^\nu_{\mu\rho}(p)X^\mu X^\rho = \Gamma^\nu_{\mu\rho}(p)X^\mu X^\rho,$$

by (i). Since X was arbitrary, it follows immediately that $\Gamma^\nu_{\mu\mu}(p) = 0$ for all μ. By the symmetry of $\Gamma^\nu_{\mu\rho}(p)$ in its lower two indices, one can also show that $\Gamma^\nu_{\mu\rho}(p) = 0$ for all μ, ρ (Exercise). To conclude that partial derivatives of $g_{\mu\nu}$ vanish, we use the following identity at p:

$$\begin{aligned} 0 &= g_{\mu\sigma}\Gamma^\sigma_{\nu\rho} + g_{\nu\sigma}\Gamma^\sigma_{\mu\rho} = \frac{1}{2}g_{\mu\sigma}(g^{-1})^{\sigma\alpha}(\partial_\nu g_{\rho\alpha} + \partial_\rho g_{\nu\alpha} - \partial_\alpha g_{\nu\rho}) + \mu \leftrightarrow \nu \\ &= \frac{1}{2}(\partial_\nu g_{\rho\mu} + \partial_\rho g_{\nu\mu} - \partial_\mu g_{\nu\rho}) + \mu \leftrightarrow \nu \\ &= \partial_\rho g_{\nu\mu}. \end{aligned}$$

□

Remark B.2. *One may compare the existence of normal coordinate in a neighbourhood of every point to Darboux's Theorem in symplectic geometry, which states that on a symplectic manifold $(\mathcal{M}, \omega)$ of dimension $2n$ and $p \in \mathcal{M}$, here exist coordinates $(x^1, \ldots, x^n, y^1, \ldots, y^n)$ in neighbourhood of p, such that $\omega = \sum_{i=1}^n dx^i \wedge dy^i$. The key difference for $(\mathcal{M}, g)$, however, is that the identities $g_{\mu\nu} = m_{\mu\nu}$ and $\partial_\rho g_{\mu\nu} = 0$ only necessarily hold <u>at</u> p and not in a full neighbourhood of p. Related to this, we will see that second-order derivatives of $g_{\mu\nu}$ can <u>not</u> all be made to vanish in general.*

Remark B.3. *Let $\gamma : (-\epsilon, \epsilon) \to \mathcal{M}$ be a geodesic with $\gamma(0) = p$. With a little bit more work and for $\epsilon > 0$ suitably small, it is possible to construct a coordinate chart in a neighbourhood U of $\gamma(-\epsilon, \epsilon)$, such that $g_{\mu\nu}(q) = \Gamma^\nu_{\mu\nu}(q) = \partial_\rho g_{\mu\nu}(q) = 0$ for all $q \in \gamma(-\epsilon, \epsilon)$. The corresponding coordinates are called* Fermi normal coordinates.

If we interpret timelike geodesics in Lorenzian manifolds as representing free falling observers in spacetime, carrying orthogonal measuring rods, the existence of Fermi normal coordinates may be thought of as providing a precise mathematical formulation of the equivalence principle *in general relativity. This, loosely formulated, states that non-gravitational experiments performed by free-falling observers at sufficiently small time and length scales cannot be distinguished from those performed by non-accelerating observers in Minkowski spacetime.*

B.2 Curvature

Let $(\mathcal{M}, g)$ be a Riemannian or Lorentzian manifold with Levi–Civita connection.

We introduce a map that describes an *intrinsic*[56] notion of curvature of $(\mathcal{M}, g)$.

Definition B.4. *The* Riemann curvature endomorphism *is defined as the map:*

$$\begin{aligned} \mathrm{Riem} &: \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \to \mathcal{T}(\mathcal{M}) \\ \mathrm{Riem}(X, Y)Z &:= \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z. \end{aligned}$$

The proposition below shows that we can interpret Riem as a $(1, 3)$-tensor field, which we will call the *Riemann curvature tensor*.

[56] Independent of how the manifold may be embedded in some larger manifold.

Proposition B.8. $\mathrm{Riem} \in \mathcal{T}^{(1,3)}(\mathcal{M}) = \Gamma(T^*\mathcal{M} \otimes T^*\mathcal{M} \otimes T^*\mathcal{M} \otimes T\mathcal{M})$

Proof. We can characterize $(1,3)$ tensor fields as maps on $\mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \to \mathcal{T}(\mathcal{M})$ that are multilinear over $C^\infty(\mathcal{M})$. It is immediate that the map is multilinear over $\mathbb{R}$, by the linearity of the operations in its definition. Exercise: Show that

$$\begin{aligned}\mathrm{Riem}(fX, Y)Z =& f\mathrm{Riem}(X, Y)Z,\\ \mathrm{Riem}(X, fY)Z =& f\mathrm{Riem}(X, Y)Z,\\ \mathrm{Riem}(X, Y)(fZ) =& f\mathrm{Riem}(X, Y)Z,\end{aligned}$$

for all $f \in C^\infty(\mathcal{M})$ and $X, Y, Z \in \mathcal{T}(\mathcal{M})$. □

We denote the components of Riem with respect to a basis on $T_p\mathcal{M}$ by $R^\sigma{}_{\mu\nu\rho}$, with $\mathrm{Riem}(X, Y)Z = R^\sigma{}_{\mu\nu\rho}Z^\mu X^\nu Y^\rho \partial_\sigma$. In particular, with respect to a coordinate basis we have that:

$$\begin{aligned}\mathrm{Riem}(\partial_\nu, \partial_\rho)\partial_\mu &= \nabla_{\partial_\nu}\nabla_{\partial_\rho}\partial_\mu - \nabla_{\partial_\rho}\nabla_{\partial_\nu}\partial_\mu\\ &= \nabla_{\partial_\nu}(\Gamma^\sigma_{\rho\mu}\partial_\sigma) - \nabla_{\partial_\rho}(\Gamma^\sigma_{\nu\mu}\partial_\sigma)\\ &= \partial_\nu\Gamma^\sigma_{\rho\mu}\partial_\sigma - \partial_\rho\Gamma^\sigma_{\nu\mu}\partial_\sigma + \Gamma^\sigma_{\rho\mu}\Gamma^\alpha_{\nu\sigma}\partial_\alpha - \Gamma^\sigma_{\nu\mu}\Gamma^\alpha_{\rho\sigma}\partial_\alpha\end{aligned}$$

Hence,

$$R^\sigma{}_{\mu\nu\rho} = \partial_\nu\Gamma^\sigma_{\rho\mu} - \partial_\rho\Gamma^\sigma_{\nu\mu} + \Gamma^\kappa_{\rho\mu}\Gamma^\sigma_{\nu\kappa} - \Gamma^\kappa_{\nu\mu}\Gamma^\sigma_{\rho\kappa}.$$

In particular, with respect to normal coordinates at $p \in \mathcal{M}$, we have that

$$R^\sigma{}_{\mu\nu\rho}(p) = \partial_\nu\Gamma^\sigma_{\rho\mu}(p) - \partial_\rho\Gamma^\sigma_{\nu\mu}(p).$$

We will also refer to the $(0,4)$-tensor field $\mathrm{Riem}^\flat$ with components $R_{\mu\nu\rho\sigma}$ as the Riemann curvature tensor.

Definition B.5. *The* Ricci tensor *is defined as the* $(0,2)$*-tensor field* Ric *with components* $R_{\mu\nu}$*, defined as follows:*

$$R_{\mu\nu} = R^\sigma{}_{\mu\sigma\nu}.$$

Definition B.6. *The* Ricci scalar $R \in C^\infty(\mathcal{M})$ *is defined as follows:*

$$R = (g^{-1})^{\mu\nu}R_{\mu\nu}.$$

The Riemann tensor and its contractions are natural quantities to consider on $(\mathcal{M}, g)$ because they are invariant under isometries, which are diffeomorphisms $\Phi : \mathcal{M} \to \widetilde{\mathcal{M}}$, where $(\mathcal{M}, g)$ and $(\widetilde{\mathcal{M}}, \widetilde{g})$ are Riemannian/Lorentzian manifolds, such that $\Phi^*\widetilde{g} = g$.

Proposition B.9. *Let* $(\mathcal{M}, g)$ *and* $(\widetilde{\mathcal{M}}, \widetilde{g})$ *be Riemannian/Lorentzian manifolds with Riemann curvature tensors* Riem *and* $\widetilde{\mathrm{Riem}}$*, respectively. Let* $\Phi : \mathcal{M} \to \widetilde{\mathcal{M}}$ *be an isometry. Then*

$$\begin{aligned}\widetilde{\mathrm{Riem}}(\Phi_*X, \Phi_*Y)\Phi_*Z &= \Phi_*(\mathrm{Riem}(X, Y, Z)),\\ \Phi^*\widetilde{\mathrm{Riem}} &= \mathrm{Riem}.\end{aligned}$$

Proof. Exercise. □

The Riemann curvature tensor enjoys various symmetry properties.

Proposition B.10. *With respect to an arbitrary coordinate chart, the components of the Riemann tensor satisfy:*

(i) $R^{\sigma}{}_{\mu(\nu\rho)} = 0$,

(ii) $R^{\sigma}{}_{[\mu\nu\rho]} = 0$,

(iii) $\nabla_{[\alpha} R_{\sigma\mu]\nu\rho} = 0$ *(Bianchi identity)*,

(iv) $R_{\sigma\mu\nu\rho} = R_{\nu\rho\sigma\mu}$,

(v) $R_{(\sigma\mu)\nu\rho} = 0$.

Proof. Exercise. □

From the above symmetry properties, we can deduce symmetry properties at the level of the Ricci tensor and Ricci scalar. We first introduce the Einstein tensor.

Definition B.7. *The Einstein tensor G is a $(0,2)$-tensor field on $\mathcal{M}$ defined as follows:*

$$G = \mathrm{Ric} - \frac{1}{2} R g.$$

Corollary B.11. *The Ricci and Einstein tensors are symmetric and, with respect to an arbitrary coordinate chart, G satisfies the* contracted Bianchi identity

$$\nabla^{\mu} G_{\mu\nu} = 0.$$

Proof. Symmetry of G follows immediately from symmetry of Ric. Recall,

$$R_{\mu\nu} = R^{\sigma}{}_{\mu\sigma\nu} = (g^{-1})^{\sigma\alpha} R_{\alpha\mu\sigma\nu}.$$

By property (iv) of Proposition B.10, we have that

$$(g^{-1})^{\sigma\alpha} R_{\alpha\mu\sigma\nu} = (g^{-1})^{\sigma\alpha} R_{\sigma\nu\alpha\mu} = R_{\nu\mu}.$$

To derive the contracted Bianchi identity, we combine (i) and (iii) to infer that $R^{\sigma}{}_{\mu\nu\rho} = R^{\sigma}{}_{[\mu\nu]\rho}$ and

$$\nabla_{\alpha} R_{\sigma\mu\nu\rho} + \nabla_{\sigma} R_{\mu\alpha\nu\rho} + \nabla_{\mu} R_{\alpha\sigma\nu\rho} = 0.$$

Now we contract the above identity with $(g^{-1})^{\sigma\nu}(g^{-1})^{\mu\rho}$ to obtain:

$$\begin{aligned} 0 &= (g^{-1})^{\sigma\nu}(g^{-1})^{\mu\rho} \nabla_{\alpha} (R_{\sigma\mu\nu\rho} + \nabla_{\sigma} R_{\mu\alpha\nu\rho} + \nabla_{\mu} R_{\alpha\sigma\nu\rho}) \\ &= \nabla_{\alpha} R - \nabla^{\nu} R_{\alpha\nu} - \nabla^{\rho} R_{\alpha\rho} \\ &= \nabla_{\alpha} R - 2\nabla^{\nu} R_{\alpha\nu} \\ &= -2\nabla^{\nu} G_{\alpha\nu}. \end{aligned}$$

□